

Technics Panasonic Quasar®

Technical Guide

**Compact Disc
Player
Technology**

SL-P150 Series
SL-P405C Series

Audio Division
Matsushita Electric Industrial Co. Ltd.

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Today, our music sources consist of FM/AM tuners, recorded discs and tape. Recently, a new source has been added on PCM (pulse code modulation) technology. The new source that is attracting is

Digital Audio. This technical guide will cover the general principles of digital audio dealing with new compact disc format including objective circuit descriptions.

1. Analog VS Digital

1-1. Quality

Analog recordings have undergone remarkable improvements to data, but they are still limited by distortion and in dynamic range. Digital recording, on the other hand, is a revolutionary technique which eliminates these limitations.

Digital disc typically have a flat frequency response throughout the entire audible frequency range of 20 through 20,000 Hz. Their wide dynamic range exceeds 90dB, as the digital signal can always be recovered and is identical to the original signal. Totally different from present analog turntable system, the digital discs never exhibit deterioration in sound reproduction quality no matter how many times it is played. Even more remarkably, they can include flexible, convenient automatic play and program search features that were never possible with conventional analog player designs.

1-2. Advantages of Optical System

a) Exceptional Durability

With the mechanical system as in the current analog player system, the stylus must be in physical with the disc. Therefore both the stylus and the disc will wear out after a certain period of time, causing serious deterioration of sound quality.

With the optical system, however, the lack of contact between the disc and the pick-up means that there is no sonic deterioration no matter how many times the disc is played.

b) Harmful Effect of Dust and Scratches Minimized

Mechanical and variable capacitance system are easily affected by dust and scratches, since signals are impressed directly in the disc surface. The compact disc, on the other hand, has a 1mm thick protective layer over the signal pits. More importantly, the spot size of the laser beam is only about 1mm in diameter at the surface of the disc, but is as small as 1mm at the signal layer. This means that a dust particle or a scratch on the disc surface is actually only a millionth in size on the signal surface.

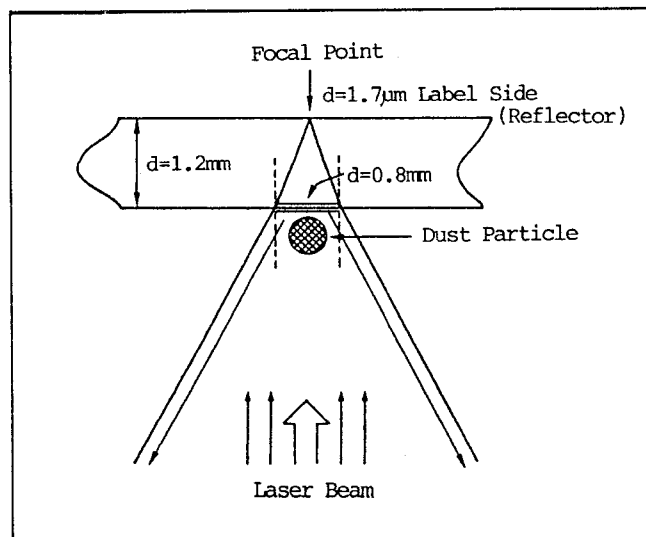


Figure 1-1

In addition, since the built in error correction circuit (MN6617 or MN6622) can cope with a burst error of 6,468 bits on the signal surface, most dust and scratches become insignificant.

2. Digital Systems

In recent years, various committees have convened to discuss the standardization of both hardware and software for the new digital audio systems. The following chart2-1 is outlines of the digital audio

systems that have been accepted and are being manufactured, and also the table2-1 is a compares some features of the two playback only systems like a CD (compact disc) and AHD (audio high density).

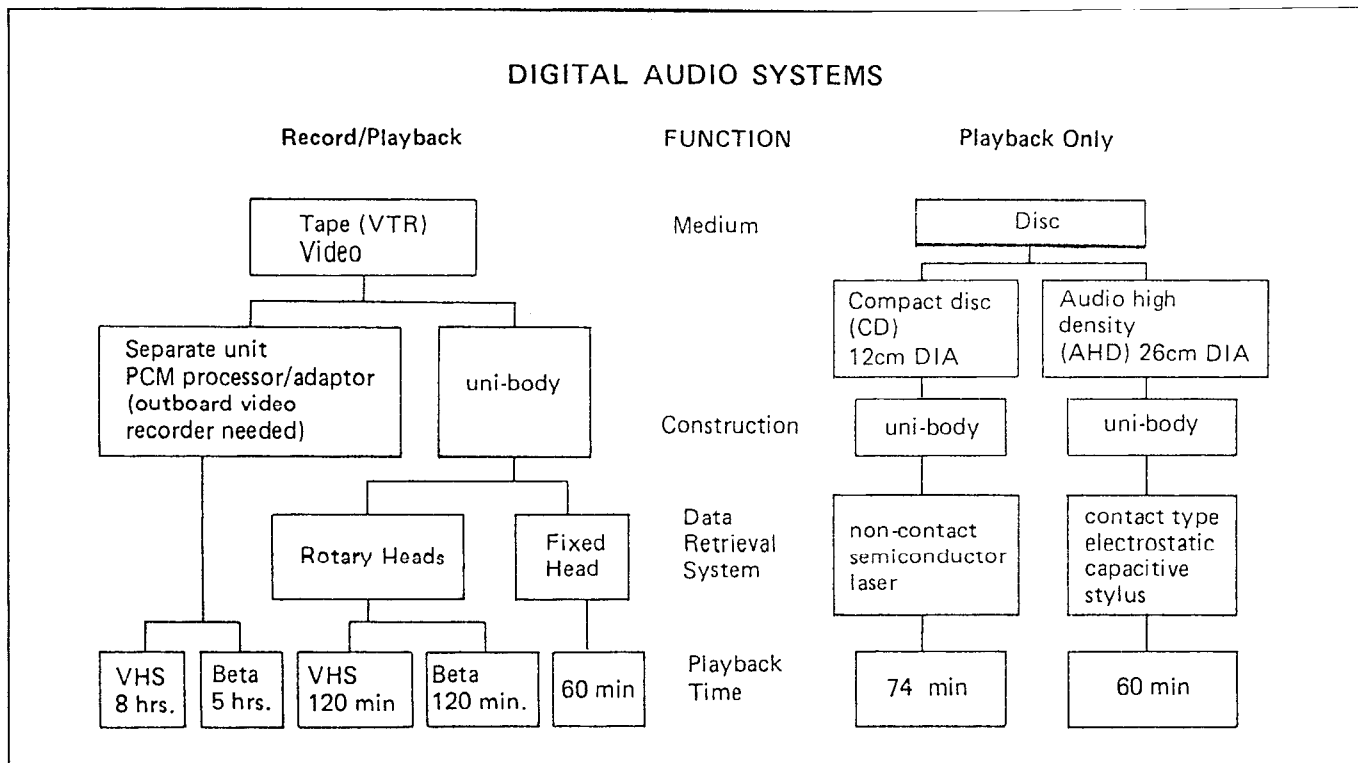


Chart 2-1 Comparison of Digital Audio System

Name of system	Compact Disc (CD)	Audio high density (AHD)
Proposed by	Philips, Holland	Victor, Japan
Disc diameter	12 cm	26 cm
Channels	Stereo 2ch or 4ch	4ch (3ch & color still image)
Play time	74 min. (2ch)/side	60 min./side
Disc Speed	200~500 r.p.m. Constant Linear velocity	900 r.p.m. Constant angular velocity

Table 2-1 Comparison of CD and AHD System

3. Compact Disc Features

1) Superior Sound Features

To maintain master recorder quality, the information on the disc is recorded with 16 bits linear quantization.

To improve recording density (playback time) eight to fourteen bits modulation (EFM) is used. This allows 2 channel playback for max 74 minutes.

To realize stable data reproduction a special error correcting code called the Cross Interleave Reed-Solomon Code (CIRC) is used.

2) Virtually Indestructible

The non contact laser playback system allows repeated use with no degradation in quality.

The plastic coating over the signal information protects it against scratches, dirt and fingerprints.

3) Small Size

The small size allows for easy compact storage (12cm and 8cm CD).

4) Random Access

The control information recorded on the disc allows random access and character display such as names of tunes, artists etc.

4. PCM Features

Using a PCM system for audio reproduction has certain advantages.

- 1) Not affected by noise of the recording medium, The digital system only has detect two types of signals (0 and 1). This is much easier than trying to detect the infinite number of possible amplitudes and frequencies of an analog signal.
- 2) Eliminates effects of uneven rotation of the playback equipment due to a quartz controlled reading clock.

- 3) Able to correct errors, besides including the recorded signal, the digital data has error correction information. This allows the digital circuit to calculate what the information should be if the music signal can not be read properly because the disc surface is marked by dirt or scratches etc.

5. PCM Process for CD

The PCM process has several fundamental steps regardless of whether the application is satellite data transmission or simply audio reproduction etc. A general description is given of the recording and playback procedures followed by an explanation of each steps.

6. Disc Mastering

6-1 Compact Disc Cutting

In a Compact Disc Cutting System, the Digital Audio signal is sent through a CIRC encoding modulator unit. In this way, EFM modulation is performed and control and sync bits are added. The modulated signal is then sent to a laser cutter to produce the metal master.

The cutting of the metal master is not really a burning of the layer by the laser. In figure 6-1 the real cutting is represented.

Step 1:

A glass plate is polished for optimum smoothness of the surface.

Step 2:

The photo-resist liquid is coated on the surface. Thoroughness of the glass surface and the thickness of the coating determine the depth of the pits on the compact disc.

Step 3:

The photo-resistive coating is then, exposed to a laser beam, modulated with digitized audio information.

Step 4:

When developing the exposed photo-resist layer the pits of information are revealed.

Step 5:

The surface is silvered to protect the pits.

Step 6:

The surface is plated with nickel to make a metal master.

Step 7:

The metal master is used to multiply mother plates. These mothers are in turn used to multiply metal master, or stampers.

Step 1. • A glass plate base is (cleaned and polished)

Step 2. • A photo resistive coating is applied.

Step 3. • The digital pulse train modulates the laser beam.

Step 4. • The coating is developed and a metal is deposit. The exposed part is removed.

Step 5. • A nickel master is made.

Step 6. • A mother is made from the master.

• A stamper is made from the mother.

Step 7. • The disc is made from the stamper.

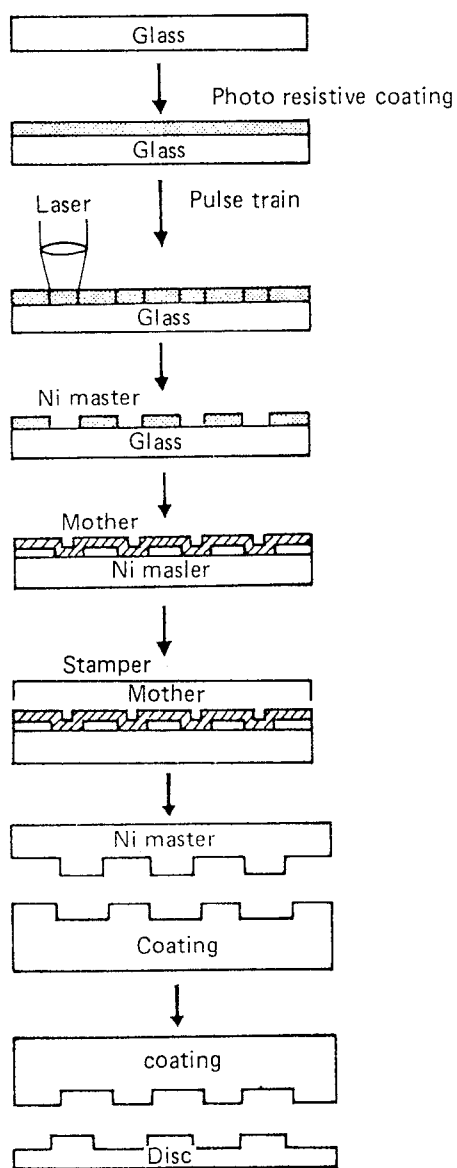


Chart 6-1 Disc Mastering Process

6-2 Compact Disc Stamping

The stamper is now ready to stamp compact discs. The stamping process can be injection molding, compression molding or polymerisation to make the impressions in the disc.

The signal surface is then coated with a reflective material (by vaporizing aluminium) to enable optical read-out, and further protected with an extra coating which will also support the disc table. Due to the dimensions of the pits and the track pitch, special care has to be taken in stamping compact discs in order to limit tolerances.

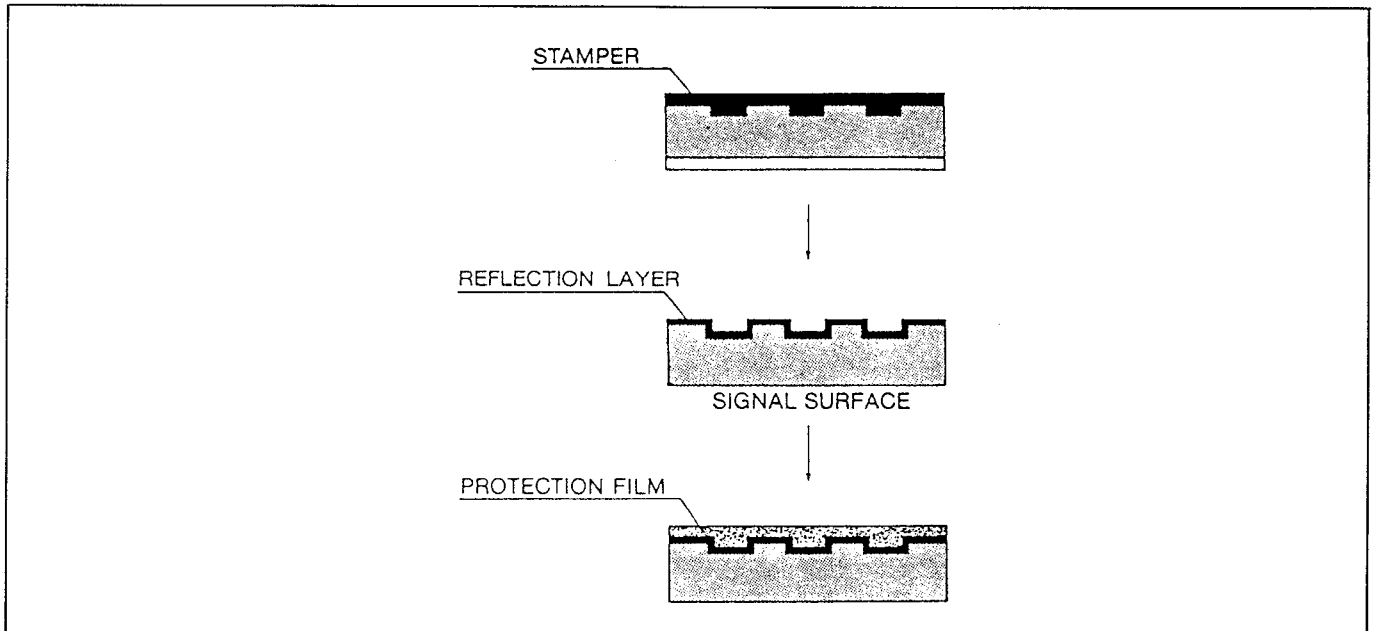


Chart 6-2 Disc Stamping Process

7. Disc Structure and Specification

The digital signal is recorded as pits on the glass plate surface, but as

viewed from the playback laser they are projections.

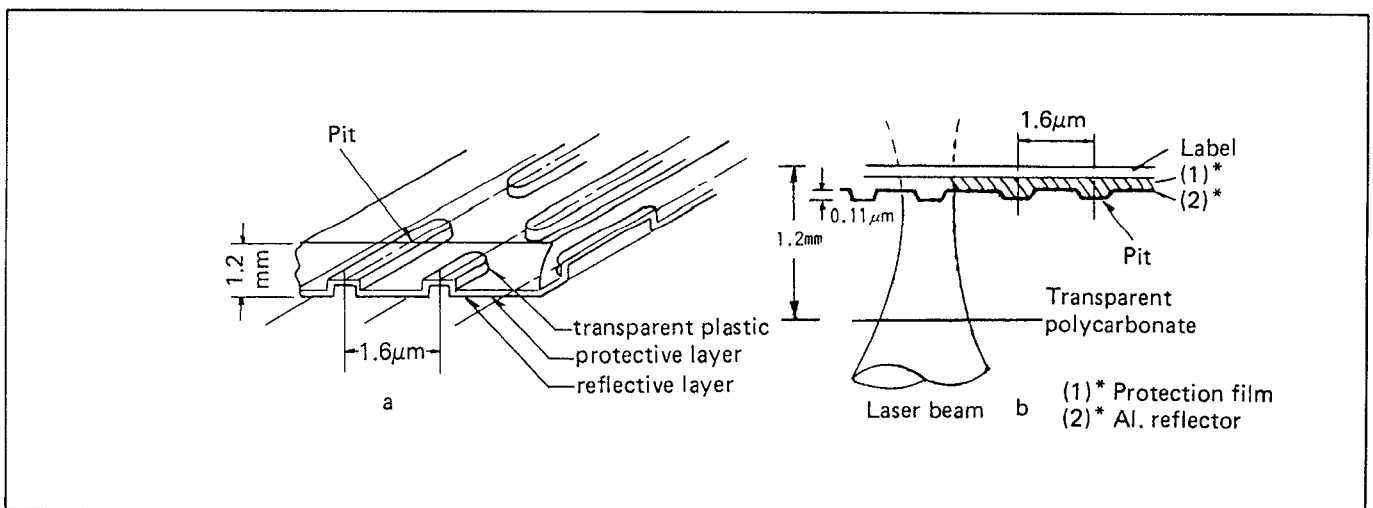


Chart 7-1 Disc Structure

The pits are $0.5\mu\text{m}$ wide, 0.9 through $3.3\mu\text{m}$ long and $0.11\mu\text{m}$ deep. The distance between each track is $1.6\mu\text{m}$.

An average disc may have as many as 20,000 tracks with about 6 through 7 billion pits.

And the compared to a 30cm LP disc, there are approximately 1,000 grooves with a average groove equal to $100\mu\text{m}$ in width.

The aluminum surface of the disc is protected by a polycarbonate layer with a refractive index of 1.5, This is used as part of the playback optical system to focus the beam on the pits.

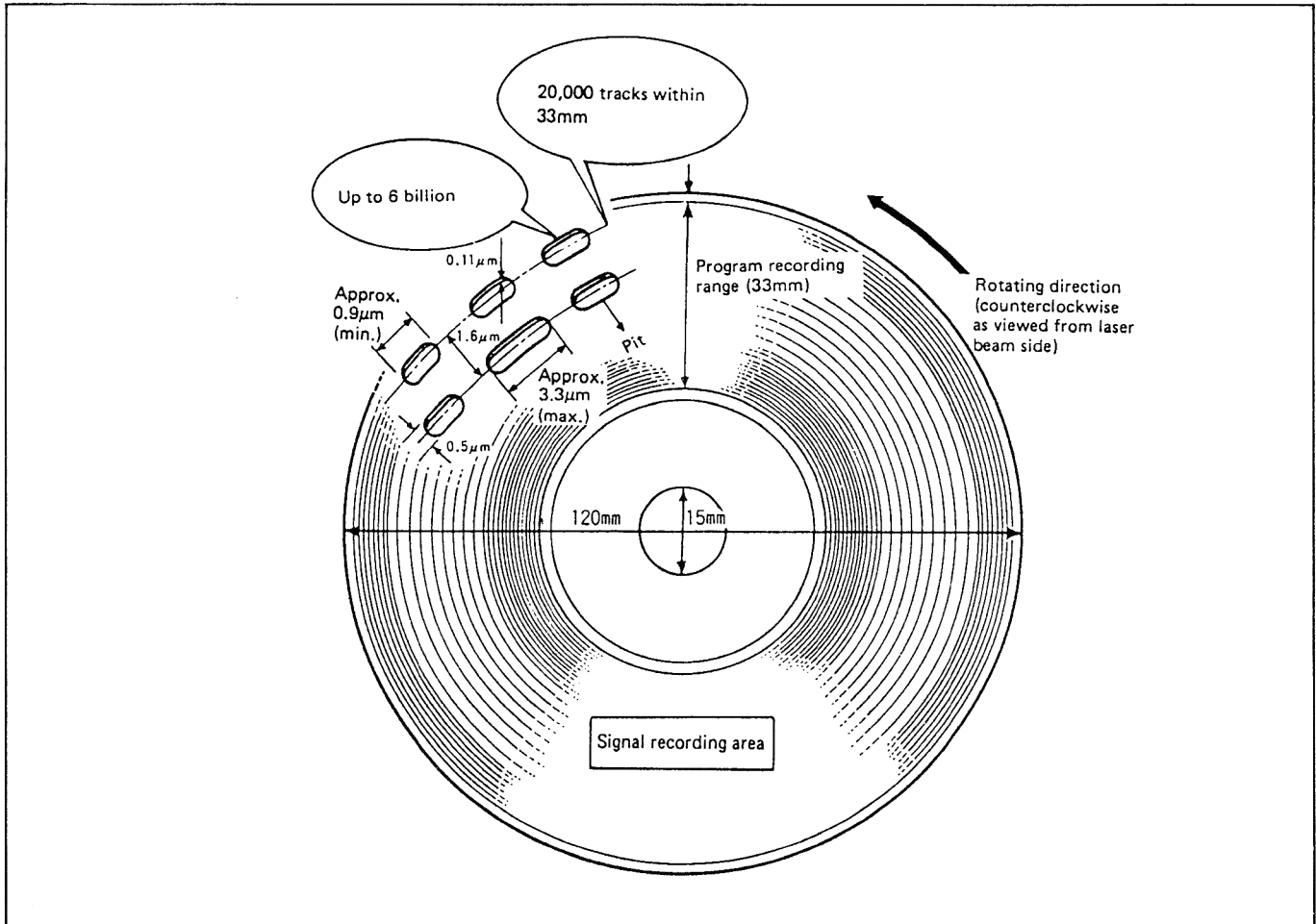


Chart 7-2 Compact Disc

7-1 Compact Disc Parameters

- Constant Linear Velocity:
1.2 m/s or 1.4 m/s
- Playing Time: 63 through 74 min.
- Track Spacing: $1.6\mu\text{m}$
- Diameter of the record: $d = 120\text{mm}$
- Lead-in diameter: $d = 47\text{mm}$
- Lead-in diameter: $d = 117\text{mm}$
- Recorded width: 35mm
- Disc Speed:

- lead-in track = 486 RPM at 1.2 m/s
568 RPM at 1.4 m/s
- lead-out track = 196 RPM at 1.2 m/s
228 RPM at 1.4 m/s
- Number of Track: $20625 = 35\text{mm}/1.6\mu\text{m}$
- Total Track Length: 5300m

Because of the constant linear velocity principle (constant scanning velocity) the disc speed has to be adjusted during

play. This is done by the CLV servo system.

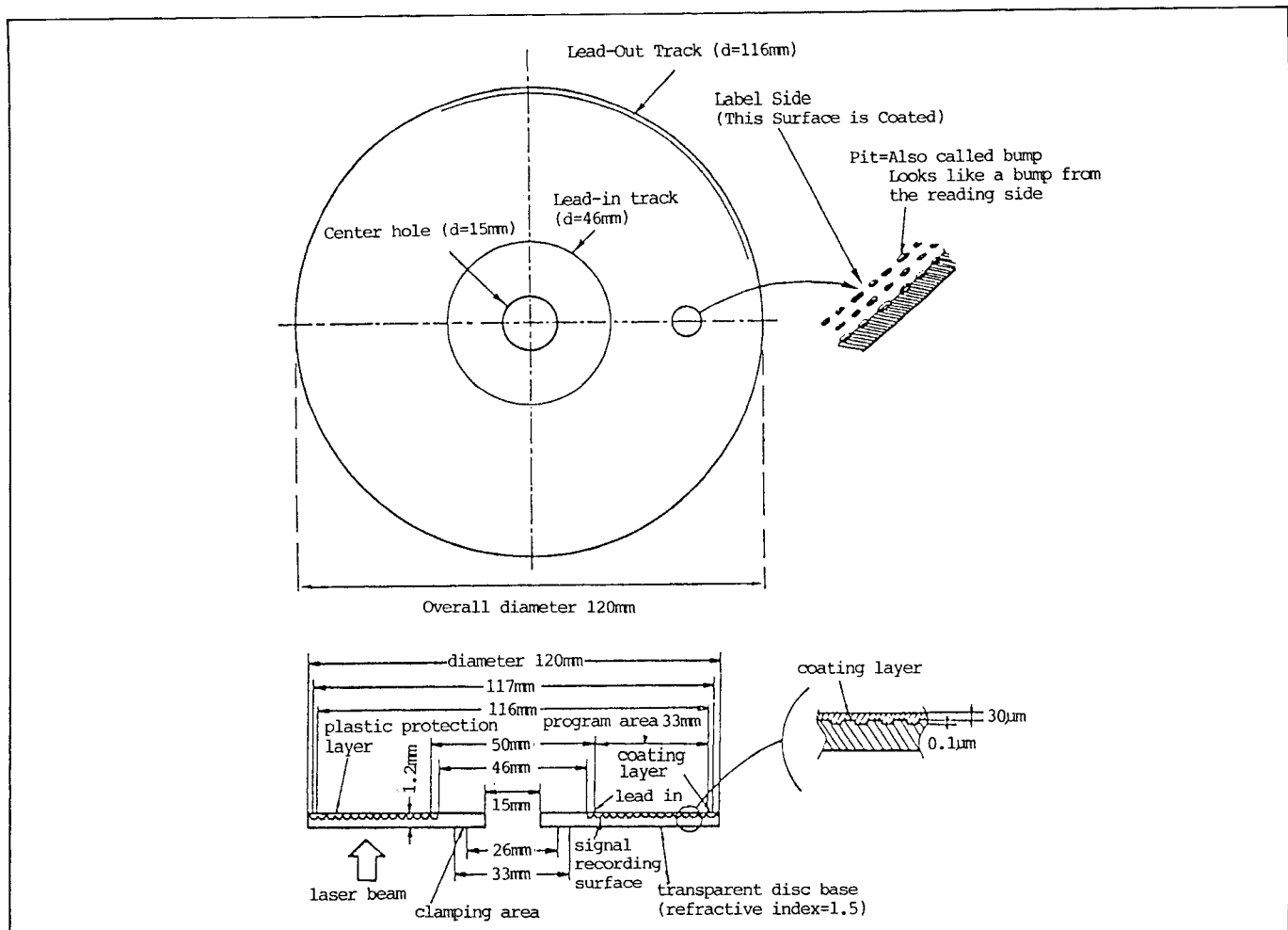


Chart 7-3 Compact Disc Specification

7-2 Audio Parameters

The information area on the CD is divided in three parts:

- The lead-in track;
- The audio program area;
- The lead-out track;

The audio program area consists of 16-bits wide words, encoded as 2's complement numbers. The lead-in and lead-out tracks consist of digital zero's.

The audio information can be recorded with a linear frequency characteristic (no emphasis), or a first-order pre-emphasis characteristic. In the latter case, the time constants are 50 microsec and 15 microsec.

The player detects via 4 control bits in the data stream whether or not emphasis is applied and switch the de-emphasis circuit accordingly.

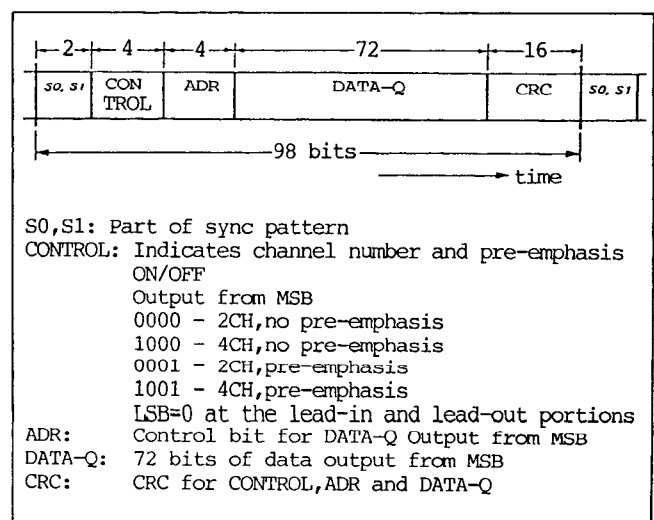


Table 7-1 Channel Q Structure

8. Advantages of the CD Player

Table 8-1 is a comparison chart of the LP and CD player system. It proves that the CD player is far superior to the LP player in every aspect of dynamic range, distortion, frequency response, and wow and flutter specifications. Especially, the CD player exhibits a remarkably wide 90dB of dynamic range throughout the entire audible frequency spectrum. In contrast, the dynamic range of the LP player is 70dB at its best.

In the low and high frequency range, this figure becomes even worse and only 40 - 50dB of dynamic range is reported. Harmonic distortion of the CD system is never worse than 0.01%, which is less than one hundredth of that of the LP player. In addition, wow and flutter are simply too minute to be measured. This is because, in playback, digital data

are once stored in a RAM and then released in perfect, uniform sequence determined by a reference clock of quartz precision.

The CD player also offers a number of convenience features. Though manual access to a piece of music on the disc is simply impossible since the track pitch is as fine as 1.6 μ m, the CD player enables total electronic control using the built-in CPU and the eight control bits inserted after the sync bits on the disc. The control bits are equivalent to approximately 2.7% of the total digital codes recorded on the disc. Presently, using two of these control bits, the information described below can be recorded for increased operating convenience.

CD System		Conventional LP Player
Specifications		
Frequency response	20 Hz - 20 kHz $\pm$ 0.5 dB	30 Hz - 20 kHz $\pm$ 3 dB
Dynamic Range	More than 90 dB	70 dB (at 1 kHz)
S/N	90 dB (with MSB)	60 dB
Harmonic distortion	Less than 0.01 %	1-2 %
Separation	More than 90 dB	25-30 dB
Wow & Flutter	Quartz precision	0.03 %
Dimensions		
Disc	12 cm (dia.)	30 cm (dia.)
Playing Time (on one side)	60 minutes (max. 74 min.)	20-25 minutes
Operation/Reliability		
Durability Disc	Semi-permanent	High-frequency response is degraded after being played several tens of times
Durability Stylus	Over 5,000 hours	500-600 hours
Operation	- Quick and easy access due to micro computer control - A variety of programmed play possible - Increased resistivity to external vibration	- Needs stylus pressure adjustment - Easily affected by external vibration
Maintenance	Dust, scratches, and fingerprints are made almost insignificant	Dust and scratches cause noise

Table 8-1 Comparison Between CD and LP System

8-1 Lead-in and Lead-out Signals

The lead-in signal is located just in front of where the signal area starts, while the lead-out signals appears right after the end of the signal area. These two signals are used to control the movement of the optical pick-up.

8-2 Table of Contents (TOC)

Written in the lead-in area is the time information on control codes including the start time of each selection as well as the total number and playing time of selections, which can be read out before performance starts for the convenience of program search and so on. Such information can also be displayed.

8-3 Control Codes

These codes distinguish between 2-channel recording and 4-channel recording. They can also detect whether preemphasis is given to a particular recording. Furthermore, they can even change the circuit connections of the player automatically according to the detected conditions of the disc.

8-4 Music Start Flag

The music start flag is provided in the blank space between selections. By counting this flag, therefore it is possible to locate any desired selection on the disc. Even when the end of a selection fades in to the beginning of the next, the flag can be inserted a minimum of two seconds in front of the second selection so that such selections can be recognized not as a single piece of music but as two separate selections.

8-5 Track Number and Index

Selections recorded on the disc can be numbered from 1 to 99. Each bar in a piece of music can also be addressed up to 99.

8-6 Time Code

With the time code function, you can tell the time lapse from the beginning of each selection in minutes, seconds, and 1/75 of a second. During the blank space between selections, the time is counted down.

As exemplified above, the control bits can be used in a variety of ways. They not only let you locate any portion of music exactly and quickly but also allow random playback in any order you set. And despite the long playing time of an hour on one side, access to the desired point can be achieved almost instantly. This is one of the greatest advantages of the CD system which cannot be found in any other system.

9. Analog Signal Versus Digital Signal

An analog signal is a continuous signal in two respects: value and time.

A digital signal is discrete, both in value and in time. Discrete in time means that the signal is defined on discrete intervals of time. Discrete in value means that the signal has finite number of possible values.

A digital signal is a sequence of words, each representing the value of a sample. The value of a word or digital signal, the binary code, is a combination of "1" (on) and "0" (off). The composing parts of the binary code are the bits.

Bit is an abbreviation for "binary digit". It is the smallest unit that can have a meaning in binary notation, and that can only have two values, "ON" or "OFF", represented as "1" or "0".

The higher the number of bits used in the binary code, the higher the resolution and the combination of bits, which means the higher the accuracy to describe the value of the sample.

The number of combinations can be calculated by the involution of 2, as each bit has only 2 conditions.

- 1 bit : 2 combinations
- 2 bits : 4 combinations
- 3 bits : 8 combinations
- 16 bits : 65,536 combinations

The larger the number of bits, the larger the content which can be expressed. An additional bit doubles the number of combinations and, in case of digital audio, offers 16dB extra of dynamic range. A 14 bits more than assures a dynamic range of 85dB, while a 16 bits system offers more than 90dB.

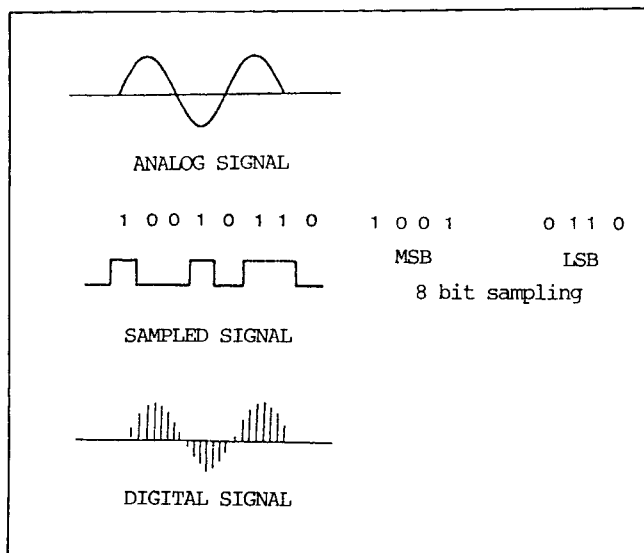


Figure 9-1

10. Analog to Digital Conversion

For a better understanding of the conversion process in a compact disc player, the conversion from analog to digital, which is performed during the recording process, is described briefly.

This conversion involves two distinct operations:

- 1). The sampling process, which converts a continuous-time signal into a discrete-time signal.
- 2). The quantization process, which converts a continuous-value signal into a discrete-value signal.

The next block diagram shows these operations:

- 1 : Analog signal = Continuous in time
Continuous in value
- 2 : Sampled signal = Discrete in time
Continuous in value
- 3 : Digital signal = Discrete in time
Discrete in value

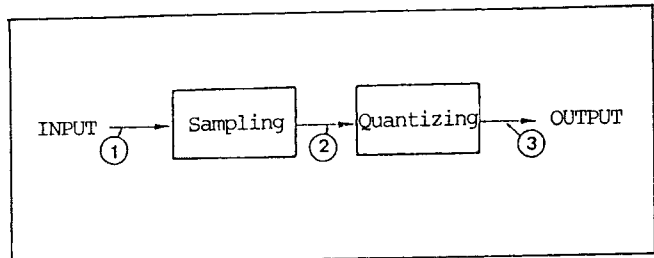


Figure 10-1 A/D Conversion

The sampling process is performed by a sample and hold circuit. The quantization process is done by an analog to digital converter. The digital output signal at 3 will then be coded to obtain a signal mutable for recording.

10-1 Sampling

By definition, an audio signal varies continuously in time. To enable it to be converted into a digital signal, it is necessary that the signal is first sampled, i.e. at certain points in time a sample must be taken of the input value which can be converted subsequently. The fixed time intervals between each sample are called sampling intervals (t_s).

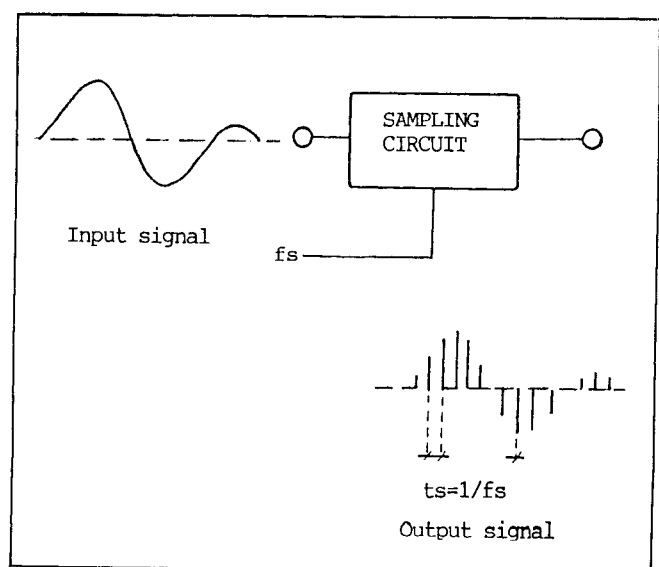


Figure 10-2 Principle of Sampling

Although the sampling operation may seem to introduce a rather drastic modification of the input signal:

- Since it ignores all the signal changes that occur between the sampling times.
- It can be shown that the sampling process itself, in principle, removes no information whatsoever, as long as the sampling frequency is at least twice the highest frequency that is present in the input signal.

This is the famous Nyquist criterion on sampling (also called Shannon criterion or Shannon theorem).

Since the highest frequency found in an audio signal is approximately 20KHz, and as selecting an unnecessary high sampling frequency would increase both complexity and cost of the conversion circuitry, a frequency of 44.1KHz has been chosen as the standard for the compact disc.

The correctness of the Nyquist criterion can be understood when we consider the spectra of the input and output signals, see figure 10-3.

If we consider an input signal $i(t)$ which has a maximum frequency $f_{\max}$, its spectrum may have any form between 0 Hz and $f_{\max}$ (1); the sampling signal $s(t)$, having a fixed frequency f_s , can be represented by one signal line at f_s (2). The sampling process is equivalent to multiplication of $i(t)$ and $s(t)$, and the spectrum of the output signal, as shown in figure 10-2, can be seen to contain the same spectrum as the input signal, together with the repetition of the same spectrum, modulated around multiples of the sampling frequency.

As a consequence, proper low-pass filtering can completely isolate and thus completely recover the input signal.

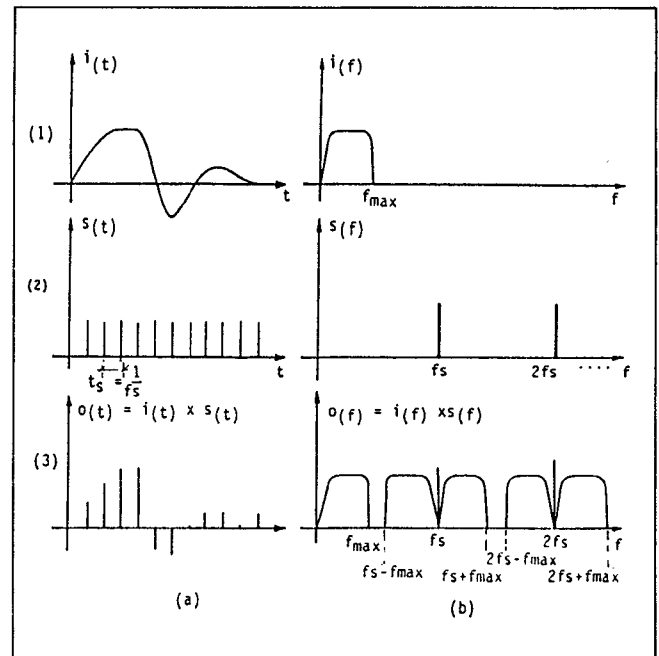


Figure 10-3 Time Domain (a) and Frequency Domain (b) in Representation of Sampling Process

From figure 10-3 it can also be understood easily that f must be greater than $2 \times f_{\max}$. If this would not be the case, the original spectrum would overlap with the modulated part of the spectrum, and consequently be inseparable from it, see figure 10-4.

For example, a 20KHz signal sampled at 35 KHz would produce a 5 KHz difference frequency. This phenomenon is known as aliasing and must be avoided by all means.

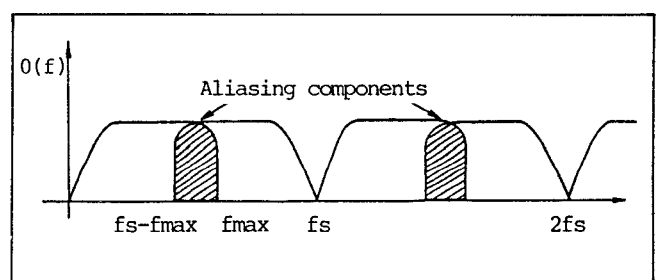


Figure 10-4 Principle of Aliasing

For this reason, a very sharp cut-off filter (anti-aliasing filter) is absolutely required in the signal path to remove all the unwanted harmonics out of the input signal before the sampling is performed. If this is not done, aliasing will cause unremovable distortion components.

10-2 Sample Hold Circuit

In the practice of analog to digital conversion, the sampling operation is performed by so called sample hold circuit which, once the sample has been taken, store the sample analog voltage for the time during which it can be converted into a digital code by the A/D converter. The principle of the sample hold circuit is relatively simple.

A basic sample hold circuit is in fact a voltage memory device that stores a given voltage in a high-quality capacitor. To sample the input voltage, switch S closes momentarily; when S opens, capacitor C holds the voltage until S closes again and passes the next sample, see figure 10-5.

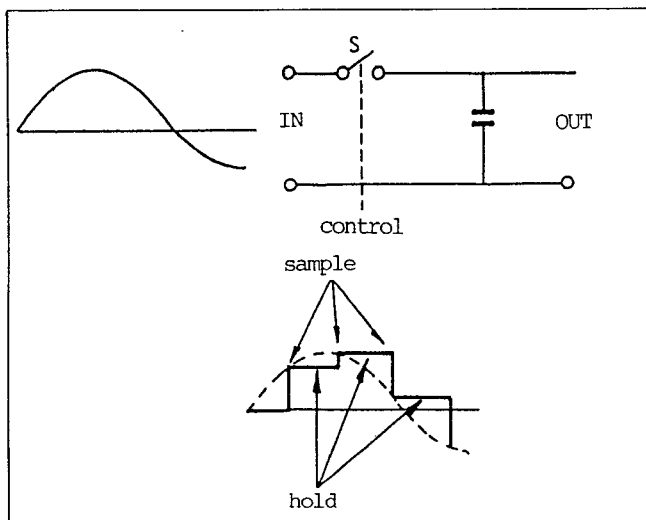


Figure 10-5 Basic Sample-Hold Principle

The sample hold circuit are not only used in A/D conversion to sample an analog input signal, but also in the reverse stage, after the D/A conversion, to remove transients from the output of the D/A converter and thus facilitate the task of the low pass filter. In this case, it is often called deglitcher.

10-3 Filtering

In the foregoing, we have seen that low pass filtering is required at two places in a digital audio system:

- 1) During recording, in the A/D stage before the input sampler, we need an anti-aliasing filter that drastically suppresses all input frequencies higher than the Nyquist frequency.
- 2) During playback, i.e. in the compact disc player, we need a reconstruction filter after the sample hold in the D/A stage to separate the sample audio signal from its images around the harmonics of the sampling frequency.

Consequently, both filters must be very linear within the passband, and have a very high rejection out of it; this will unavoidably bring along phase and group delay distortions within the audio passband as well as ripple and ringing. Apart from that, the noise and THD performance of these filters has to be better than the limits posed by the quantization process itself. In practice this means that 100dB S/N ratio have to be attained.

For the practical realization of such low pass filter, there are several approaches possible:

1) Passive Filter:

7th to 9th order filters must be used to obtain the required rejection. The coils which must be used give problems because of non-linearity, size and cost. Noise, however, is hardly a problem in this type of filters.

2) Active Filter:

Can also be used but the specifications are more stringent than usually encountered with active-filter designs; therefore, a very careful design is necessary. The distortion can be made better than with passive filters, but the noise worse.

3) Digital Filter:

Digital filter techniques allow to design almost ideal characteristics with relative ease. Their use for digital audio coding, however, implements an increased complexity of the related circuitry (initial gentle analog filtering with linear phase response followed by conversion and then digital filtering).

In practical CD players, active analog filters are used in most players; however, the latest generation deck players use digital filters combined with an oversampling D/A conversion circuit.

10-4 Principles of Quantization

1. Outline

Even after sampling an audio signal, we are still in the analog domain: indeed, the amplitude of each sample can vary infinitely between minus and plus a certain maximum value.

The decisive step to the digital domain is now taken by quantization, see figure 10-6. Replacing the theoretically infinite number of sample by a finite number of values.

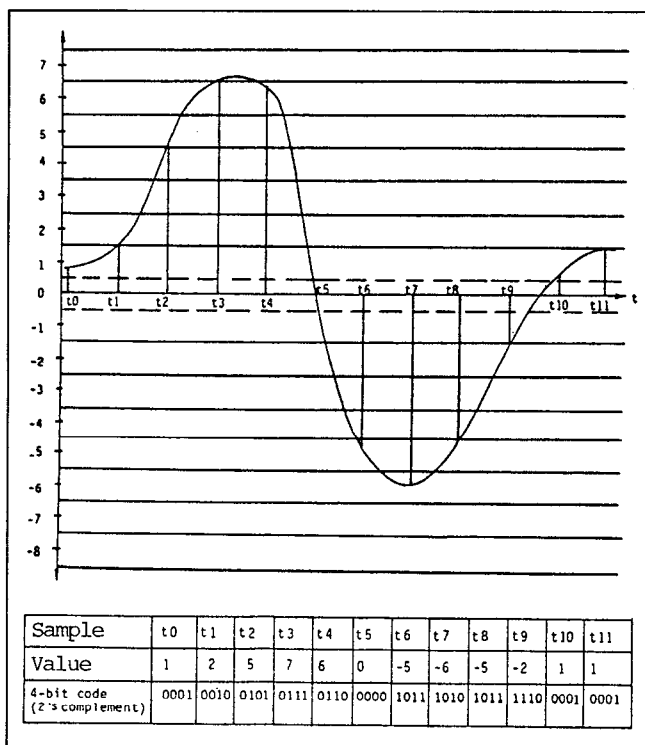


Figure 10-6 Principle of Quantization

In a practical system the analog signal range is divided (quantized) in a number of regions, and the samples of the signal are assigned a certain value according to the region in which they fall. These values are then converted into digital (binary) numbers; in our example, 4 bits are needed to indicate the 16 regions (16 combinations).

The example shows a bipolar system in which the input voltage can be either positive or negative (the normal case for audio). In this case, the coding system is often the so called 2's complement code (LSB= 2⁰, 1SB= 2¹, 2SB=2², etc), while negative numbers are indicated by simple complementing the positive codes (i.e. changing the state of all bits) and adding 1LSB.

In such a system, one bit (the MSB) is used as a sign bit: generally, it is zero for positive values, but one for negative values. More on the several coding system is given in further chapter.

The regions in which the signal range is divided are called quantization intervals, mostly represented by the letter Q. A digital word assigned to a certain quantization interval is assumed to represent the voltage at the center of this quantization interval. The figure 10-7 shows a typical step-wise quantizer characteristic.

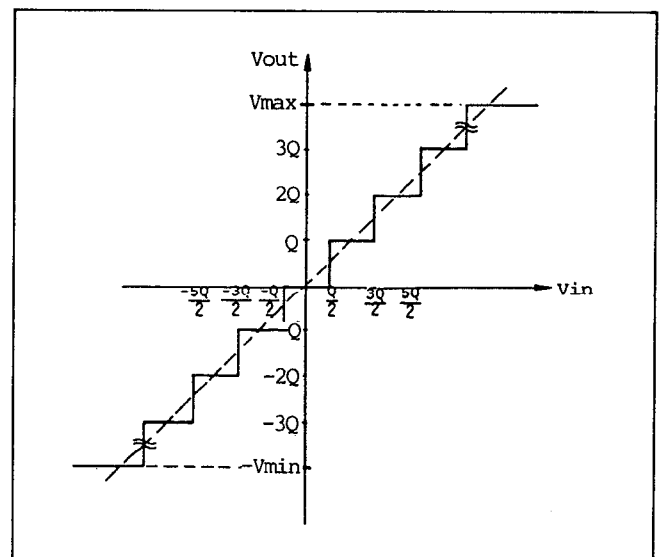


Figure 10-7 Quantizer Characteristic

The following table shows the output code for a 2's complement 6-bits quantization.

0111111111111111	= +32767
0111111111111110	= +32766
0111111111111101	= +32765
0111111111111100	= +32764
:	
:	
:	
0000000000000011	= +3
0000000000000010	= +2
0000000000000001	= +1
0000000000000000	= 0
1111111111111111	= -1
1111111111111110	= -2
1111111111111101	= -3
1111111111111100	= -4
:	
:	
:	
1000000000000011	= -32765
1000000000000010	= -32766
1000000000000001	= -32767
1000000000000000	= -32768

Table 10-1

2. Quantization Error or Noise

in fact, quantization can be regarded as a mechanism in which some information is thrown away, keeping only as much as is required to retain a certain accuracy as is needed in a certain application.

By definition, since all voltages in a certain quantization interval are represented by the voltage at the center of this interval, the process of quantization is a non-linear process and creates an error, called quantization error.

This error can be perceived as a kind of noise, and therefore the quantization error is also called quantization noise. It is clear that, the smaller the quantization intervals, the smaller this quantization noise will be.

With a 16 bits system, as used in Compact Disc, a theoretic 98dB S/N ratio possible as following calculation:

$$S/N \text{ (dB)} = 6.02 \times n + 1.76$$

* n = number of bits

11. Architecture of D/A Converter

1. Introduction

Obviously, one of the most important components in a D/A conversion system is the converter itself. From the foregoing it has become clear that very high resolution is needed to obtain a satisfactory signal to noise ratio.

Compact Disc system use a 16 bits resolution (compared to e.g. 8 bits for digital video applications).

2. D/A Converters

The principle of the best known, so called weighted current, D/A converter is shown in figure 11-1.

Each bit of the input word controls a switch in the circuit that turns on or off a specified current. The problem in this apparently simple circuit is the number of precision resistors needed : an n-bit converter requires n precision resistors over a 1 : 2ⁿ⁻¹ value range.

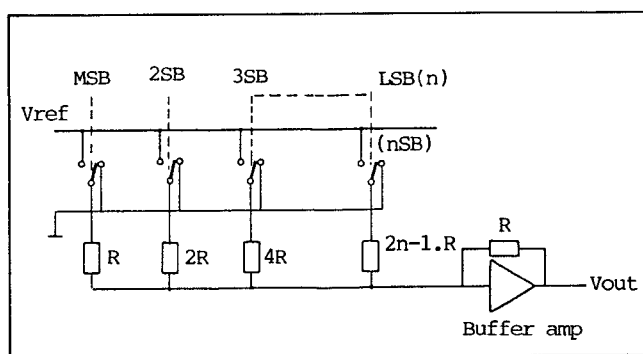


Figure 11-1 Weighted Current D/A Converter

Also the required accuracy in the resistors for the highest bits is extremely high as the accuracy of its current switched by the LSB (which, in a 16 bits converter, is 1/32,768 of the current switched by the MSB; this means an accuracy better than 0.003%). Also, this accuracy must be maintained over a long periode of time and temperature range.

Therefore, in practice, other types of networks are used for high resolution converters, called ladder networks by which some of these tremendous problems can be simplified. As example of a D/A converter using a ladder network is shown in figure 11-2.

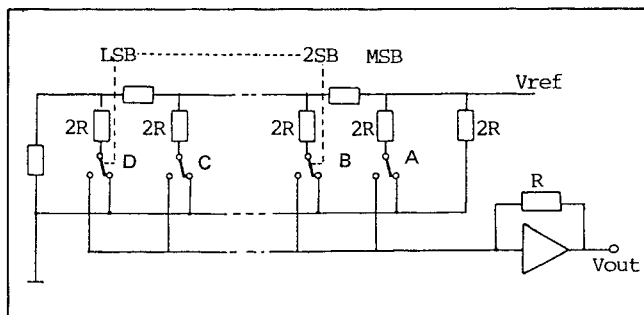


Figure 11-2 Ladder Network D/A Converter

The following is an example of D/A conversion using a 4 bits word with ladder network. When the digital pulse train is converted to a digital word, switch A (MSB) through D (LSB) correspond to each bit. Incase of 1 is represented by a closed switch and 0 by an open switch. The sum of the currents of the closed switches flows through R/L and corresponding voltage is developed at the output. These voltages are used to construct the pulse amplitude modulation (PAM) signal.

The ladder network is used two types of resistors (R and 2R), if $I=24\text{mA}$ the following conditions exist.

Digital Quantity				Current Flowing to each switch				Analogue Quantity
A	B	C	D	A	B	C	D	Total
1	1	1	1	8	4	2	1	= 15mA
1	1	1	0	8	4	2	0	= 14mA
:	:	:	:	:	:	:	:	
1	0	1	0	8	0	2	0	= 10mA
1	0	0	1	8	0	0	1	= 9mA
:	:	:	:	:	:	:	:	
0	0	0	1	0	0	0	1	= 1mA
0	0	0	0	0	0	0	0	= 0mA

Note

A is ON, then $I_L = 1/3 \times I = 8\text{mA}$

B is ON, then $I_L = 1/3 \times I \times 1/2 = 4\text{mA}$

C is ON, then $I_L = 1/3 \times I \times 1/4 = 2\text{mA}$

D is ON, then $I_L = 1/3 \times I \times 1/8 = 1\text{mA}$

Table 11-1

12. Cannel Modulation for CD Recording

1. Basic Requirements

After the sampling and quantization circuitry in the A/D converter a 16 bits sample signal is obtained.

These signal are in the NRZ format (NON-Return to Zero); they may have a high DC content and are not self-clocking, as the run length (distance between transients in the signal) is not limited. Therefore, they are not suited for direct recording and have to be converted into another code which should meet special requirements:

A. Correct Reabout at High Information Densities

The modulation transfer function or frequency characteristic of the CD-reabout system can be modelled as a low-pass filter with linear phase.

The cut-off frequency (ideal optics) is simply given by $2NA/\lambda \times V$ (where λ : wavelength, NA : numerical aperture of the objective lense, V : liniar scanning velocity).

A typical frequency characteristic is :

$$f_c = 2 \times NA / \lambda \times V = 1.4423\text{MHz}$$

Note

$$V = 1.25 \text{ m/s}$$

$$NA = 0.45$$

$$\lambda = 0.78\text{m}$$

Non ideal optics, due to defocusing or disc skewing with respect to the optical axis, will degrade the phase and amplitude charactristic. A nodulation system should satisfy the requirement of low sensitivity to tolerances of the optical light path even at high information densities.

B. Clock

The bit clock is one of the most important signals in the CD system. It is used for the synchronization of the digital data and the motor control. The bit clock must be regenerated from the reabout signal itself by detecting the pit edges. The signal must, therefore have a sufficient number of transitions and T max (the maximum distance between transitions pit/ land or land/pit) must be as small as possible.

C. Low Frequency

As the signal of the Compact Disc is covered by a 1.2 mm thick transparent plastic layer, it is protected against dirt change the envelope of the reabout signal., giving rise to low frequency noise. These low frequency disturbances can be filtered out provided that the signal contains no low frequency components of its own. Another reason for imposing the low frequency constraint is the use of servo system in the Compact Disc.

The low frequency components cause interference in the servo system making them unstable.

D. Error Propagation

The error propagation of the modulation system must be as small as possible and should match the CIRC error correction

system that treats 8 data bits as a unit. The error propagation should consequently be limited to the eight consecutive data bits forming a symbol. In this way, error correction can work in an optimal way.

The above requirements have led to the adoption of the Eight to Fourteen Modulation (EFM) method for the CD. This code converts eight consecutive data bits into fourteen channel bits which are then sent to the cutting laser. In the CD player, the reversss takes place with the detected signal.

2. Explanation of EFM Modulation

Starting from the 16 bits code, the sample word is split in two 8 bits symbols. Each symbol consists of 8 data bits, and these 8 bits are converted into 14 channel bits.

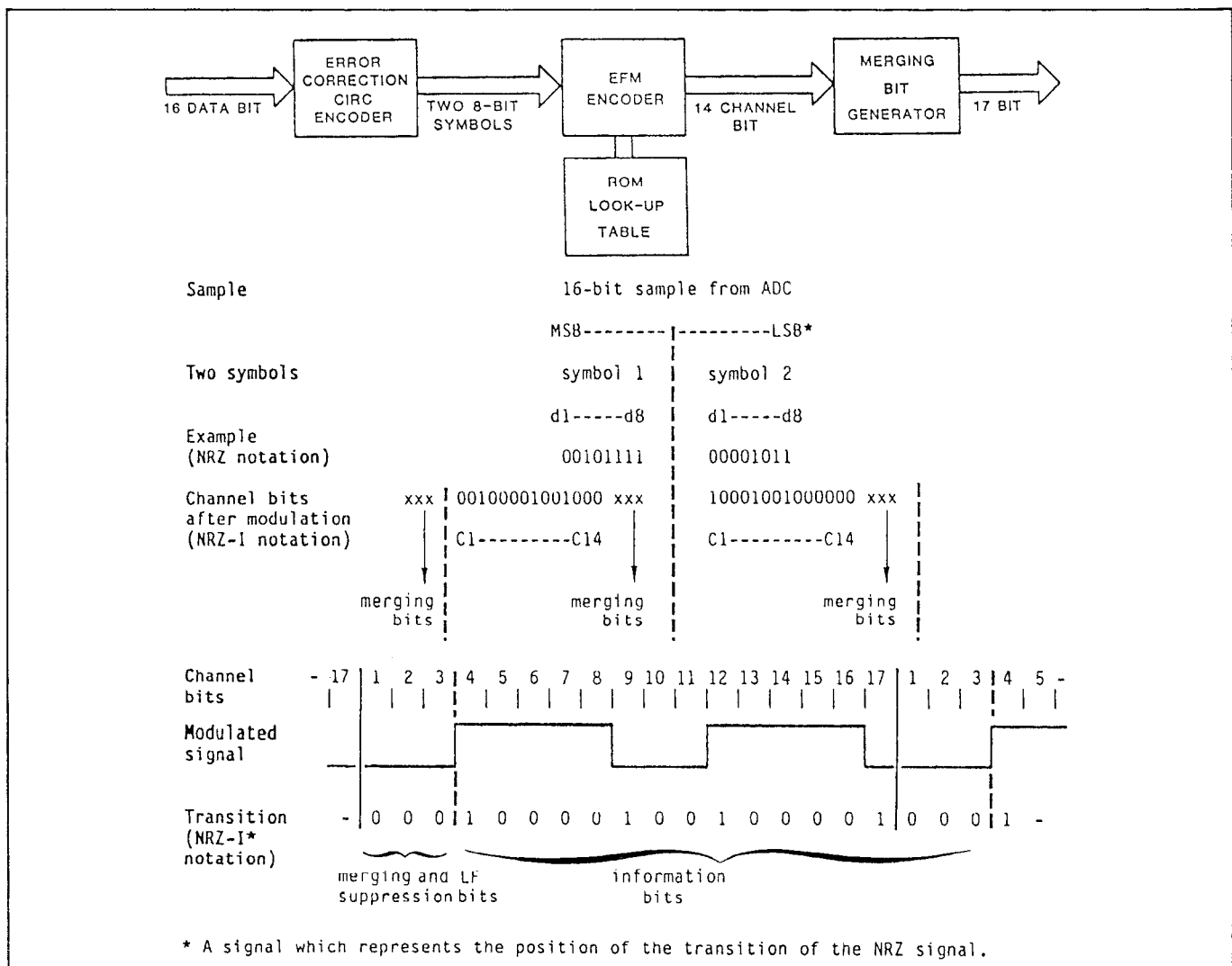


Figure 12-1 Block Schematic

The idea behind this technique is as follows:

An 8 bits code can represent 256 ($=2^8$) possible combinations, as shown in the following table 12-1.

MSB	7SB	6SB	5SB	4SB	3SB	2SB	1SB	No
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	1	1
0	0	0	0	0	0	1	0	2
:	:	:	:	:	:	:	:	
:	:	:	:	:	:	:	:	
1	1	1	1	1	1	1	0	254
1	1	1	1	1	1	1	1	255

Table 12-1

A 14 bits code, on the other hand, can represent 16,384 ($=2^{14}$) different combinations, as shown in the following table 12-2.

MSB	13SB	12SB	::	3SB	2SB	1SB	No
0	0	0	::	0	0	0	0
0	0	0	::	0	0	1	1
0	0	0	::	0	1	0	2
:	:	:	::	:	:	:	
:	:	:	::	:	:	:	
1	1	1		1	1	0	16382
1	1	1		1	1	1	16383

Table 12-2

Out of these 16,384 possible patterns, the patterns are selected in which more than two, but less than ten zeros appear continuously (there must be at least two "0" and no less than ten "0" between each "1").

That means, only 267 patterns satisfy this condition without 16,384 possible patterns.

The 11 more patterns out of these 267 are eliminated and the 256 remaining combinations are used.

A look-up table (in a ROM for instance) is used to assign the 256 combinations (of the 8 bits code) to the 156 combinations (of the 14 bits code).

Examples

8-bit word	14-bit word
00000011	00100100000000
01001110	01000001001000
10101010	100100010000100
11110010	00000010001001

Table 12-3

The advantages of converting from 8 bits to 14 bits are followings:

1. The number of bit inversions (1 to 0, 0 to 1) is reduced. This implies that the frequency range (bandwidth) is reduced.
2. The DC content is reduced, which is imperative for tracking and clock extraction.
3. The number of clock elements is increased (at least one clock element for one symbol of 8 bits is required).
4. The data processing can be done easier.

3. Junction Bits

To each block of 14 channel bits, 3 extra bits are added, 2 bits for junction the blocks and 1 redundancy bit for LF suppression.

As these three merging bits do not contain any data, one of them can be used to reduce the DC component in the signal. To select the merging bits, the DSV (Digital Sum Value) is used. The DSV is calculated as follows:

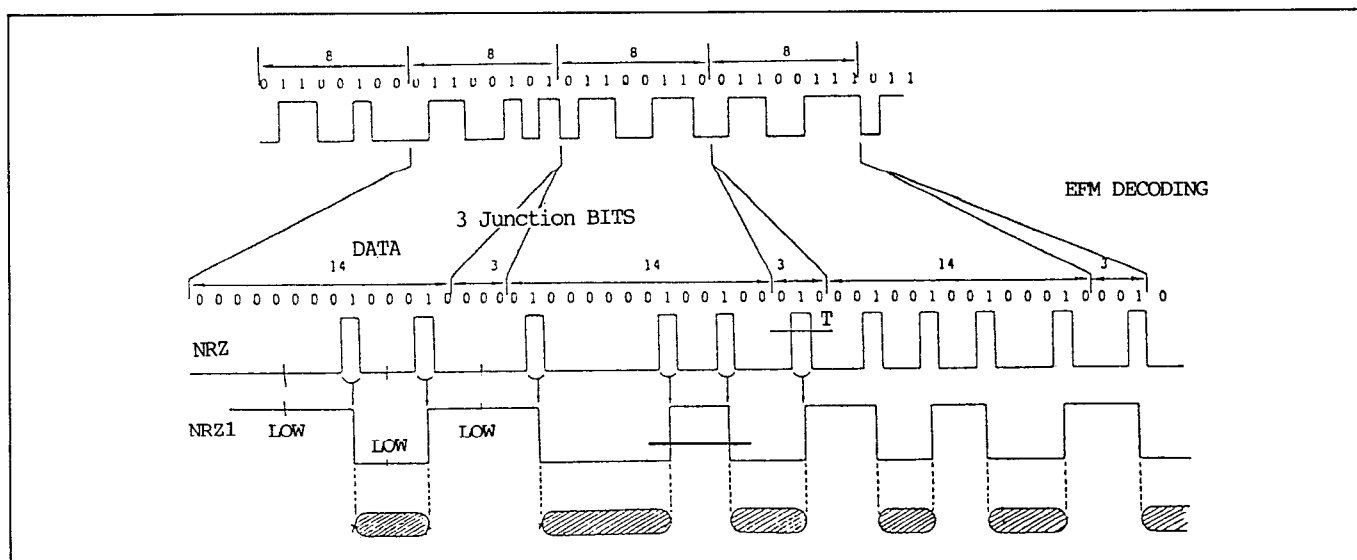


Table 12-4

In the 17 channel bits (three junction bits and 14 data bits) each channel bit counts +1 or -1, depending on the signal polarity.

By making the sum of +1 and -1, the DCV is obtained for one data block. The DCV should be as close to 0 as possible, as this indicates a DC free signal.

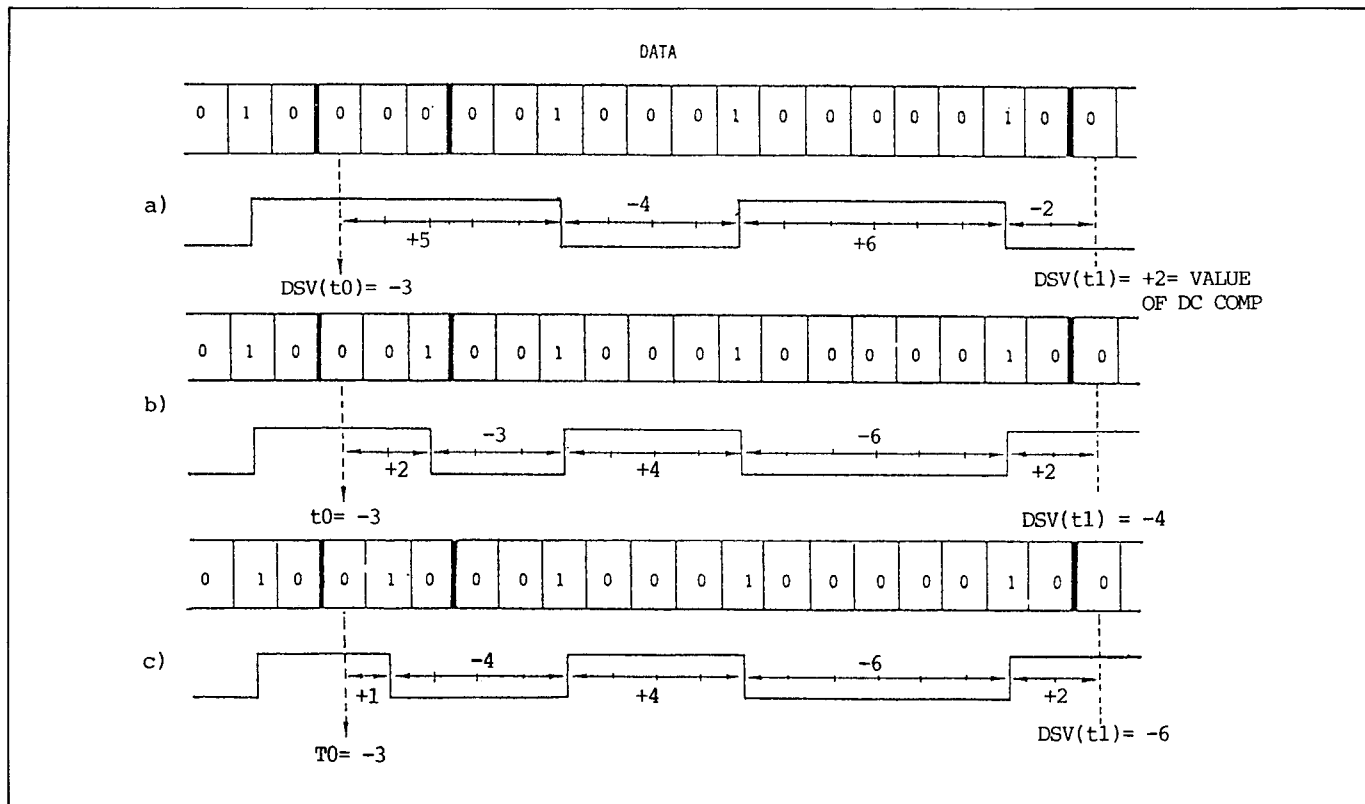


Figure 12-2 Selection of the Merging Bits

The figure 12-2 gives an example of a signal with different DC contents, depending on the selection of the junction bits. Of course, this selection requires at least 2 zeros (3 channel bits) and a maximum of 10 zeros (11 channel bits) between two transients.

The table 12-4 gives all valid possibilities for one channel block. As a combination of 3 bits gives 8 possibilities, the valid combination has to be selected, corresponding to the rules in EFM i.e at least two zeros between two ones, and maximum 10 zeros.

0 0 0 ***
0 0 1 ***
0 1 0 ***
0 1 1
1 0 0 ***
1 0 1
1 1 0
1 1 1

Table 12-5

Only the combinations indicated by an marked *** are valid in this example, as they allow two zeros between two ones. In the example it is assumed that the DSV of the previous block, DSV = -3.

If three zeros 000 are selected as junction bits (see figure 12-2 a), counting starts with 5 positive channel bits, at transient 4 negative channel bits. The DSV (t1) of the data block is $DSV(t1) = (-3) + (+5) + (-4) + (+6) + (-2) = +2$

In this case there is a positive DC level in the output signal.

By adding the junction bits 001, and starting with the same DSV (t0) = -3, the result become $(-3) + (+2) + (-3) + (+4) + (-6) + (+2) = -4$ (see figure 12-2 b). Thus, the result becomes negative, so the DC content in the output signal will be negative too : $DSV(t1) = -4$.

Incase of the figure 12-2 c, the junction bits combination 010 is added. The counting of the DSV will give : $(-3) + (+1) + (-4) + (+4) + (-6) + (+2) = -6$. The result become more negative which means a worser case, as a more negative DC component is present in the output signal. This example shows that the DC level of the output signal can be influenced by the junction bits. During the EFM modulation, the computer will add the three junction bits so, that an optimum result will be obtained. In the example no other combination is possible as 100 dose not allow two zeros between to transients, i.e. the last bits of the previous symbol and the actual junction bits.

4. Modulated Signal

The signal now obtained after "Eight to Fourteen" modulation is still not usable, as for 1 bit there are 2 transients. This signal, the most simple and basic from of modulation is called NRG (NON-Return to Zero) see figure 12-3. In order to fulfil the requirements of the EFM code, pratically the minimum of three bits (3T) and the maximum of 11 channel bits (11T), between two transients, the NEZ signal is converted into a NRZI modulated signal.

NRZI stands for "Non-Return to Zero Inverted". This signal has only one transition for a 1 bit plulse, which means the polarity of the signal changes for each 1 encountered. As a result, the polarity of the signal is not important, as only transients indicate the presence of a 1 in the signal. It is this signal which is reflected on the disc by the pits.

13. Error Correction Basics

1. Nature and Causes of Code Error

The error correction scheme is an essential part of the CD system. With analog techniques, the effort of error such as dropouts is that the sound is merely skipped, and the listener may not hear it, unless he is very attentive. However, with digital audio, the dropouts are heard in the form of irritating pulse noise.

Roughly categorized, there are two types of code error:

- The first type are random error, caused in disc production by inaccurate photo-resist coating, cutting, microscope bubbles, etc.
- The second type are the burst error resulting from scratches, dust and fingerprints on the disc surface; when these error occur, some bits or even bytes may be missing, this means dropouts are produced and this leads to critical reabout error in the digital data.

2. Error Detection

First of all, it is necessary to detect whether there is a code error. The error detection code is used for this purpose: extra bits are added to the data bits under certain rules, and then, by finding out whether this rule is being kept or not in playback, error can be detected.

a) Parity Check:

The simplest way for error detection, a single bit is added to a word during recording so that the sum of 1 in the word and its corresponding parity bit. Let us assume that there is an error in one of the bits during playback as shows figure 13-1 (a). Then, the number of "1"s in total become an odd number and this indicates that there is an error.

However, in the next example as shows figure 13-1 (b), there are two error but the total number of "1"s is an even number. As shown in the second example, the parity method cannot be used to detect multiple error which occur within a short period of time.

	Word	Parity	Totale of "1"s
(a)	1 1 0 1 0 0 1 0	0	even
	↓		
	1 0 0 1 0 0 1 0	0	odd (ertoneous)
(b)	1 1 0 1 0 0 1 0	0	even
	↓ ↓		
	1 0 0 1 0 1 1 0	0	even (but ertoneous)

Figure 13-1

By adding more parity bits (extended parity checking), misdetection probability can be further reduced (for n parity bits, it is $1/2^n$).

b) CRCC:

A CRCC code (Cyclic Redundancy Check Code) is a more complicated but more effective way of error detection than parity checking. It would lead us too far to explain its operation principles in detail; the principle is to multiply the data bits by another digital signal, and to transmit the resulting code. A subsequent division at the receiving end can then show us whether an error has occurred or not. Although this may seem rather complicated. Such an operation can easily be incorporated in an LSI such as used in CD decoding.

3. Error Correction and CIRC

An error correction code not only detects the code errors but also perfectly corrects them. This type of code is extremely complex, and so a simplified description will be given in figure 13-2. To correct code errors, it is essential to find out where and how the code errors have occurred.

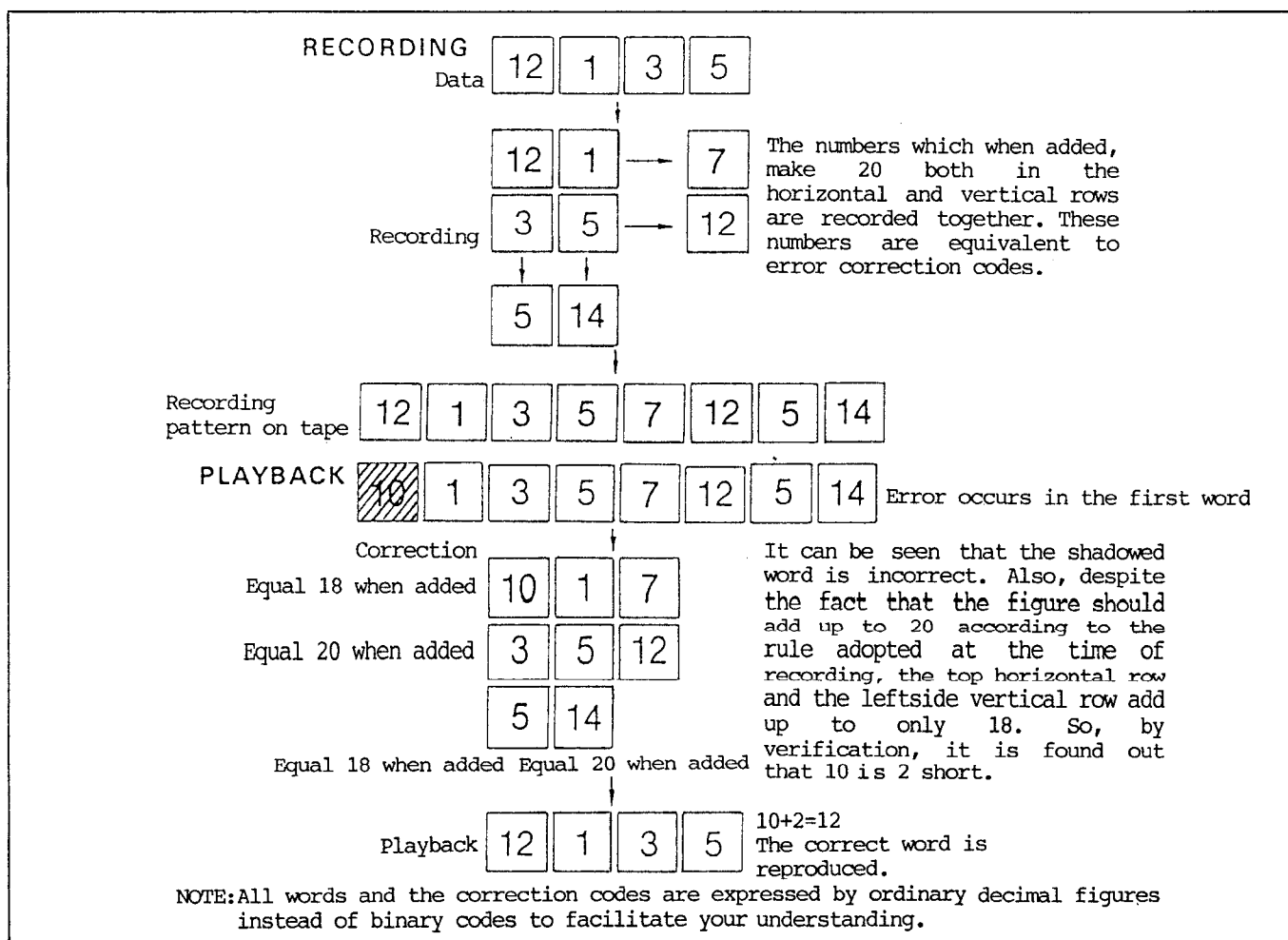


Figure 13-2

It can be seen that most error correction schemes (which are much more complex than the example) also entail a rather large redundancy. This, however, is unavoidable.

The error correction system used in the CD system is CIRC (Cross Interleave Reed Solomon Code). In specifying the performance of a coding system for disc applications both the burst error correcting and the random error correcting capability are important.

4. Error Correction CIRC

The CIRC is a code developed for efficient data error detection and correction on the CD.

Most of the error which occur on the CD are random, but during playback burst errors may be generated due to fingerprints, dust or scratches on the disc surface. To recover the data efficiently, the CIRC is used.

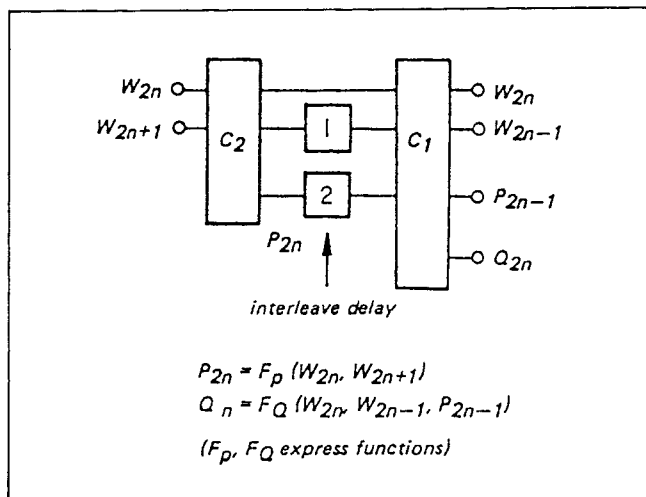


Figure 13-3 Cross Interleave Encoder

The actual operation content of CIRC has not been made public, so only the outline is summarized as follows; The CIRC is a type of cross interleave code,

as can be derived from its name. The Cross-interleave coding was developed by other manufacture and it has been used already in many digital audio products. The following figures are some examples.

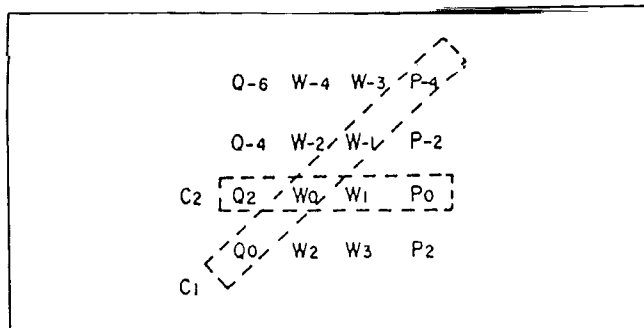


Figure 13-4 Cross Interleave Code Series

C2 and C1 are optional encoders, and on this example test words P2n and Q2n are generated by (3,2), (4,3) codes. By decoding C1, C2 alternately, error can be corrected successively during playback.

The CIRC generate 4 test symbols each for Pm, Qm by (32,28), (28,24) codes.

The error correction process by CIRC uses generating polynomial, multiplication and subtraction, the same as that introduced in the PCM operation description, but this has not been publicly released either, also compared to PCM, the processing is much more complicated.

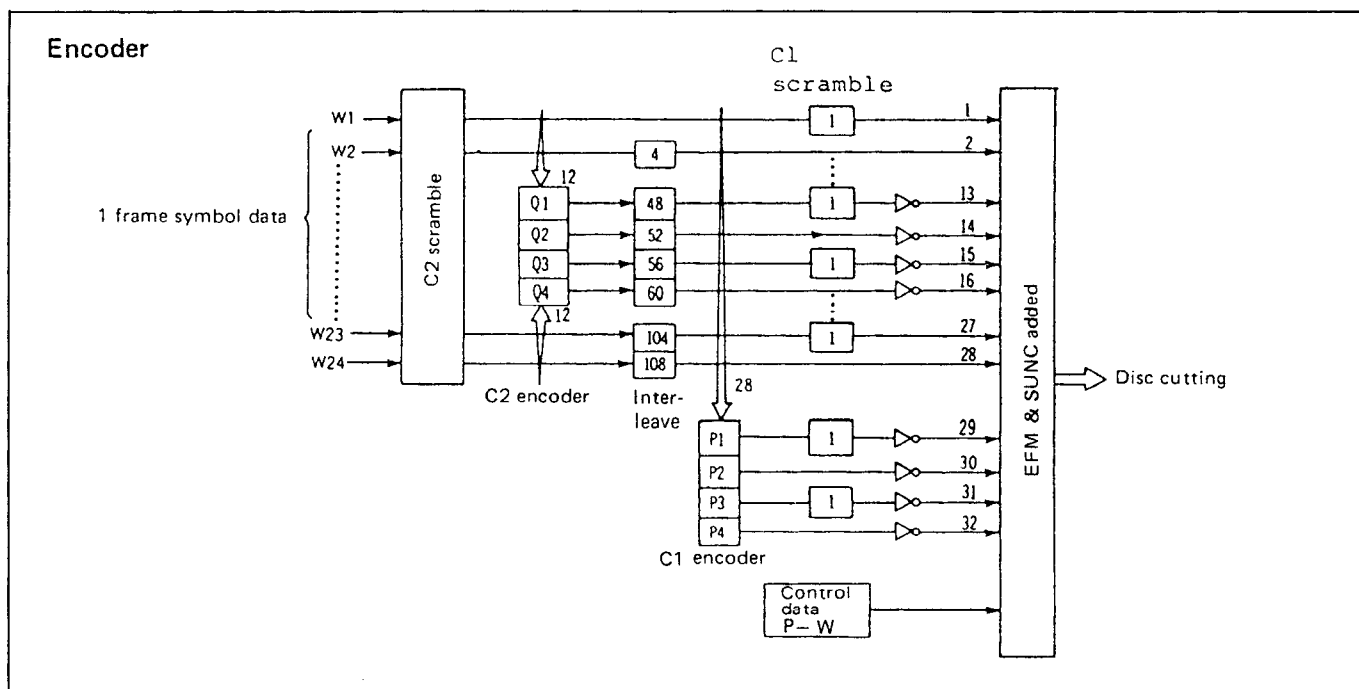


Figure 13-5 Signal Processing in Recording System

The CIRC principle is as follows;

- a) The audio signal is sampled at the A/D converter and these samples of 16 bits are split up into 2 words of 8 bits.
- b) 6 symbols of 16 bits, from both the left and the right channel, are applied to the CIRC encoder. This means 12 symbols of 16 bits, which means 24 words of 8 bits are stored into a RAM memory.
- c) The first operation in the CIRC encoder is "scrambling". This scrambling operation consists of a 1-symbol delay 16 bits insertion for the even words and a mixing up of the connections to the C2 encoder.
- d) The 24 scrambled 8 bits words are then applied to the C2 encoder which generates 4 Q words (control codes of 8 bits). The C2 encoder inserts these 4 Q words between the 24 incoming words, so that at the output of the C2 encoder 28 words are available.
- e) Between the C2 and the C1 decoder, 28 8 bits delay lines, with unequal length, are inserted. This generates

the "interleaving" of the data, which means that due to the processing delays, the sequence of the data words will be changed completely according to a determined pattern, so that the result will be a random pattern.

- f) The next step in the CIRC encoding is the C1 encoder. The C1 encoder generates 4 new control words of 8 bits from the 28 input 8 bits words, the P words issuing 32 8 bits words.
- g) After the C1 encoder another 1-word delay (8 bits) is inserted between the even words, and the control words, P and Q are inverted. The result, 32 x 8 bits words, is a CIRC encoder signal and is applied to the EFM modulator. At the playback side, the CIRC decoding circuit has to restore, from the demodulated signal leaving the EFM demodulator, the original 16 bits samples which are then applied to the D/A converter.

The random data must be lined up properly.

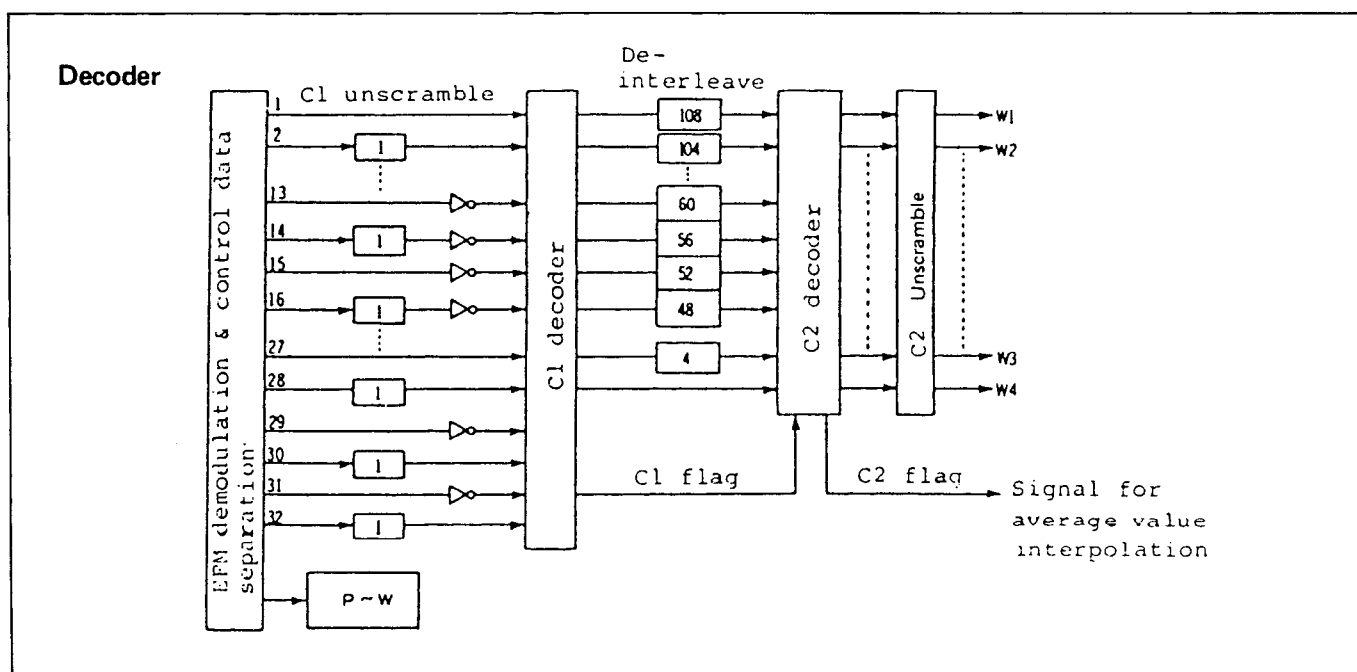


Figure 13-6 Playback Signal Processing

The DECODER operates as follows;

- a) From the EFM demodulator, 32 words of 8 bits are applied to the CIRC decoder.
- b) The first step delays the odd data words over 1 word (8 bits) and inverts the control words P and Q, so that the C1 decoder input corresponds to the C1 encoder output (at the recording side).

- c) The C1 decoder accepts 32 words of 8 bits each, 4 parity words of which are used for C1 decoding. The parity has been generated at the recording side, according to the rules of the Reed Solomon encoding, and because of these parity words, the C1 decoder is able to correct a word error in every block of 32 words. If there is more than one

erroneous word, the C1 decoder detects, regardless of the number of error, that it has received an uncorrectible block. In that case, it will let all 28 words pass through uncorrected, but an error flag is set for each symbol to mark that all those from C1 are unreliable at that moment.

Because the internal delays between the C1 and C2 decoders are of unequal length, the symbols marked with an error flag at a specific moment, enter the C2 decoder input at different moments.

Thus the C2 decoder has an indication for every word, whether it is an error or not. If a word does not carry an error flag it is error free. If no more than 4 words carry an error flag,

the C2 decoder can correct a maximum of 16 frames.

In cases where even the C2 decoder can not correct, it will let the 24 words pass through uncorrected, but marked an error flag originally set by the C1 decoder.

d) Next, the C2 output signals are unscrambled and the output words carrying the uneven words are given 2 words delay.

e) Finally, the CIRC decoder output sequence will correspond with the original input signal before CIRC encoding.

The following table lists the most relevant specification of the CIRC system;

ASPECT	SPECIFICATION
MAX. FULLY CORRECTIBLE BURST LENGTH (BURST ERROR CORRECTIBILITY)	UP TO 4000 BITS (2.5mm)
SAMPLE INTERPOLATION RATE (RANDOM ERROR CORRECTIBILITY)	1 SAMPLE/10 HRS AT BER = 10^{-4} 1000 SAMPLE/MIN. AT BER = 10^{-3} (WHICH IS 1 OUT OF 2.6 MILLION ON THE AVERAGE)
UNDETECTED ERRONEOUS SAMPLES (CLICKS)	LESS THAN 1/750 HRS AT BER = 10^{-3} NEGLECTIBLE AT BER = 10
CODE-RATE (EFFICIENCY)	3 DATA BITS WILL RESULT INTO 4 BITS AFTER ENCODING, ON THE AVERAGE
COMPATIBILITY WITH FUTURE 4 CHANNEL VERSIONS	NO CHANGES IN THE DECODER CHIP REQUIRED

Table 13-1

5. Error Concealment

When errors are so big that they can not be corrected, some type of "concealment" is necessary.

a) Muting:

The erroneous word is muted (set to zero). This is a rather rough method which is not employed in CD player.

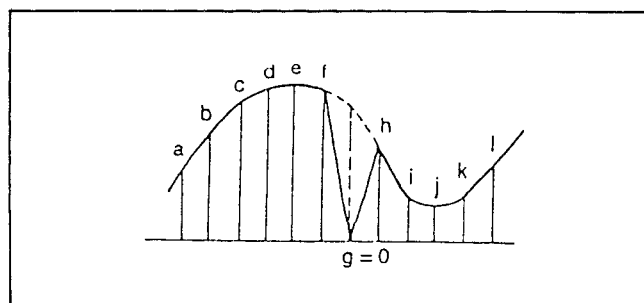


Figure 13-7

b) Previous word holding:

As its name implies, the value of the word before the erroneous word is held so that there is almost no audible difference. The previous word holding is a memory operation since the preceding value is repeated. This method is used in combination with the linear interpolation in case of two missing subsequent words.

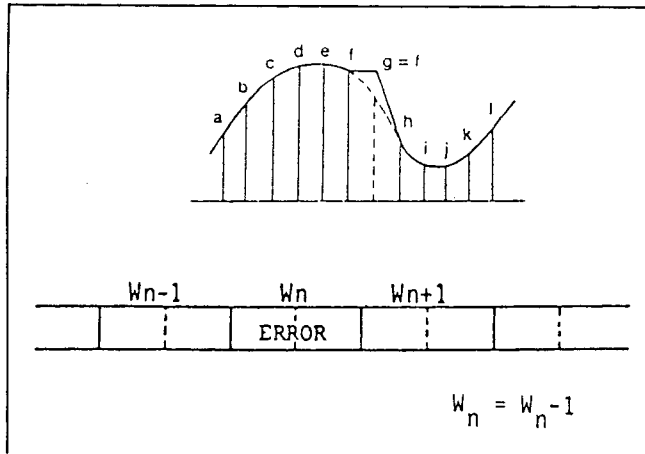


Figure 13-8

c) Linear interpolation:

This is the most accurate method of concealing errors. The word including errors is replaced by the average value of the preceding and succeeding words which are supposed to be correct. The compensation capability of this interpolation is of course superior to that of the previous word holding.

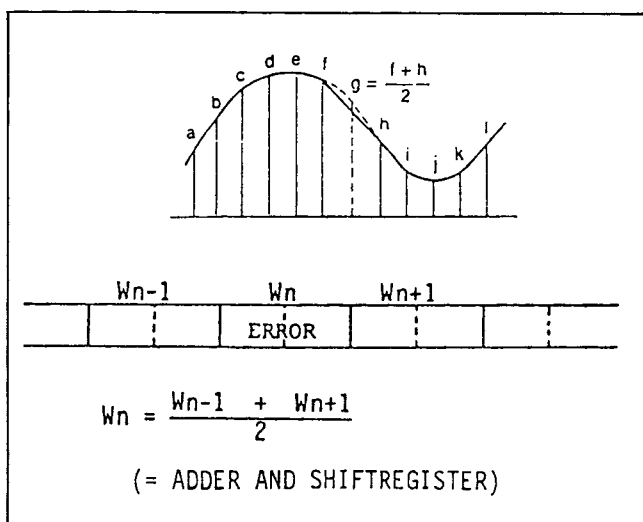


Figure 13-9

The linear interpolation requires an "adder" circuit and a shift register. The sum $W_{n-1} + W_{n+1}$ is made in an adder circuit. The result is stored in a shift register. By shifting the bits (word) 1 bit to the right, the value is divided by two as shown in figure 13-9. Linear interpolation is used in case one word is missing or erroneous, this means when correction is not possible in the C2 decoder.

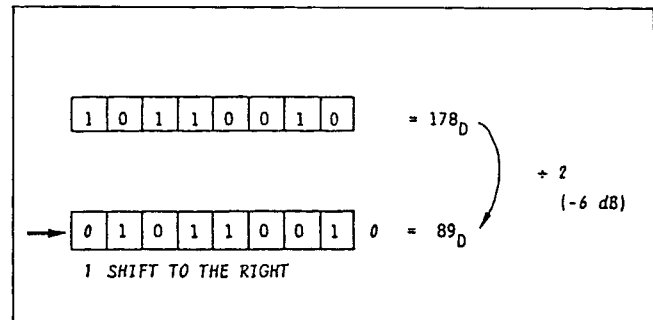


Figure 13-10

d) Previous word hold with interpolation:

In this method previous value is held and replaces the first erroneous word. This new value will be used to calculate the average value of the preceding and succeeding words.

This operation needs two circuits. The first step is a memory operation, and the next step requires an "adder and shift register". This method is used in case 4 successive symbols, which means 2 successive words can not be corrected.

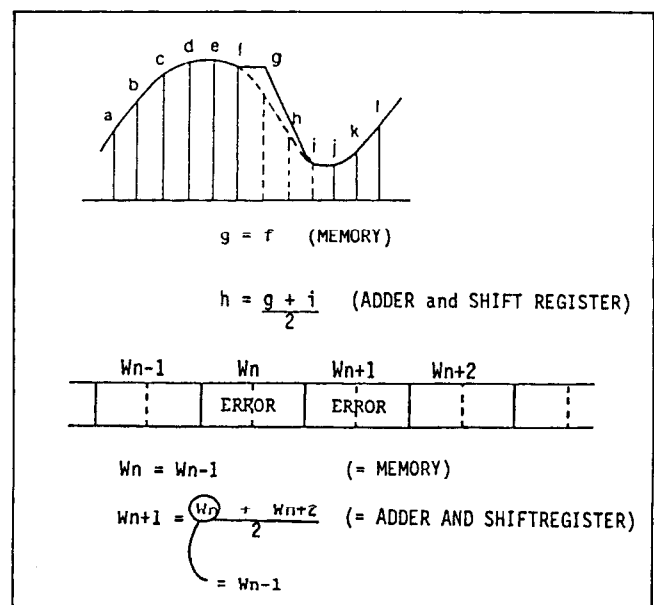


Figure 13-11

14. Description of the CD Subcode

1. What is the Subcode

The information on the CD is contained in the ca. 20000 tracks, whereas each track is subdivided in frames. The "framerate" equals 7350 Hz, which means that every second, 7350 of these frames are processed in the CD player. After EFM demodulation, a frame look as follows: (again, data are expressed as 8 bits words)

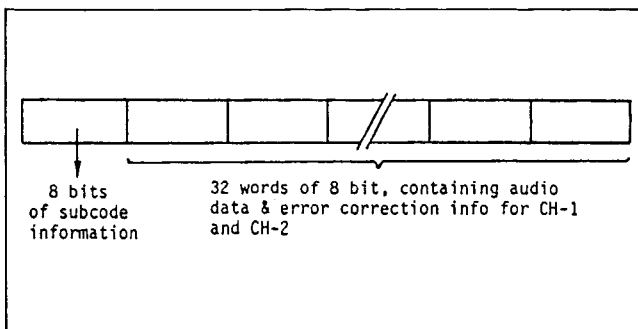


Figure 14-1

NOTE

A 60 minutes CD contains: $60 \times 60 \times 7350 = 26,460,000$ frames

The each frame holds 8 bits of subcode information, this means the subcode area of CD stores 211,680,000 data bits (ca. 200 Megabits).

The 8 bits of the subcode word are used for 8 different subcode channels (P, Q, R, S, T, U, V and W). Presently, only channel P and channel Q are defined. The content and the format of the channels R, S, T, V and W is still under discussion.

The 8 bits of the subcode word have no meaning at the frame level, Only when a block of 98 frames is read by the player, the 8 subcode channels (98 bits each) get a specific content. The content of the channels is refreshed on updated for each new block of 98 frames (or 75 times a second).

2. Description of the P and Q Channels

- Channel P is a simple music track separator flag, Simple search systems can be implemented using only this P flag.
- Channel Q is used for control purposs of more sophisiticated player, using an 8 bits microromputer. The channel Q stores data such as track number, time, and others.

3. The Frame Structure

One subcoding block consists of 98 subcoding symbols. The repetition frequency of one block is 75 Hz. The first two subcoding symbols are the subcoding sync patterns S0 and S1.

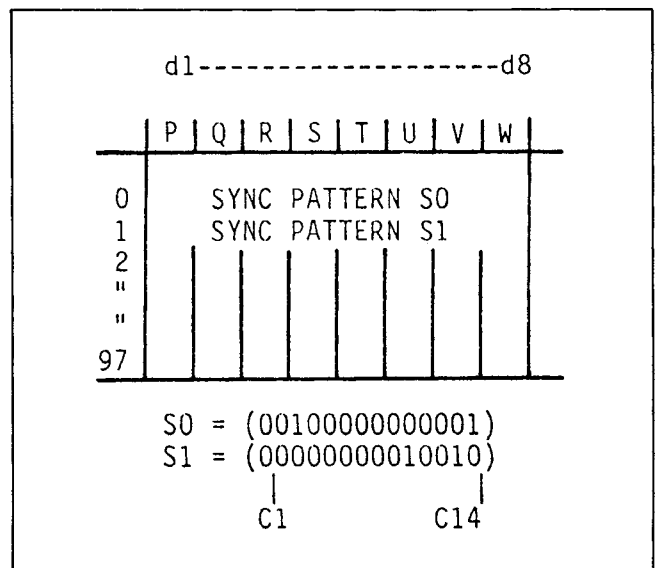


Figure 14-2

4. Channel P

The channel P is a flag bit that indicates the start of a track with the following code rules;

Music : P = 0

Start flag : P = 1

The minimum length of the encoded start flag in channel P is 2 seconds, the end of the encoded start flag gives the start of the next track. If the actual pause exceeds 2 seconds, the length of the start flag gives the actual pause length.

a) Lead-in track:

In the lead-in track channel P is encoded as music. The first piece of music is preceded by a start flag of 2-3 seconds.

b) Lead-out track:

The lead-out track is preceded by a start flag of 2-3 seconds (during the last music track on the disc). The end of the start flag gives the begin of the lead-out track. The channel zero for 2-3 seconds after the start of the lead-out track, next P switches between 0 and 1 in a $2\text{Hz} \pm 2\%$ rhythm (duty cycle $50 \pm 10\%$).

The channel P can only change after the subcoding sync pattern S0 and S1. The encoding of channel P is one subcoding block delayed to the encoding of channel Q.

5. Channel Q

The general data format of channel Q is as follows;

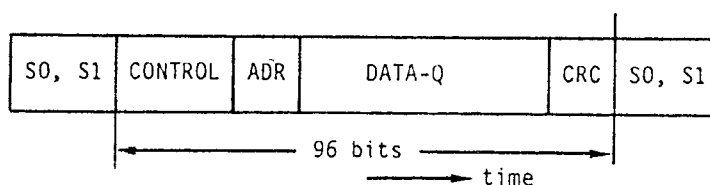


Figure 14-3

a) Control:

4 flag bits for the number of channels and preemphasis on/off, MSB is first out.

0000 - 2 audio channels is without preemphasis.

1000 - 4 audio channels is without preemphasis.

0001 - 2 audio channels with preemphasis of 50/51 μs .

1001 - 4 audio channels with preemphasis of 50/51 μs .

The preemphasis flag (bit 4 of the CONTROL block) can only change during an actual pause of at least 2 seconds.

b) ADR:

4 control bits for DATA-Q, MSB is first out.

c) DATA-Q:

72 bits data, always MSB first out.

d) CRC:

A 16 bits CRC on CONTROL, ADR and DATA-Q, MSB first out. On the disc the parity bits are inverted. The remainder have to be checked at zero.

e) The following three modes are defined for DATA-Q:

(1) Mode-1

ADR = 1 = (0001)

Mode-1 occupies at least 9 out of 10 successive subcoding blocks.

Two different data formats are possible in mode-1, and during the lead-in track, the data format as follows;

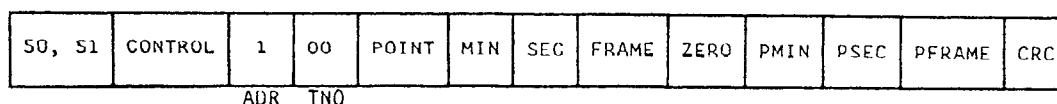


Figure 14-4

During the music and lead-out tracks on a disc, the data format as

follows;

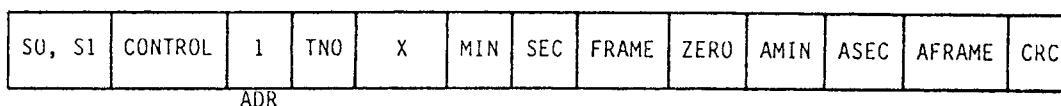


Figure 14-5

TNO: Track number expressed in 2 digits BCD (8-4-2-1 code).

00: Lead-in track

The end of the lead-in track is at the starting diameter of the program area.

01-99: Track numbers

A track can be preceded by a pause with the same track number. The track numbering has to start the value 01 and has to increment by one. In case a program is stored on several discs, the numbering may be continued.

AA: Lead-out track

The lead-out track starts at the end of the last music track on a disc, without a preceding pause encoding.

X: Index to TNO, 2 digits BCD.

During the lead-in track, the index X is not available.

00: Pause encoding

The pause encoding in channel Q gives the pause length like it is in the music. The first piece of music is preceded by a pause encoding of 2-3 seconds. The lead-out track is encoded as music.

01-99: Subdivision number

During the lead-out track X is 01. without a music track (TNO=01-99 and X 00) the first value of X is 01. The value of X can only increment by one.

ZERO: These 8 bits are zero.

MIN Running time within a track SEC expressed in 6 digits. BCD MIN, FRAME: SEC and FRAME each 2 digits.

The time is set to zero at the start of track. Time increases in the music and decreases in the pause, ending with the value zero at the end of the pause. In the lead-in and the lead-out tracks the time increases. The minutes are stored in MIN, the seconds in SEC. One second is subdivided into 75 FRAMES (running from 00 - 74)

AMIN Running time on the disc ASEC expressed in 6 digits BCD. AMIN, AFRAME: ASEC, and AFRAME each 2 digits.

At the starting diameter of the program area, the running time is set to zero and TNO takes the value of the first track on the disc. The minutes are stored in AMIN, the seconds in ASEC. One second is subdivided into 75 AFRAME (running from 00 - 74).

POINT During the lead-in track, a table of contents (TOC) is stored into PMIN these locations. This table of PFRAME: contents is continuously repeated in the lead-in area (TNO = 00).

In each table of contents, the items are repeated three times (see table 14-1). At the end of the lead-in area, the table of contents can be ended with any value of POINT.

The value of PMIN, PSEC, and PFRAME gives the starting point of the track number pointed to by POINT. These values give the start position of the track on the absolute time scale (AMIN,

ASEC, and AFRAME) with an accuracy of +/- one second. The start position of a track is the first position with the new track number and X 00.

If POINT= A0;
The value of PMIN gives the track number of the first piece of music on the disc, PSEC and PFRAME are zero.

If POINT= A1;
The value of PMIN gives the track number of the last track on the disc, PSEC and PFRAME are zero.

If POINT= A2;
In PMIN, PSEC and PFRAME the starting point of the lead-out track is given.

Frame number	Point	PMIN	PSEC	PFRAME
n	01	00	02	32
n+1	01	00	02	32
n+2	01	00	02	32
n+3	02	10	15	12
n+4	02	10	15	12
n+5	02	10	15	12
n+6	03	16	28	63
n+7	03	16	28	63
n+8	03	16	28	63
n+9	04	:	:	:
n+10	04	:	:	:
n+11	04	:	:	:
n+12	05	:	:	:
n+13	05	:	:	:
n+14	05	:	:	:
n+15	06	49	10	33

n+16	06	49	10	33
n+17	06	49	10	33
n+18	A0	01	00	00
n+19	A0	01	00	00
n+20	A0	01	00	00
n+21	A1	06	00	00
n+22	A1	06	00	00
n+23	A1	06	00	00
n+24	A2	52	48	41
n+25	A2	52	48	41
n+26	A2	52	48	41
n+27	01	00	02	32
n+28	01	00	02	32
:	:	:	:	:
:	:	:	:	:

Table 14-1

- This point can be determined by the software maker.
- The stop point of track in the player is the location where the TNO changes.
- The start point of track in the player is the first location with the new TNO and X 00. The accuracy of the

start and stop points depends on the player design. The switch delay of the deemphasis depends on the player design.

During the lead-in track, ATIME and X are not available.

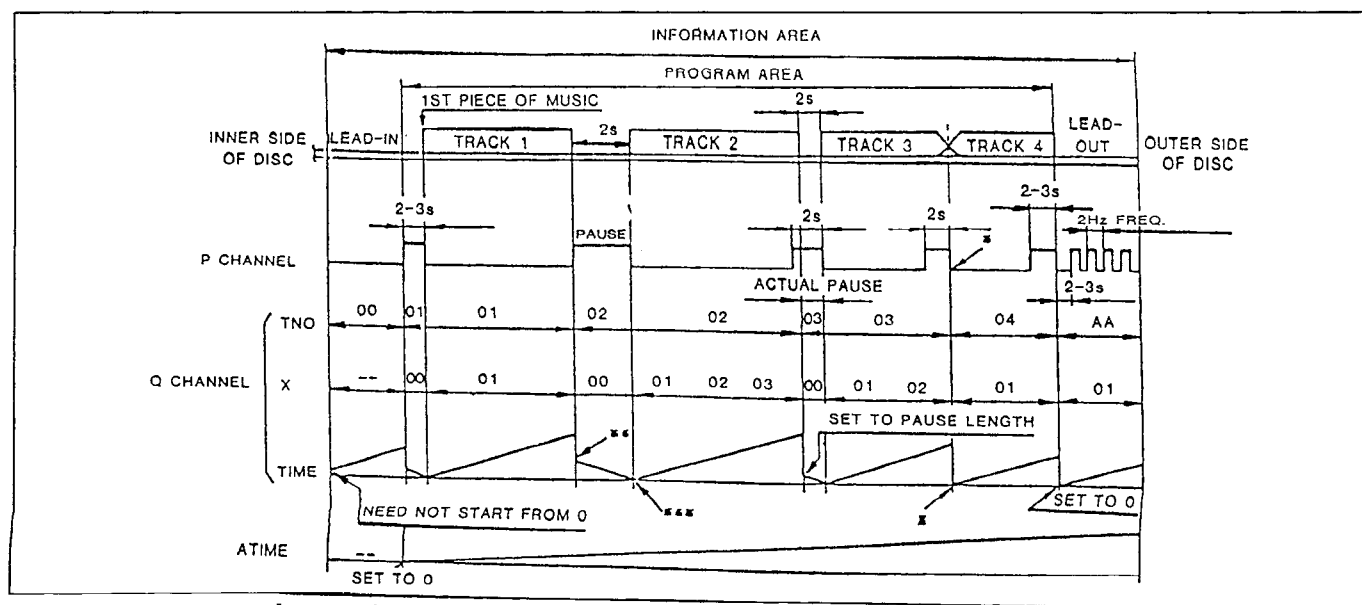


Figure 14-6 Example of Encoding in the P and Q Channels

(2) Mode-2

ADR = 2 = (0010)

If mode-2 is present, it occupies at

least 1 out of 100 successive subcoding blocks.

The data format in mode-2 as follows;

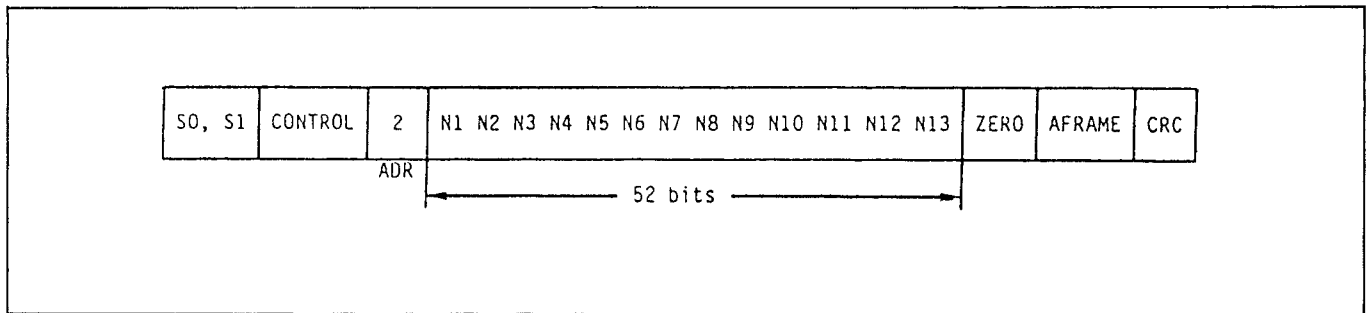


Figure 14-7

N1-N13: Catalogue number of the disc expressed in 13 digits BCD (Binary Code Decimal). Used is the UPC/EAN code (BAR coding). The catalogue number does not change on a disc. In case no catalogue number is encoded according to the UPC/EAN code, N1-N13 are all zero, or mode-2 can be deleted from the disc.

ZERO : These 12 bits are zero.

AFRAME: The continuation of AFRAME in mode-1 (two digits BCD running from 00 to 74). During the lead-in area (TNO= 00), these 8 bits are zero.

Note : UPC ... Universal Product Code
EAN ... European Artical Number

(3) Mode-3

ADR = 3 = (0011)

If mode-3 is present, it occupies at least 1 out of 100 successive subcoding blocks. The mode-3 used to give a unique number to a music track. This is done by means of the International Standard Recording Code (ISRC). The ISRC is defined in DIN-31-621.

If no ISRC code is used, mode-3 must be deleted. During the lead-in and lead-out tracks, mode-3 is not present on the disc. The ISRC code can only change immediately after the track number (TNO) has been changed.

The data format in mode-3 as follows;

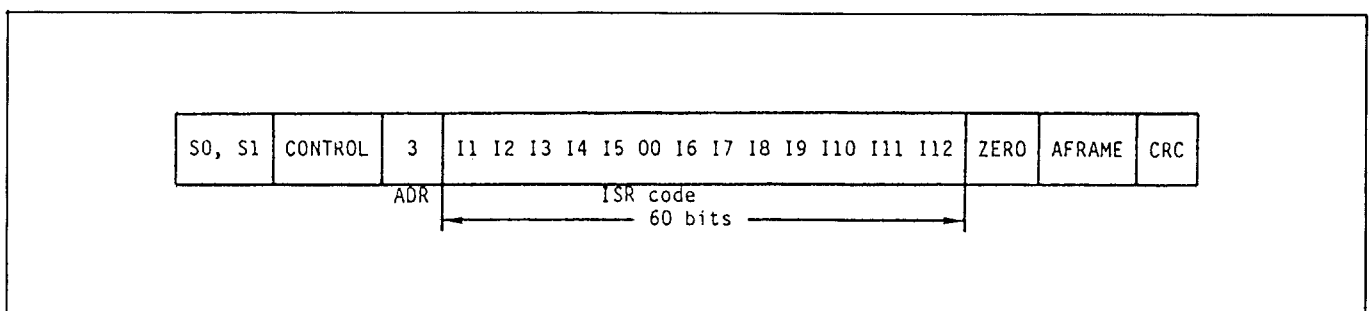


Figure 14-8

15. Calculation of the Different Bit Rates

1. Analog to Digital Conversion

$F_s = 44.000 \text{ KHz}$
 $q = 16 \text{ bits}$
 $n = 2 \text{ Channels}$

Bits Rate = $F_s \times q \times n = 1.411200 \text{ MHz}$

2. CIRC Encoding

6 words (of 16 bits) of the L-channel A/D converter and 6 words (of 16 bits) of the R-channel A/D converter are fed to the "C2" encoder. These 12 words of 16 bits are broken into 24 words of 8 bits. This is the input of the encoder

The "C2" encoder adds 4 words of 8 bits, and its output is 28 words of 8 bits. The "C2" encoder is called a (24, 28) encoder. and The "C1" encoder receives these 28 words at the input and adds again 4 words of 8 bits. Its output is 32 words of 8 bits and the "C1" encoder is called a (28, 32) encoder.

The extra words (8 words x 8 bit = 64 bits) are called redundant words and are used for the error correction.

The new bit rate now equals:
 $1.411200 \times 32/24 = 1.881600 \text{ MHz}$

3. The Control Word

One control word (1 word x 8 bits) is added to every 32 words of the encoder.

The new bit rate becomes:
 $1.881600 \times 33/32 = 1.940400 \text{ MHz}$

4. EFM Encoding

Every word of 8 bits is replaced by a "symbol" of 14 bits. To these 14 bits 3 extra junction bits are added. The resultant operation transforms 8 data bits into 17 (14 + 3) channel bits.

The new bit rate is:
 $1.940400 \times 17/8 = 4.123350 \text{ MHz}$

5. The Sync Word

The 33 symbols of 17 bits (= 561 bits) are preceded by one sync word (24 bits) and its 3 junction bits. Every 561 bits are complemented with 27 (24 + 3) bits, giving 588 bits in total. These 588 bits make one "frame".

The final bit rate (on the CD) become:
 $4.123350 \times 588/561 = 4.321800 \text{ MHz}$

NOTE

(1).The frame frequency equals:

$F_{\text{frame}} = 4.321800/588 = 7350 \text{ Hz}$
 and the period $T_{\text{frame}} = 136 \mu\text{s}$

(2).The subcode is divided in blocks of 98 frames, The subcode frequency is:

$F_{\text{sc}} = 7350/98 = 75 \text{ Hz (13.33 ms)}$

(3).The playing time is calculated by counting these blocks:

75 blocks = 1 sec (00-74)

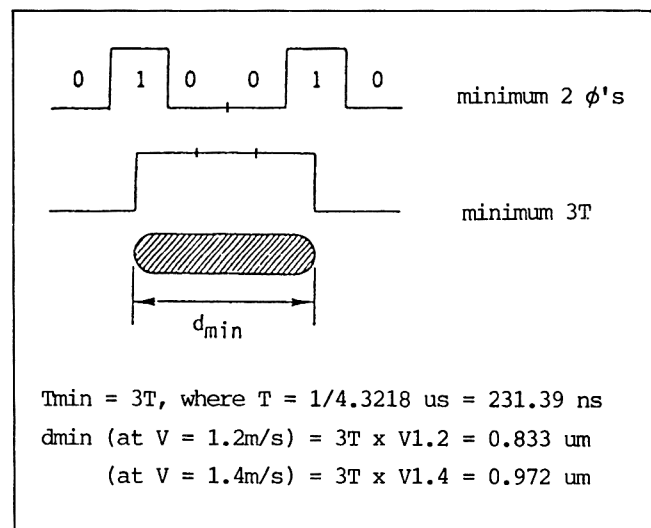


Figure 15-1 Minimum Pit Length

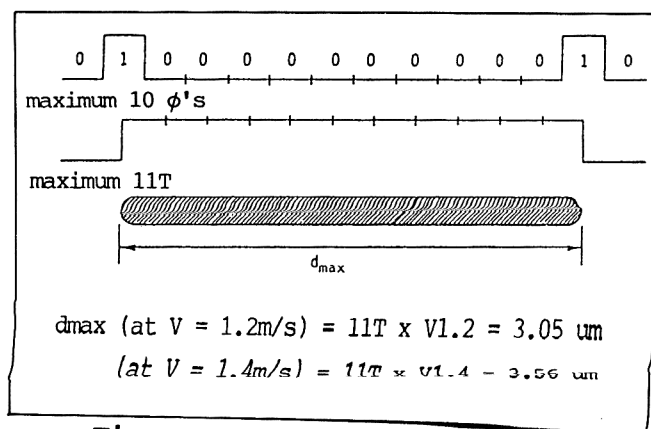


Figure 15-2 Maximum Pit Length

16. Optical Pickup

1. Basic Operation of the Laser Diode

The basic operation of any laser consists of pumping the atoms into a stimulated state from which the electrons can escape and fall to the lower energy ground state by giving up a photon of the appropriate energy. In a solid-state laser such as the laser diodes used in CD player, this principle is applied to a PN-junction, as will be explained below.

First, let us review the energy band diagrams as shown in figure 16-1.

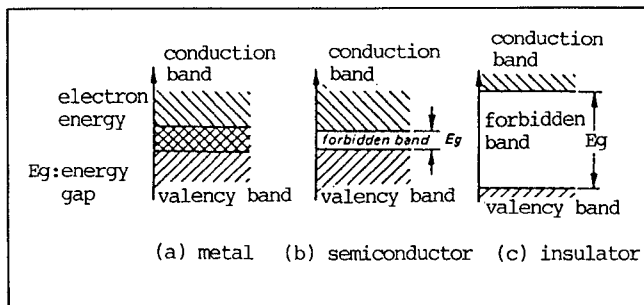


Figure 16-1 Energy Band

As is commonly known, the energy state of electrons falls within a certain range or energy band.

The energy bands required for electrical conductivity are:

- (a) The "valency band", indicating the energy level of the valency electron and:
- (b) The "conduction band", indicating the energy level of the free electron.

In a metal, the valency band and conduction band overlap each other in part, so that the valency electron can enter the conduction band and move freely, which causes conductivity.

In a semiconductor or insulator, there is an energy gap between the valency band and the conduction band; this band is called the forbidden band. In case of an insulator, the forbidden or energy gap " E_g " is large.

For a semiconductor, the energy gap is rather small, so that it can become a conductor by applying some external energy.

P-type and N-type semiconductors can be made by inserting impurities into the semiconductor crystal. In the P-type semiconductor, positive holes are generated, which can move freely within the valency band, and in the N-type semiconductor, electrons are generated which can move freely in the conduction band.

If both types of semiconductors are combined into one crystal, i.e. in a PN junction, a natural electrical field is created in this PN-junction section, but the free electrons and positive holes are kept in the respective areas, and do not move to the other area, as they do not have enough energy to cross the E_g level. However, by applying energy from outside, e.g. a voltage or current, to the PN junction, the electrons and positive holes start moving into the other area as shown in figure 16-2. As a result, electrons and positive holes, coexisting in the PN junction, are now recombined, and natural light is emitted.

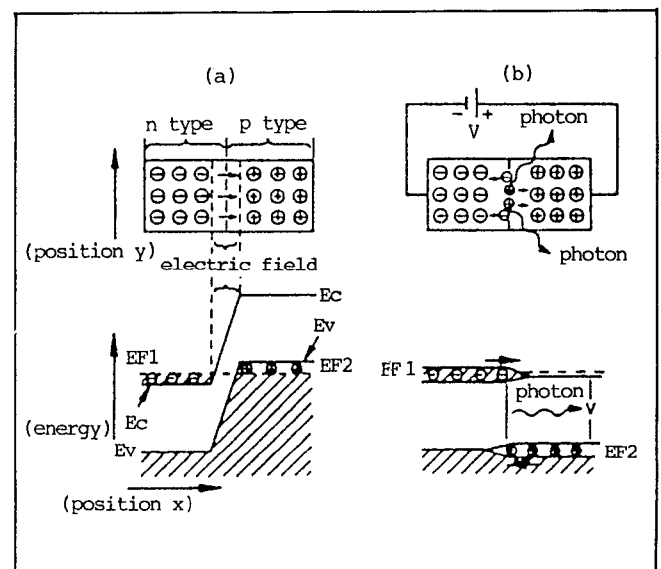


Figure 16-2 Light Emission

The principle of the injection laser diode as shown in figure 16-3. The natural light emitted due to the recombination of the electrons and positive holes is caught between two mirror plates; so photons can excite other photons, by injecting their light energy in the PN junction, thus creating more electrons and positive holes etc.

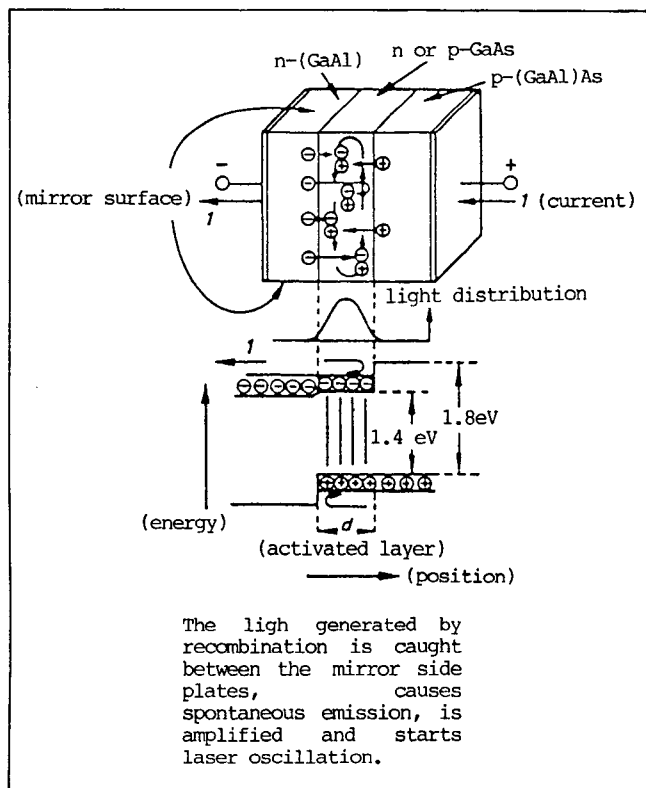


Figure 16-3 As Double Hetero Laser Diode

The light generated by recombination is caught between the mirror side plates, causes spontaneous emission, is amplified and starts laser oscillation. This stimulated emission can be so rapid as to represent nearly an avalanche, completely draining the high-energy states in a subnanosecond burst of very great energy.

On the other hand, certain types of lasers can be produced which will operate in an equilibrium condition, giving up photons at just the input energy pumping rate. The injection laser diode operates in this mode.

The material in an injection laser diode is rather heavily doped so that, under forward bias, The region near the junction has a very high concentration of holes and electrons. This produces the inversion condition with a large population of electrons in a high-energy band and a large population of holes in the low-energy band.

In this condition, the stimulated emission of photons can overcome the photon absorption and a net light flux will result. In operation, the forward current must reach some threshold, to achieve lasing operation. Just beyond this threshold, the laser operates with a single "thread" of light, and the output is relatively stable but low.

As the current is increased, the light output climbs extremely rapidly. The light output is highly temperature sensitive. A small current variation or modest temperature change can cause the output to rise so rapidly that it destroys the chip.

A photodiode, monitoring the light output, is commonly used in a feedback loop to overcome this problem. Like the LED, the ILD must be driven from a current source, rather than a voltage source as the unit can go into thermal runaway.

As a device begins to draw current, the junction begins to warm and forward drop decreases, which tends to drive the voltage higher and in turn decrease the forward drop until the current tends toward infinity and the device is destroyed.

In addition, the ILD has a tendency to deterioration with operation, and this deterioration is greatly accelerated when operating the ILD outside of its optimum limits. Therefore, in an optical pickup assembly the operating point set at the factory should not be touched without the proper measuring equipment.

A typical operating curve is shown as follows figure 16-4.

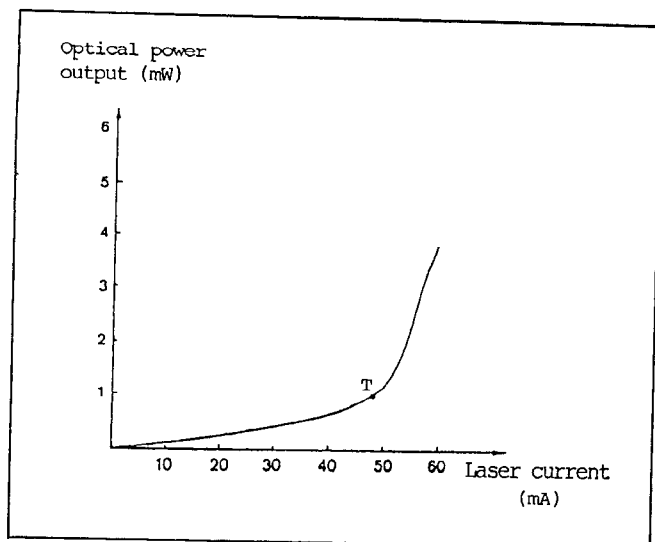


Figure 16-4

In an (Ga Al) As ILD with a double heterostructure (DH), a very narrow PN-junction layer (Ga As) is sandwiched between layers of a different semiconductor (Al Ga As). The properties of this semiconductor confine the electrons and holes to the very narrow junction layer, leading to an inverted population at a low input current. It oscillates at a 780nm wavelength, generating a monochromatic lightbeam.

2. Opto-Electronics

The technological marriage of the fields of optics and electronics is called "opto-electronic". It is a remarkable diverse field involving such topics at the nature of optical radiation, the interaction of light with matter, radiometry, photometry and the characteristics of various sources and sensors.

1) The Optical Spectrum

By convention, electromagnetic radiation is generally specified according to its metric wavelength (λ). The frequency of a specific electromagnetic wavelength can be found from:

$$f = c/\lambda$$

Where f is frequency in Hz:
 c is frequency of light ($=3 \times 10^8$ m/sec)
 λ is wavelength in m

The optical portion of the electromagnetic spectrum extends from 10 nm (100 Å) to 10^6 nm and is divided into three major categories: ultraviolet (UV), visible and infrared (IR).

Ultraviolet are those wavelengths, falling below the visible spectrum and above x-rays.

UV is classified according to its wavelength as extreme or short wave UV (10 - 200 nm), far (200 - 300 nm) and near or long wave UV (300 - 370 nm).

Visible are those wavelengths between 370 - 750 nm and they can be perceived by the human eye. Visible light is classified according to the various colours its wavelengths elicit in the mind of a standard observer.

The major colour categories are: violet (370 - 455 nm), blue (456 - 492 nm), green (493 - 577 nm), yellow (578 - 597 nm), orange (598 - 622 nm) and red (623 - 750 nm).

Infrared (IR) are those wavelengths above the visible spectrum and below microwaves. IR is classified according to its wavelength as near (750 - 1500 nm) middle (1600 - 6000 nm), far (6100 - 40000 nm) and far-far (41000 - 10^6 nm).

2) Optical Waves with Matter

An optical wave may interact with matter by being reflected, refracted, absorbed or transmitted. The interaction normally involves two or more of these effects.

a) Reflectance

Some of the optical radiation impinging upon any surface is reflected away from the surface. reflectance varies according to the properties of the surface and the wavelength and in real circumstances may range from more than 98% to less than 1% (lampblack body).

Reflection from a surface may be either diffuse, specular or both. A diffuse reflector has a surface which is rough when compared to the wavelength of the impinging radiation.

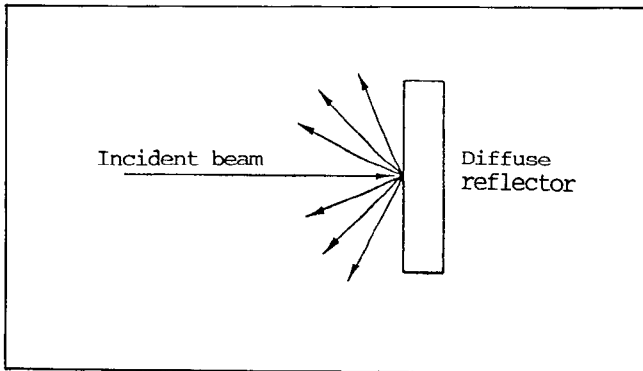


Figure 16-5 Diffuse Reflector

A specular reflection has a surface which is smooth when compared to the wavelength of the impinging radiation. A perfect specular reflector will reflect an incident beam without altering the divergence of the beam. A narrow beam of optical radiation impinging upon a specular reflector obeys two rules:

- (1) The angle of reflection is equal to the angle of incidence.
- (2) The incident ray and the reflected ray lie in the same plane as normal line extending perpendicularly from the surface.

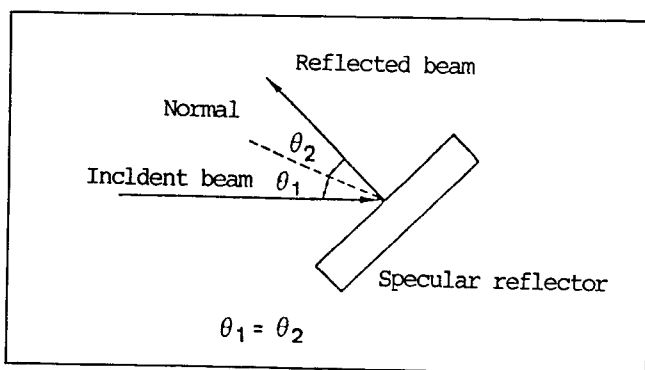


Figure 16-6

b) Absorptance

Some of the optical radiation impinging upon any substance is absorbed by the substance. Absorptance varies according to the properties of the substance and the wavelength and in real circumstances may range from less than 1% to more than 98% ("lampblack" body).

c) Transmittance

Some of the optical radiation impinging upon a substance is transmitted into the substance. The penetration depth may be slight (transmittance = 0) or deep (transmittance more than 75%).

The reflectance (ρ), the absorptance (φ), and the transmittance (γ) correspond to :

$$\rho + \varphi + \gamma = 1 \text{ (100\%)}$$

d) Refraction

A ray of optical radiation passing from one medium to another is bent at the interface of the two mediums if the angle of incidence is not 90 degree. The index of refraction is the sine of the angle of incidence divided by the sine of the angle of refraction. Refractive index varies with wavelength and ranges from 1.0003 to 2.7.

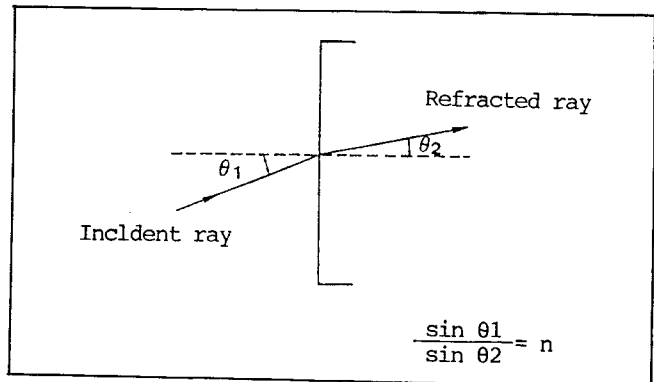


Figure 16-7 Refraction

3) Optical Components

Optical components are used to manipulate and control optical radiation and to provide optical access to various sources and sensors.

Glass is the most common optical material at visible and near-infrared wavelengths, but other wavelengths require more exotic materials such as calcium aluminate glass, lithium fluoride etc.

a) The Thin Simple Lens

A thin simple lens refracts an optical ray passing through it as follows:

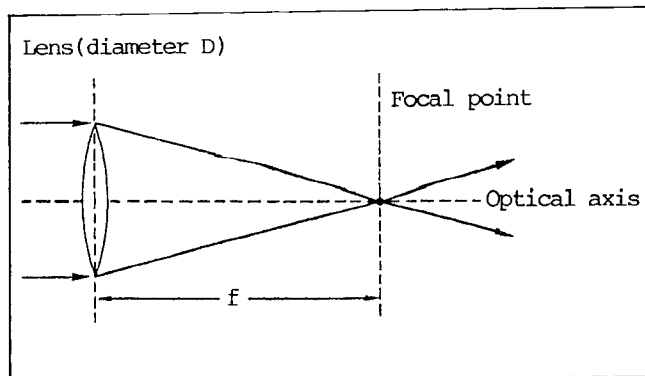


Figure 16-8 Thin Lens

A lens may be either positive (converging) or negative (diverging). The focus point of a lens is that point at which the image of an infinitely distant point source is reproduced.

Both the source and the focus point lie on the lens axis. The focus length (f) is the distance between the lens and the focus point. The f/number of a lens defines its light collecting ability and is given by:

$$f/\text{number} = f/D$$

Where f is the focus length and D is the diameter of the lens.

A small f/number denotes a large lens diameter for a specified focus length and a higher light collecting ability of faster "speed" than a large f/number .

Numerical Aperture (NA) is a measure of the acceptance angle of a lens and is given by:

$$NA = n \cdot \sin \theta$$

Where n is the refractive index of the object or image medium and θ is half the maximum acceptance angle.

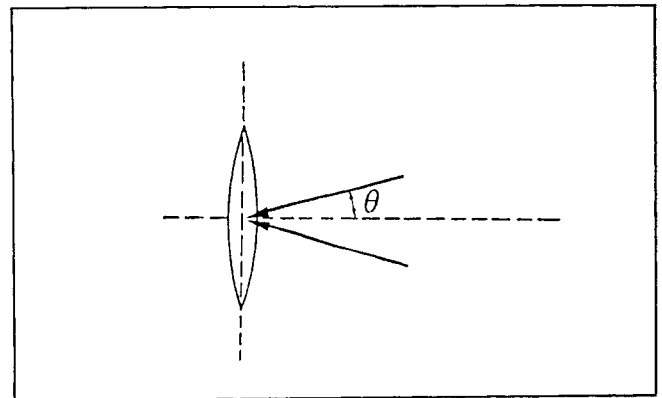


Figure 16-9

The relation of the focus length (f) to the distances between the lens and the object being imaged (s) and the lens and the focused image (s') is given by the gaussian form of the thin lens equation:

$$1/s + 1/s' = 1/f$$

The combined focus length for the thin lenses in contact or close proximity and having the same optical axis is given by:

$$1/f = 1/f_1 + 1/f_2$$

The relationship between the focus length (f) and the refractive index (n) is :

$$1/f = (n-1) \times (1/r_1 - 1/r_2)$$

Where r_1 is the radius on the left lens surface and r_2 is the radius on the right lens surface.

b) The Cylinder Lens

A cylinder lens is a section of a cylinder and therefore concentrates in only one plane.

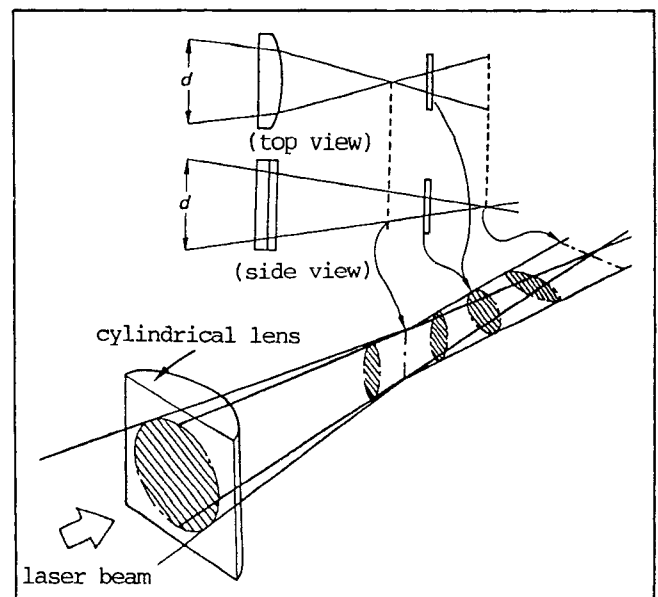


Figure 16-10 Cylindrical Lens Operation

c) The Half Mirror

prism is an optical transparent body used to refract, disperse or reflect an optical beam. The simplest prism is the angle prism.

An optical ray striking one of the shorter faces of the prism at the normal angle is totally internally reflected at the hypotenuse, undergoes a 90 degree deviation, and emerges from the second shorter face.

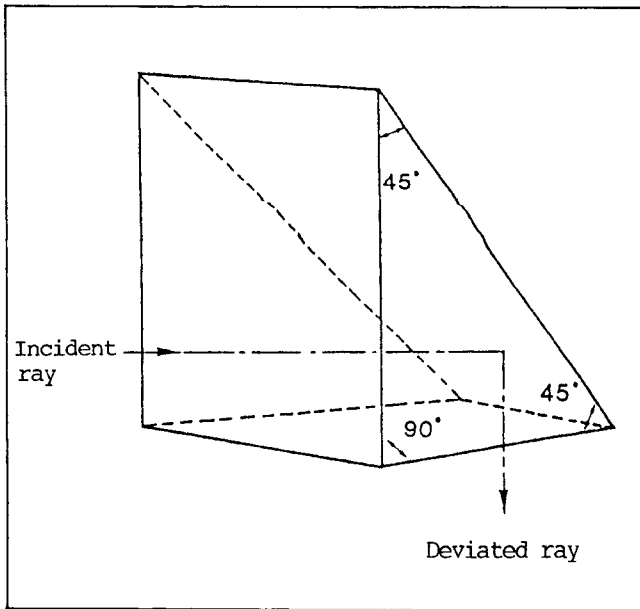


Figure 16-11

Operation of a totally internally reflecting prism is dependent upon the fact that a ray impinging upon the surface of a material having a refractive index (n) smaller than the refractive index (n') of the medium in which the ray is propagating, will be totally internally reflected when the angle of incidence is greater than a certain critical angle (θ_c) given by:

$$\sin \theta_c = n'/n$$

d) The Critical Angle Prism

When light goes over from a medium with a high refractive index to a medium with a low refractive index, the relationship between the angle of incidence and the angle of reflection at the transition plane is as shown in figure 16-12.

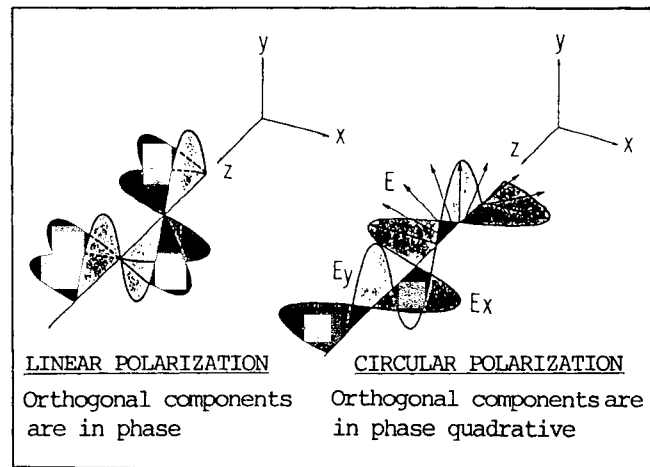


Figure 16-12

The next figure gives the relationship between the reflection coefficient of P-polarized light, R_p , and the reflection coefficient of S-polarized light, R_s . The shows in figure 16-13 it can be seen that in the case where the incident angle is slightly smaller than the critical angle (42 degree), the reflection coefficients will change drastically and there will be an important difference between the reflection coefficients R_p and R_s .

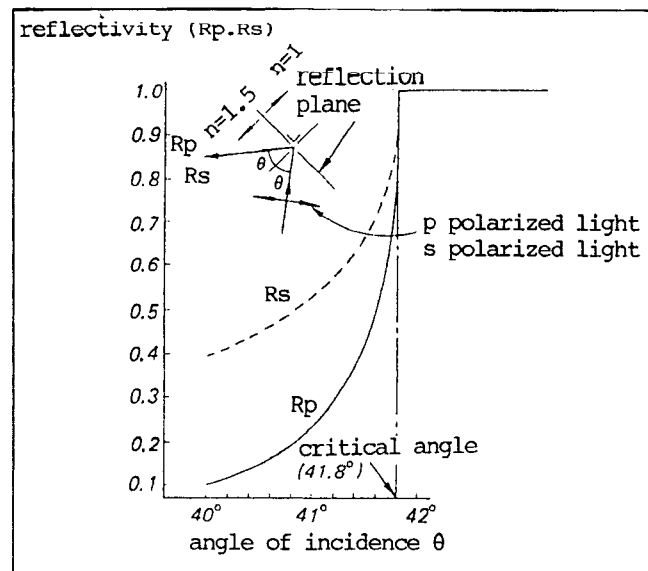


Figure 16-13

This means that, with a small deviation of the incident angle, an important change in reflection can be detected, and an important difference in reflection of P-polarised and S-polarised light. As can be seen in the figure, the change in reflectivity becomes more important for P-polarized light (R_p).

The whole lens system is designed so, that the angle of the incident light in the center of the beam has a critical value. If a parallel beam enters, the incident angle for the whole beam will be the critical angle, so all light will be

reflected. However, when a convergent or divergent light beam enters, only part of the light will be reflected. The critical angle prism is designed so, that the light beam is reflected three time in the prism as shows in figure 16-14.

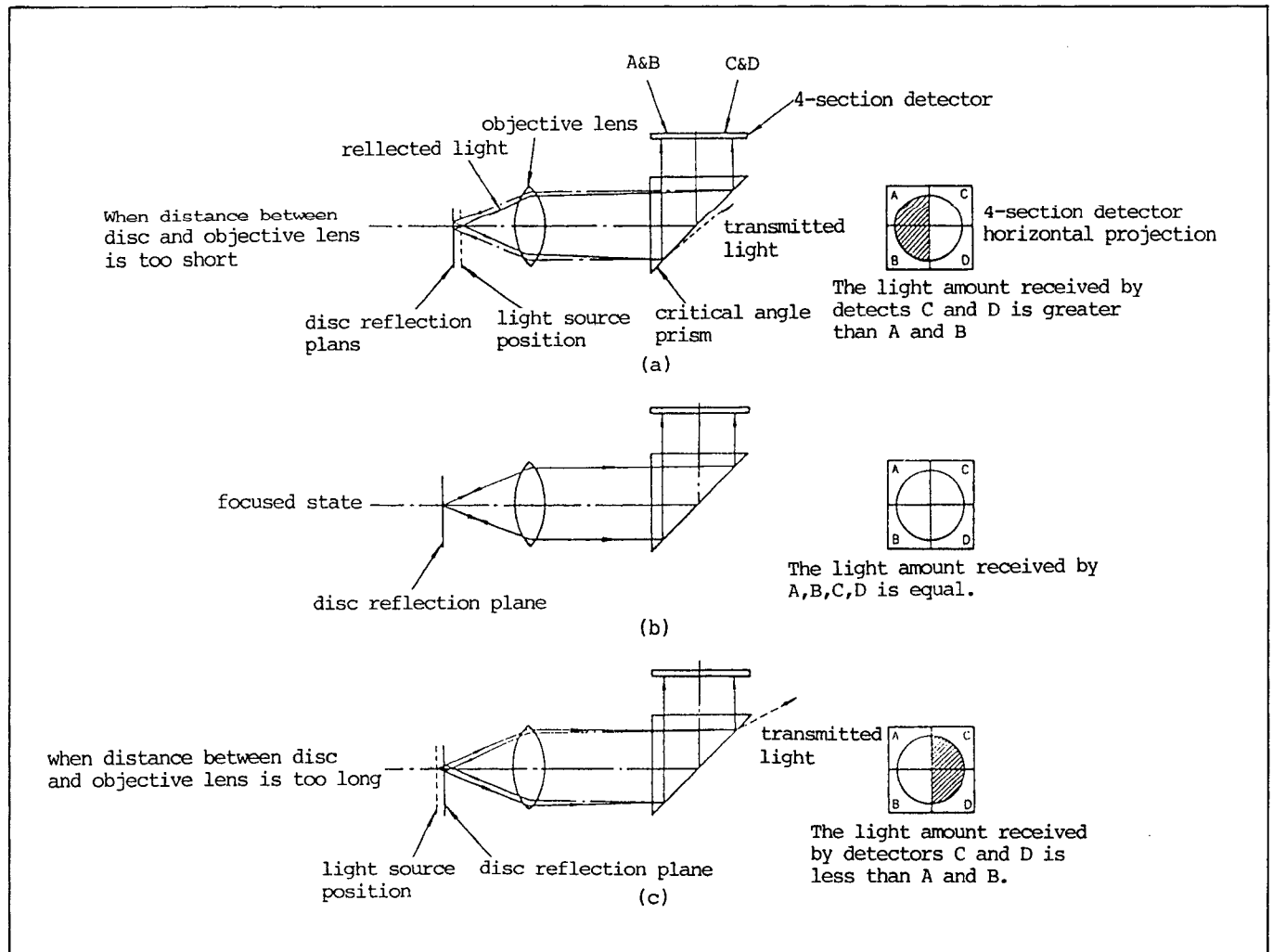


Figure 16-14 Focus Error Detection

As the whole optical system is designed to obtain only P-polarised light, and as the beam is reflected three time, the disc surface position can be detected with extremely high precision.

3. Optical Pickup Construction and Operation

The optical pickup is the heart of the CD player. It pickup modulated laser beam bounced off the disc pits and converts it into electrical data. The pickup (including the traverse mechanism) requires special precision design. On the Technics CD player series, the pickup is integrated onto a single assembly board and is mechanically isolated from the player chassis by a suspension mechanism to prevent floor vibrations and acoustic feedback from the loudspeakers.

The following optical components in particular require careful design:

- Single Beam System
- 2-Dimension Actuator
- Focusing Lens
- Focus and Tracking Error Detection

The detail of each sections of these components as follows:

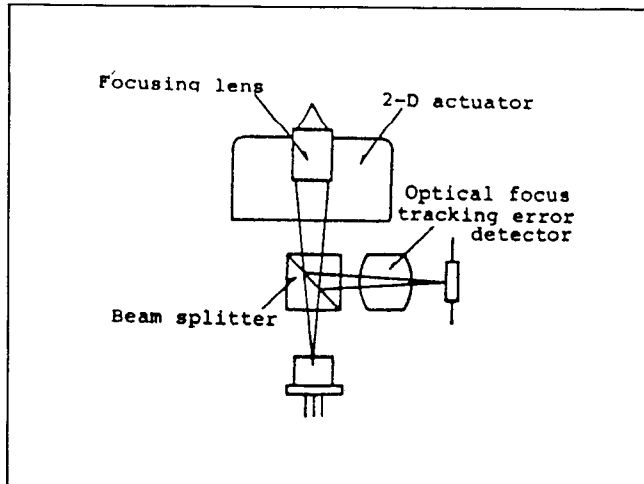


Figure 16-15 Optical Pickup Construction

1) Single Beam System

The single beam detection system includes the push-pull, phasedifference, heterodyne, and others. The push-pull type treats the pits string on the disc as a single track and detects tracking error from variation of the far-field pattern of the diffracted ray, whereas the phase-difference and heterodyne types detect the variation of far-field patterns of the beam defracted from individual pits.

The push pull type requires a simpler optical system than the three beam technique, but is sensitive to the disc's tilt angle or pit depth and the error signal is affected by the focus lens position. For these reasons, it is rarely used for practical application.

The phase difference and heterodyne types eliminate the drawbacks found with the push pull technique. While the optical system is similar to that used in the push pull technique (except for a 4-split photodetector), they have different circuit configurations.

The single beam technique requires a servo circuit a little more complex than that required for the three beam system, but depending on servo circuit design provides better playability than the latters.

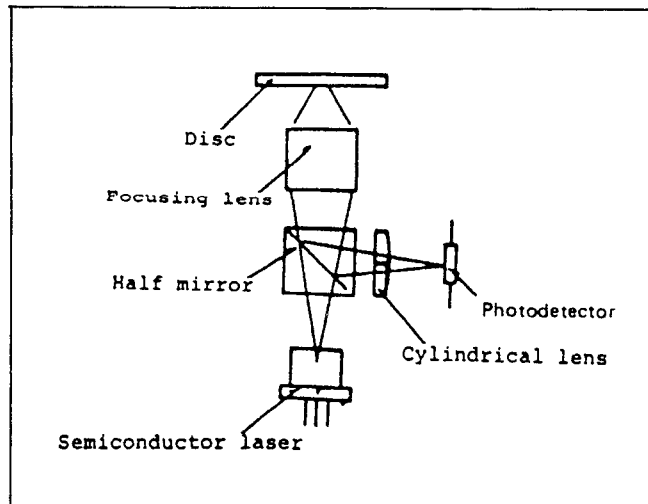


Figure 16-16 Single Beam Construction

2) 2 Dimension Actuator

The 2-D actuator is the only movable component of the optical pickup. It requires precision design because it drives the focusing lens and some other pickup components requiring extreme positional accuracy. Recent 2-D actuators include three types: 4 wire, rotary, and swing arm types. In here will explain the 4 wire actuator as follows;

4 Wire Actuator:

The 4 wire actuator uses four metallic wires to suspend a moving coil bobbin. When the coil is energized, the wires are slack, which lets the coil bobbin move in the focusing and tracking directions. The main point is to drive the bobbin's center of gravity and this gives the four wires and even deformation and keeps the optical axis of the lens straight.

This type of actuator is insensitive to vibration because it is almost free from friction or play. In addition, since the four wires also serve as electrical leads for the moving coil, the probability of wire break failure is minimized. These features in combination ensure a high reliability for the assembly.

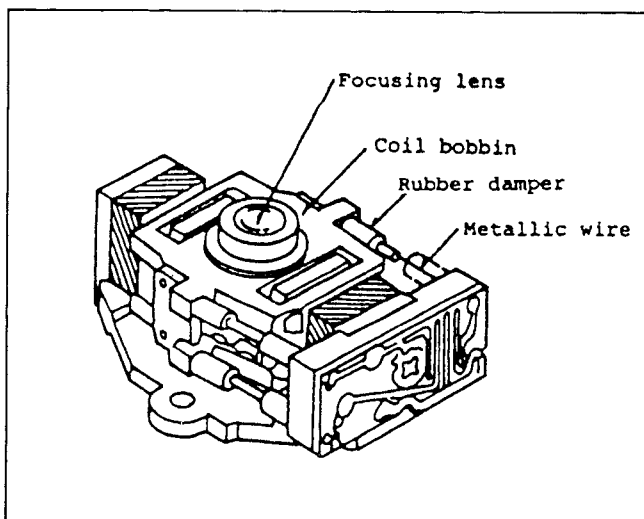


Figure 16-17 Wire Actuator Construction

3) Focusing Lens

The focusing lens availability include the infinite magnification type for use in parallel beams, and the finite magnification type for use in divergent beams emitted from a semiconductor laser.

a) Infinite Magnification type:

This type has no limitation in the designed optical path length.

b) Finite Magnification type:

This type has a fixed distance from the semiconductor laser to the disc surface, and hence requires a lens that matches the construction of the individual optical pickup in which it is used. However this type of focusing lens requires no collimate lens, and renders stable performance due to simple optical system configuration. In addition, the distance from the semiconductor laser to the disc surface can be considerably reduced when compared with the infinite magnification type. However, to compensate for variations of magnification or view angle resulting from focusing lens movement, the finite magnification

type normally requires stricter specifications than required for the finite magnification type.

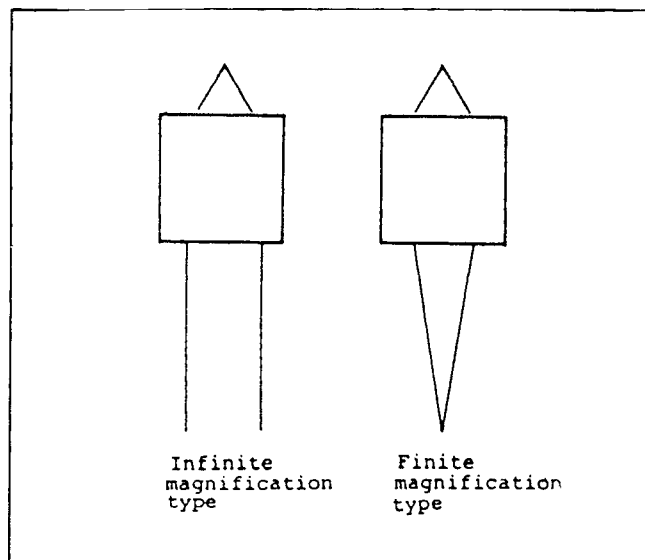


Figure 16-18

4) Focus and Tracking Error Detection

The optical pickup (photodetector section) yields two basic output signals. One is the musical data signal (RF Signal) which is coupled to the signal processing circuit. The other one is the servo signal used for optical servo control (focusing and tracking servos). This is coupled to the servo circuit. These signals are affected differently by contamination or scratches on the disc surface, as described below:

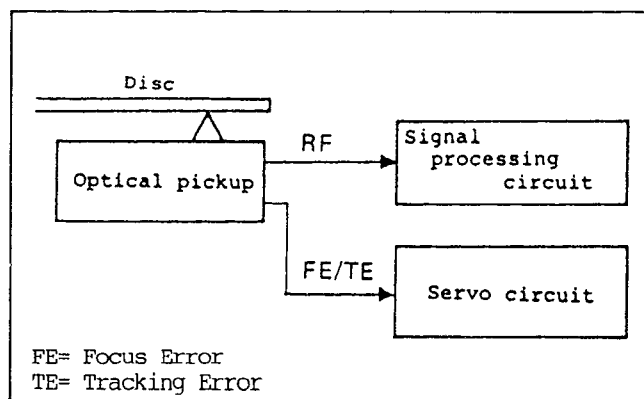


Figure 16-19 Servo Detection

a) Fingerprints and Other Contaminations on disc:

If the disc surface is contaminated with fingerprints or any other contaminations, the laser beam bounced off the disc surface diverges as shown in figure 16-20, this also leads to diverged signal energy.

The beam, no matter how diverged, contains the RF signal component, and must be picked up completely and fed to the signal processing circuit. This requires the photodetector (PD) to have as large a surface area as possible. The larger PD elements, A5

and A6, used in the 6-segment PD (as shows figure 16-20) pick up the signal energy which has not been picked up by the inner 4 segment PD elements, thus serving to prevent reduced RF signal level.

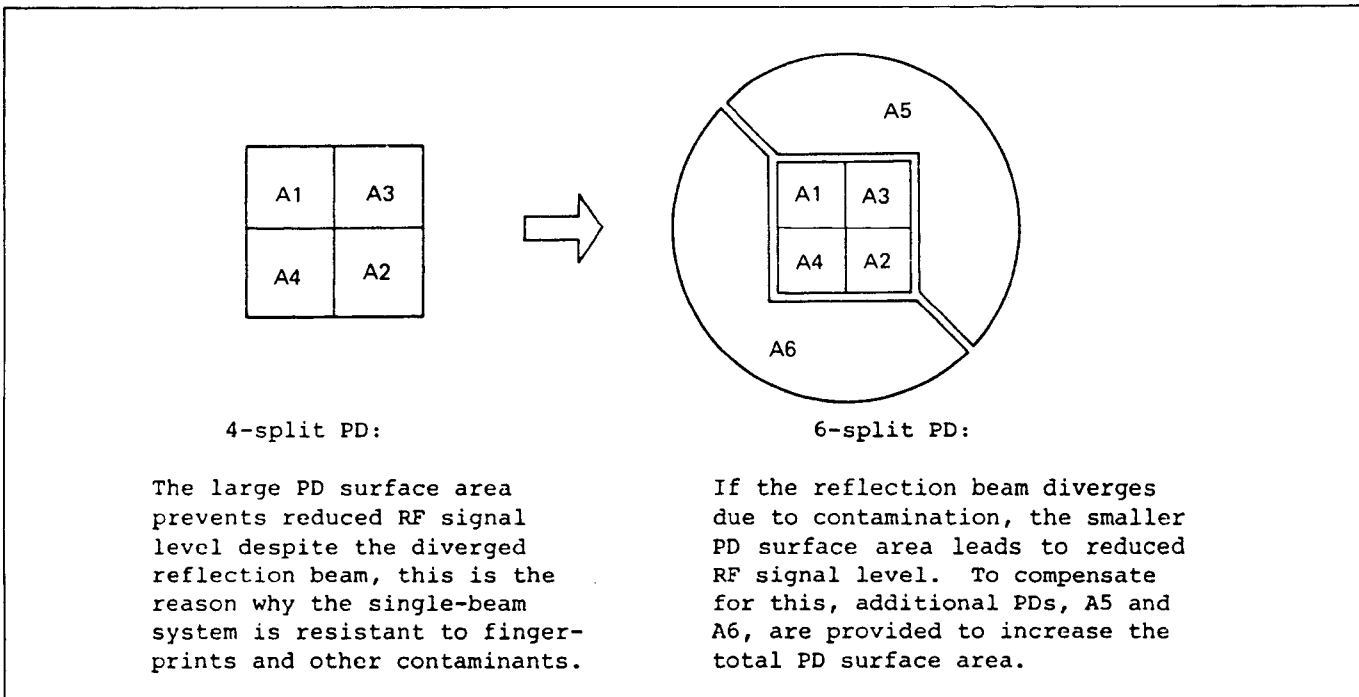


Figure 16-20 Photodetector Construction

b)Scratches and Foreign Objects

The beam spot diameter on a normal disc surface is approximately 120 μm . If the beam is irradiated at a scratch or other defects on the disc surface, whisker-like noise occurs around the beam spot as shows figure 16-21.

This noise has a very random nature, and the conventional 4 segment PD with large surface area pickup all of this noise components added to the servo signal, leading to servo errors and hence degraded play performance.

Whith the 6 segment PD, on the other hand, its 4 segment PD area (for servo signal pickup) is one fourth (250 μm) compared to the conventional pickup detector area which has an area of 1000 μm , and so the affect of whisker-like noise can be considerably lessened as illustrated as shows in figure 16-21. For this reason, the 4 segment PD (A1, A2, A3, A4) for servo signal pickup should preferably have as small an area as possible. Incase of music data should be detect all 6 segment (A1, A2, A3, A4, A5, A6).

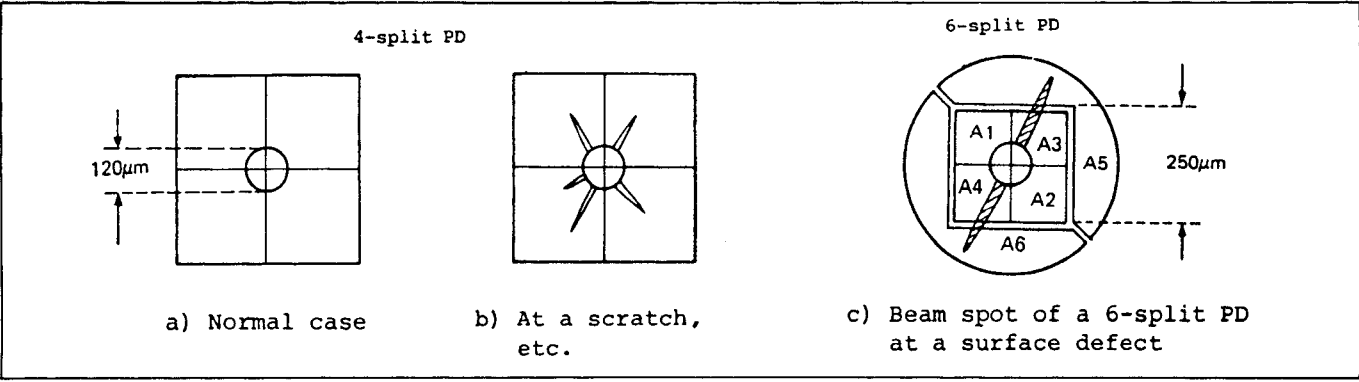


Figure 16-21

The above section a) and b) suggests that a larger photodetector is preferable for RF signal pickup, whereas a smaller photodetector is preferable for servo signal pickup. In order to satisfy these incompatible requirements, the most CD player has replaced the conventional 4 segment PD

with a 6 segment PD to successfully achieve more stable play performance.

The figure 16-22 shows the size comparison between the 4 segment PD and 6 segment PD for the single beam system.

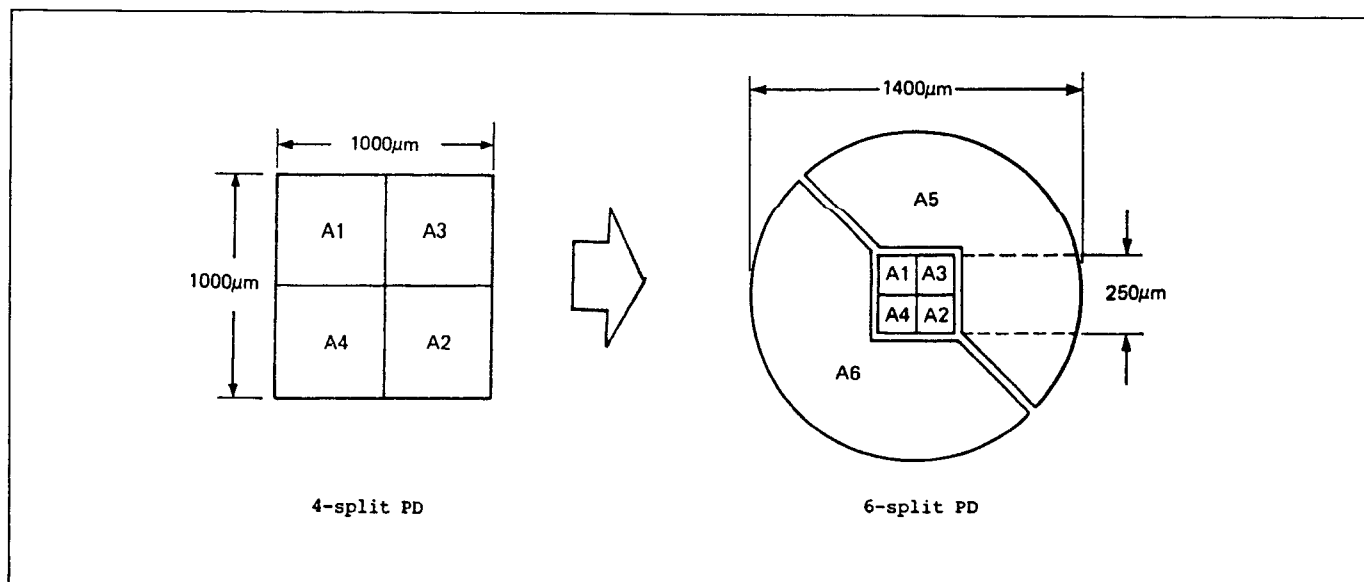


Figure 16-22

c)Signal Output of the 4 Segment and 6 Segment Photodetector.

	Signal name	4-SEGMENT PD	6-SEGMENT PD
1	RF signal	$A1 + A2 + A3 + A4$	$A1 + A2 + A3 + A4 + A5 + A6$
2	FE signal	$(A1 + A2) - (A3 + A4)$	$(A1 + A2) - (A3 + A4)$
3	TE signal	$\Delta t1 + \Delta t2 \begin{pmatrix} \Delta t1=A1+A2 \\ \Delta t2=A3+A4 \end{pmatrix}$	$\Delta t1 + \Delta t2 \begin{pmatrix} \Delta t1=A1+A2 \\ \Delta t2=A3+A4 \end{pmatrix}$

Figure 16-23

CONCLUSION

- 1.The PD section for RF signal pickup should preferably have a larger detection surface area.
- 2.The PD section for servo signal pickup should preferably have a smaller detection surface area.

The PD used in the most CDplayer has achieved the superior RF signal pickup performance of the single beam system for discs with fingerprints, and a stable servo performance comparable to the three beam system for scratched discs.

5) Reading the Disc Information

The figure 16-24 shows a cross sectional view of the disc and laser beam.

When the laser beam hits the surface of the disc it is about 1mm in diameter. The transparent layer has a refractive index of 1.5 and used as a lens to reduce the size of the beam to 1 μm to focus on the pits.

The ratio of the areas of the 1 mm spot on the surface compared to the 1 μm spot on the pit area 10 1,000,000 to 1. Therefore excessive dirt or scratches on the disc surface may affect the reading accuracy.

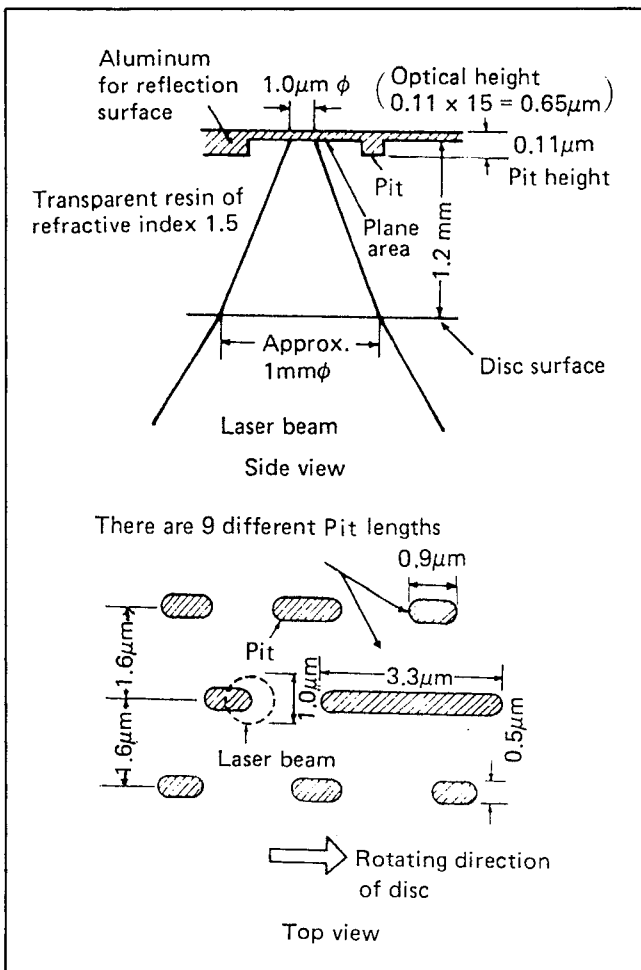


Figure 16-24 Disc

The actual height of a pit is 0.11 μm . Looking from outside the disc, through the polycarbonate layer (refractive index = 1.5), the perceived height is 165 μm , this corresponds to about 1/4 wavelength. Therefore, the light is 180 degree out of phase (1/2 wavelength) with the light reflected from the non-pit area as shows in figure 16-25.

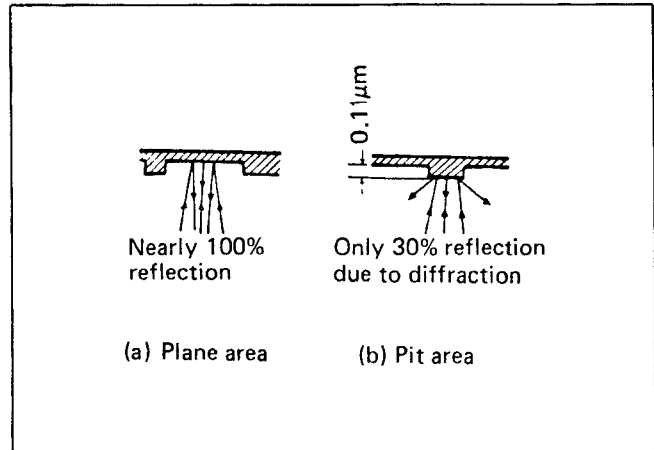


Figure 16-25 Pit vs Output Signal

Figure 16-26 shows the relationship between the disc surface and the waveform generated.

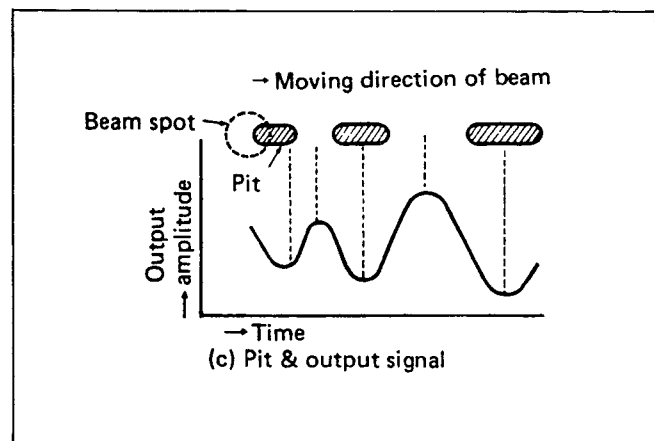


Figure 16-26 Reflection from Disc

When the beam hits the plane (non-pit) area, the light is reflected 100% and returns to the photo detector. This waveform is actually a continuous analog waveform from the digital signal of 4.4218 MHz (bits/sec) recorded on the disc and is called an RF signal.

6) Focus Error Signal

The laser must always be focused on the pit area. Since it is impossible to make a disc that has less than $0.11 \mu\text{m}$ warp

(pit size), the focus lens must move up and down (following the disc) to maintain proper focusing.

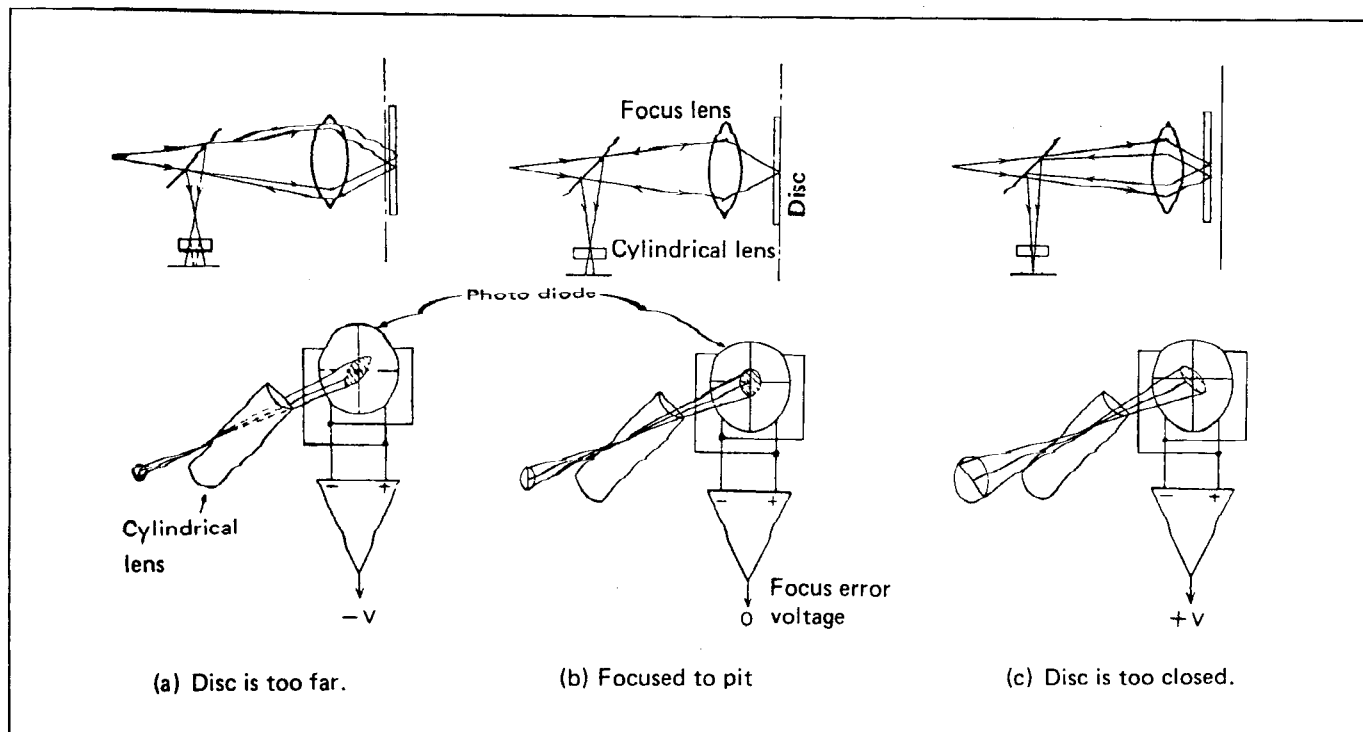


Figure 16-27 Focusing Condition

They cylindrical lens affects the laser beam in such a way that if it is not focused properly it produces an

elliptical shaped spot, The spot falls on a 4 segment photodetector (A1, A2, A3, A4).

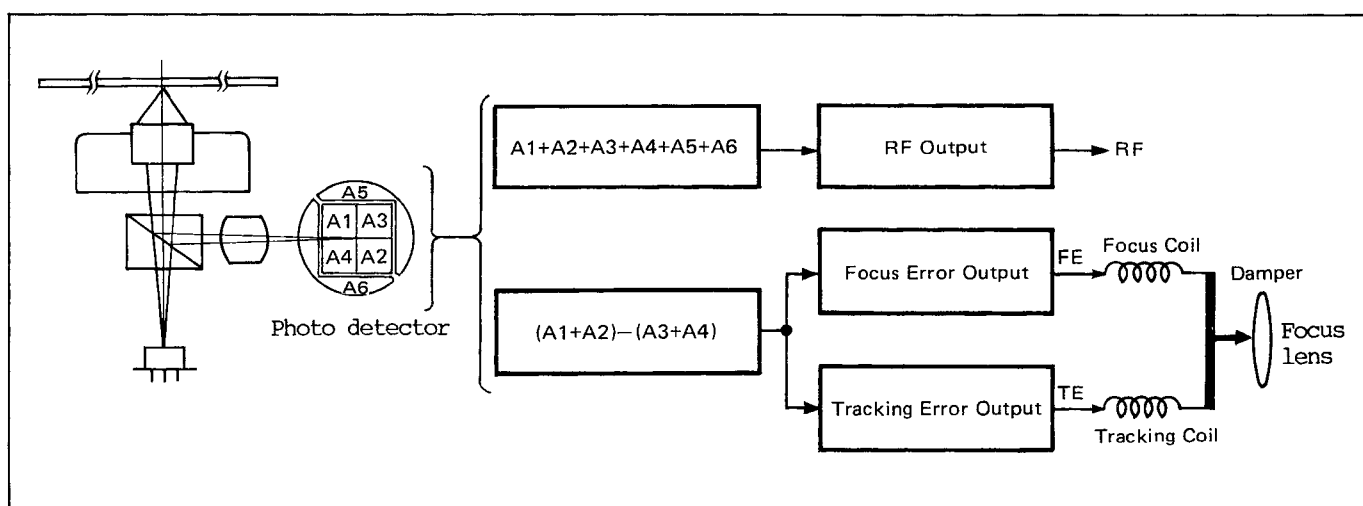


Figure 16-28 Head Amplifier Circuit

The voltage from opposite sections of the detector is added and subtracted from the other sections $(A1 + A2) - (A3 + A4)$. The result is a focus error voltage that is used to drive the focus lens to the correct focusing point.

7) Tracking Error Signal

The figure 16-30 shows the basic arrangement of the one beam method (the phase difference detection method). Suppose that the image of a pit in a track runs over the four-segment PD (A1, A2, A3, A4) deviating rightwards or leftwards.

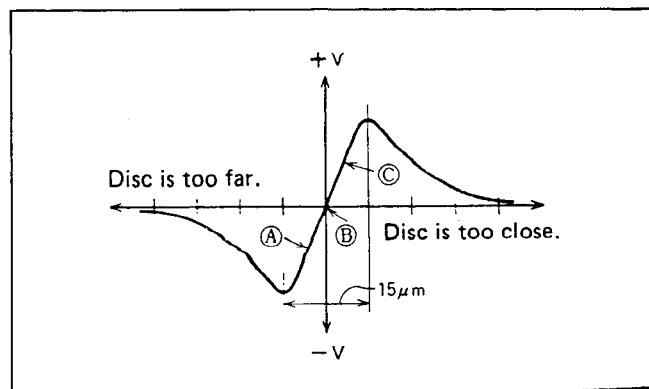


Figure 16-29 Focus Servo Response Curve

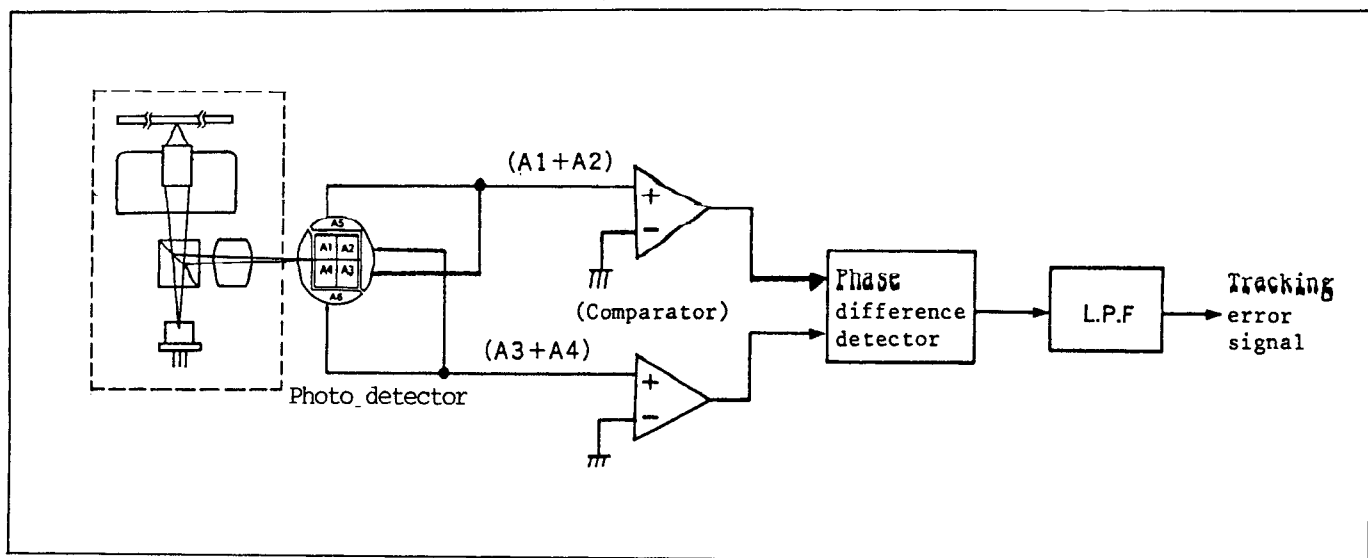


Figure 16-30 Basic Arrangement

A phase difference is produced between the output signals of $(A1 + A2)$ and $(A3 + A4)$ in proportion to the degree of deviation of the track. The phase difference is detected by the phase

detector and passed through the low pass filter for generating the tracking error signal, as shows figure 16-31 how a phase difference is detected.

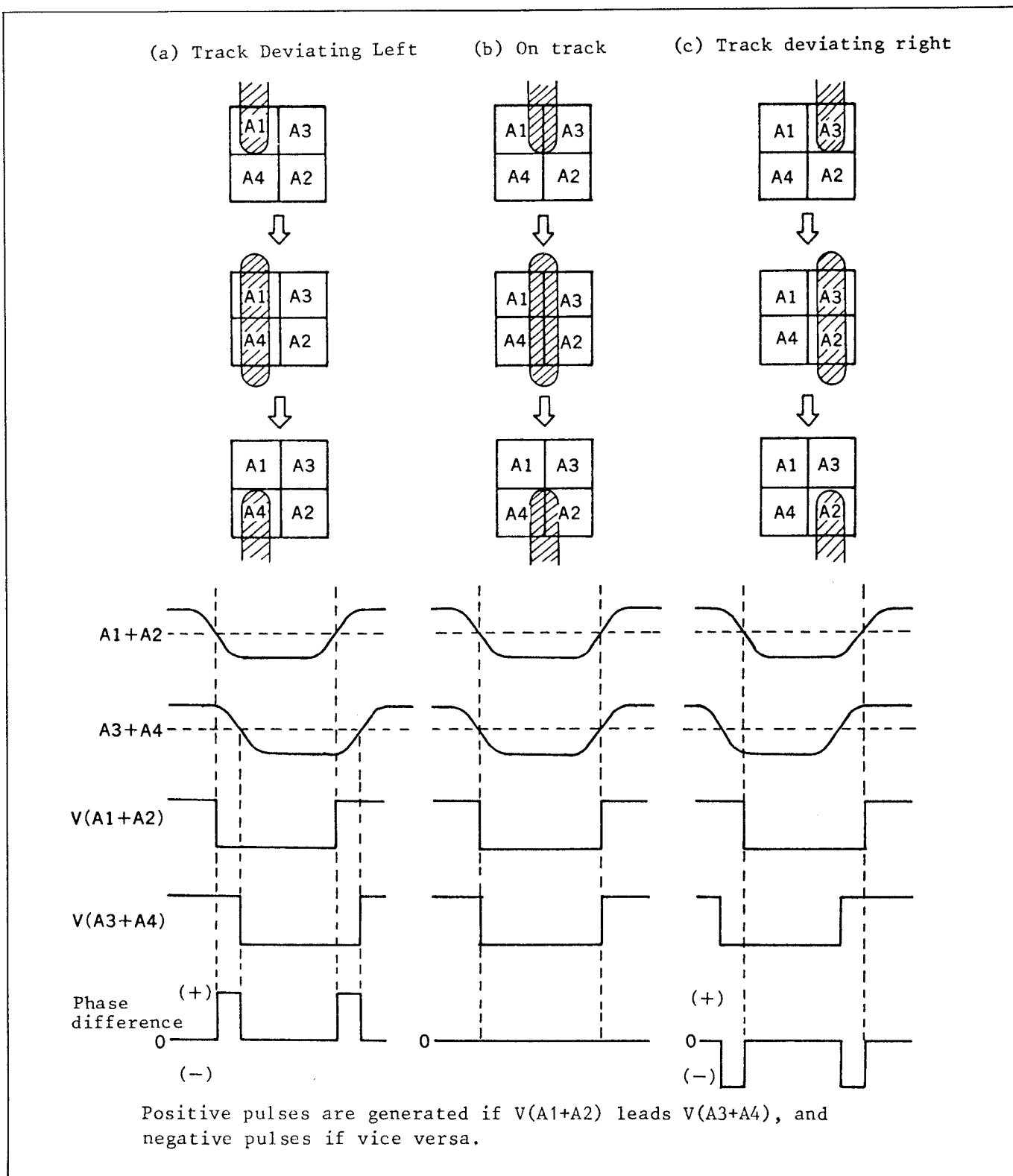


Figure 16-31 Detection of Tracking Error

17. Servo Circuit

The servo IC AN8370S must satisfy the proper control needs for the optical pickup functions. In addition, it must also include the following functions, which are not directly related to the optical servo circuit functions.

- 1) Focus servo pull-in
- 2) Trick play
- 3) Laser power control

Before we begin a description of the servo circuit operation, let's take a look at the servo IC (AN8370S) which integrates all the related servo functions (see the block diagram figure 17-1). And the servo IC contains the following functional blocks:

- 1) Laser power control
- 2) RF amplifier
- 3) Focus servo head amplifier
- 4) Focus servo pull-in logic
- 5) Tracking servo head amplifier
- 6) system logic control
- 7) Trick play logic
- 8) Traverse speed control

***The focus servo pull-in logic is a completely local function of the servo IC and can drastically reduce the load on the system controller CPU.

***The tracking servo head amplifier, which is based on the digital accu-servo system is comprised of high-speed ECL devices, and is capable of detecting phase differences of approx, 20 through 30 ns.

***The trick play logic controls the high-speed access operation characteristic of the CD player. It consists of a KICK pulse generator, tracking actuator holding circuit, servo pull-in stabilizer, cross-track count detector, and some other related logic.

***The traverse speed controller regulates the optical pickup track crossing speed. It is compatible with high-speed traverse mechanism that use a linear motor (most CD player used linear motor operation).

1) Block Diagram of AN8370S

The optical servo reads the signal recorded on the disc by the laser beam, And the AN8370S is the main IC of the optical servo system and comprises a laser APC, RF signal amplifier, Focus servo (including automatic focus function), Tracking servo (including KICK function), Drop out detector, and also Control logic for controlling the optical servo of IC. The block diagram of IC (AN8370S) is shows figure 17-1.

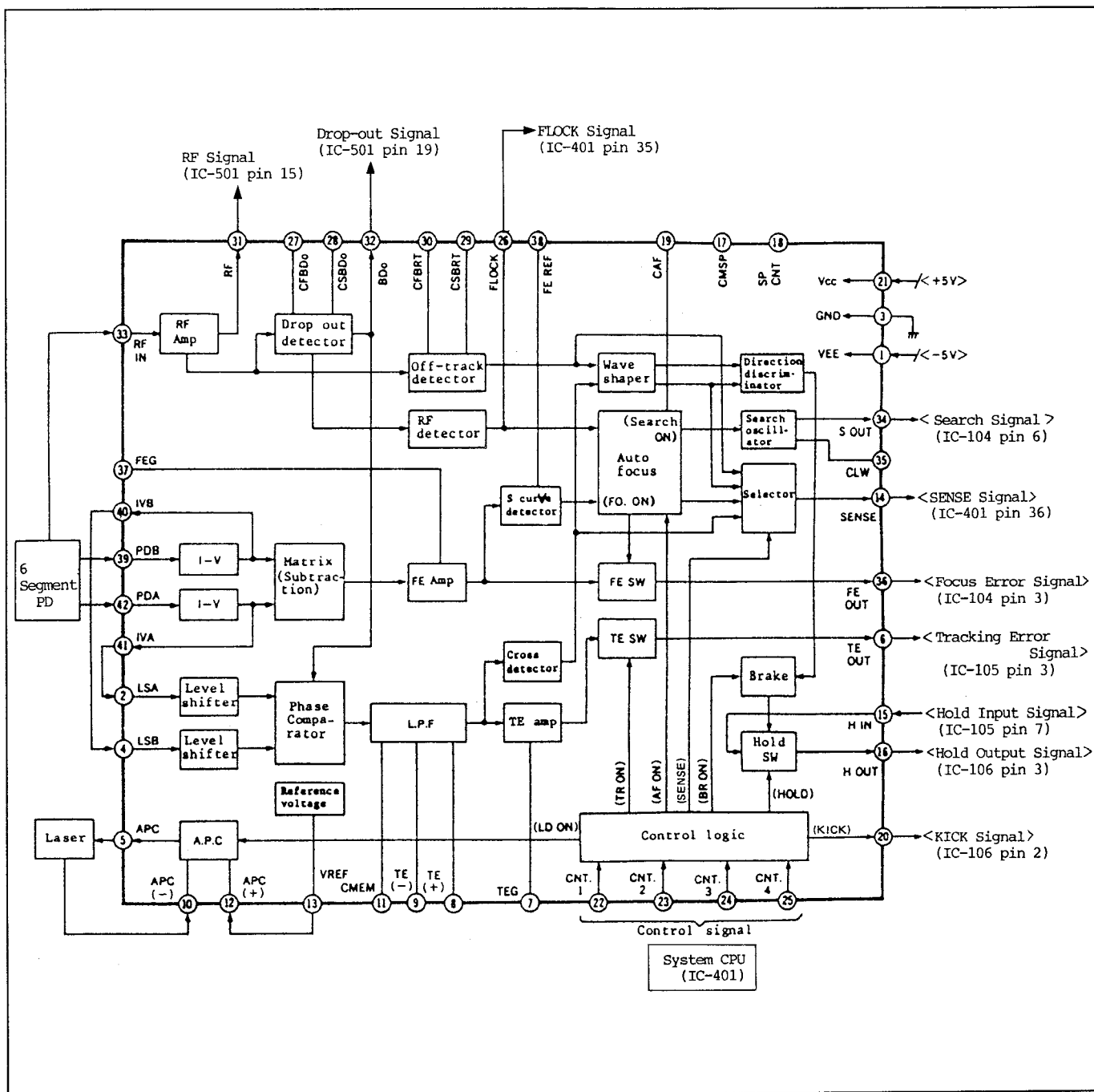


Figure 17-1 Block Diagram of AN8370S (Optical Servo IC)

2) Control Logic Circuit

The control data needed for optical servo system control is supplied from the system CPU in 4 bits parallel lines (CNT1 -CNT4). The control logic within

the servo IC-101 decodes the input control data to provide control commands to the optical servo system. The control logic block diagram shows in figure 17-2

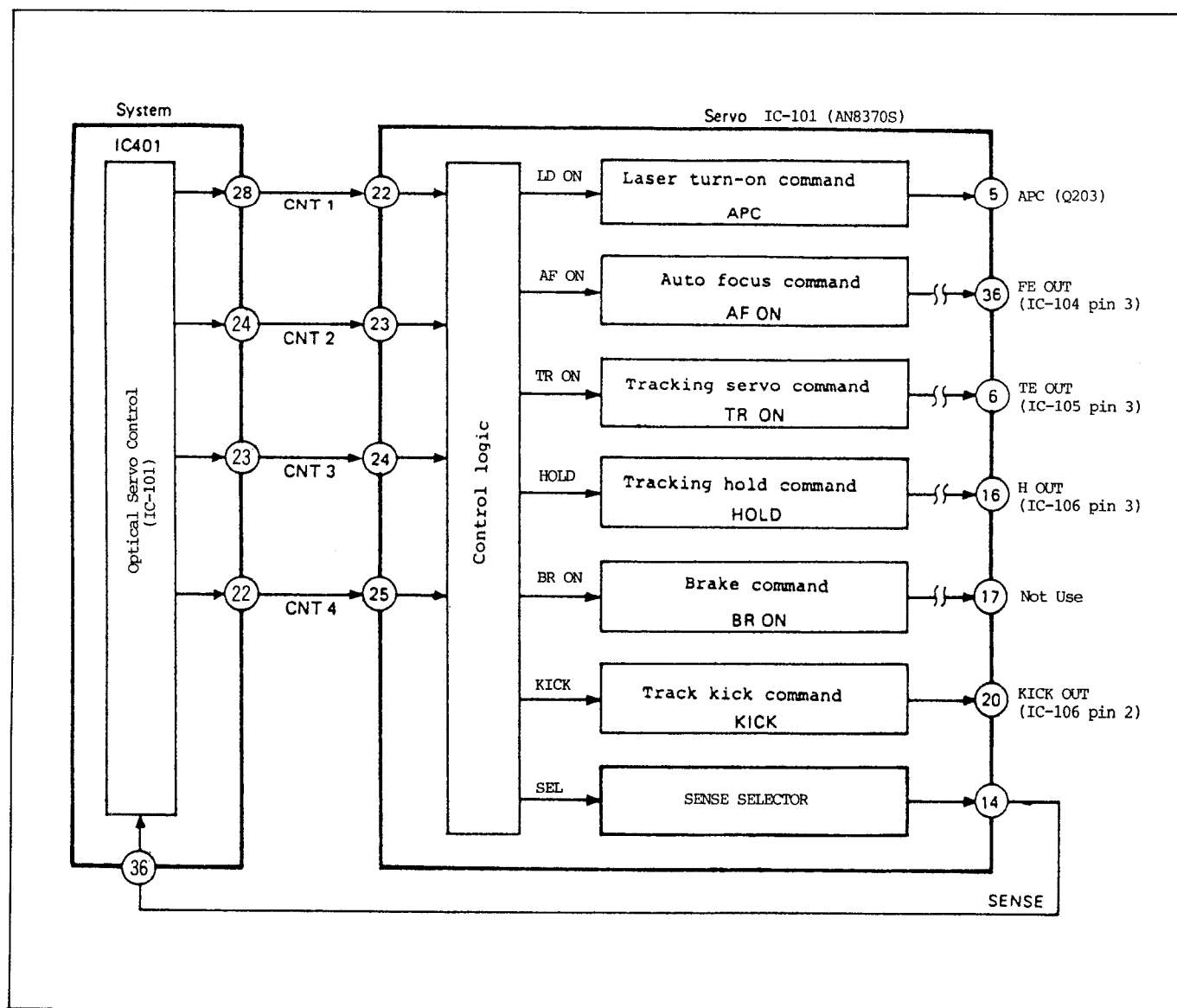


Figure 17-2 Control Logic Block Diagram

The command name and functions as follows:

- (1)LD ON: Laser Ignition Command
This command controls the turning on and off of the laser APC circuit.
- (2)AF ON: Auto Focus Command
This command controls the opening and closing of the focus servo loop, and the on and off of the search oscillation switch.

- (3)TR ON: Tracking Servo Command
This command controls the switched of the tracking servo loop open and close.
- (4)HOLD : Tracking Hold Command
The preceding tracking error is hold in tracking.
- (5)BR ON: Tracking Brake Command
The tracking actuator is braked after quick traverse or KICK for eliminating unwanted vibration in a short time and making looking easy.

- (6)SEL : SENSE Signal Change Command
This command changes the SENSE signal (Confirm the status of the optical servo) transmitted from AN8370S to the system CPU.
The content of the SENSE signal is changed according to the mode of the system.
- (7)KICK : Track KICK Command
This command controls the track KICK output signal. The

KICK is classified to KICK FWD (inside to outside) and KICK REV (outside to inside). They are further divided to one track, 10 tracks, and 100 tracks mode according to the distance of KICK.

The logic circuit is provided with 16 modes of decoding function ($2^4 = 16$) but only 8 modes are used by object model as shows in table 17-1.

(1..."H", 0..."L")

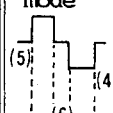
HEX NO	MODE	Input Logic				Control Logic							Remarks
		CNT 1	CNT 2	CNT 3	CNT 4	LD ON	AF ON	TR ON	HOLD	BR ON	KICK	SENSE	
1	PLAY	0	0	0	0	ON	ON	ON	OFF	OFF	OFF	off track	Play mode
2	BRAKE	0	0	1	1	ON	ON	ON	OFF	ON	OFF	off track	Brake mode
3	KICK CROSS	0	1	0	0	ON	ON	OFF	ON	OFF	OFF	Cross	KICK mode 
4	KICK REV	0	1	0	1	ON	ON	OFF	ON	OFF	ON (REV)	Shaped CROSS	
5	KICK FWD	0	1	1	0	ON	ON	OFF	ON	OFF	ON (FWD)	Shaped CROSS	
6	ACCESS	0	1	1	1	ON	ON	OFF	OFF	OFF	OFF	Shaped CROSS	Coarse feed
7	DISC DETECTOR	1	0	0	0	ON	ON	OFF	OFF	OFF	OFF	S-shape curve latch	Disc sense
8	STANDBY	1	1	1	1	OFF	OFF	OFF	OFF	OFF	OFF	Shaped CROSS	Forced return

Table 17-1 4 Bits Parallel Control Data Functions

3) Auto Power Control (APC) Circuit

The APC circuit is difficult to check due to its circuit configuration. This section only describes the theory of its operation.

The system CPU IC-401 yields a parallel 4 bit command signal(see table 17-1) to IC-101 which ultimately controls the APC circuit.

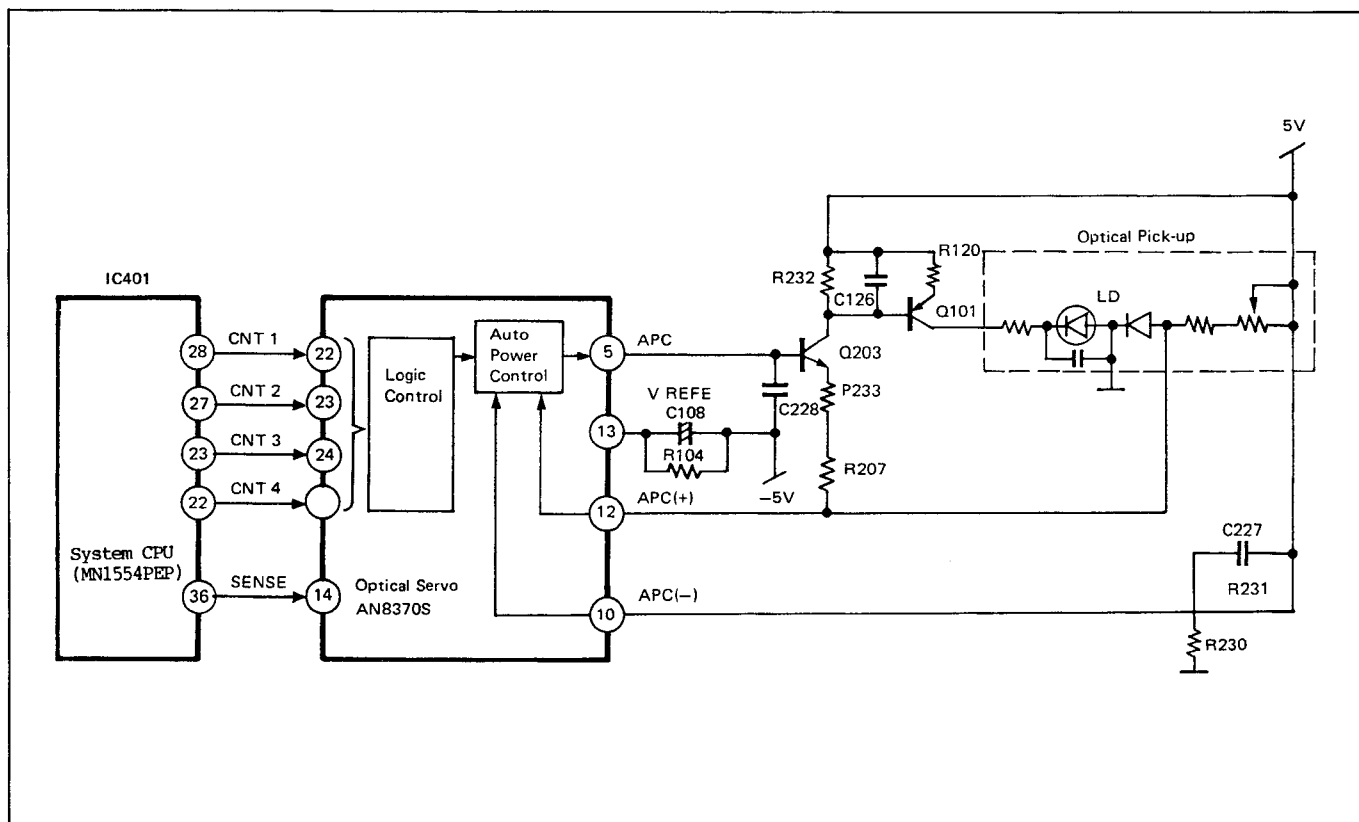


Figure 17-3 Auto Power Control Circuit

When IC-101 receives a Laser On (LD ON) command (H,L,L,L), the logic controller issues a corresponding command to the internal APC block. As a result, pin 5 of IC-101 is set High, which turns Q203, and Q101 "on" and hence activates the LD (laser diode).

A loop circuit is formed by Q101, LD (a pair of the laser and PIN diode), and APC of IC-101. APC is carried out by detecting the quality of laser by the PIN diode. If the beam of laser is intensified, the resistance of the PIN diode is lowered and the voltage at the

pin 10 of IC-101 is raised (the voltage at the pin 12 of IC-101 is the reference voltage) to diminish the intensity of the laser beam.

The APC regulates the LD's output power by controlling the supply voltage according to the output light intensity. The LD's output light intensity is preset at the factory with the potentiometer (VR) in the laser power control circuit. Therefore do not adjust this potentiometer as it requires precision adjustment (or the life of the LD may be shortened).

4) RF Amplifier and Dropout Detector Circuit

(1) RF Amplifier

The RF signal is a summation ($A1+A2+A3+A4+A5+A6$) of the output of all the 6 segment photodetector elements. Summation is performed within the optical pickup. The resulting RF signal is applied to

pin 33 of the servo IC-101, via transistor, where it is amplified by an RF amplifier having gain of approx, 34dB. The output of the RF amplifier appears at pin 31 of IC-101 and couples to the PLL/DATA SLICE circuit IC-501 pin 15.

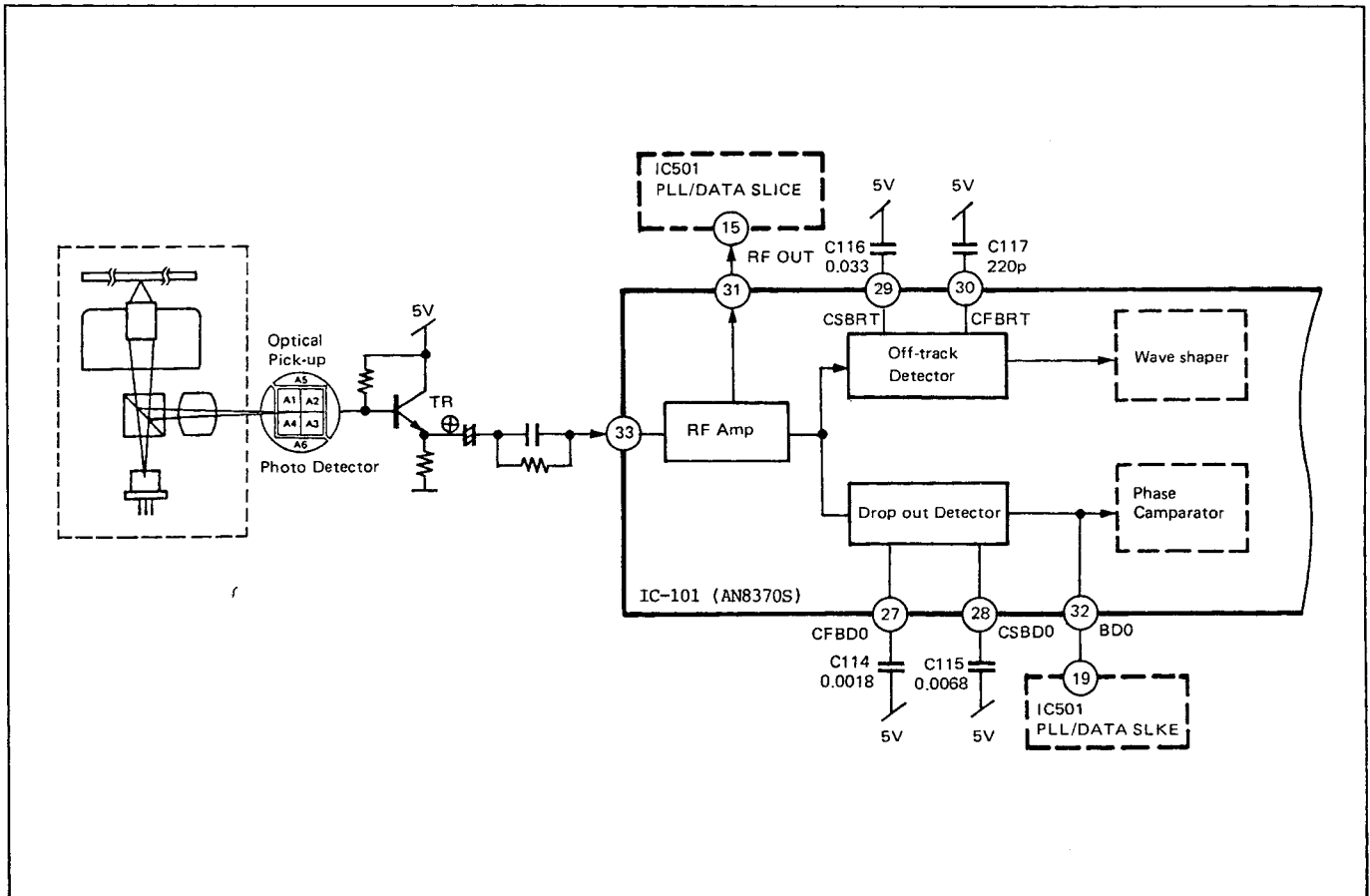


Figure 17-4 RF Amplifier Circuit

(2) Dropout Detector

The dropout detector detects signal dropouts in the RF signal, stabilizes the tracking servo against dropouts, and provides the detection signal to pin 32 of IC-101.

If a dropout occurs in the RF signal, the dropout detector detects it using different time constants condensers connected to pin 27 ($0.0018 \mu F$) and pin 28 ($0.0068 \mu F$). The dropout signal level is "H" to hold error signal before dropout. The dropout signal is applied to pin 32 of IC-101 to IC-501 pin 19.

The Process of detecting dropout is shows in figure 17-5.

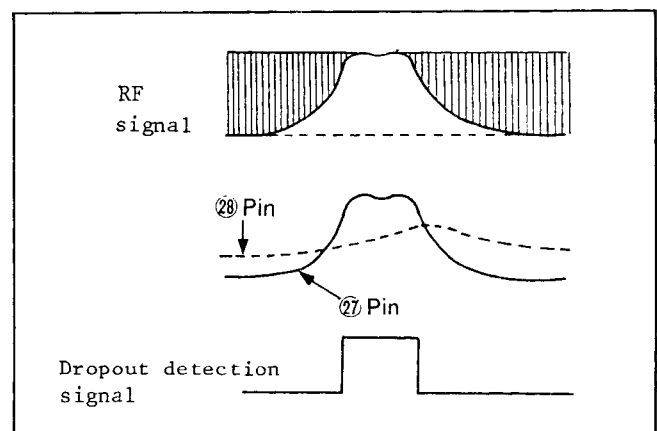


Figure 17-5 Dropout Detection Signal

(3) Off-Track Detector

The off-track detector detects the laser beam being off track. When tracking servo is turned off, the laser beam makes a traverse movement across the tracks during SEARCH, SKIP, mode even track-jump due to condition of the disc surface etc.

As the beam crosses track, the off-track signal is turned on and off in relation to the RF signal, as shows in figure 17-6.

The off-track detector detects off-track using differents time constants condenser connected to pin 29 (0.033 μ F) and pin 30 (220 pF).

The off-track signal is used for the

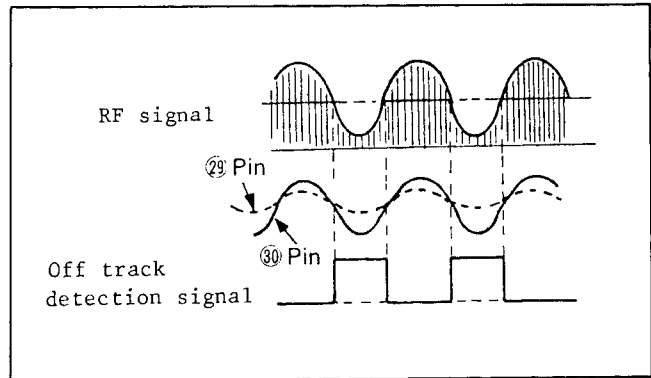


Figure 17-6 Off-Track Detection Signal

servo IC-101 internal control and as the sense signal for the system CPU.

5) Focus Servo Circuit

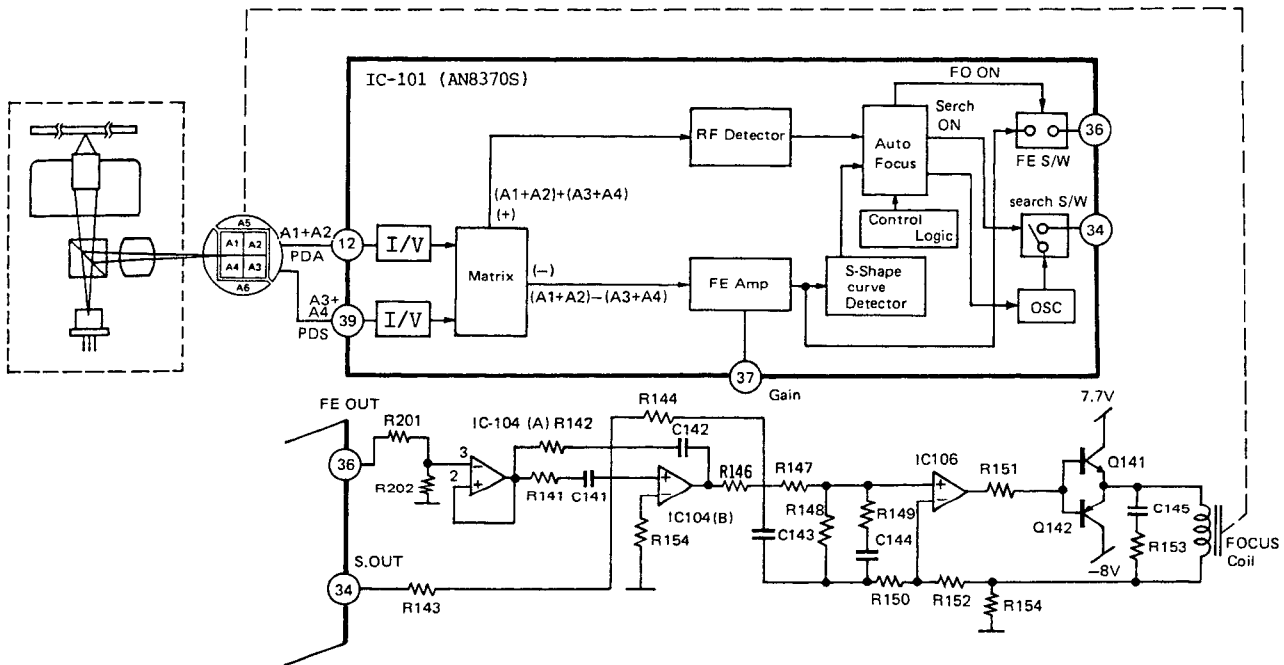


Figure 17-7 Focus Servo Circuit

This servo circuit controls the vertical movement of the 2-axis device and guarantees that the focus point of the laser beam is precisely on the mirror surface of the compact disc.

the main spot detector. This photodetector actually is an array of four photo-diodes, which are A1, A2, A3, and A4. The reflected laser beam hits these four diodes and generates a four error signal:

$$\text{Focus Error (F.E)} = (A1+A2)-(A3+A4)$$

Figure 17-8 shows the three possible situations and the resultant output from

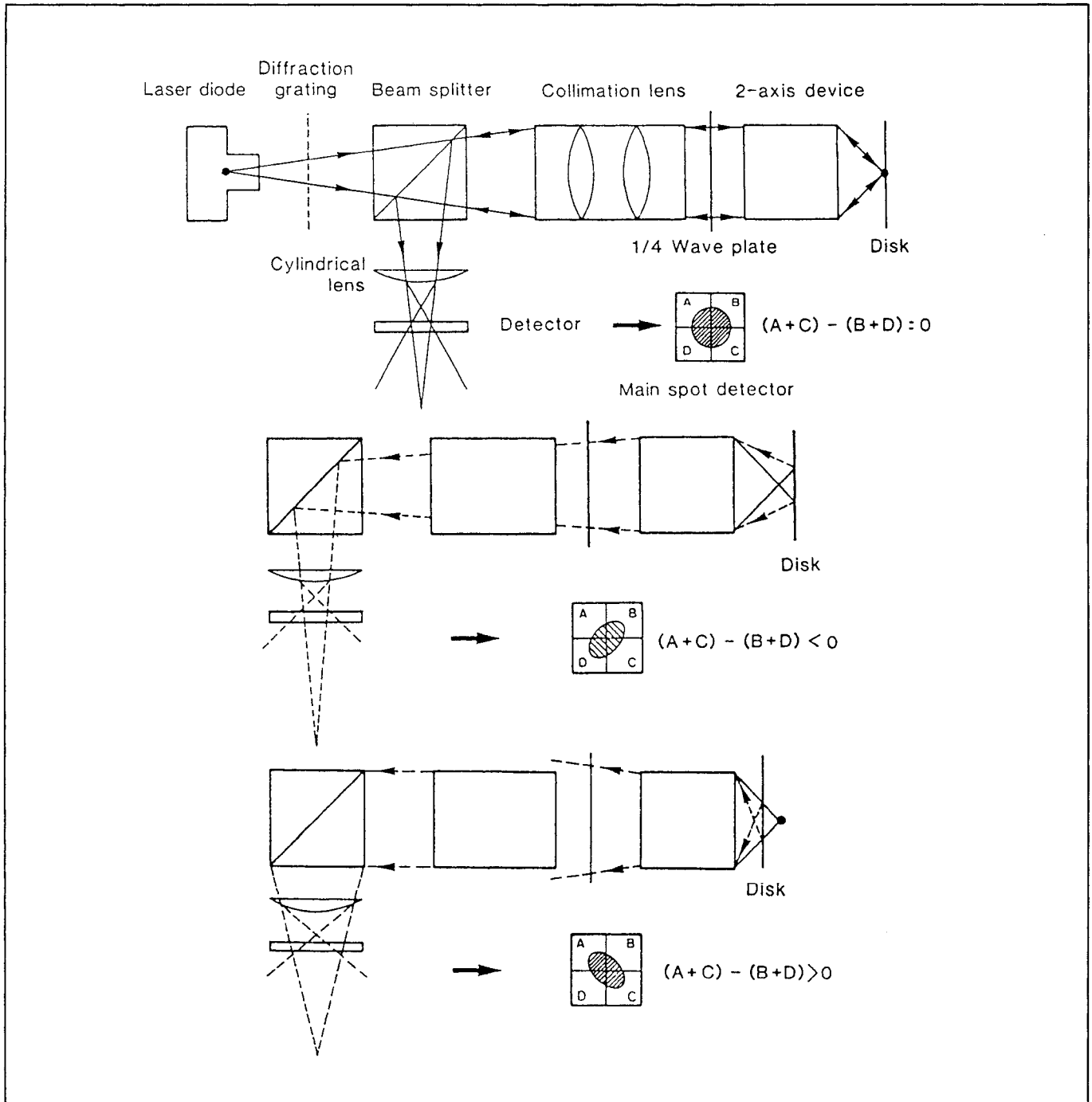


Figure 17-8 Detection of Focus Error

(1) Autofocus Circuit

The autofocus circuit is operated by the AF ON (autofocus command) signal output from the control logic circuit. Two signals (RF detection signal and S-curve detection signal) are input to the autofocus circuit which regulates the search switch and FE (focus error) switch.

1 At detection of the disc

On the receipt of the AF ON (autofocus on) signal, at first the search switch is turned on and next the focus error switch is turned off. In this situation, the search signal (sawtooth waves) drives the focus coil forcedly to move the lens up and down.

When the distance between the disc and the lens nears the focus length (lock-in focus range), the FE signal shows an "S" curve, with the search switch is turned off and the focus error switch is turned on to close the focus servo loop, as shows in figure 17-9.

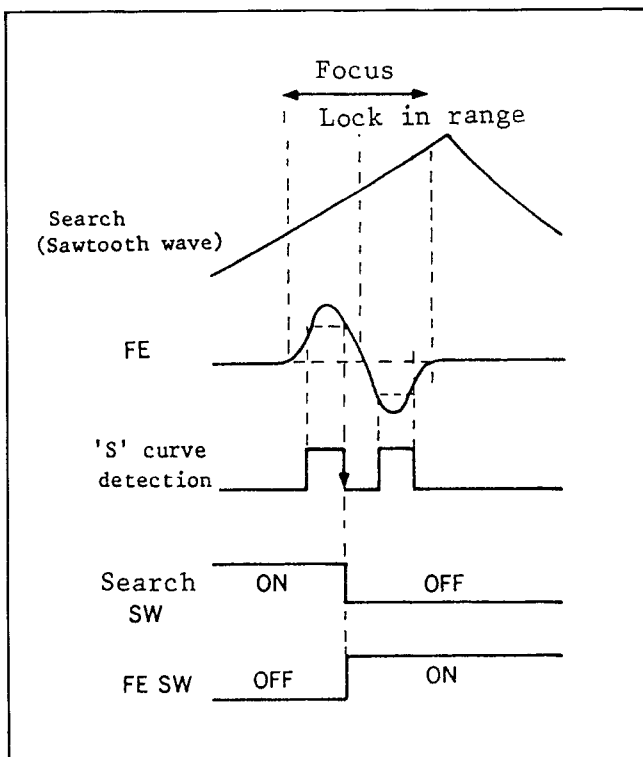


Figure 17-9 Focus On Timing

2 Re-locking in focus

The autofocus circuit checks the servo status by RF detection when the focus servo loop is closed. If the disc deviates from the focal point and RF search oscillation is started. The search oscillation signal is input to the system CPU. Whether or not autofocus is working is known by the voltage at the pin 26 (FLOCK) of IC-101. The voltage level is active "L" when locked in focus.

(2) Focus Error Circuit

The sums of output signals from pair of diagonal segments each ($A1+A2$) and ($A3+A4$) of the 4 segment PD are applied to the pin 39 and 42 of IC-101. The I/V converter converts the output currents of PD to voltages.

The gain is set by an exterior resistor, and VR101 is the best eye adjuster for setting off the sensitivity difference of PD.

The difference between the converted two signal ($A1+A2$) - ($A3+A4$) is focus error signal (FE). The FE signal applied to the pin 36 (FE OUT) of IC-101 via the FE amplifier and FE switch, and applied to the pin 3 of IC-104 (A).

VR104 is the adjuster of focus gain (FO GAIN) and VR105 is the adjuster of focus off set (FO OFFSET).

The FE signal applied to the focus drive circuit from pin 36 of IC-101, The first subjected to phase compensation by R142, R141, and C141. It then goes through a low-pass filter comprised of IC-104 (B), R144, C142, and C143, and then an amplifier (with low frequency compensation) comprised of IC-106, R149, C144, and R148.

The output of the amplifier is applied to current-feedback focus drive, Q141 and Q142, which then drives the focusing lens drive coil as required, as shows in figure 17-10.

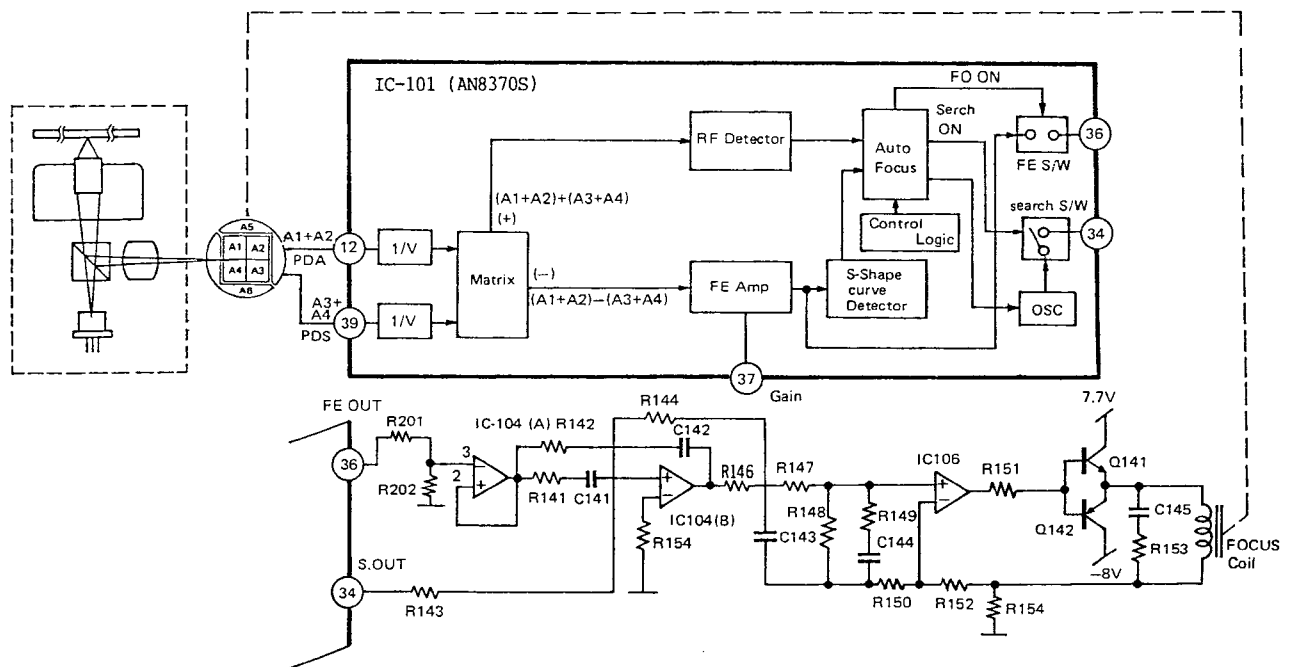


Figure 17-10 Focus Servo Circuit

6) Tracking Servo Circuit

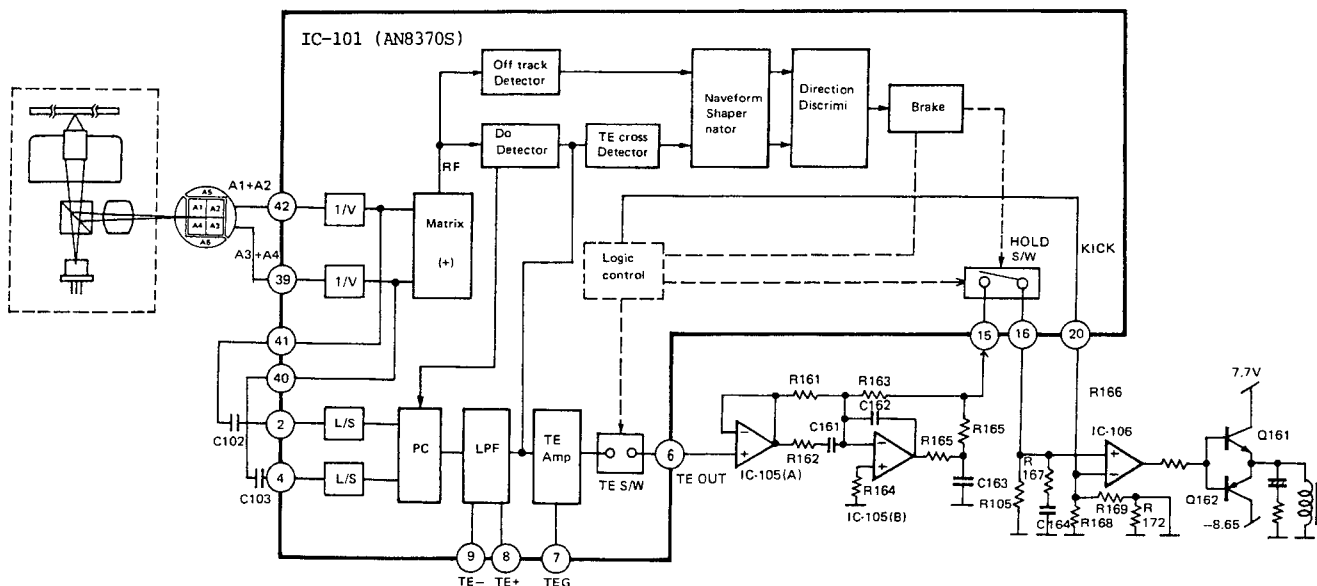


Figure 17-11 Tracking Servo Circuit

The tracking servo circuit accepts parallel 4 bits control data transferred from the system CPU of IC-401 and drives the entire tracking servo system, according to the CPU commands, after focus servo is established.

After focus servo is established, the tracking error (TE) signal is created by applying the diagonal PD output signals (A1+A2), (A3+A4), into the servo system IC-101 pin 39 and pin 42 (same as focus servo).

With this, the tracking servo circuit identifies whether the laser beam is on track or off track, by picking up an error signal that corresponds to the off-center span of the disc. The internal control logic closes the tracking servo loop switch (TE Switch) when the beam is on track.

The specified track is sought with the track signal. Since the TE signal is not needed during the track KICK sequence, the circuit closes the Hold switch to temporarily hold the previous tracking error signal.

(1) Tracking Error Circuit

The tracking error circuit applies the I/V converted signal as in the preceding paragraph to the pin 2 and pin 4 of IC-101 via C102 and C103, and shapes the waveforms of the signals by the level shifter.

As outlined in the para. , the difference in phase between the two signals is detected and pulses in proportion to the phase difference are generated as the tracking by the polarity of the pulses.

The analog tracking error (TE) signal is obtained by smoothing the above mentioned generated pulses with LPF. The above processes are carried out by the phase comparator and LPF.

VR102 is the adjuster of the tracking gain (TR GAIN) and VR103 is the adjuster of the tracking offset (TR OFFSET).

The TE signal appears at pin 6 via TE switch. The output TE signal is first subject to phase compensation by R61, R162, and C161, then goes through a low-pass filter comprised of IC-105 (B), R165, R163, C162, and C163, after that finally couples to pin 15 of the servo IC-101, where it is applied to the KICK and HOLD circuit. As shows figure 17-11.

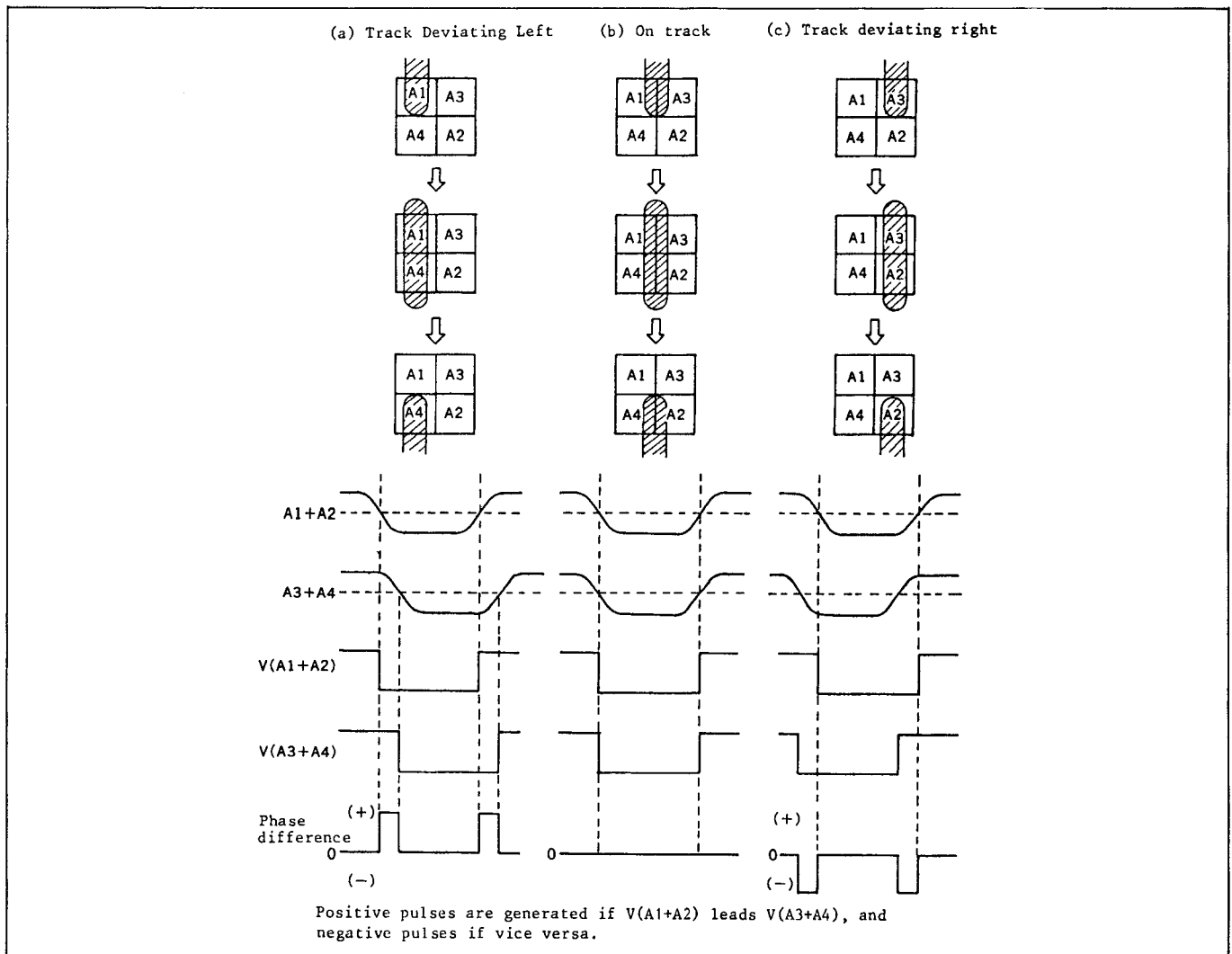


Figure 17-11

(2) KICK/ HOLD Circuit

The KICK signal is to skip the laser beam to other track in the pause or the manual search mode.

The timing of KICK is set by the logic control signal from the system CPU IC-401, and the KICK signal appears at the pin 20 of IC-101. The waveform of KICK signal are as shows figure 17-12.

The HOLD circuit retains the preceding TE signal during KICK mode, also the TE switch is turned off during KICK mode, Figure 17-13 shows the waveform of signal in pause KICK.

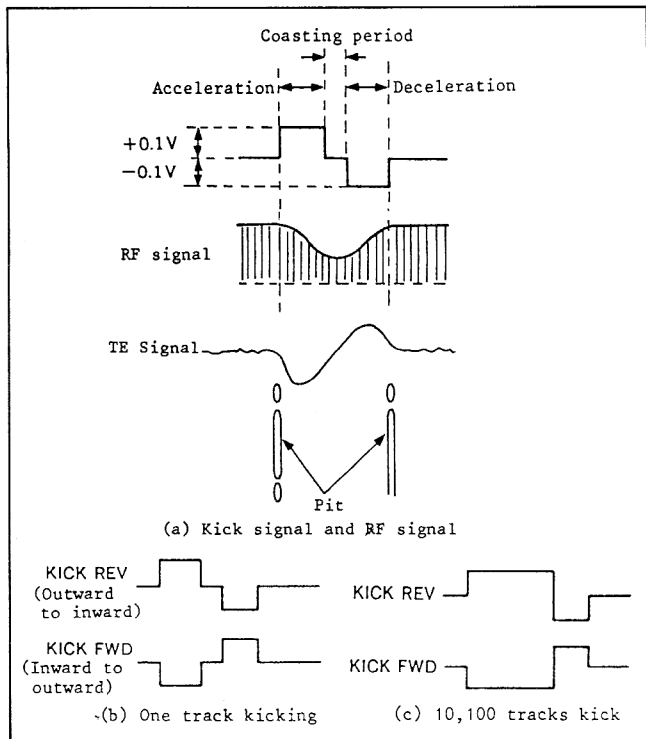


Figure 17-12 KICK Signal and RF Signal

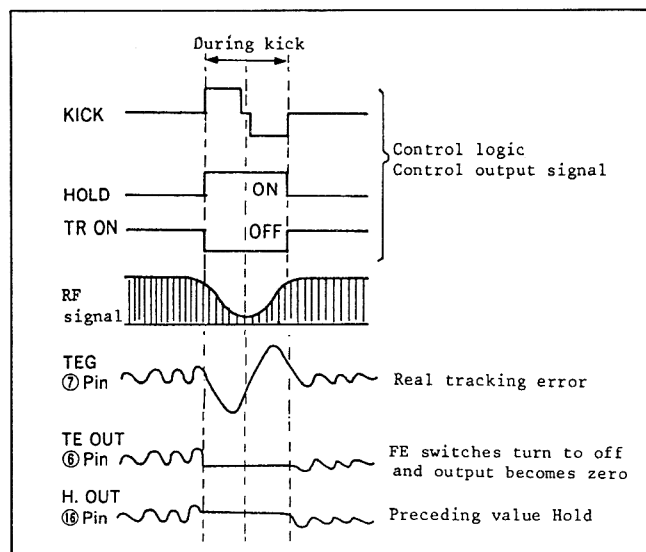


Figure 17-13 PAUSE KICK Mode

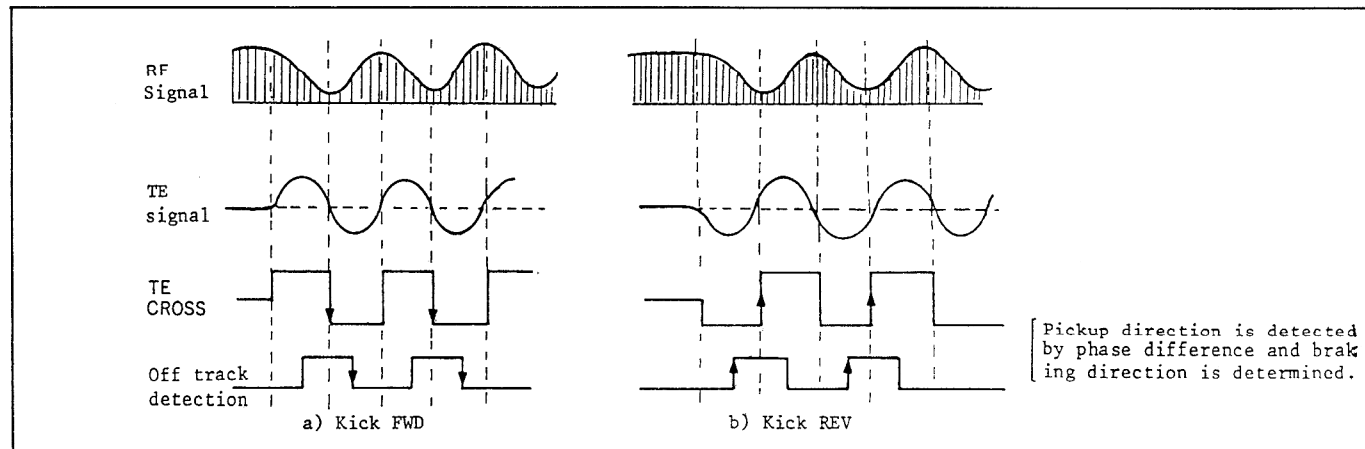


Figure 17-15 KICK Direction Discrimination

(3) Braking Circuit

The brake circuit brakes the tracking actuator after quick traverse or KICK (10 or 100 track) for rapidly locking the tracking servo. The principle of the brake circuit as shows in figure 17-14.

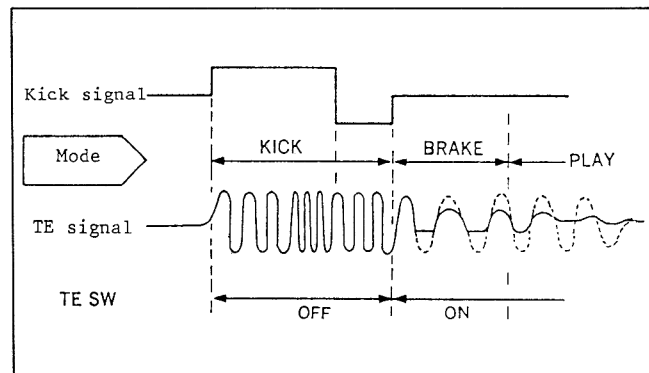


Figure 17-14 Brake Timing

In braking, the TE signal is limited on one side (the continuous line shows in figure 17-14) to absorb energy (of the actuator) after KICK.

The figure 17-14 shows braking in the KICK REV mode. In the KICK FWD mode, the braking is made by reversing the polarity of the sideband limiter. Whereas, the brake circuit requires a signal for discriminating the direction of KICK.

The KICK direction discriminating signal is formed by the tracking cross (CROSS) signal and off-track signal as shows in figure 17-15.

The phase of the CROSS signal leads in the KICK FWD mode and lags in the KICK REV mode. The brake direction is determined by the detection of the above phase difference.

(4) Tracking Drive Circuit

The TE signal appearing at pin 16 of IC-101 is applied to an amplifier (with low-frequency complifier) comprised of IC-106, R105, C164, and R167. The output of the amplifier couples to the current feedback type tracking drive circuit

comprised of Q161, and Q162 which then drive the tracking drive coil.

7) Relation with Servo IC and System CPU

The figure 17-16 shows the interconnection diagram of servo IC-101 (AN8370S) and system IC-401. There are four buses for carrying signals from the system IC to AN8370S and two buses for carrying signal from AN8370S to the system IC.

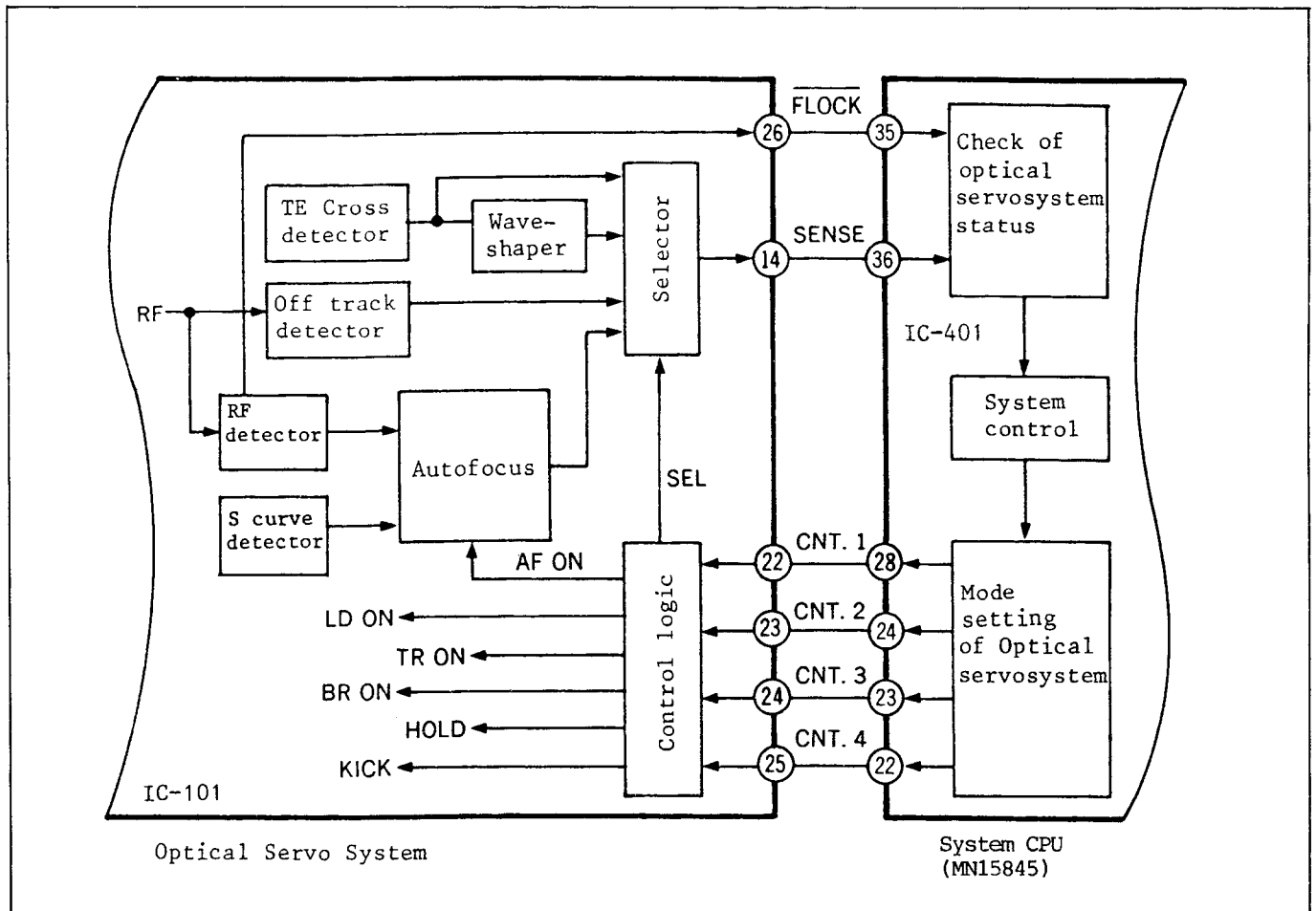


Figure 17-16 Connection Diagram of System CPU and Optical Servo System

- (a) Output signal from system:
-Control logic signals (CNT1 -CNT4)
- (b) Input signal from system:
-RF detection signal (FLOCK).....
...Active "L"

- Servo system status signal (SENSE)..
Type of SENCE signal:
(1)Off-track signal
(2)Tracking cross signal
(3)Waveshaped tracking cross signal
(4)Focus S curve detection signal.

8) Traverse Servo Circuit

The 2-axis device can move horizontally and thus follows a limited number of tracks (fine tracking), Another servo circuit, which called the traverse servo circuit, is used to move the complete optical unit across the disc surface (course tracking).

This servo circuit uses the same tracking error signal due to normal playing mode (instead of the SKIP mode only) as the feedback input, the output of the tracking servo circuit is linearly related to the tracking error signal, whereas the output of the traverse servo circuit has a built-in hysteresis.

Only when the TE signal exceeds a fixed threshold level, the traverse servo drives the traverse linear motor. This is shown in figure 17-17.

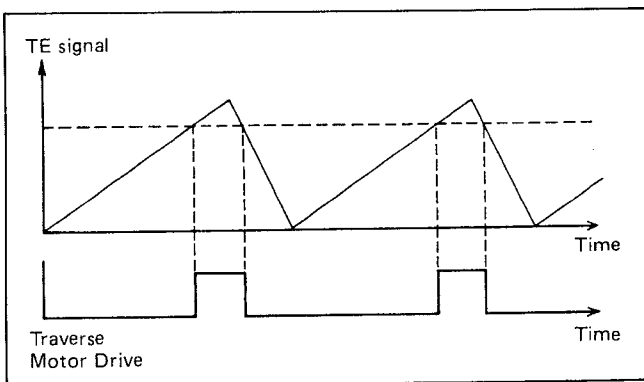


Figure 17-17

(1) Traverse Servo Operation in Play Mode

- The TE signal is supplied from the tracking drive circuit to pin 2 of IC-102.
- The TE signal selected by IC-102 due to play mode, in case of the search mode, the TE signal is ground by IC-102 pin 2 through pin 3.
- In case of play mode, the TE signal is sent into the IC-103 (C) pin 9 via IC-102 pin 1.
- The output of IC-103 (C) pin 8 is used to set the base potential of the traverse driving transistors Q181 and Q182 to determine the direction of the traverse movement.

- The base potential of Q181 and Q182 is set to positive or negative, according to the TE signal polarity to control the traverse drive current, so that in turn tracking error is minimized.
- The traverse drive mechanism has a speed position sensor (potentiometer) which tracks the optical pickup movement to provide pickup position information. The voltage range which varies with the speed position sensor, is set between 0 volt corresponding to the beginning of track one and 5 volts corresponding to the end of the last track.
- The speed position sensor signal (0 to 5 volts) that tracks the optical pickup movement is applied to pin 3 of IC-103 (A) and then to pin 6 of IC-103 (B), where it is voltage-differentiated. This voltage is used to brake for traverse movement, due to mis-calculation of the tracks.

(2) In KICK Mode

The control signal coming from pin 20 of IC-101 either KICK REW or KICK FWD depends on KICK plus like 10 kick or 100 kic etc.

The control signal applied pin 2 of IC-102 via tracking servo circuit, when KICK mode is necessary, after that operation same as above c) through e).

(3) In SKIP Mode

The control signal coming from either pin 20 (TRV REV) or pin 21 (TRV FWD) of the system CPU IC-401, at that time even this signal selected IC-102 analog switch to ground the TE signal in normal play mode and, instead, applies either the FWD or REV SKIP signal to IC-103 (D) to drive the traverse mechanism. The subsequent operations of the traverse mechanism are identical to above step c) through e) in the previous paragraphs.

The figure 17-18 shows traverse servo circuit.

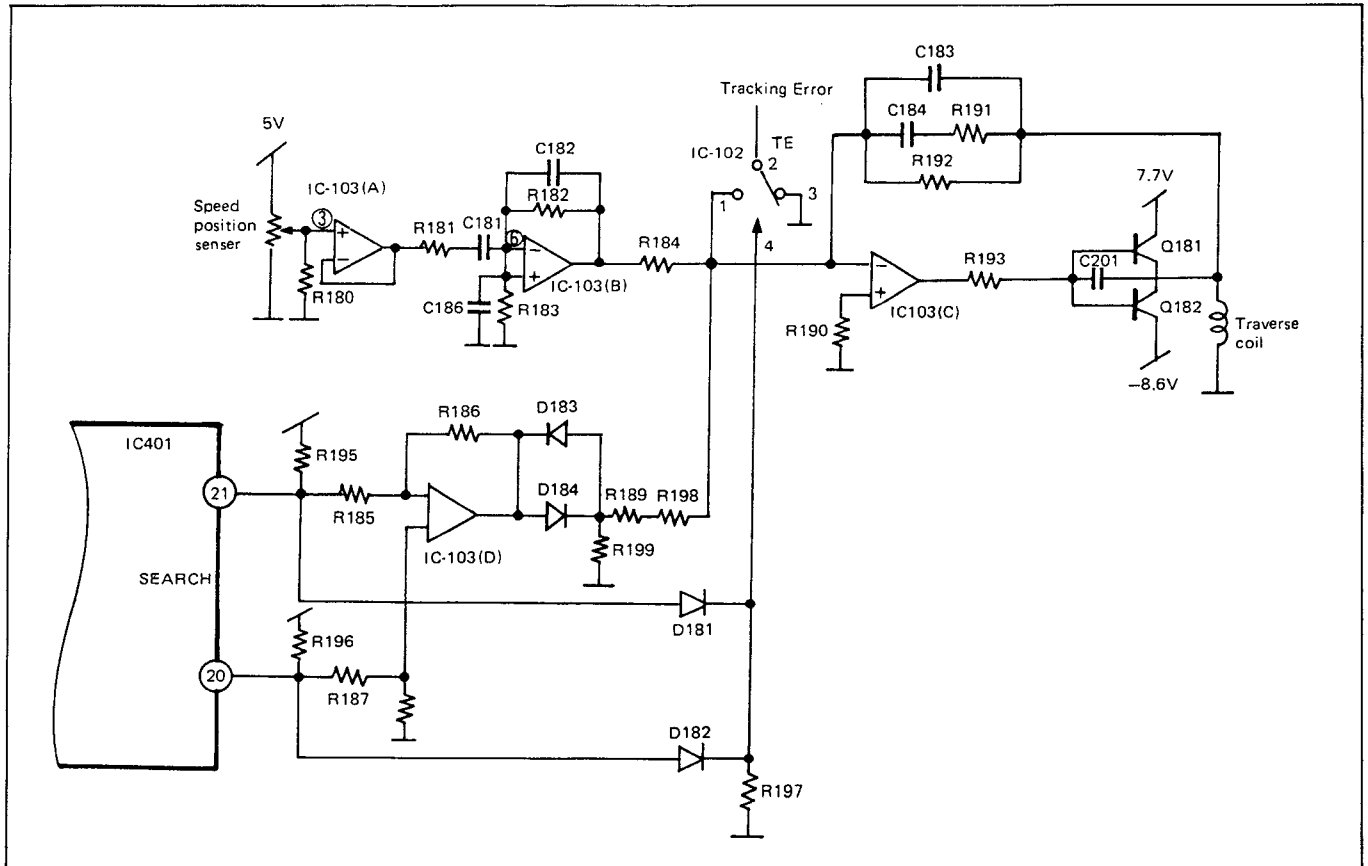


Figure 17-18 Traverse Servo Circuit

9) Linear Motor Operation

A linear motor is used for the optical pickup transportation mechanism in the new CD players. The figure 17-19 shows

the optical pickup transportation mechanism. The linear motor drive coil (traverse coil) is a part of the optical pickup and performs transportation operation.

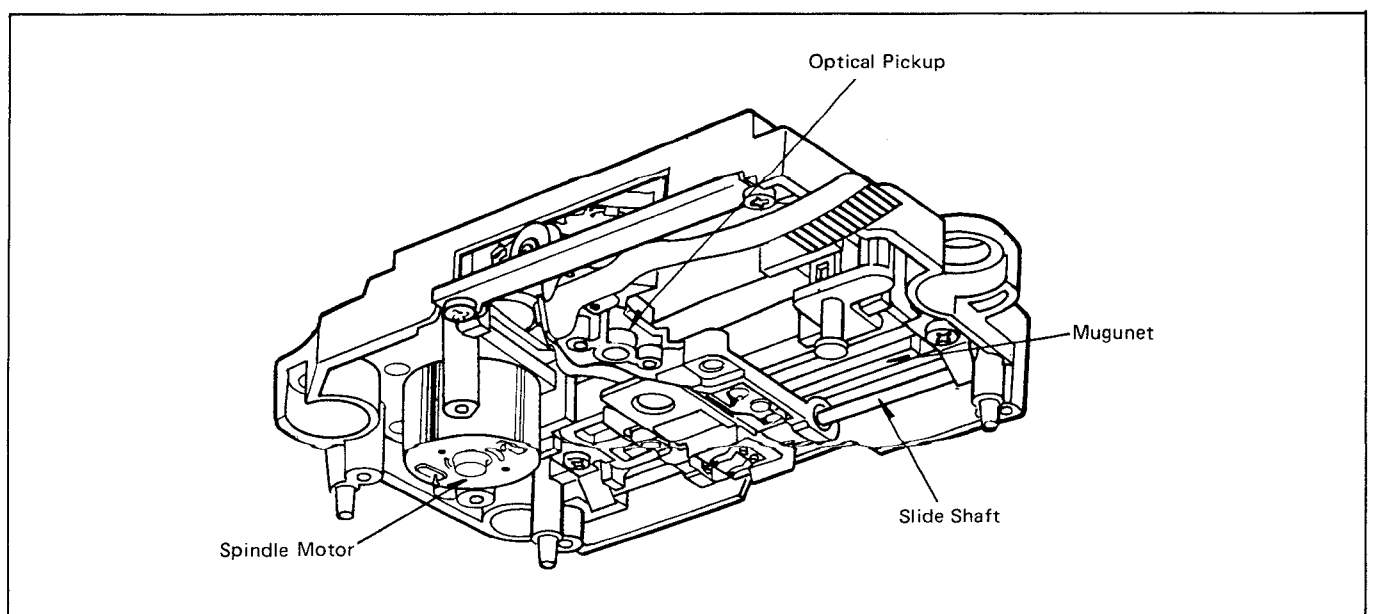


Figure 17-19 Optical Pickup Transportation Mechanism

The linear motor consists of a coil wound around a yoke plate and a magnet at both end. The coil undergoes an influence from a force field and will

move across the yoke depending on the direction of the current through the coil.

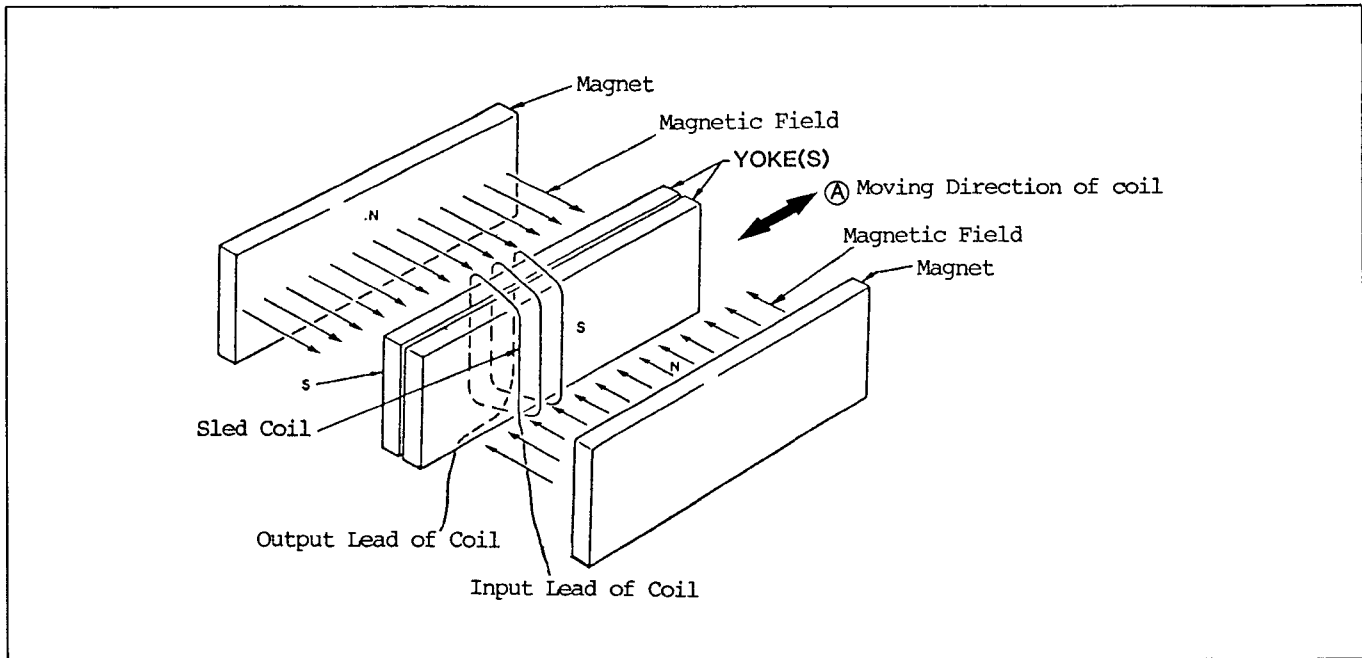


Figure 17-20 Liner Motor Principle

The traverse servo circuit receives the tracking error signal at pin 2 of IC-102. The servo circuit will output a traverse signal to correct the position of the optical pickup to keep it on the pit track.

To perform a 1, 10, or 100 track jump, (SKIP) the command from the system IC-401 will be controlled at IC-103, the servo control, and this circuit will drive the traverse motor. This allows a 1, 10, or 100 track jump to be executed very fast and smooth.

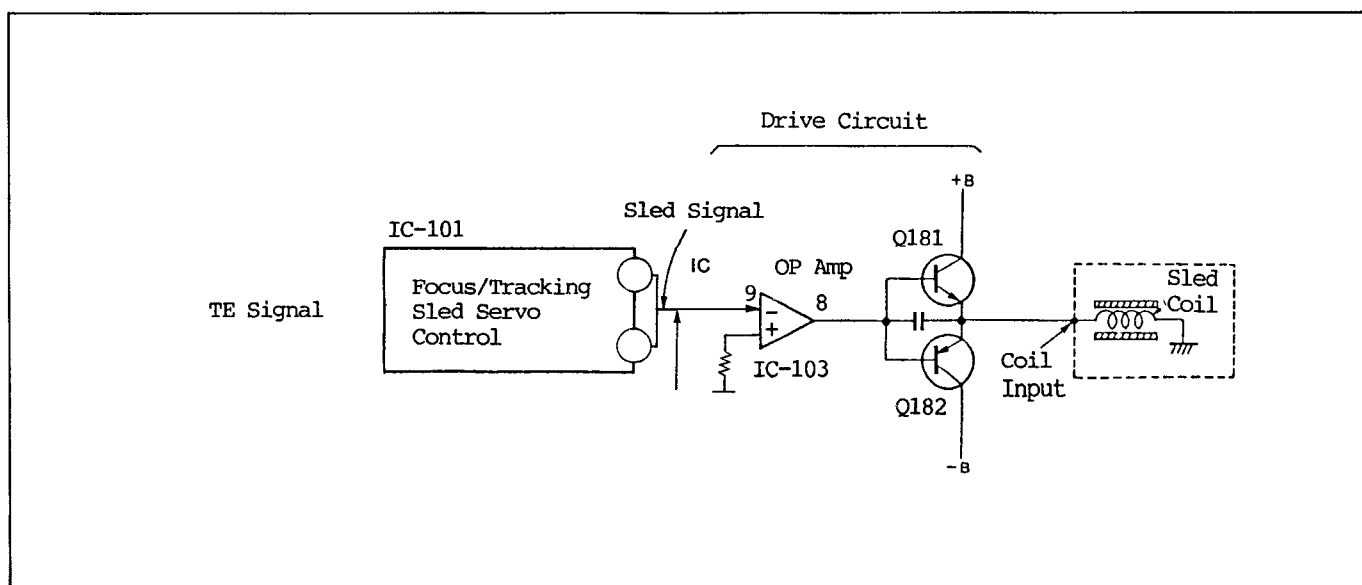


Figure 17-21 Linear Motor Servo Circuit Principle

(1) Principle of the Linear Motor

As can be seen from figure 17-20, a linear DC motor is in fact a coil suspended in a magnetic field.

When a current I is sent through the coil, a force will be applied onto the coil and the latter will move accordingly. The direction of the motion according to the force applied, is determined by Fleming's left hand rule.

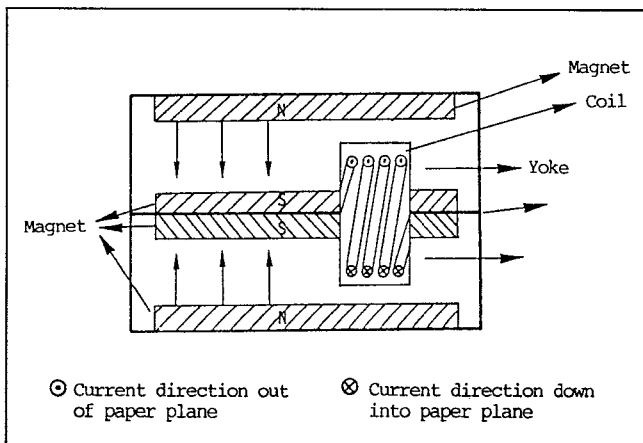


Figure 17-22 Linear Motor Principle

The index finger points in the direction of the magnetic field, whereas the middle finger points in the direction of the current through the coil, and the thumb indicates the direction of the motion of the coil as shown in figure 17-22 and figure 17-23.

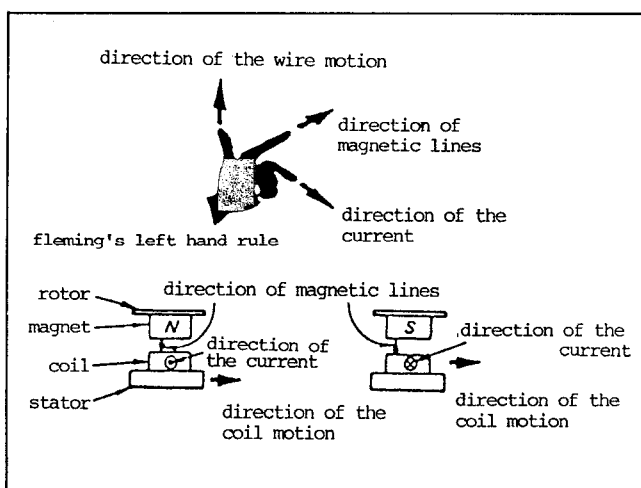


Figure 17-23

10) Spindle (CLV) Servo Circuit

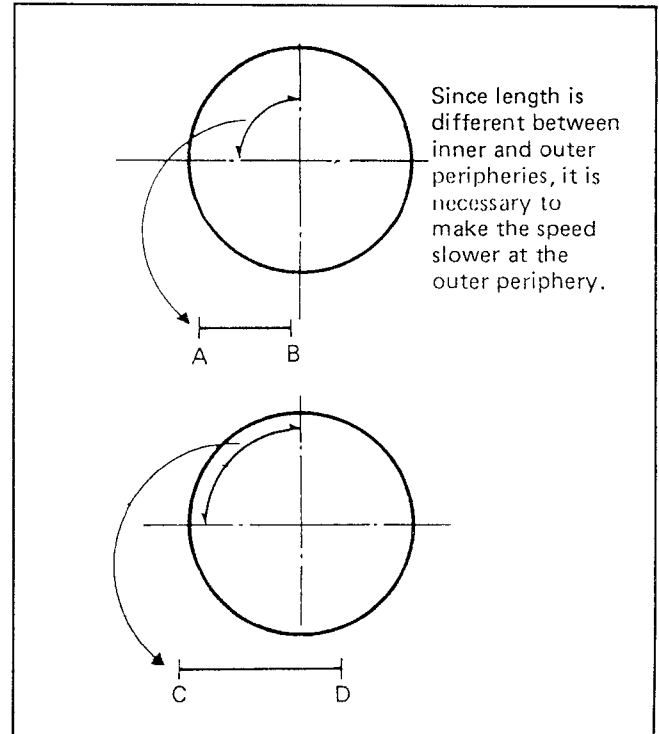


Chart 17-1 Arc Length at Different Radii

What is a Linear Velocity?

The compact disc uses a constant linear velocity (CLV) drive system. This means that the disc speed changes to maintain a constant reading speed approx 1.3 m/s. In figure 17-1, shows the distance from A to B is shorter than from C to D.

To read this data at a constant speed the disc must rotate slower at the outer periphery. The range is 500 rpm through 200 rpm from inner to outer periphery. The proper speed is maintained by reading the synchronizing signal (7.35KHz) and comparing it to a reference signal.

As pointed out before, the data on the disc is recorded in blocks, which we called "frame". Each frame starts with a unique sync word, which is used to control the spindle motor.

The frequency of these sync words is compared against a fixed frequency (derived from a crystal oscillator) in a phase comparator and the motor is driven according to any frequency or phase difference.

As the length of the tracks increases linearly from the inner (lead-in) track to the outer (lead-out) track, the number of frames per track increases in the same respect. This means that the frequency of the sync words also increases, which causes the motor speed to decrease, resulting in a constant linear velocity. The angular velocity typically decreases from 500 RPM (lead-in track) to 200 RPM (lead-out track).

A run-up circuit is activated when the disc is initially located to set the motor speed at about 500 RPM. This makes it possible to read the sync words and hence to enable the servo circuit.

(1) Correlation with IC-301, IC-401, IC-501

The CLV servo circuit controls disc rotation to a constant linear velocity. It includes three servo modes, i.e. CLV (fine), the 11T, and 6T.

- ** 6T = Traverse Tracking
- ** 11T = Coarse Servo Control
- ** CLV = Provides Fine Servo Control

The major block of the Spindle Servo Circuit are implemented in the Digital Signal Processor IC-301, System CPU IC-401, Data Slice & PLL IC-501 as shows in figure 17-24

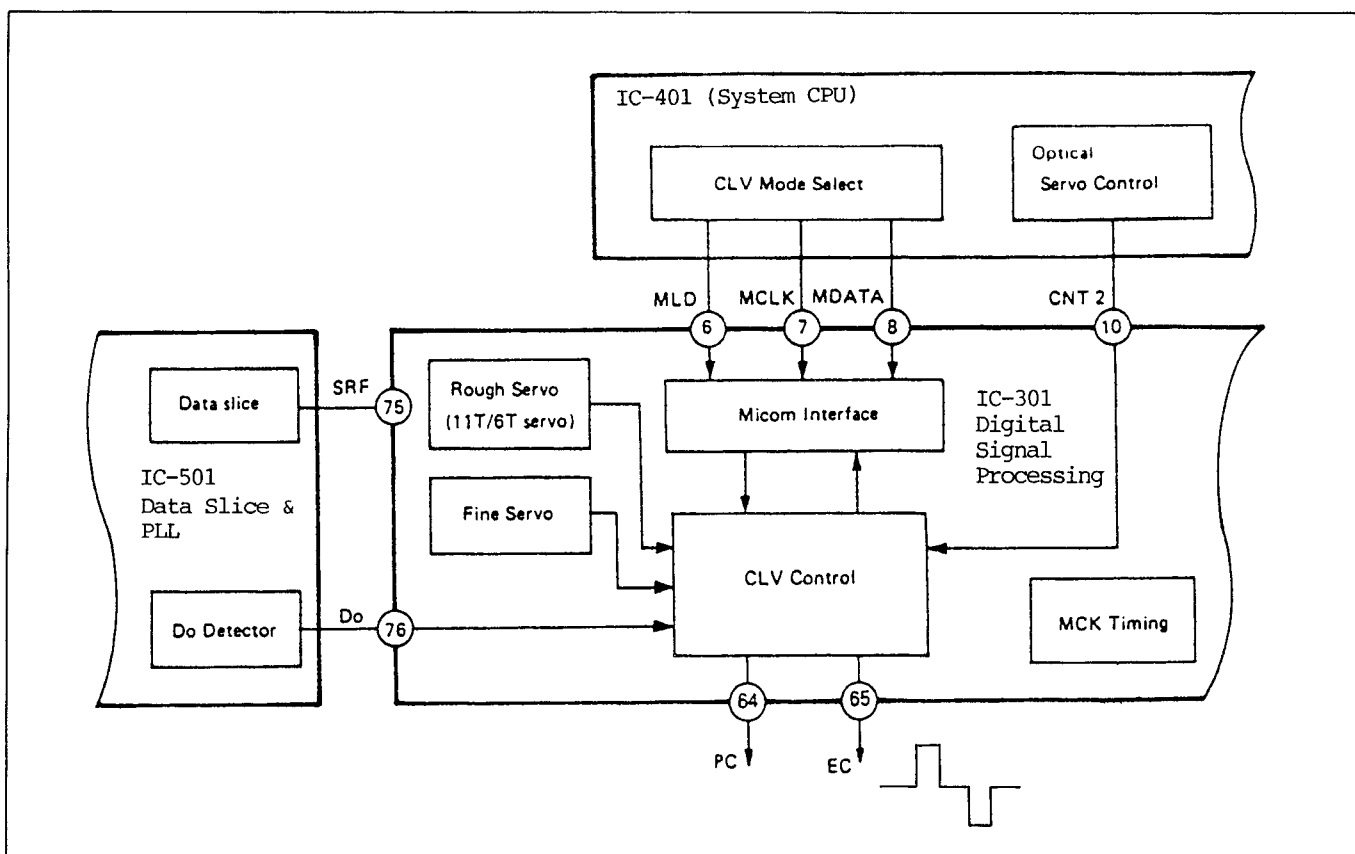


Figure 17-24 Connection Diagram of System CPU, Digital Signal Processing and PLL/Data Slice

(2) CLV Servo Stepping with Play Mode

(a): 11T Servo

The 11T servo controls the SRF signal (data-sliced RF signal) so that the signal's maximum period is 11T time the reference clock (1T=4.3MHz) period.

(b): CLV Servo

The CLV servo initiates its operation after IC-301 is made ready for internal digital processing by the 11T servo.

(3) CLV Servo Stepping with SKIP Mode

(a): 6T Servo

The 6T servo regulates the linear velocity by tracking traverse position during SKIP traverse mode.

(b): 11T Servo

Same as above (a)

(c): CLV Servo

Same as above (b)

(4) Spindle Motor Drive Circuit

Figure 17-25 shows the actual spindle motor drive circuit.

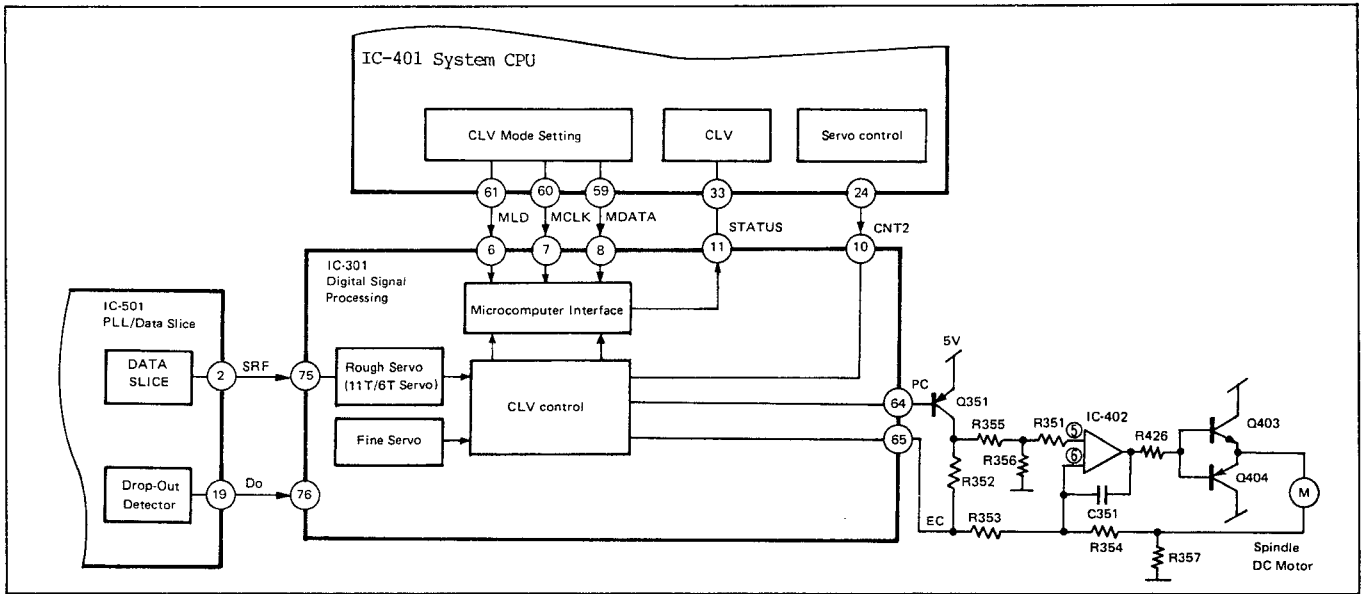


Figure 17-25 Spindle Motor Drive Circuit

During the process, from start-up to CLV, the servo steps through three stages:

- (a) Constant Rotation (Start-up)
- (b) 11T Servo (Coarse Control)
- (c) CLV Servo (Fine Control)

1). Operation from Start-Up to Play:

- (a) The MDATA line from pin 59 of IC-401 issues a command (TTON) to pin 8 of IC-301 to start rotation of the spindle motor via the CLV servo within IC-301.

The MDATA signal is a serial data line consisting of a 4 bits address and 4 bits data.

Mode = CLV

Address = A4 A3 A2 A1
0 0 1 0

Date = D4 D3 D2 D1
H:W/FG H:TTIN 00 STOP
L:W/O FG L:TTOF 01 ACC
10 BRAKE
11 NO-Ope

- (b) Pin 64 of IC-301 is set "L" activating the spindle motor drive circuit, then Q351 turned on, the 5 voltage will be divided 2.5 voltage by R355 and R356 and into the pin 5 of IC-402. The 2.5 volts is reference voltage for the spindle motor drive circuit.

The 11T servo mode is selected when the tracking servo is active (TRON signal/CONT2 is "Low") and the PLL is unlocked.

- (e) Until the frequencies are synchronized, the EC signal at pin 65 of IC-301 provides acceleration or deceleration commands to the spindle motor drive circuit as shows in figure 17-26.

- (c) The synchronization circuit within IC-301 compares the disc's sync frequency clock CLDCK generated from EFM and the PCK signal from pin 5 of IC-501.

If the PCK frequency is higher or lower than that of EFM, the FPC signal (pin 78 of IC-301) is set "High" or "Low" respectively.

(The PLL circuit within IC-501 adjust the PCK frequency according to the FPC level)

- (d) The "Rough Servo" within IC-301 switches from the constant rotation servo to the 11T servo. Depending on the status of the SRF signal derived from pin 2 of IC-501.

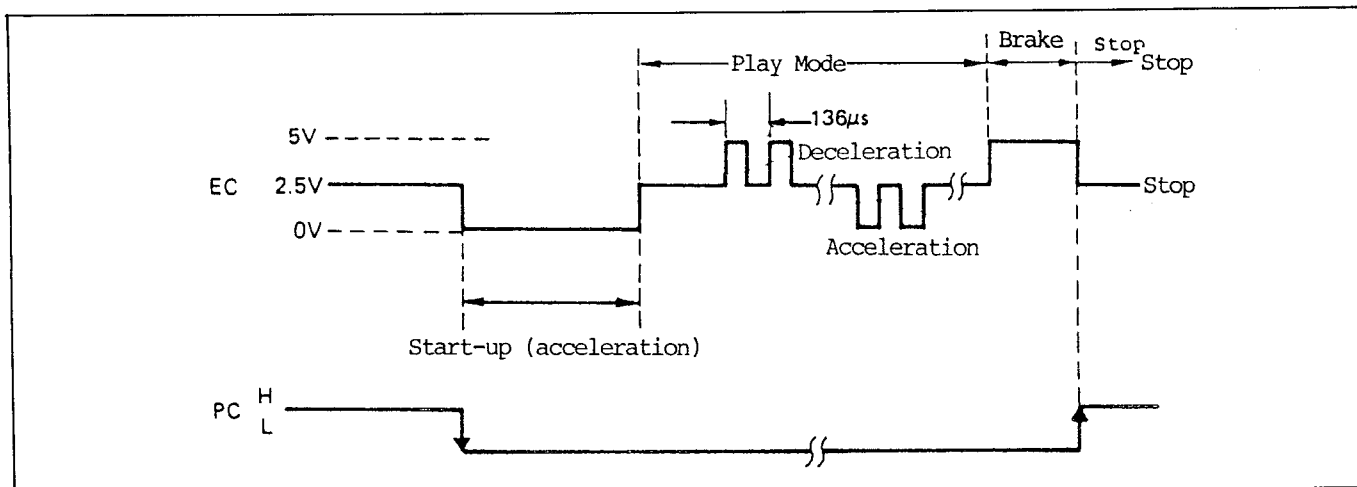


Figure 17-26

When positive pulses are generated, the deceleration mode is selected, and when negative

pulses are generated, the acceleration mode is selected.

11) PLL Servo Circuit

A clock component is extracted from the RF signal which is read off the disc. The clock is used to regenerate the digital musical signal. (The CLV servo controls the spindle motor so that the digital signal read off the disc is synchronized with a reference quartz-controlled clock. It includes a coarse servo which starts up until the PLL is locked, and a fine servo which brings the read signal into phase

synchronization with the reference clock)

The PLL and CLV servo are in a parent-child relationship. The CLV servo absorbs slow speed variation resulting from the disc being off-center or other factors, whereas the PLL absorbs finer jitter, so that an uninterrupted digital signal is regenerated from the disc pits.

Figure 17-27 shows block diagram of PLL circuit.

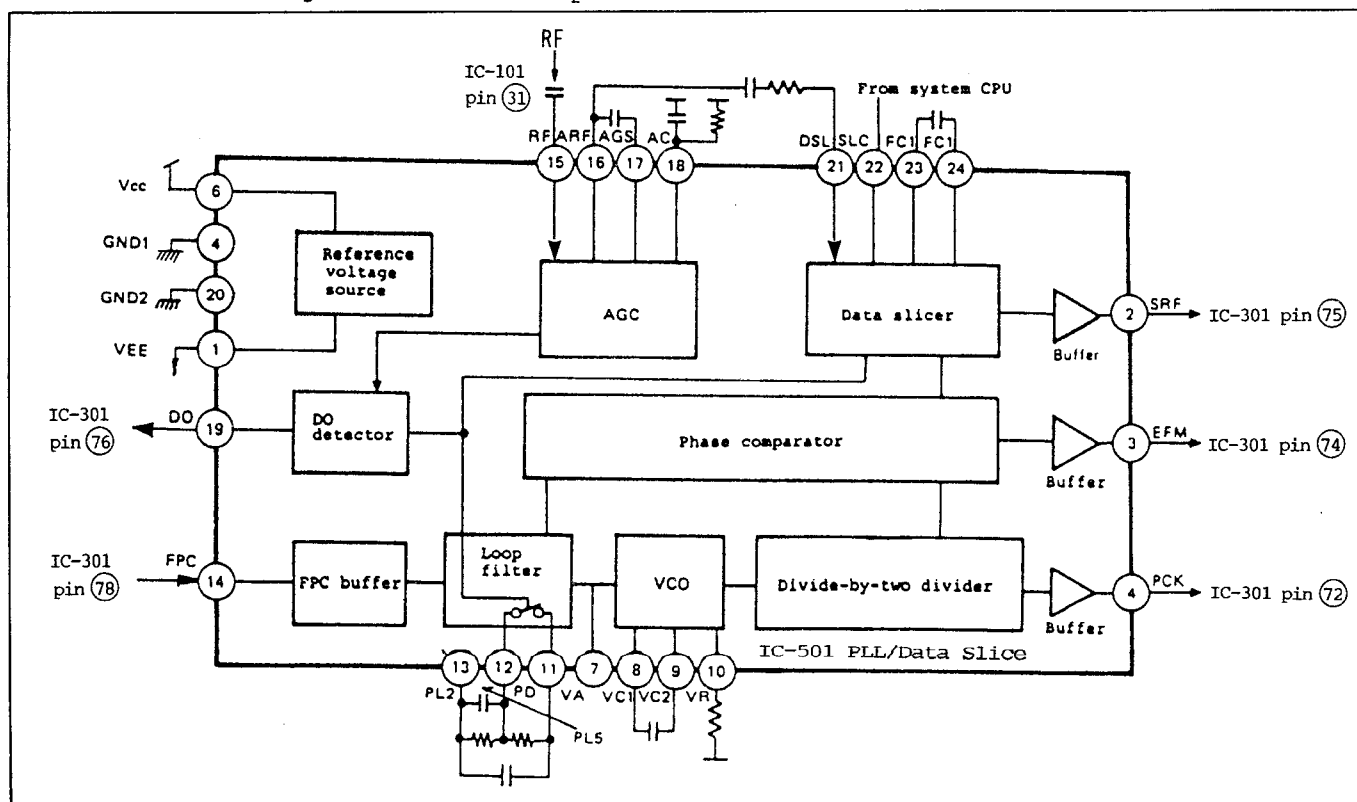


Figure 17-27 Block Diagram of AN8371S

(a) EFM Signal

The RF signal is an analog, continuous signal as shows in figure 17-28. For succeeding digital signal processing, the RF signal is wave-shaped by the data slicing circuit to a digital form signal. The signal output from the data slicing circuit is the EFM signal

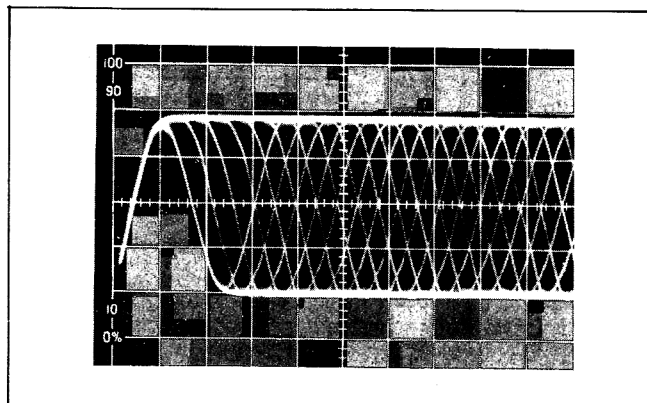


Figure 17-28 RF Signal

(b) PCK Signal

The PCK signal is a clock signal in synchronous with the EFM signal. The EFM signal includes a jitter component to some degree. Therefore the clock signal in synchronous with the EFM signal is required for transmitting the EFM signal exactly to the digital signal processing circuit.

The PLL circuit receives the EFM signal and forms the PCK signal. Figure 17-29 shows the timing chart of the EFM signal and the PCK signal.

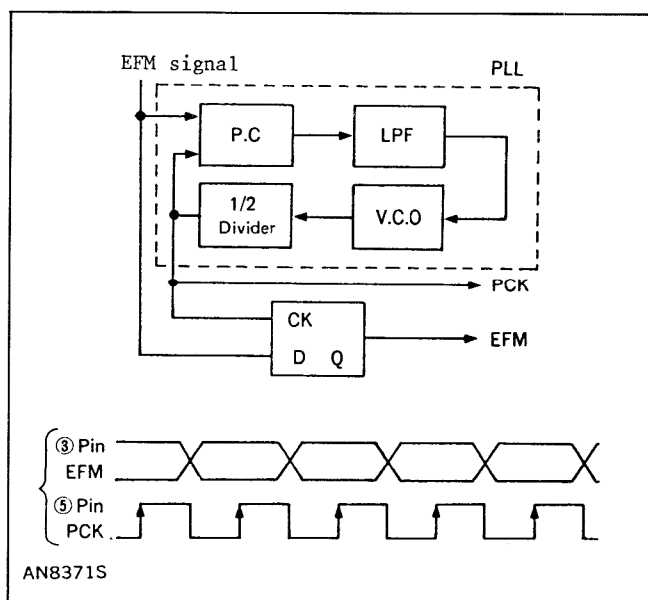


Figure 17-29 EFM and PCK Signal

(c) SRF Signal

The SRF signal is a signal obtained by data-slicing the RF signal. The DO signal is the detection signal of dropout in the RF signal (DO signal is active "High").

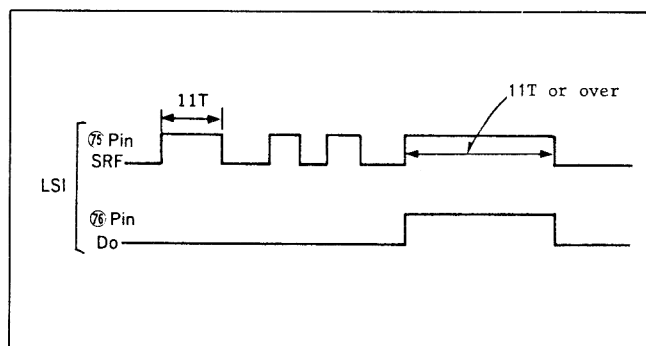


Figure 17-30 SRF Signal

(d) FPC Signal

The FPC signal is the VCO control signal of the PLL circuit and is fed back pin 78 of IC-301.

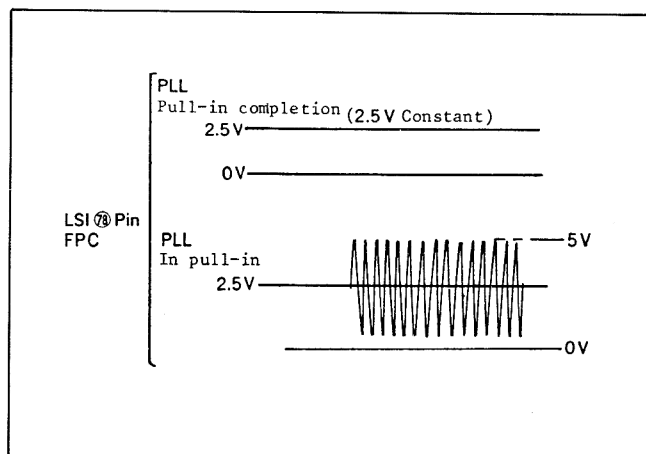


Figure 17-31 FPC Signal

12) Pin Description of AN8370S

● IC101: AN8370S (Optical Servo Control)

Pin No.	Mark	I/O Division	Function
1	VEE	I	Power supply (connected to -5V)
2	LSA	I	Phase difference input (A)
3	GND	I	GND terminal
4	LSB	I	Phase difference input (B)
5	APC	O	Auto laser power control output
6	TEOUT	O	Tracking error signal output
7	TEG	I	Tracking error gain adjusting input
8	TE(+)	I	Phase difference to voltage conversion (+)
9	TE(-)	I	Phase difference to voltage conversion (-)
10	APC(-)	I	Laser power inversion input
11	C.MEM	I	Capacitor connection for phase difference memory
12	APC(+)	I	Laser power non inversion input
13	VREFE	O	Reference current generation
14	SENSE	O	Selector output (track-crossed)
15	HIN	I	Tracking hold circuit input
16	HOUT	O	Tracking hold circuit output
17	SPCNT	O	Trackcrossing speed control output (Not used, open)
18	C.MSP	I	Trackcrossing reference speed setting capacitor connection (Not used, open)
19	C.AF	I	Auto focus timer capacitor connection
20	KICK	O	Track kick signal output
21	VCC	I	Power supply (connected to +5V)
22	CNT1	I	Control input (FOON Focus servo "ON" signal)

Pin No.	Mark	I/O Division	Function
23	CNT2	I	Control input (TRON Tracking servo ON signal)
24	CNT3	I	Control input (KICKF Kick direction [Foward] command)
25	CNT4	I	Control input (KICKR Kick direction [Reverse] command)
26	F.LOCK	O	Focus lock signal output
27	C.FBDO	O	Capacitor connection for inversion RF high detection
28	C.SBDO	O	Capacitor connection for inversion RF low detection
29	C.SBRT	O	Capacitor connection for non-inversion RF slow detection
30	C.FBRT	O	Capacitor connection for non-inversion RF fast detection
31	RF OUT	O	RF signal output
32	BDO	O	Drop-out detection output
33	RF IN	I	RF signal input
34	S.OUT	O	Focus search signal output
35	C.LW	I	Capacitor connection for triangular wave generation
36	FE.OUT	O	Focus error signal output
37	FEG	I	Focus error gain adjusting input
38	FE.REF	I	Focus error comparison voltage generation
39	PDB	I	Photo detection current input (B)
40	IVB	O	Current/voltage conversion (B)
41	IVA	O	Current/voltage conversion (A)
42	PDA	I	Photo detection current input (A)

13) Pin Description of AN8371S

● IC501: AN8371S (Data slice and PLL)

Pin No.	Mark	I/O Division	Function
1	VEE	I	Power supply (connected to -5V)
2	SRF	O	RF signal output data-sliced into digital value
3	EFM	O	EFM signal output synchronized with PCK
4	D.GND	I	GND terminal (digital system)
5	PCK	O	Clock output extracted from SRF
6	VCC	I	Power supply (connected to +5V)
7	VA	I	VCO free run frequency adjusting current input (not connected)
8, 9	VC1, 2	I	Capacitor connection for VCO oscillator frequency
10	VR	I	Resistor connection for VCO oscillator frequency
11	PD	I	Capacitor connection for PLL DO protection
12	PL1	I	PLL loop filter connection

Pin No.	Mark	I/O Division	Function
13	PL2	I	PLL loop filter connection
14	FPC	I	Frequency comparison error signal input
15	RF	I	RF signal input
16	ARF	O	RF signal output with AGC output
17	AGC	I	ARF signal input for AGF drop-out detection input
18	AC	I	Loop filter for AGC connected
19	DO	O	Drop-out detection pulse output
20	A.GND	I	GND terminal (analog system)
21	DSL	I	RF signal input for data slicing
22	SLC	I	Slicing level control signal input
23	FC1	I	Filter capacitor for data slicer connected
24	FC2	I	Filter capacitor for data slicer connected

14) Pin Description of MN1554PEP

● IC401: MN1554PEP (System Control)

Pin No.	Mark	I/O Division	Function
1	BRECV	—	(Not used, open)
2	BSEND	—	(Not used, open)
3	SYNC	O	(Not used, open)
4	SIRQ	I	Not used (connected to +5V)
5	BLKCK	I	Sub-code block (Q data) clock input (75Hz)
6	CLDCK	I	Sub-code block (Q data) clock input (7.35kHz)
7	SBO	I	(Not used, open)
8	SUBQ	I	Sub-code (Q data) input
9	RST	I	Reset signal input
10 ┌ 13	P20 ┌ P23	O	Not used (connected to +5V)
14	CLOSE	O	Loading motor "Close" command
15	OPEN	O	Loading motor "Open" command
16	SLOW	O	(Not used, open)
17	MUTE	O	Muting control
18	SEEK	O	Traverse servo control (Not used, open)
19	NC	—	Not connected
20	TRV.R	O	Traverse "Reverse" command signal
21	TRV.F	O	Traverse "Forward" command signal
22	CNT4	O	Optical servo IC control signal (KICKR: Kick direction [reverse] command)
23	CNT3	O	Optical servo IC control signal (KICKF: Kick direction [forward] command)
24	CNT2	O	Optical servo IC control (TRON: Tracking servo)
25	VDD	I	Power supply (connected to +5V)
26	DOWN	O	(Not used, open)
27	UP	O	(Not used, open)
28	CNT1	O	Optical servo IC control signal (FOON: Focus servo)

Pin No.	Mark	I/O Division	Function
29	CLOSE	I	Disc holder "Open" detection
30	OPEN	I	Disc holder "Close" detection
31	BCLK	I	(Not used, connected to GND)
32	BDATA	I	(Not used, connected to GND)
33	STAT	I	Processing status input from signal processing LSI
34	COMP	O	TOC reading control (ON at "L") (connected to GND)
35	FLOCK	I	Optical servo condition (focus) input
36	SENSE	I	Optical servo condition (track cross) input
37	RECV	I	Data receipt command signal
38	SEND	I	Data transmission command
39	ACK	I	Data discrimination signal
40	CLK	I	Data lock signal
41 ┌ 44	DATA0 ┌ DATA3	I	Key scan signal
45 ┌ 52	NC	I	Not connected
53	OSC2	I	Clock terminal
54	OSC1	I	Clock input
55	X1	I	Optical servo condition input
56	X0	O	(Not used, open)
57	GND	I	GND terminal
58	DMUTE	O	Muting control
59	MDATA	O	Command data output
60	MCLK	O	Data clock output (command clock)
61	MLD	O	Data output (command load)
62	DOUTON	O	Optical output control signal (Not used, open)
63	EMPH	O	Emphasis signal output
64	NC	—	Not connected

15) Description of Modified Servo Circuit (MN8373S and MN8374S)

1. General

The servo circuit in CD player with the standard mechanism (SL-P150, SL-P200, SL-P250, etc) has been entirely modified during production. The units containing modified servo circuit can be identified as follows:

- (1). Serial number suffix: Refer to supplement.
- (2). A caution label sticking: THE ORIGINAL DESIGN HAS BEEN CHANGED.

2. Circuit Configuration

The servo circuit used in the conventional FF-1 CD players comprised a servo IC (AN8370S) and discrete circuit. In contrast, the modified servo circuit consists of three IC chips with on discrete circuit.

- (1). IC-101 (servo amplifier)
=Processes analog servo signals.
- (2). IC-102 (servo controller)
=Processes digital servo signals.
- (3). IC-103 (servo driver)
=Drives the focus, tracking, and traverse coils.

The block diagram of IC-101 (AN8373S) and IC-102 (AN8374S) are shown in figure 17-33.

3. Departures from the Conventional Servo Circuit and some Cautions

While the new servo circuit replaces discrete components with integrated circuits, the basic circuit configurations and operations are almost identical to those of the conventional servo circuit, with the exception of the following details:

- (1). PLL adjustment eliminated
=Servo signals are automatically pulled into the PLL lock range by using the CLK (88.2KHz) signal.
- (2). 2.5V reference for tracking error and focus error signals
=Because the new circuit operates on a signal +5V supply, the reference voltage for the error signals has been changed to 2.5V as shown in figure 17-32.

NOTE

The 2.5V reference is also used for turntable height adjustment.

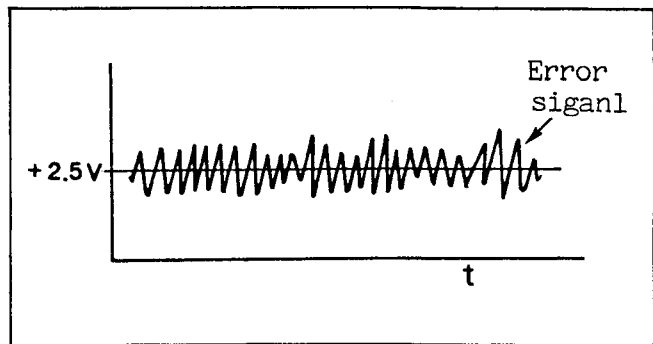


Figure 17-32

Focus and Tracking Error Signal

- (3). RF signal polarity inverted
=An inverted RF signal appears at J101. As a result, the RF and dropout waveforms have a polarity opposite to that of the conventional, as shown in figure 17-34.
- (4). Servo gain circuit and adjustment
=Focus: 825 Hz 150 mV
=Tracking: 1.2 KHz 150 mV
=Tracking balance: 1.1 KHz 600 mV
These voltages are input voltages required for adjustment.

4. Servo Circuit Block Diagram

Note) → P.B. signal

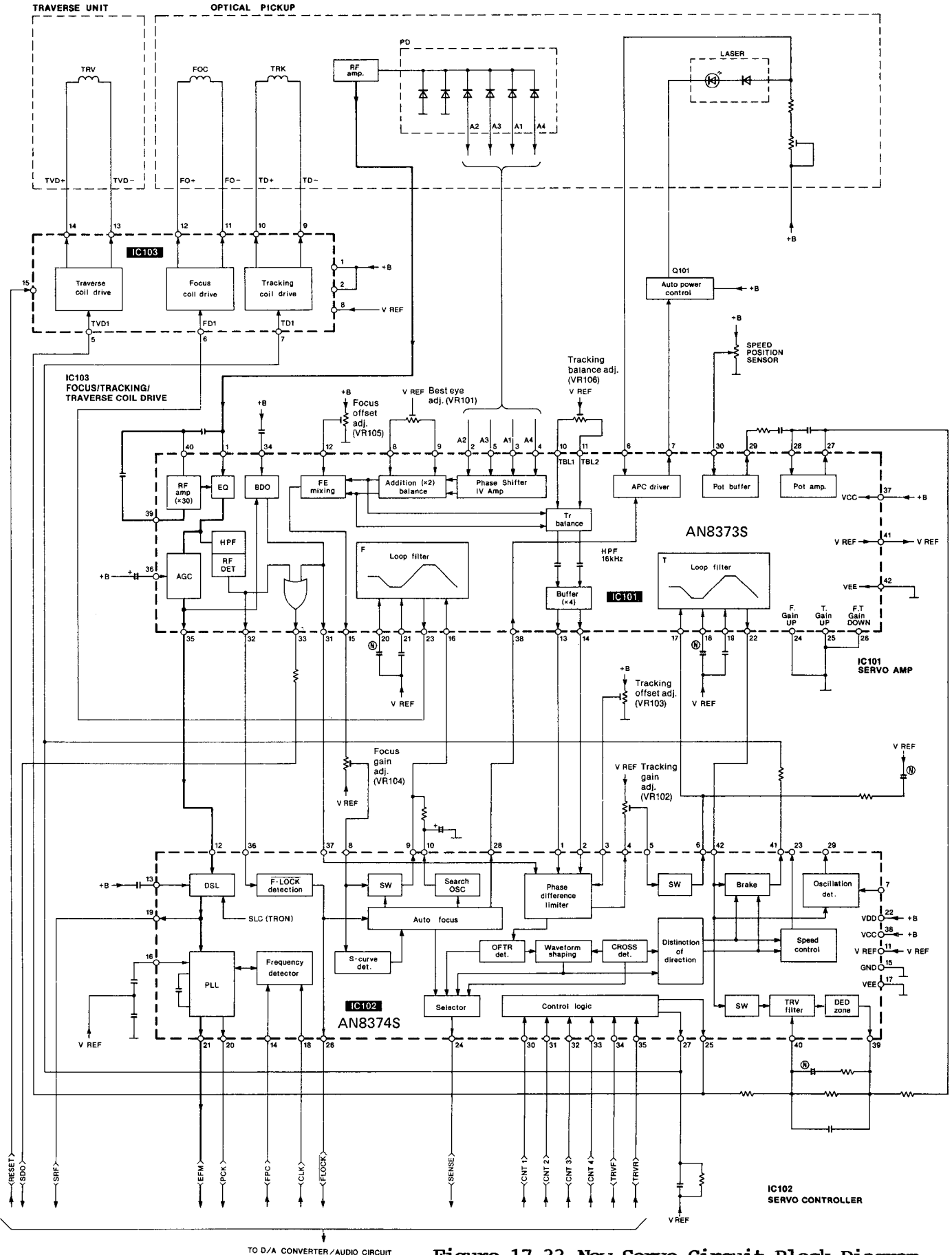


Figure 17-33 New Servo Circuit Block Diagram

5. Data Signal Flow

(1). RF Signal Flow

- * The photo detectors A1 - A6 output is applied to IC-101's RF IN INPUT (pin 39) via an RF amplifier within the Optical Pick-up.
- * After being amplified by another RF amplifier within IC-101, the signal passes through an equalizer for 3T compensation, goes into the AGC (Auto Gain Control) circuit where it undergoes automatic gain control, and appears at the ARF (RF signal controlled by auto gain control) output (pin 35) of IC-101.

(2). EFM Data Generation

- * The RF signal applied to IC-102's ARF input (pin 12) is sliced by the DSL (data sliced) into a pulse signal EFM (eight to fourteen bit modulation), and then coupled to the PLL circuit.
- * The PLL is synchronized with the EFM signal and then coupled to the signal processing block.

6. PLL Operations

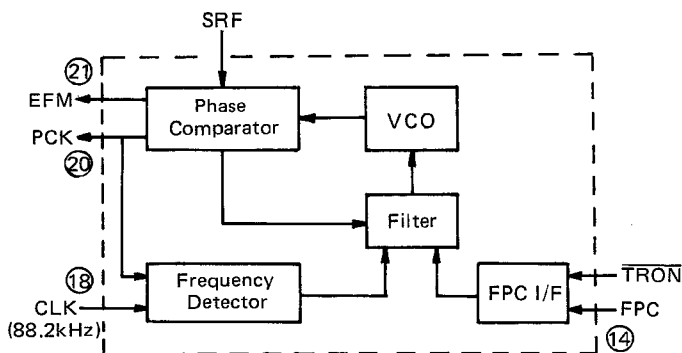


Figure 17-35 PLL Block

- (1). The new PLL circuit has replaced the PLL adjustment with a frequency comparator, one of whose input is clocked by the CLK signal (88.2KHz/11.34us) supplied from the digital signal processor.

The PLL circuit shows figure 17-35.

- (2). If PLL lock is lost, the frequency comparator's error output, this error signal will be controlled VCO's free-running frequency. The large error output of the frequency comparator drives the VCO until range. When the PLL is locked (TRON active), the FPC (frequency comparison error signal) is active and the VCO is controlled by the phase comparator's error output.

7. Control Logic (IC-102)

Commands used to set the servo circuit status are supplied from the system CPU in parallel 4 bit data (CNT1 - CNT4).

SENSE is the status read signal for IC-102, which is transferred to the system CPU. All other control logic operations are identical to those of the conventional control logic.

8. Focus Servo Circuit Operation

- (1). The outputs of the photo detector are coupled to pin 2 - 5 of IC-101, where they are phase corrected by the A1 + A2 and A3 + A4 signals and converted from current to voltage signals. They are transferred to the focus balance circuit and subjected to photo detector position error correction by means of VR101 (best eye adjustment).

- (2). The difference between the A1 + A2 and A3 + A4 signals, produced within the FE mixing circuit, serves as the Focus Error signal which appears at IC-101 pin 15.

The signal is then applied to IC-102 pin 8. VR105 is used for offset voltage injection control.

- (3). Since the Focus switch is closed in the PLAY mode, the FE signal appears at IC-102 pin 9 and couples to the loop filter within IC-101, where it is used to determine the focus servo's frequency response. The output of the loop filter at IC-101 pin 23, is transferred to the focus driving circuit and drives the focus actuator.

9. Tracking Servo Circuit Operation

- (1). The focus balance circuit output, $A1 + A2$ and $A3 + A4$, are also transferred to the tracking balance circuit. The output of the tracking balance circuit are applied to the phase difference limiter within IC-102, where they are phase compared to produce a tracking error signal.

The tracking error signal at IC-102 pin 4 is subjected to gain adjustment, then applied to the loop filter, within IC-101 via the tracking switch.

- (2). The loop filter output appearing at IC-102 pin 22, is coupled to pin 42 (BR IN/ tracking error input) of IC-102, where it is fed into a braking circuit. The output of the braking circuit appears at pin 41 (BR OUT/ tracking drive control signal) and couples to the tracking coil driver to drive the tracking actuator's.

10. Traverse Servo Circuit Operation

- (1). PLAY mode:

The tracking error signal applied to IC-102 pin 42 goes through the traverse switch and TRV (traverse circuit) filter (for response compensation) and appears at pin 39. This output is used to control the traverse actuator.

- (2). KICK mode:

The KICK signal, decoded from the system CPU command by the control logic, appears at pin 27. It is used to control the tracking actuator which KICKs the pick-up by a span of 1, 10, or 100 tracks according to the system CPU command.

- (3). Forced traverse mode:

The TRVF, TRVR signals supplied from system CPU to pins 34 and 35, respectively, are decoded to produce a traverse drive signal. The signal appears at pin 25 and drives the traverse coil.

- (4). Speed sensor operation:

The varying voltage output of the potentiometer connected to IC-101 pin 30, is differentiated by the circuit at pin 27 to 29 and fed into the traverse coil. Smooth traverse action is achieved by feeding back a control signal which reflects the traverse movements of the optical pick-up. (The potentiometer yields a negative potential when the optical pick-up moves outward).

11. RF Signal Detection and Auto Focus Circuit

The auto focus circuit within IC-102 performs focus search. If it detects an FE signal S-Curve it closes the focus switch contacts to close the focus servo loop. At this point the RF detector within IC-101 detects the RF signal and supplies it to the F.LOCK DET. input at IC-102 pin 36.

The F.LOCK signal is used to indicate to system CPU that focus is "IN". If focus goes "OUT" and the RF signal is no longer detected, the auto focus circuit opens the focus loop switch and starts focus search again.

12. Dropout Detection Circuit

- (1). Black Band Dropout (BDO)

Dropouts are detected from the RF signal within IC-101. In order to stabilize the tracking servo against dropout, the BDO signal is supplied from IC-102 pin 37 to the phase difference limiter and used to hold the tracking servo operation.

- (2). Signal Dropout (SDO)

If an RF signal dropout is detected by the RF detector, or a BDO is detected, the SDO signal is supplied to the signal processor to stabilize the CLV servo operation.

The SDO signal is supplied to the signal processor to stabilize the CLV servo operation. The SDO signal, produced within IC-102, is used to stabilize PLL circuit operation.

13. Braking Circuit

The braking circuit applies a brake to the tracking actuator so as to speed the tracking servo pull-in sequence required after a SKIP or KICK operation.

The direction of actuator movement is detected by the OFTR DET (off track detection), CROSS (tracking cross) and DIRECTION signals from IC-102 and a brake signal is added to the tracking error signal.

16. Pin Description of AN8373S

Pin No.	Mark	I/O Division	Function
1	AMP1	I	RF signal input (X30 amp.)
2	PDAD	I	Photo detector current input (A2)
3	PDA	I	Photo detector current input (A1)
4	PDBD	I	Photo detector current input (A4)
5	PDB	I	Photo detector current input (A3)
6	LPD	I	Non-inverting laser power input
7	LD	O	Laser power auto control output
8	FBL1	I	PD balance adjustment
9	FBL2	I	
10	TBL1	I	Tracking balance adjustment
11	TBL2	I	
12	FOOFS	I	Focus offset adjustment
13	IVA	O	Current/voltage conversion output (A)
14	IVB	O	Current/voltage conversion output (B)
15	FE	O	Focus gain adjustment output
16	FPI	I	Focus error signal input
17	TPI	I	Tracking error signal input
18	C. TPL	I	Tracking error filter capacitor input
19	C. TPH		
20	C. FPL	I	Focus error filter capacitor input
21	C. FPH		

Pin No.	Mark	I/O Division	Function
22	TPO	O	Tracking error signal output
23	FPO	O	Focus error signal output
24	FGC	I	Focus gain up signal input (Not used, connected to GND)
25	TGC	I	Tracking gain up signal input (Not used, connected to GND)
26	GD	I	Focus/tracking down signal input (Not used, connected to GND)
27	PTO	O	Position detecting amp. output
28	PTI	I	Position detecting amp. input
29	PBO	O	Position detecting buffer output
30	POT	I	Position detecting buffer input
31	BDO	O	Dropout detection output
32	RFDET	O	RF detection signal output
33	SDO	O	Dropout detection pulse output
34	C. SBDO	I	Dropout detecting capacitor input
35	ARF	O	RF signal output
36	C. AGC	I	AGC detecting capacitor input
37	VCC	I	Power supply (+5 V input)
38	LDON	I	Laser power control input
39	RF IN	I	RF signal input
40	AMPO	O	RF signal output
41	VREF	O	Reference voltage output
42	GND	I	Ground terminal

17. Pin Description of AN8374S

Pin No.	Mark	I/O Division	Function	Pin No.	Mark	I/O Division	Function
1	LSA	I	Phase difference input (A)	22	VDD	I	Power supply (+5 V input)
2	LSB	I	Phase difference input (B)	23	SPCNT	O	Track crossing speed control output (Not used, open)
3	TEOFS	I	Tracking offset adjustment	24	SENSE	O	Selector output (track crossing state)
4	TE	O	Tracking gain adjustment	25	TRV	O	Traverse servo control output
5	TEG	I		26	$\overline{\text{FLOCK}}$	O	Focus lock signal output
6	TE OUT	O	Tracking error signal output	27	KICK	O	Track kick signal output
7	TE BPF	I	Tracking error gain detecting filter (Not used, open)	28	LDON	O	Laser power control output
8	FEG	I	Focus gain adjustment	29	VDET	O	Focus/tracking gain up output (Not used, open)
9	FE OUT	O	Focus error signal output	30	CNT1	I	Control input (FOON: Focus servo ON signal)
10	CLW	O	Triangular wave oscillator capacitor output	31	CNT2	I	Control input (TRON: Tracking servo ON signal)
11	VREF	I	Reference voltage input	32	CNT3	I	Control input (KICKF: Kick direction (forward) command)
12	ARF	I	RF signal input	33	CNT4	I	Control input (KICKR: Kick direction (reverse) command)
13	CDSL	I	Data slice filter capacitor input	34	TRVF	I	Traverse forward command signal
14	FPC	I	Frequency difference signal input	35	TRVR	I	Traverse backward command signal
15	GND	I	Ground terminal	36	$\overline{\text{RFDET}}$	I	RF detection signal input
16	C. PLL	I	PLL loop filter constant	37	BDO	I	Dropout detection input
17	VSS	I	Ground terminal	38	VCC	I	Power supply (+5 V input)
18	CLK	I	Frequency pull-in clock signal (88.2 kHz) input	39	TVPO	O	Traverse position detecting resistor/capacitor inputs
19	SRF	O	Sliced and digitized RF signal output	40	TVPI	I	
20	PCK	O	Clock output extracted from SRF	41	BROUT	O	Tracking drive control output
21	EFM	O	EFM signal output synchronous with PCK	42	BRIN	I	Tracking error signal input

18. Pin Description of AN8377

Pin No.	Mark	I/O Division	Function	Pin No.	Mark	I/O Division	Function
1	PVCC	I	Driver power supply (+5 V input)	9	TD-	O	Inverting output of tracking driver
2	VCC	I	Power supply (+5 V input)	10	TD+	O	Non-inverting output of tracking driver
3	TB	O	External transistor base driving output	11	FD-	O	Inverting output of focus driver
4	VMON	O	Voltage (+5 V) output	12	FD+	O	Non-inverting output of focus driver
5	TVDI	I	Traverse error signal input	13	TVD-	O	Inverting output of traverse driver
6	FDI	I	Focus error signal input	14	TVD+	O	Non-inverting output of traverse driver
7	TDI	I	Tracking error signal input	15	$\overline{\text{RESET}}$	O	Reset signal output
8	VREF	I	Reference voltage input	16	PC	I	PC input (connect to GND)

18. Digital Circuit

The digital processing circuit is the heart of the CD player. It accepts EFM data transferred from the RF data punch circuit (IC-501: data slice & PLL) and provides EFM demodulation, data interleave, error correction, and other

signal processing functions. The following figure 18-1 shows a digital processing circuit's internal block diagram and correlations between the blocks.

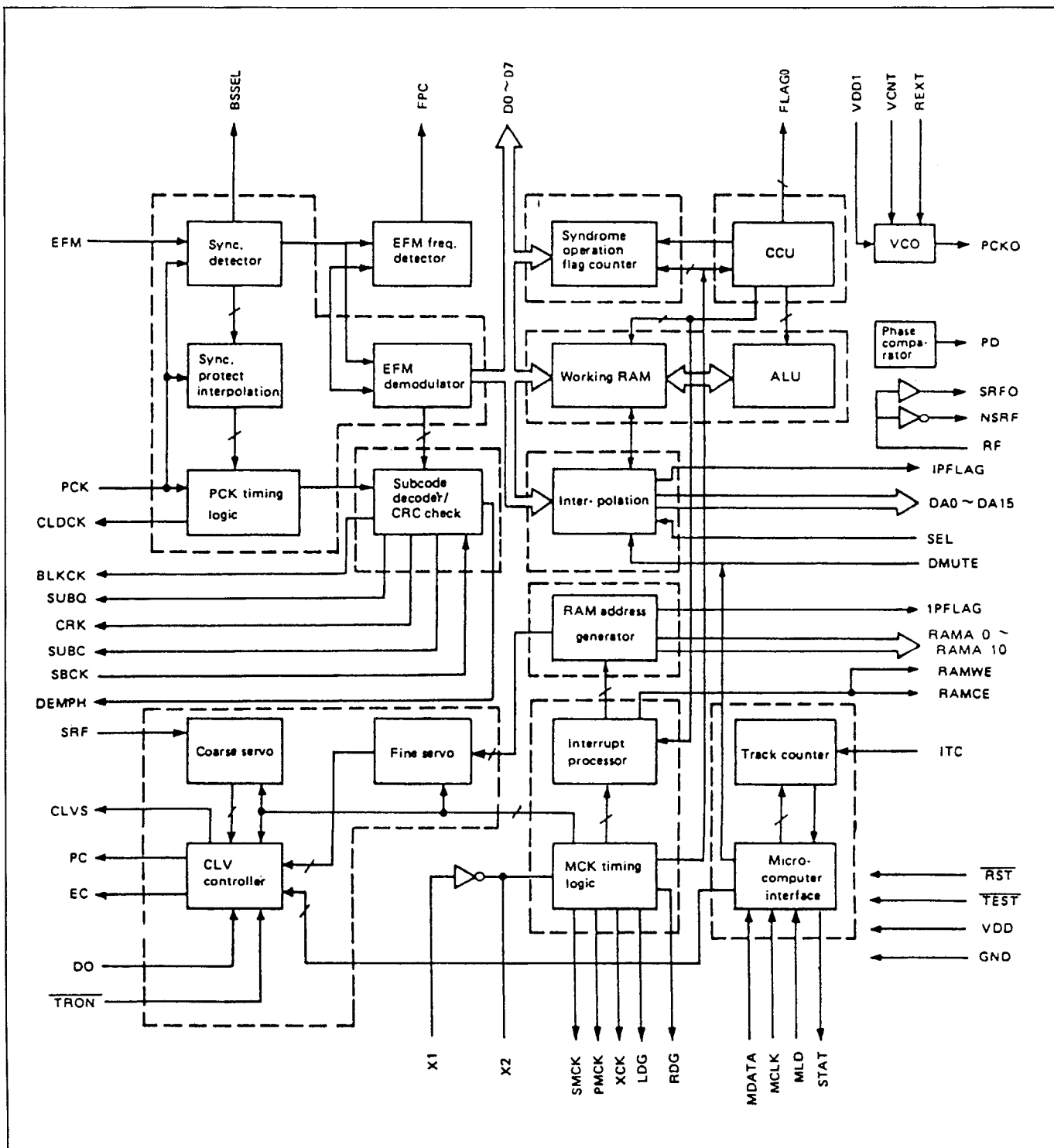


Figure 18-1 Block Diagram of MN6617

The PCK (bit period clock= 4.3218 MHz) is used to time signal processings during 16K RAM write. RAM read and subsequent operations are timed by a clock which is derived from the crystal-controlled master clock.

The jitter contained in the play signal is absorbed by the RAM. The basic internal operations of the digital processing circuit are explained on the following pages.

1) EFM Demodulation

(1) Sync signal detection, protection, and interpolation:

The EFM signal read off the disc (under optical system servo control) is coupled to the data slice/clock sampling circuit, where it is converted into a logic-level signal, separated into the EFM and PCK signals, and applied to the EFM demodulator.

The input EFM signal is first converted into NZRZI data (inverted for every one; non-inverted for every zero), then is applied to a 24 bits shift register, where the sync signal is extracted from the data.

Interpolation is required to cope with sync, signal dropout or pseudo sync, signal generated due to defects on the disc. In order to prevent the synchronization system being disturbed by the pseudo sync, signal a small sync. Detection window is provided each time the sync signal should be found, so that sync detection is performed only within the window.

If no sync signal is detected in the window, interpolation is done by an internal 588 increment counter (which counts 588 PCK pulses). Also the number of sync detection failures is counted, and the width of the detection window is automatically controlled according to the count value.

The FCLV signal status changes depending on whether the sync signal has been detected or not:

*In every frame where on sync signal was detected, the FCLV is set low.

*In every frame where the sync signal was detected, the FCLV is set high. (the sync detection failure counter is reset at a high transition of the FCLV)

If no sync signal has been detected for eight consecutive frames, it is considered an off-synchronization. The RESY output of the device is set low, which fully opens the window to attempt synchronization:

**If a step out occurred, the RESY is set low.

**When the device attempts a synchronization, the RESY is set high.

The MN6617 can use the TRON (tracking on) signal for external synchronization control. When the tracking servo is off (TRON = high), the device always fully opens the window, so that it is ready for synchronization as soon as the tracking servo is turned on:

**When tracking servo is on, TRON is set low.

**When tracking servo is off, TRON is set high.

This enables reduced synchronization time when the player is accessing the next track.

(2) EFM demodulation

The output of the sync, detector's 24 bits shift register is coupled to a 14 to 8 decoder, where it is decoded into the original 8 bit data. Also subcodes S0 and S1 are decoded and transferred to the subcode decoder.

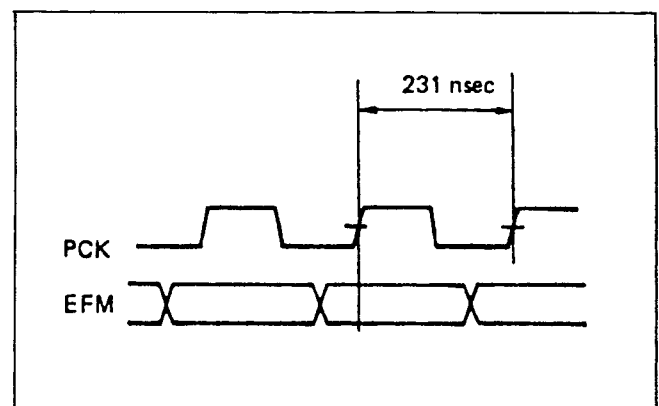


Figure 18-2 PCK and EFM Input Timing

2) Subcode Detector

(1) Q-data demodulation

The device inputs the demodulated data from the subcode-block and sync signals (S0,S1) are transferred from the EFM demodulator with the frame clock (CLDCK) timing.

The BLKCK signal indicating the block period is made active only if S0 and S1 are detected consecutively.

Q data is inverted and output as the SUBQ is synchronization with the CLDCK.

(2) CRC check on Q data

Q data is subjected to CRC check, and the check result is output to the CRC and STAT pins of the device.

If an error is detected, both CRC and STAT are set high, if no CRC error is detected, both CRC and STAT are set low.

When the TRON input is high, the CRC check result output are forced high.

(3) P to W data

The 8 bits data, (P, Q, R, S, T, U, V, and W) are subjected to subcode demodulation and loaded into an 8 bits parallel to serial register at the CLDCK timing. It is then shifted out in series through the SUBC pin at an external SBCK shift clock timing.

(4) De-emphasis detection

The device extracts one bit, which indicates if emphasis is on or off from the Q data and supplies it to the DEMPH pin. De-emphasis is on when the DEMPH is high. To suppress rapid output transistions, the DEMPH output holds the previous state if a CRC error has been detected.

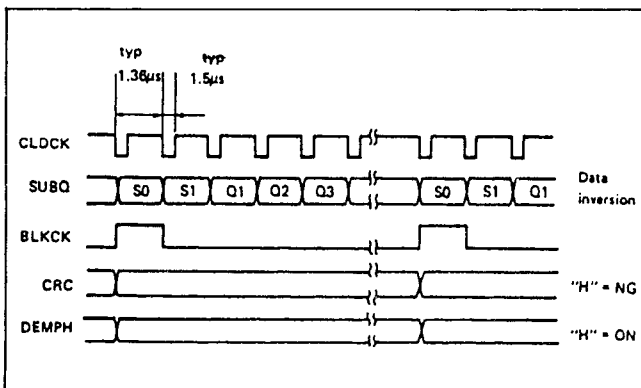


Figure 18-3 SUBQ, CRC and DEMPH Timing

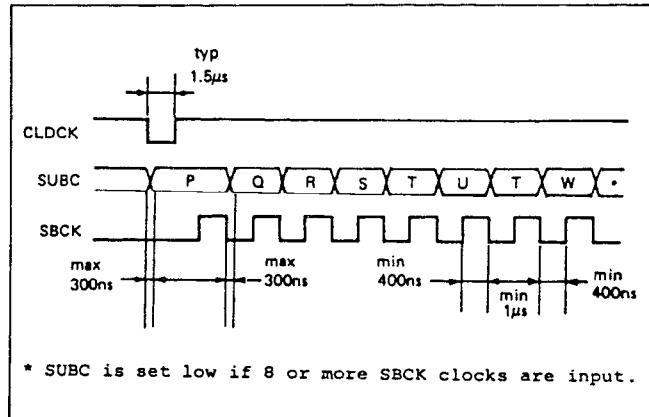


Figure 18-4 SUBC and SBCK Timing

3) RAM Interface

The MN6617 accesses the external 16K byte RAM for the following read and write functions of the modulated EFM data:

- ** C1 unscramble
- ** Deinterleave
- ** C2 unscramble
- **Error correction flags

To make these RAM accesses efficient, the RAM address controller assigns different levels of priority to RAM access. The level of priority is identified by the interrupt acknowledge logic so that the relevant address is supplied to RAM.

Each frame contains 2304 clocks of XCK (16.9344 MHz), and every RAM access cycle requires 12 XCK clocks (709 ns). So each frame contains up to 192 access slots.

(1) RAM address generation

The effective addresses for the 16K byte SRAM (static RAM) are given by an 11 bits read frame counter and an 11 bits write frame counter. The read frame counter is incremented by a crystal controlled frame clock. The write frame counter is incremented each time one frame (32 symbols) of EFM data is written into the RAM.

(2) Jitter absorption and RAM address reset

The demodulated EFM data is synchronous with the PCK signal produced by the clock extraction logic and contains disc wobble and flutter components that cause signal jitter. In order to eliminate this jitter, the EFM data is written into the 16K byte S-RAM, then read out of it at a crystal controlled clock timing.

The margin for jitter is ± 3 frames. If the difference between the contents of the 11 bits read and write frame counters exceeds ± 4 frames, a RAM address reset occurs, and the write frame counter value is loaded into the read frame counter (at the same time FLAG 6 is set).

For a forced RAM address reset, the device uses the TRON signal. This enables each track access to have the maximum jitter margin.

The figure 18-5 and 18-6 shows read and write timing.

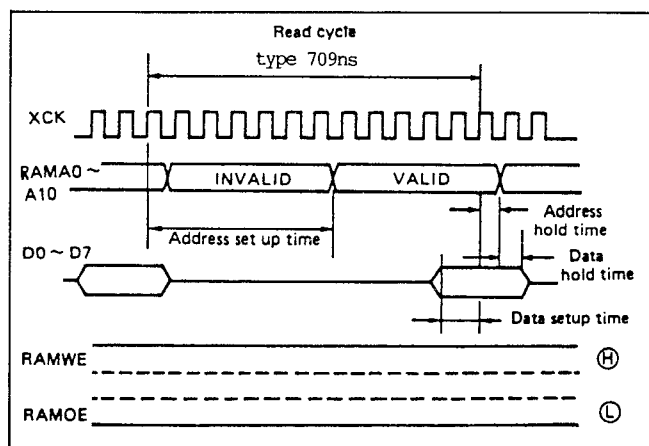


Figure 18-5

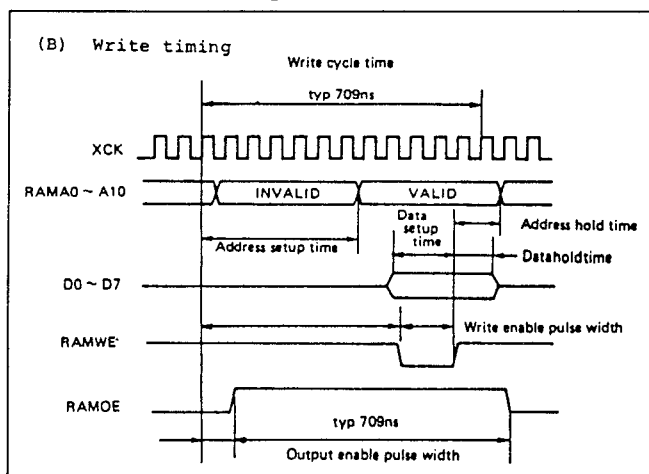


Figure 18-6

4) Interpolation and D/A Interface

Data is first subjected to error correction by the C2 decoder, then undergoes C2 unscramble. Incorrect data (16 bits) is subjected to error interpolation.

For an independent data error interpolation replaces the error with the average of the data items that precedes and succeeds the error data (average interpolation). For consecutive data errors, it replaces them with the preceding correct data (reposition interpolation).

The IPFLAG is set high for the interpolated data to indicate that the data has been interpolated. It is output in synchronously with both serial and parallel data.

The data, after being subjected to interpolation, appears at the D/A output and a serial output. The D/A output includes parallel output and a serial output, either of which is chosen by controlling the SEL input pin:

** SEL is high: Parallel output DA0 - DA15 (offset binary data).

** SEL is low: Serial D/A output appearing at SRDATA.

The serial data at SRDATA is MSB-first, two's complement data, and is interpolated under the control of the IPSEL input.

** IPSEL is low: Selects normal interpolation mode, in which both reposition and average interpolation are performed.

** IPSEL is high: All error data is interpolated by reposition interpolation.

The L/R channel discrimination for the serial output is indicated by the R/L output pin.

** L/R is low: Left channel

** L/R is high: Right channel

Note
 The following pins are exclusively used
 for CD ROM.
 SRDATAx: LSB-first, two's complement
 output.

IPBYTE : Flag indicating the reliability
 of a symbol in SRDATAx.
 BYTCK : Indicates the top of a symbol
 in SRDATAx.

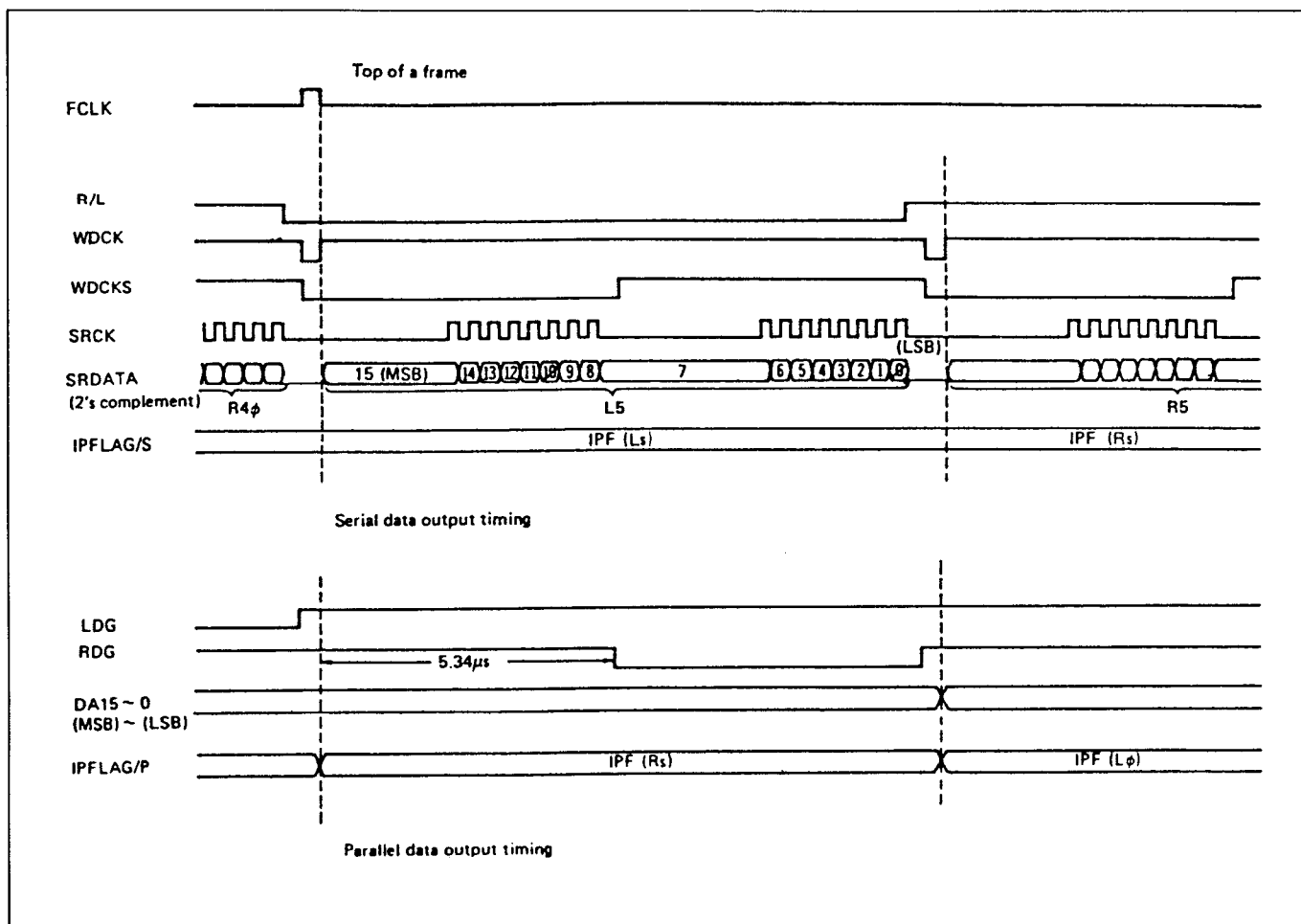


Figure 18-7

5) Error Correction

The error correction logic provides C1-double correction and C2-triple correction.

C1 correction logic produces 4 types of C1 flags (no error, single correction, double correction, and triple or more correction).

C2 correction logic selects either double or triple correction depending on the types and number of C1 flags and executes the selected correction. It also determines whether interpolation is

needed or not depending on the positions and types of C1 flags.

If the logic has indicated that interpolation is needed, it transfers to the interpolation logic the C2 flag indicating which C1 flag is to be used for the interpolation. The interpolation logic selects the type of the C1 flag according to C2 flag, and provides interpolation only for the symbol for which the C1 flag is set.

The error correction states can be monitored with flags 1 through 5.

FLAG 5	FLAG 4	FLAG 3	FLAG 2	FLAG 1	Meaning
x	x	x	0	0	C1 - No error
x	x	x	0	1	C1 - Single error corrected.
x	x	x	1	0	C1 - Double error corrected.
x	x	x	1	1	C1 - Error incorrectable.
0	0	0	x	x	C2 - No error
0	0	1	x	x	C2 - Single error corrected.
0	1	0	x	x	C2 - Double error corrected.
0	1	1	x	x	C2 - Double error incorrectable.
1	0	0	x	x	C2 - Triple error corrected.
1	0	1	x	x	C2 - Triple error incorrectable.

Table 18-1

6) Spindle Servo Control

The spindle servo control logic has four modes: CVL, 11T, 6T, and 6T break modes.

(1) 11T Mode

This mode is selected when the tracking servo is active (TRON = low) and the PLL is unlocked.

Since $XCK = 16.9344$ MHz, the SRF frame sync signal pulse width is approximately 43XCK.

- **Frame sync signal pattern:
= (10000000000100000000000100)
- **Channel bit frequency/Pulse width:
= 4.3218 MHz/ 0.32 ns
- **XCK frequency/Pulse width:
= 16.9344 MHz/ 0.059 ns
- **XCK pulse count par 11T:
= $0.23 \times 11 / 0.059 = 43$

If no dropout exists in RF, 11T becomes the maximum pulse width. In the 11T mode, this maximum pulse width is determined by counting the XCK. The logic judges whether the obtained N11 is 43XCK or not and provides the following control functions:

- ** N11 is equal to 43XCK:
= EC is set to high impedance.
- ** N11 is greater than 43XCK:
= EC provides a low for every frame.
- ** N11 is smaller than 43XCK:
= EC provides a high for every frame.

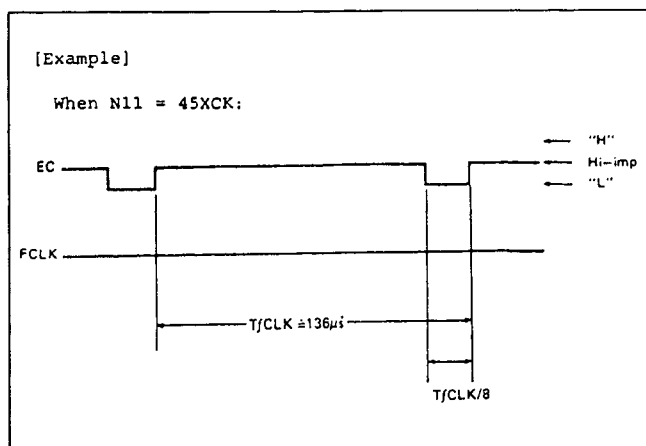


Figure 18-8

EC's pulse width varies with the deviation of 43XCK (max. $TFCLK/2$).

$$TFCLK = 1/7350 = 136 \mu s$$

When $N11 = 43XCK$, spindle motor rotation is controlled at an accuracy of approx, $\pm 1.16\%$ of the normal rotation rate.

(2) CLV Mode

The CLV mode is selected if the PLL is locked when the spindle motor rotation rate is less than approx, 4.6% of the normal rotation rate. Either the 11T or CLV mode is automatically selected provided the tracking servo is active (TRON = low).

The device compares the FCLK's phase with CLDCK's phase by comparing the value of the RAM read counter (which is synchronous with the reference frame sync signal (FCLK) with that of the RAM write counter (which is synchronous with the frame sync signal, CLDCK, extracted from the EFM pattern), and provides a phase error signal at its EC pin.

The EC signal's pulse width varies with the phase error, TPE (sec), between the CLDCK and FCLK:

- ** If CLDCK lags behind FCLK:
= EC is set low (acceleration)
- ** If CLDCK leads FCLK:
= EC is set high (deceleration)

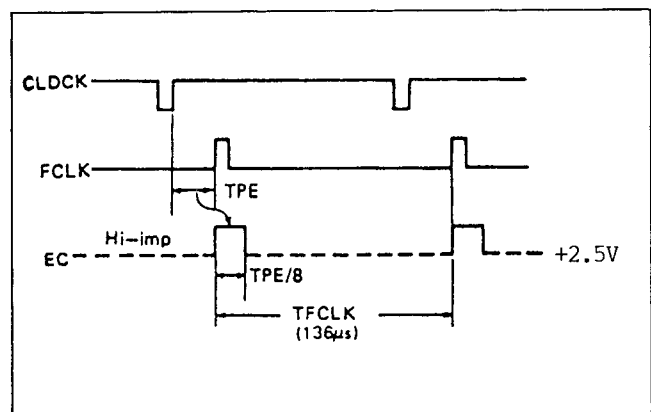


Figure 18-9

(3) 6T Mode

The 6T mode is used for coarse servo control for high-speed access. During traverse head movement, the 11T mode is not available since the RF signal is shut down for an interval of more than 11T.

Instead, the pulse width of Tmin (6T) of the divide-by-two SRF is counted with XCK/2 and the minimum value of pulse width is determined at every 8 frames. The count obtained is used as velocity information:

- ** SRF's 6T is divided by two.
- ** $XCK/2 = 0.059 \text{ ns} \times 2 = 0.118 \text{ ns}$
- ** 6T's pulse width = $0.23 \text{ ns} \times 6 = 1.38 \text{ ns}$
- ** If 6T is counted by XCK/2:
 $= 1.38 \text{ ns} / 0.118 \text{ ns} = 11.7 \text{ counts}$

Since 6T's pulse width is nearly 11 x XCK/2, the EC output provides:

- ** A low (acceleration) if N6 (obtained count) is greater than 11.
- ** A high (deceleration) if N6 is smaller than 11.

In any other case, the EC is set to high impedance.

6T divided by two:

Example

When 6T's pulse width = $12 \times XCK/2$

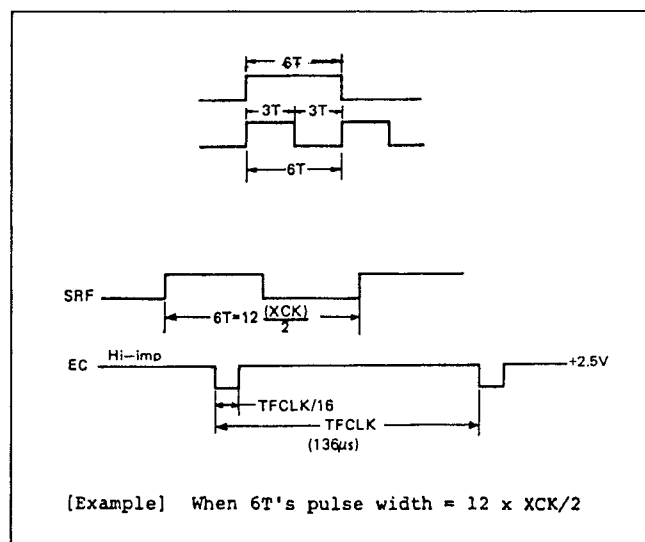


Figure 18-10

(4) 6T Brake Mode

The 6T brake mode is used to stop the turntable rotation. The device enters this mode when it receives a TTOFF command from system IC-401 by MDATA pin 59.

The device first places its EC pin in the forced deceleration mode. When the minimum pulse width (6T) of the divide-by-two SRF is reduced to approx, 7 times that at the normal rotation rate, the device sets the EC output to high impedance and set the PC output high.

If a dropout occurs, the device does not select brake release, so that the turntable continues decelerating. A dropout with a duration of less than 4.3 µs is however ignored. The figure 18-11 shows 6T brake mode.

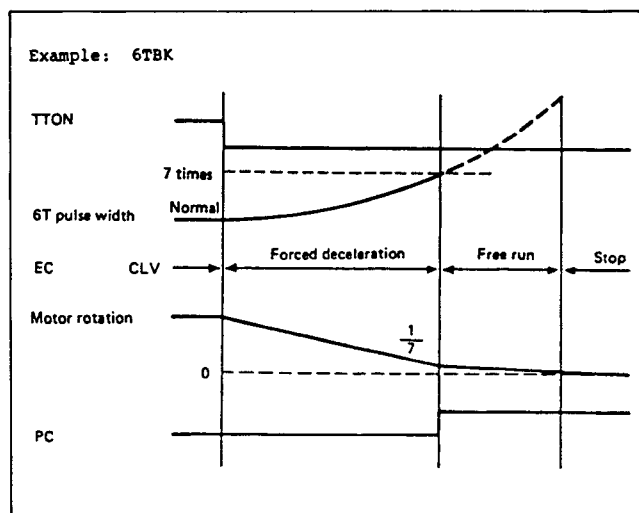


Figure 18-11

(5) Speed Limiter

When the player uses a motor with FG, the user can let the device control the upper and lower limiter of motor speed by applying the FG signal to the FG input of the device.

When it is more than 77 Hz, the EC pin is set high (forced deceleration).

If the FG generates 6 pulses per rotation:

- ** EC provides forced acceleration at 140 rpm or less.
- ** EC provides forced deceleration at 770 rpm or more.

(6) Operation upon generation of dropout

**The device ignores a dropout (DO) signal pulse width of less than 4.3μs.

**If the DO pulse width exceeds 4.3μs:

- 1.The EC is set to high impedance (free run) when in CLV mode.
- 2.The EC is set to high impedance (free run) when in 11T mode.
- 3.DO is ignored when in 6T mode.
- 4.Brake is not released when in 6T brake mode.

(7) Operation upon off tracking

If off-tracking occurs, the device dose not operate normally in any mode other than the 6T mode. It must be placed in the free-run mode by the DO signal or system control command.

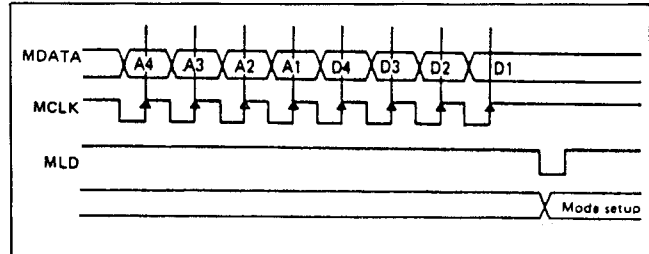


Figure 18-12

7) System CPU Interface

(1) Signal Name

The various device modes can be set from the system CPU by inputting a 4 bits address and 4 bits data (both MSB first) signal into the device with the

following timing of the MDATA, MCK, and MLD lines as shows figure 18-12.

(2)Description of MDATA commands

The details of MDATA shows in table 18-2

Mode	Address				DATA				Remarks
	A ₄	A ₃	A ₂	A ₁	D ₄	D ₃	D ₂	D ₁	
CLV	0	0	0	1	FGARI H:w/FG L:w/o FG	TTon H:TTon L:TToff	00 STOP Tree Run 01 ACC 10 Brake 11 No operation		-
Mute / ATT	0	0	1	0	-	-	ATT H: -12dB L: 0dB	MUTE H:mute L:0dB	-
TRACK COUNTER	0	0	1	1	TC ₃	TC ₂	TC ₁	TC ₀ (LSB)	Low
	0	1	0	0	TC ₇	TC ₆	TC ₅	TC ₄	Middle low
	0	1	0	1	TC _B	TC _A	TC ₀	TC ₈	Middle high
	0	1	1	0	Count enable	TC _E (MSB)	TC _D	TC _C	High
STATUS	0	1	1	1	-	-	00 CRC 01 OTC 10 CLVS 11 TTSTOP		-

Table 18-2

(a)FG ARI

The MN6617 has a motor speed limiter which uses the FG signal to be compatible also with motors having no FG, the device can deactivate the FG speed limiter by setting FG ARI is low.

**FG ARI is set high:

= FG speed limiter is activated.

**FG ARI is set low:

= FG speed limiter is deactivated.

(b)TTON

This command is used for starting and stopping turntable operation:

** TTON is set high = Turntable starts.

** TTON is set low = turntable stops.

(c)Forced Turntable Control Mode

D2	D1	Mode	Operation
0	0	STOP	Turntable forced free-run
0	1	ACC	Turntable forced acceleration
1	0	Brake	Turntable forced deceleration
1	1	No Operation	No Operation

Table 18-3

(d)ATT

This command allows the D/A output data to be attenuated. It is effective for both parallel DA0 - DA15 output and serial SRDATA output.

Output is attenuated -12dB when ATT is set high. And attenuation is released when ATT is set low.

(e)MUTE

This command allows the D/A output data to be muted in case of dropout. The D/A output is muted to logical all zeros when MUTE is set high, muting is cleared when it is set low (only when DMUTE=low)

The DMUTE pin may be used for direct muting since it is logically OR'ed with the MUTE signal.

(f)Track Counter

When an appropriate number N (the number of tracks to be counted, which is computed by the CPU) is furnished to the TCO-TCE inputs and the count enable pin is set high, the device counts the negative edge of the clock applied to the ITC pin (Track Counter Input). The TCO (internal) is set low when the ITC clock count reaches N+1.

How to set number N

The maximum number presettable for N is $2^7 - 1 = 127$ (there are 7 bits from TCO through TCE)

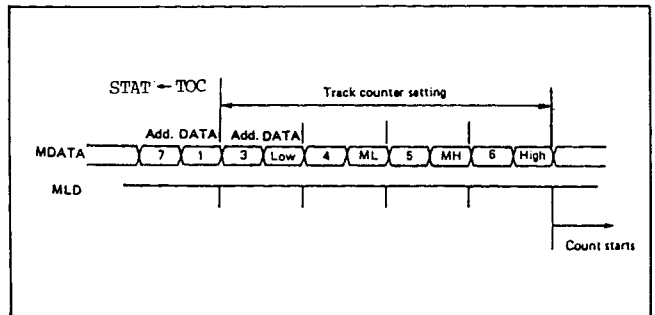


Figure 18-13

TCO output timing

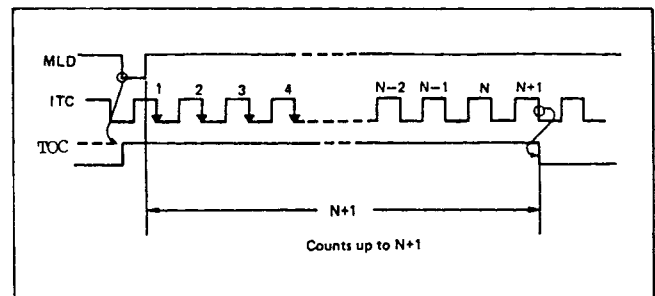


Figure 18-14

(g)STAT Signal Switching

At address 0111, the two low-order data bits provide the status signals. These signals are transferred to the CPU, which then checks contents as required.

D2	D1	Signal name	Contents
0	0	CRC	Result of the CRC Operation on the subcode H: NG L: OK
0	1	TOC	Track Counter Output
1	0	CLVS	Spindle Servo Phase Sync. Status Signal H: CLV L: Coarse servo
1	1	TTSTOP	Stop signal TURNTABLE H: Rotating; L: Stationary

Table 18-4

8) FPC Interface

The FPC interface compares the EFM signal frequency with the PCK frequency (extracted PLL clock)

- (a). If PCK frequency is higher than EFM frequency the FPC is set to high.
- (b). If PCK frequency is lower than EFM frequency the FPC is set to low.
- (c). If the two frequencies are identical the FPC is set to high impedance.

The EFM signal pulse width is counted using the PCK, and when 2T is detected, the FPC provides a low for the 16 PCK pulses interval.

When 12T is detected, the FPC provides a high for 64 PCK pulses interval.

If a dropout sense signal (DO = high) is received, the FPC is held at high impedance even while synchronization is maintained (in which BSSEL is high).

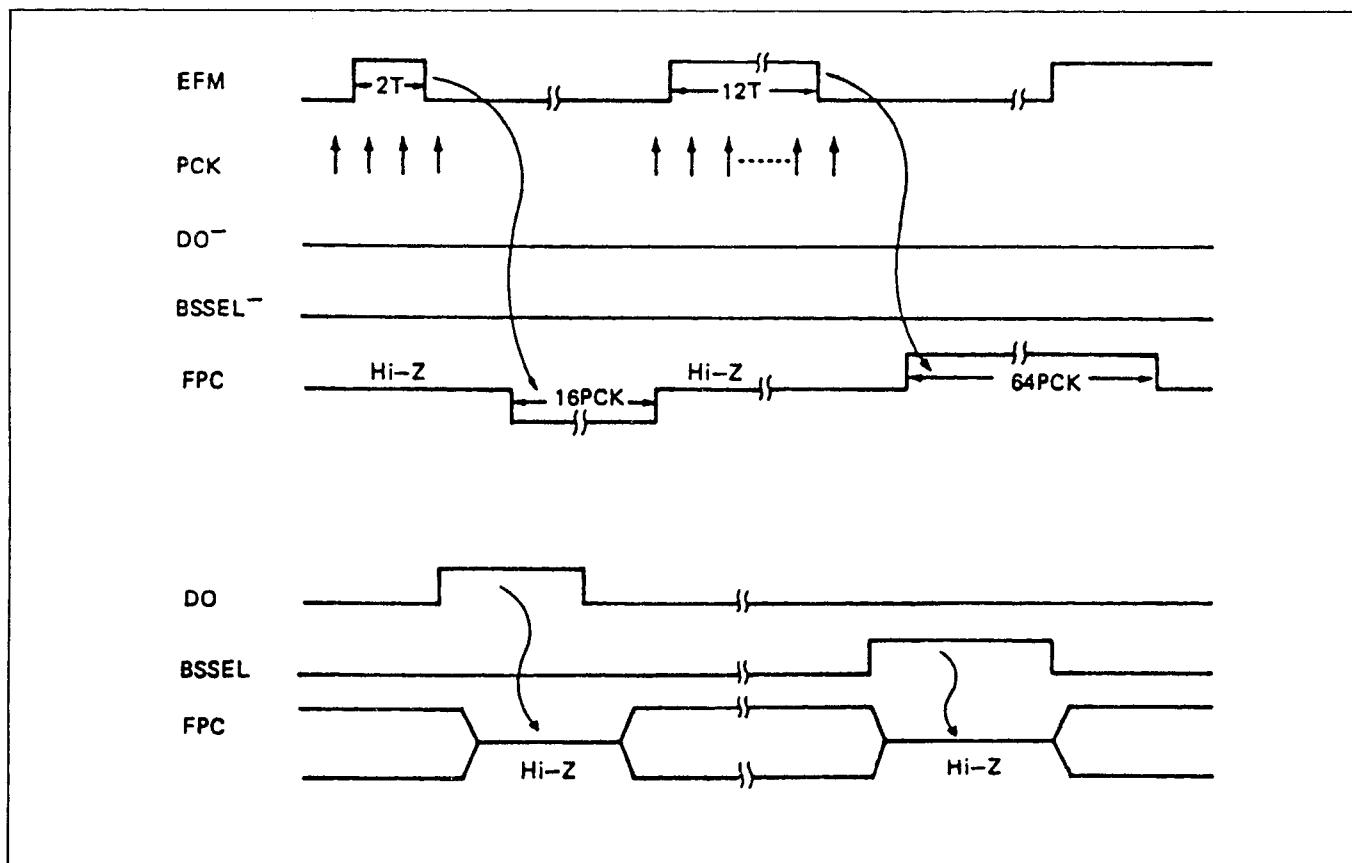


Figure 18-15

9) Pin Description of MN6617

● IC301: MN6617 (Digital Signal Processing: EFM Decoder, Error Correction, CLV Servo)

Pin No.	Mark	I/O Division	Function	Pin No.	Mark	I/O Division	Function
1	BLKCK	O	Sub-code block (Q data) clock (75Hz)	33	RESY	O	Resynchronizing signal (Not used, open)
2	CLDCK	O	Sub-code frame (Q data) clock (7.35kHz)	34	FCLV	O	Synchronizing detection signal (Not used, open)
3	SUBQ	O	Sub-code (Q data) output	35	IPBYTE	O	Interpolation flag for each byte (Not used, open)
4	CRC	O	Sub-code (Q data) CRC check (Not used, open)	36	IPSEL	I/O	Interpolation prohibited (Not used, connected to GND)
5	RST	I	Reset signal input (reset at "L")	37	FLAG5	O	C2 decoder correction flag 3 (Not used, open)
6	MLD	I	Command load input	38	FLAG4	O	C2 decoder correction flag 2 (Not used, open)
7	MCLK	I	Command clock input	39	FLAG3	O	C2 decoder correction flag 1 (Not used, open)
8	MDATA	I	Command data input	40	FLAG2	O	C1 decoder correction flag 2 (Not used, open)
9	DMUTE	I	Muting control (Not used, connected to GND)	41	FLAG1	O	C1 decoder correction flag 1 (Not used, open)
10	TRON	I	Tracking servo ON signal (tracking servo ON at "L")	42	FLCK	O	Crystal frame clock
11	STAT	O	Processing condition (CRC, OTC, CLVOK, TT STOP) output	43 } 50	D7 } D0	I/O	16K RAM data output
12	SMCK	O	Clock output (4.2336MHz)	51	RAMOE	O	16K RAM OE signal
13	PMCK	O	Pitch control clock output (Not used, open)	52	RAMWE	O	16K RAM WE signal
14	ITC	I	Track counter input signal (Not used, open)	53 } 63	RAMA0 } RAMA10	O	16K RAM address signal (RAMA0: LSB, RAMA10: MSB)
15	TEST	I	Test mode selection (Not used, connected to +5V)	64	PC	O	Spindle motor ON signal (ON at "L")
16	X2	O	Clock output (Not used, open)	65	EC	O	Spindle motor drive signal
17	X1	I	Clock input	66	FG	I	Spindle motor FG signal input (Not used, open)
18	SEL	I	DA output parallel/serial selection (serial at "L") (Not used, connected to GND)	67	VCNT	—	(Not used, open)
19	LDG/WDC	O	L channel deglitch signal/serial data word clock. (Not used, open)	68	REXT	—	(Not used, open)
20	RDG	O	Clock signal output for spindle motor control (Not used, open)	69	VDD	I	Power supply (+5V)
21	DEMPH	O	De-emphasis ON signal (de-emphasis ON at "H") (Not used, open)	70	PD	—	(Not used, open)
22	IPFLAG	O	Interpolation flag (interpolation at "H") (Not used, open)	71	PCKO	—	(Not used, open)
23	FLAGO	O	Error flag (error at "H") (Not used, open)	72	PCK	I	PLL extract clock input
24	FLAG6	O	16K RAM address reset signal (reset at "H") (Not used, open)	73	VDD	I	Power supply (connected to +5V)
25	XCK	O	Clock (16.9344MHz) output (Not used, open)	74	EFM	I	EFM signal input (PLL)
26	SRDATA	O	Serial data output (MSB first)	75	SRF	I	EFM signal input (DSL)
27	SRDATA	O	Serial data output (LSB first) (Not used, open)	76	DO	I	Drop-out signal (Drop-out at "H")
28	SRCK	O	Serial data beat clock.	77	CLVS	O	11T servo OK signal (OK at "H") (Not used, open)
29	WDCK	O	Serial data word clock (Not used)	78	FPC	O	PLL frequency comparison signal
30	BYTCK	O	Serial data byte clock (Not used, open)	79	BSSEL	O	PLL frequency in take operation signal. (Not used, open)
31	GND	I	GND terminal	80	SRFO	—	(Not used, open)
32	R/L	O	R/L signal	81	NSRF	—	(Not used, open)
				82	RF	—	(Not used, open)
				83	SUBC	O	Sub-code serial output data (Not used, open)
				84	SBCK	I	Clock for sub-code serial output (Not used, open)

Table 18-5

10) Block Diagram of Digital Circuit

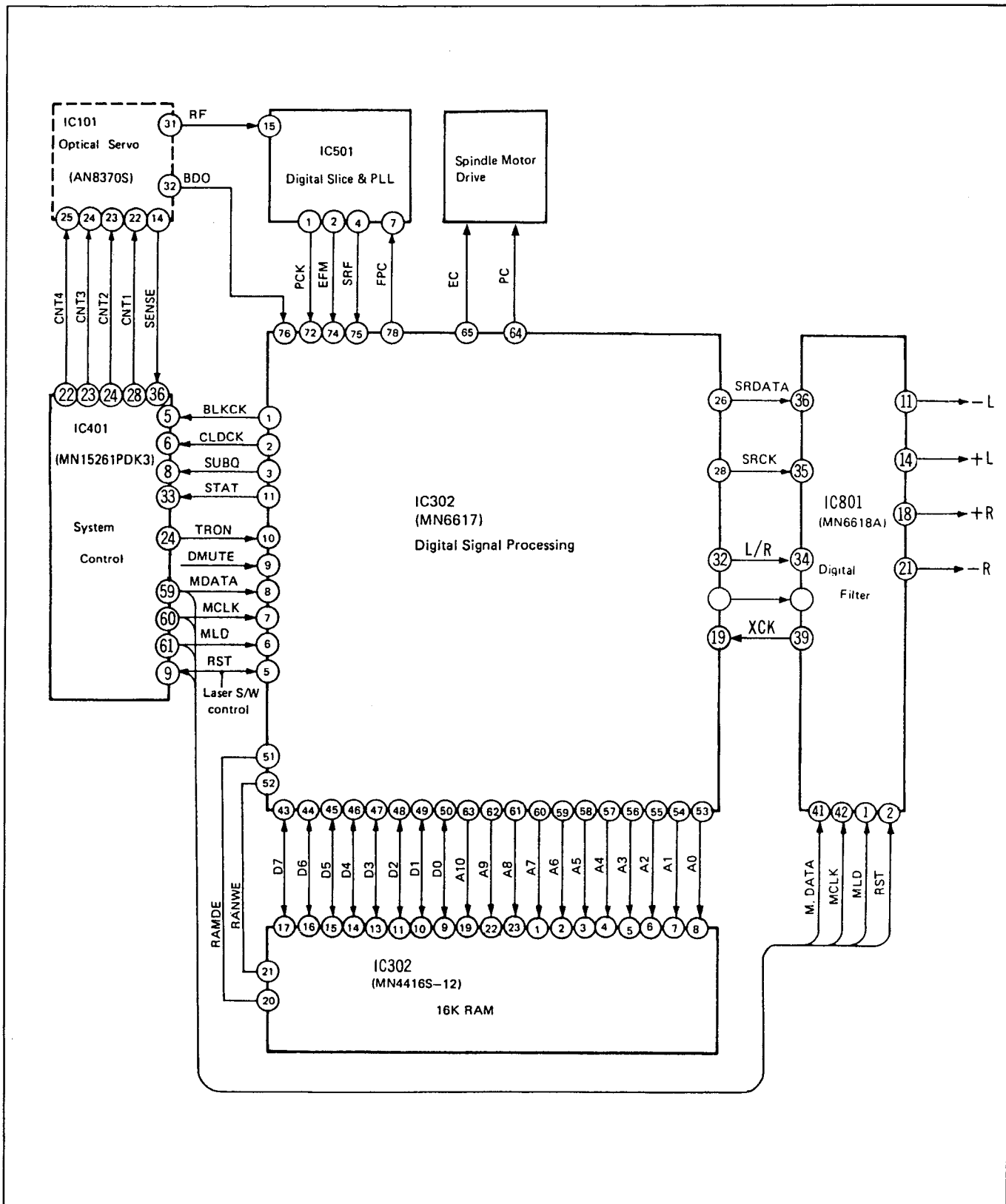


Figure 18-16

10. Digital Filtering and D/A Converter

1) Principle of the Digital Filter

The way an audio wave is converted to a digital signal, processed digitally, and then returned to a high-purity audio form, is more and more widely understood.

The conversion from analogue into digital signals is often called pulse code modulation, or PCM. The analogue waveform is first sampled (measured at regular intervals). Then it is quantized. This means that it is given the nearest numerical value in the large range of numbers used to transmit the digital sound information. The quantized value is then converted to binary form and encoded into a pulse train representing the series of numbers. This process of analogue to digital conversion is shown in figure 19-1.

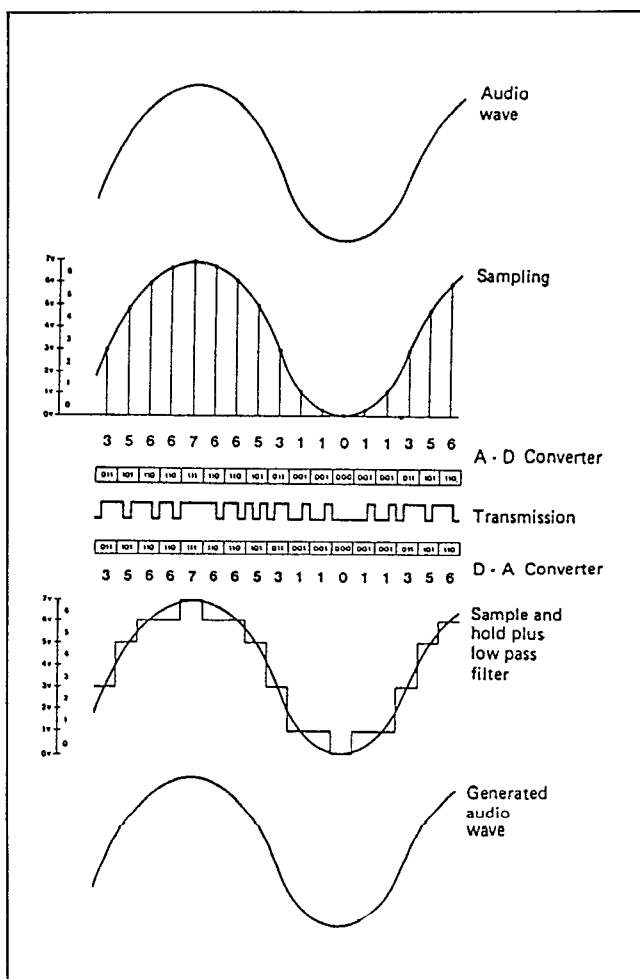


Figure 19-1

Figure 19-1 shows the digital sound reproduction process. The audio signal is sampled, quantized, converted to binary form, and encoded for recording or transmission.

Reversing the process produces a replica of the original signal. It is in this second stage that CD players use digital filtering to achieve exceptionally accurate reproduction.

The pulse train is fed into the transmission system (a cable or radio link, a tape, or a disc etc). At the reception or playback end, the pulse train is detected, decoded and regenerated in analogue form in the reverse process known as digital to analogue conversion.

2) How the Number of Bits Affects S/N

The accuracy of a digital system depends on the number of bits used for the binary number. This is because the number of bits determines the accuracy with which the samples taken from the original waveform can be quantized, i.e. expressed in digital form.

Two bits give four possible values for use in quantization, also three bits give eight possible values.

Every bit added doubles the number of possible values. In audio terms, this adds 6dB to the signal to noise ratio, because it reduces the quantization noise by a factor of 2.

The sixteen bits as used in the CD system give 65536 (2^{16}) possible values for quantization, and a signal to noise ratio of 96dB. The audio signals are encoded in the actual CD with this ratio.

96dB is an approximation. An additional factor, which depends on the type of signal, is always present, regardless of the number of bits; for sine wave signals this makes the theoretical figure 97.8dB.

3) The Effect of the Sampling Frequency

Another important factor is sampling frequency, for efficiency in digital operations, this is kept low (44.1KHz)

at a little over twice the maximum audio frequency (20.0KHz). During the sampling process this frequency and its harmonics or multiples become modulated with the original waveform in a series of frequency bands as shows figure 19-2.

The bottom end of the lowest band is rather close to the top of the audio frequency range.

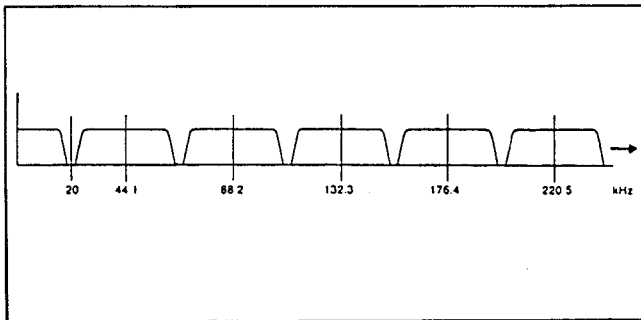


Figure 19-2

Figure 19-2 shows the frequency spectrum produced by sampling showing the periodic repetitions of the audio band. The bottom end of the lowest band is close to the audio frequency range.

4) Quantization Noise

In a digital audio system the audible noise has a different character to the familiar noise in an analogue audio system. The digital audio system background noise is called quantization noise.

The quantization noise of a system can be defined as the error signal between the original waveform before the analogue to digital conversion process, and the restored waveform after the digital to analogue conversion process. This error signal can be considered as both noise and distortion at the same time.

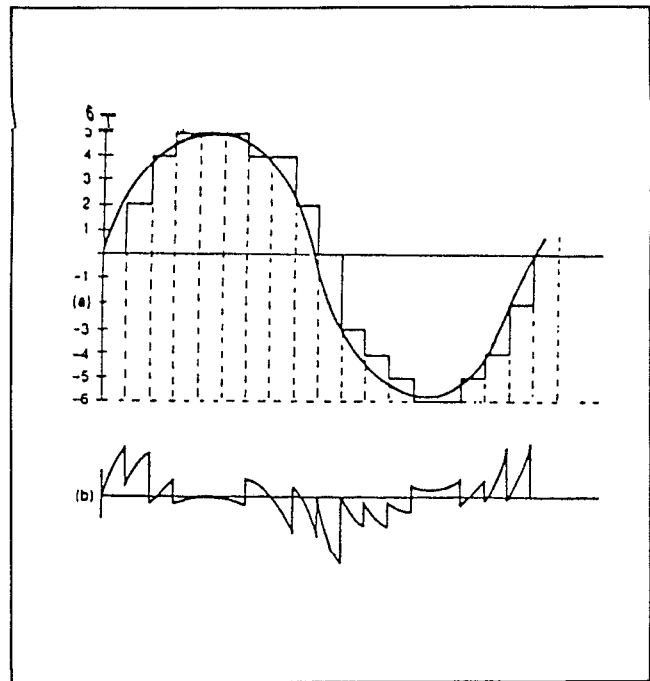


Figure 19-3

Figure 19-3 shows the quantization noise, The difference between the original audio waveform and the regenerated wave after D/A conversion (a) causes quantization noise (b).

The 16 bits analogue to digital conversion at the recording stage delivers a 16 bits digital signal with an intrinsic accuracy or signal to noise ration of maximum 96dB.

An ideal digital to analogue converter (D/A) would deliver an analogue output signal without adding noise to it. However, ideal D/A do not exist a small but finite electronic inaccuracy adds noise to the quantization noise of the digital signal.

The inaccuracy of a 16 bits D/A (the resolution limit of half the least significant bit) can be interpreted as a noise source that reduces the signal to noise ratio of the D/A to 96dB.

The effect of 16 bits analogue to digital conversion followed by 16 bits digital to analogue conversion with 1/2 LSB accuracy is to double the noise. In consequence, the maximum signal to noise ratio achievable in practice with a 16 bit system such as CD digital audio is 93dB.

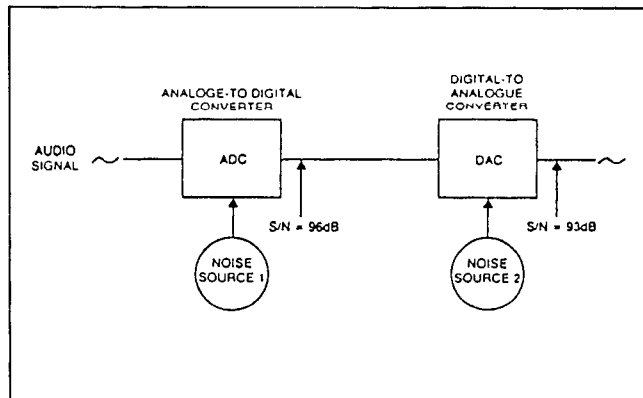


Figure 19-4

Figure 19-4 shows the signal to noise ratio deterioration in a digital chain, consisting of an A/D and a practical D/A.

5) Need for Filtering

With 16 bits (65536 possible numbers) and a sampling frequency of 44.1KHz, the CD system is theoretically capable of far higher reproduction quality than any conventional record player. But to achieve this quality, the output signal that the player derives from the disc must contain only the original audio frequencies, and not the periodic repetitions of the audio frequencies produced by the sampling (see figure 19-2).

These frequency bands can overload the power amplifier, and mix with audio frequencies or other signals to cause audible interference. They must be filtered out.

6) How D/A Conversion Operate

Figure 19-5 shows the generation process. Sampling produces only a series of sample (a) and hold (b) produces a step waveform closer to the original.

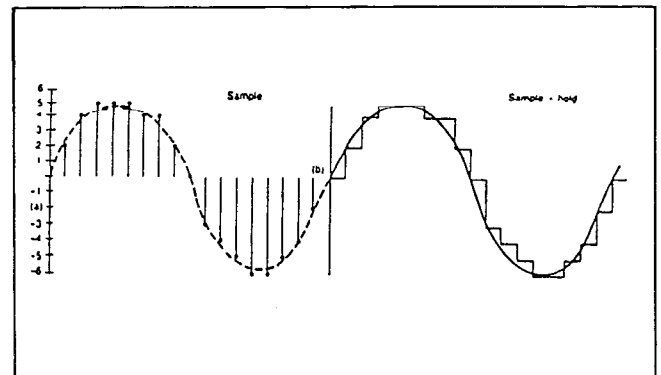


Figure 19-5

In the pulse train a series of binary number represents quantized samples of the original audio frequency. If these are simply converted to analogue values, a series of sample is produced (see figure 19-5).

The outline of this series resembles the audio wave, but the constant on and off switching produces the infinite series of frequency bands which was shown in figure 19-2. In practice, each sample is held in the converter until the next sample arrives, to produce a step waveform which is much nearer the shape of the audio waveform than the series of sample.

This sample and hold process is illustrated in figure 19-5 (b). It substantially reduces the frequency band components, as can be seen from figure 19-6. Significant components still remain, however particularly in the region near the audio band.

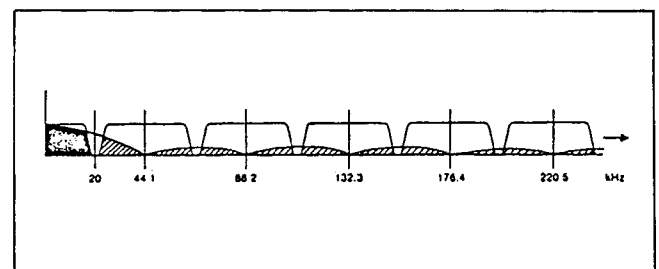


Figure 19-6

Figure 19-6 shows the effect of sample and hold. The remaining signal components are shown shaded.

7) Why Uses Digital Filtering

Most CD players suppress the frequency bands with conventional analogue filtering, after the signal has been converted back to analogue form. Such a filter is always complex, and can not avoid degrading the signal accuracy and introducing phase distortion.

Inevitable factors such as temperature changes and component ageing affect performance. Most CD players on the

other hand, carry out this essential filtering process digitally, with no loss of accuracy, and phase distortion.

Digital filtering is a pure numerical calculation process with fixed coefficients. The fixing of the coefficients determines the filter characteristics which cannot afterwards vary.

8) Digital Filtering Process

A schematic diagram of the complete our D/A conversion system including digital filtering, is shows in figure 19-7.

It can be seen that our digital filtering uses the techniques of oversampling and noise shaping. Each plays a vital role in the digital filtering and conversion process.

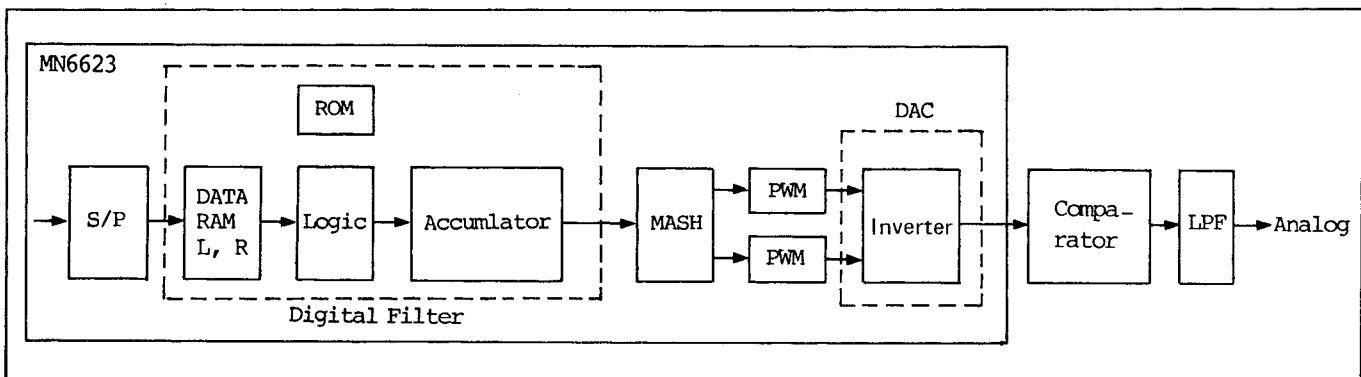


Figure 19-7

Oversampling:

Oversampling simply means multiplying the sampling frequency, in this case by four times. (44.1KHz becomes 176.4KHz.) The values of the intermediate samples are obtained by the calculation process

which determines the filter characteristic. An immediate effect of the oversampling as shows in figure 19-8 is to suppress all the lower frequency bands, centered on f_s , $2f_s$, and $3f_s$.

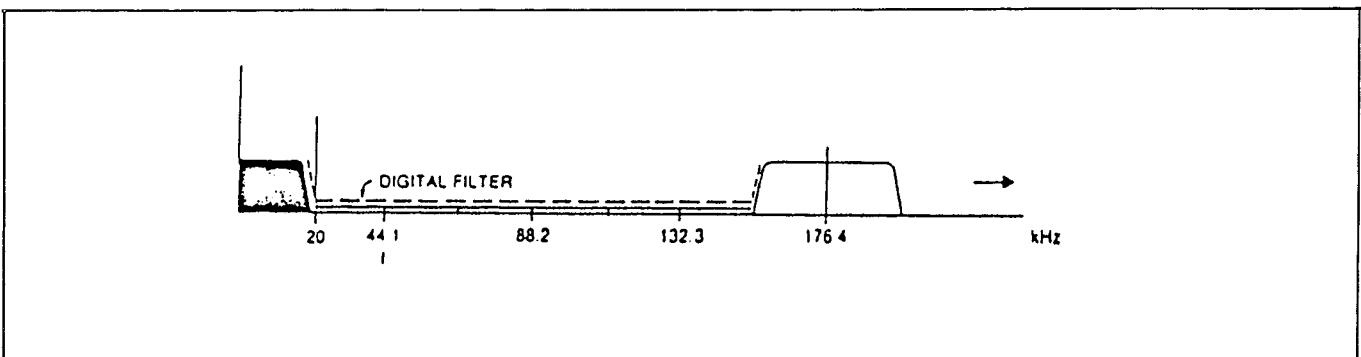


Figure 19-8

9) MN6623 Actual Operation

The MN6623 D/A converter system is composed of a digital filter, MASH, and PWM blocks as shows in figure 19-9.

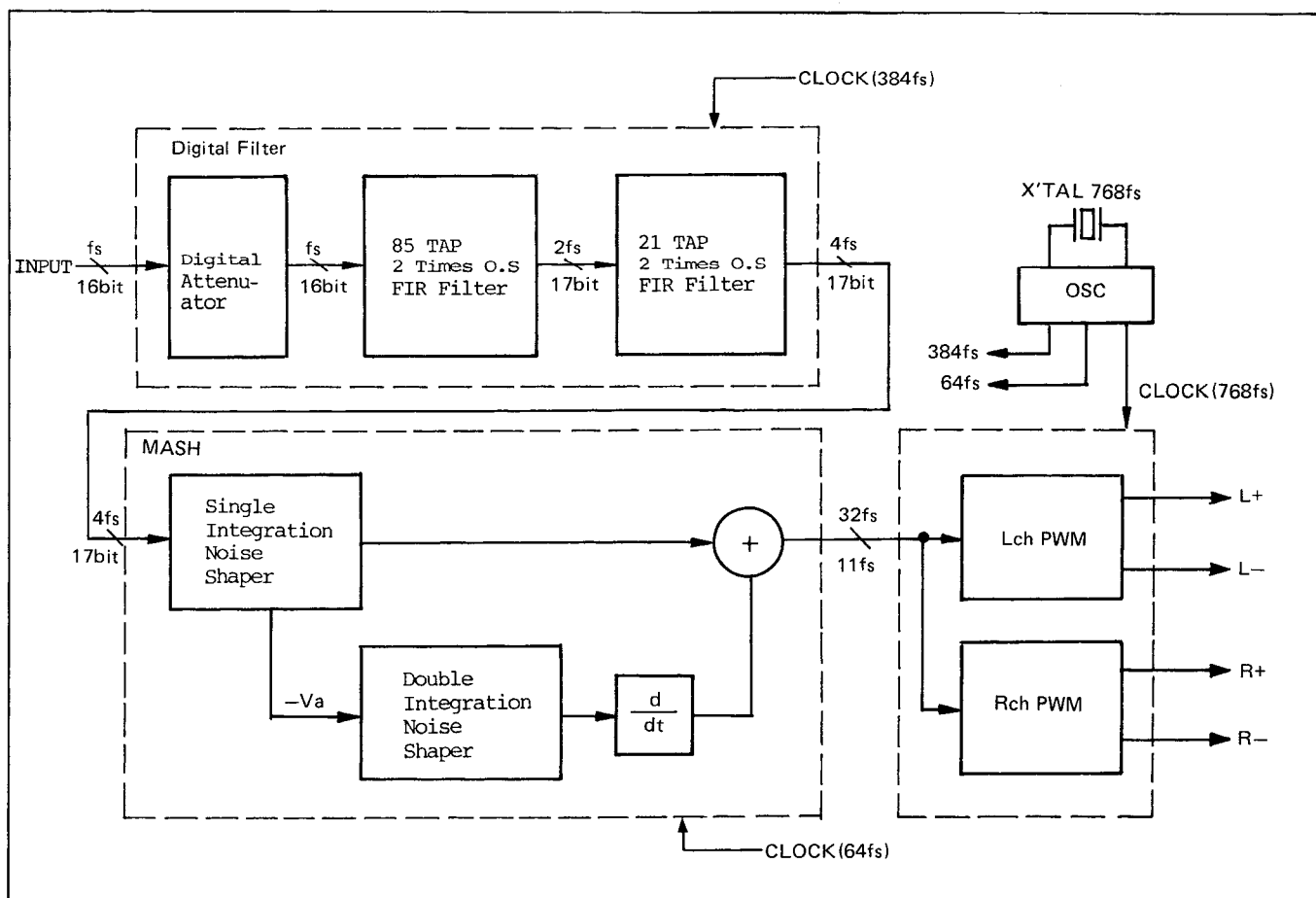


Figure 19-9 MN6623 Block Diagram

The MASH block operates at $32f_s$, where f_s is the sampled input signal frequency. This means that a 32 fold oversampling filter is required. On the MN6623, however, the first digital filter oversamples the f_s signal into a $4f_s$ signal. The following MASH block then performs zero-order holding on the $4f_s$ signal to convert it into a $32f_s$ signal.

The digital filter and MASH blocks use $384f_s$ and $64f_s$ clocks, respectively, to perform time-division multiplexing on the left and right channel signals. The PWM block uses a $768f_s$ clock to process the left and right signals independently. The MASH block converts the 16 bits, $32f_s$ signal into 11 values, which after Pulse-Width Modulation in the PWM block become analog signals. The following section describe the function of each MN6623 block.

Digital Filter Block

The digital filter block consists of three sub-blocks: ATT, FIR1, and FIR2 as shows in figure 19-10. It performs digital attenuation and oversampling to convert the input 16 bits, f_s data into a 16 bits, $4f_s$ signal.

The ATT attenuates the input data in a $1/64$ increment by multiplying the data by any 6 bits data. FIR1 and FIR2 are 2-fold oversampling digital filters with 85 and 21 taps each. Four-fold oversampling is accomplished by cascading the two fold oversampling filters.

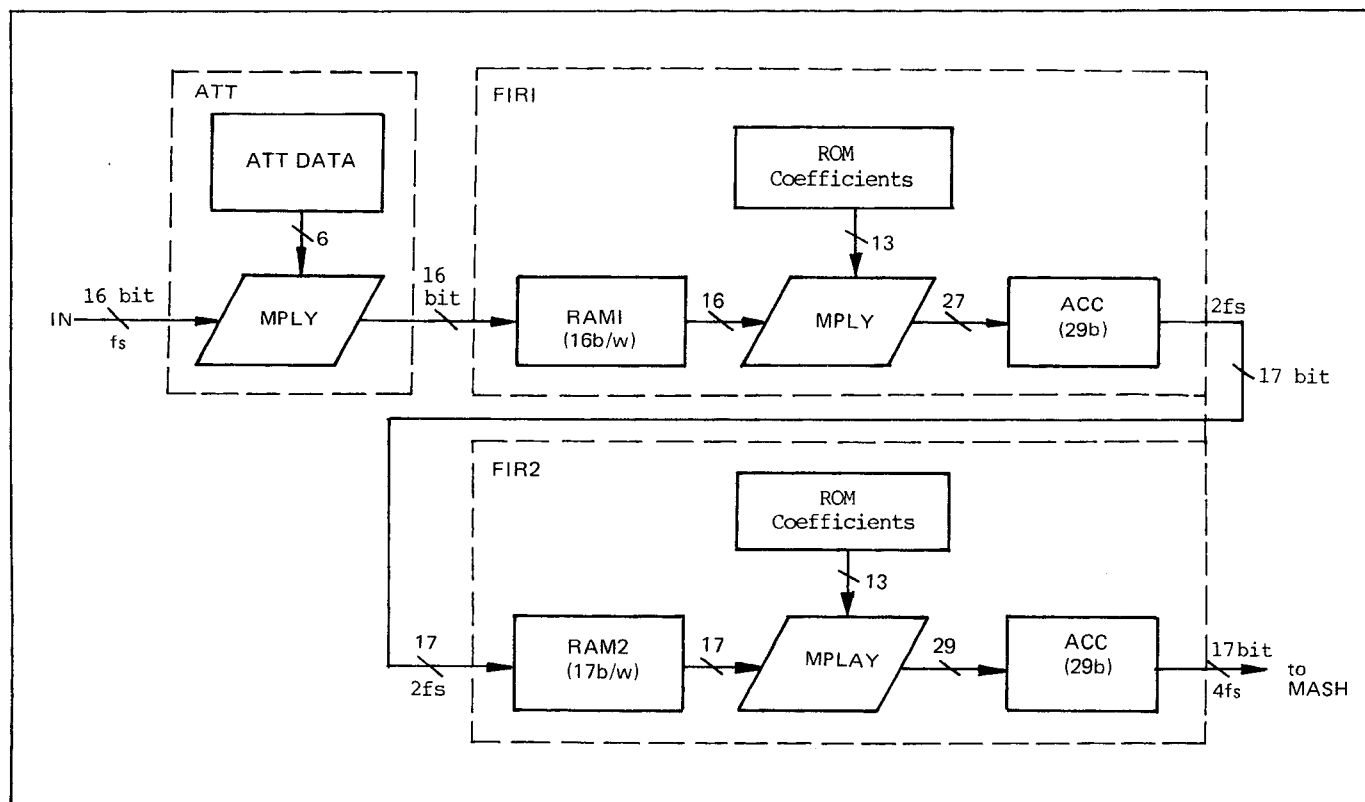
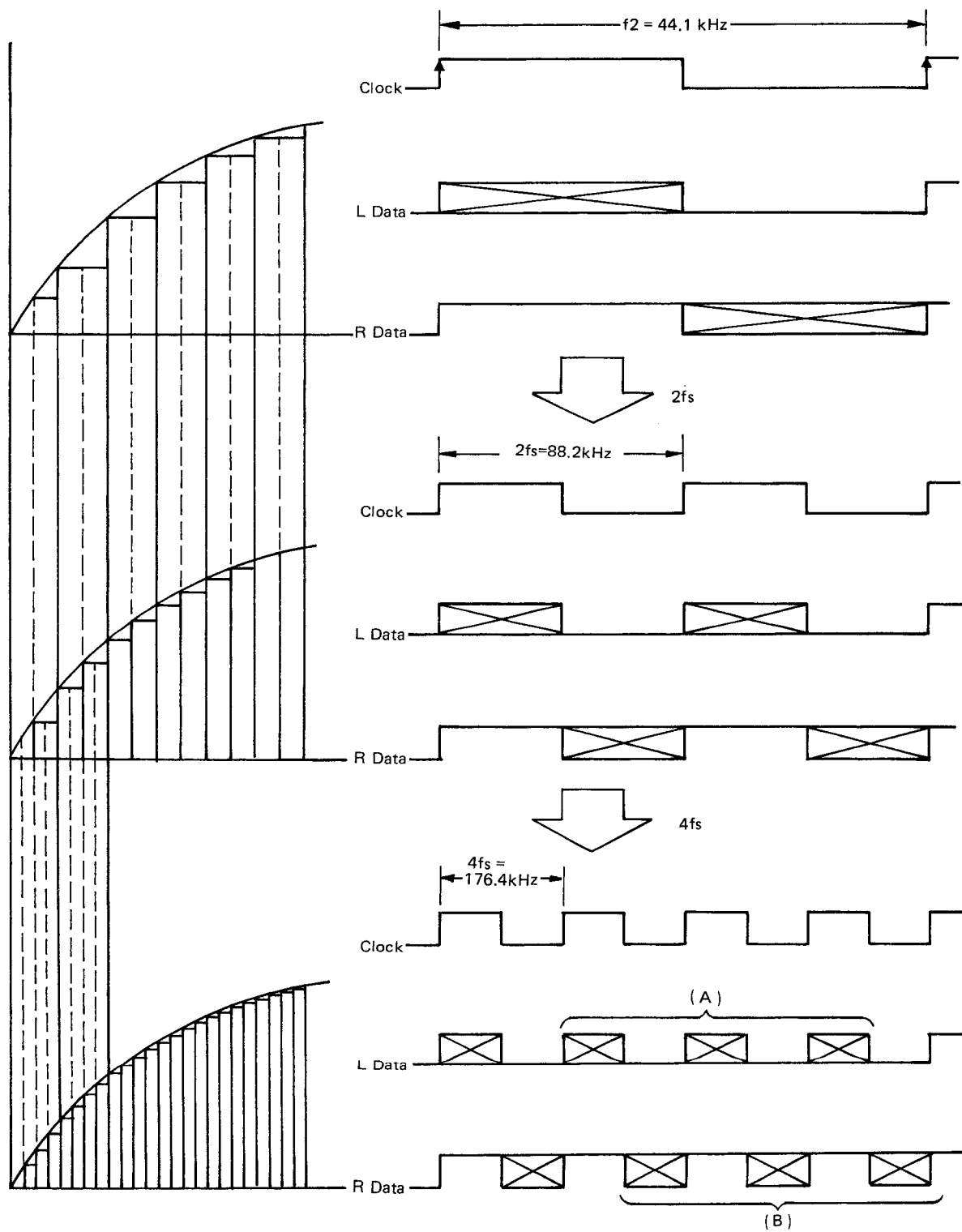


Figure 19-10 Digital Filter Block Diagram

The digital filter block contains many memories (delay logics), as shows in the block diagram above, with which it computes ROM coefficients at high speed while summing the coefficients to interpolate them. For more detail, shows in figure 19-11.



Note: (A) and (B) are the data created in the digital filter.

Figure 19-11

MASH Block

The MASH block consists of many noise shapers connected in series. The noise shapers include the signal and double integration types.

* Single integration noise shaper

The single integration noise shaper uses only a single integrator for noise shaping as shows in figure 19-12. The block marked "Q" in this figure is a local quantizer which adds quantization error to the input data. The relationship between output (Y) and input (X) for this type of noise shaper is:

$$Y = X + (1 - Z^{-1}) \cdot Vq \dots\dots\dots (1)$$

The quantization error, Vq is a random value and the factor $(1 - Z^{-1})$ in this equation represents a differentiation respons. Eq. (1) tells us, therefore, that the spectrum of the quantization error, Vq in the single integration noise shaper has an attenuation slope of -6 dB/oct toward lower frequencies as shows in figure 19-14.

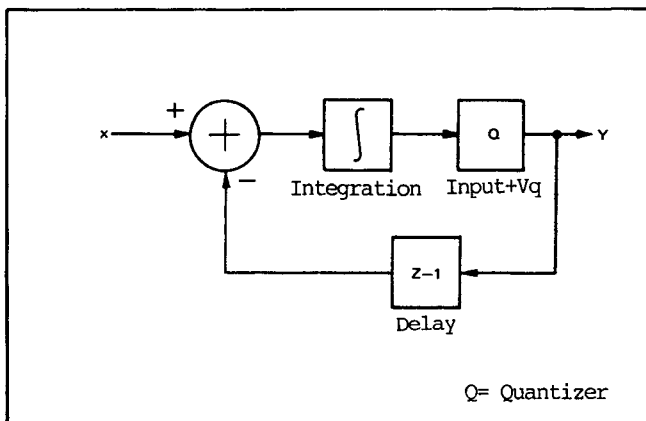


Figure 19-12 Single Integration Noise Shaper

* Double integration noise shaper

The double integration noise shaper uses two integrations in tandem as shows in figure 19-13. When the circuit is viewed at point W toward the output Y, it is

identical to the single integration noise shaper. Therefore, the relationship between Y and W is:

$$W = W + (1 - Z^{-1}) \cdot Vq \dots\dots\dots (2)$$

W expressed as a function of X is:

$$W = \frac{1}{1 - Z^{-1}} (X - Z^{-1} \cdot Y) \dots\dots\dots (3)$$

From Eqs. (2) and (3) we obtain following overall output-input relationship:

$$Y = X + (1 - Z^{-1}) \cdot Vq$$

If we compare this equation with Eq. (1), we find that the element $(1 - Z^{-1})$ is squared in the former equation.

This means that with the double integration noise shaper, the spectrum of the quantization error, Vq , has an attenuation slope of 12 dB/oct toward lower frequencies.

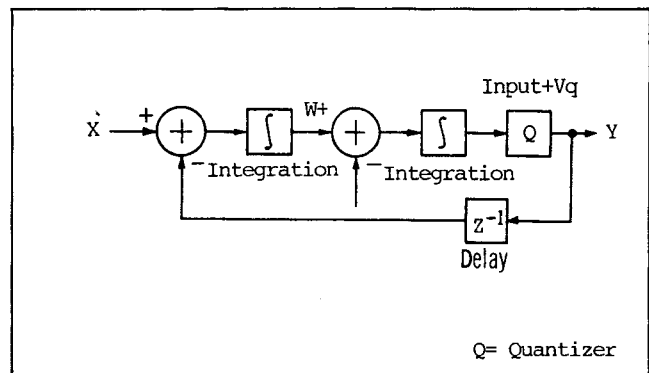


Figure 19-13 Double Integration Noise Shaper

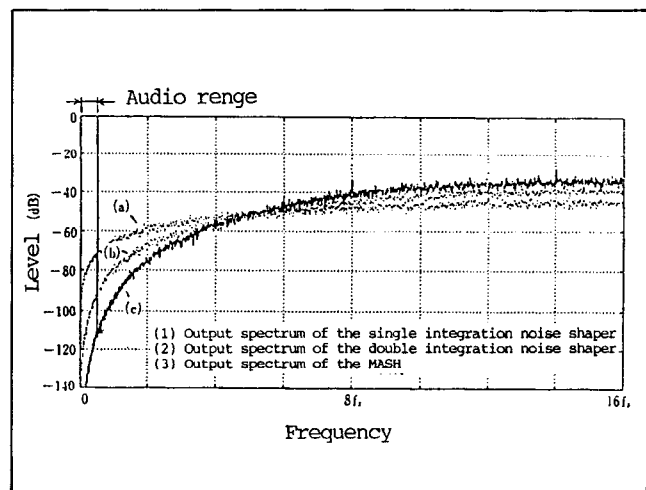


Figure 19-14 Noise Shaper Output Spectrum

The double integration noise shaper thus provides a greater attenuation of quantization noise than that obtained with the single integration type, as shows in figure 19-14.

The MASH block combines the single and double integration noise shapers in tandem so that the quantization error contained in the output of the first stage can be quantized again in the second stage further reduced.

Figure 19-15 is a block diagram of the MASH logic used on the MN6623 D/A converter. The MASH logic uses a single integration noise shaper in the first stage and a double integration noise shaper in the second stage. The outputs of the first and second stages of noise shaper, Y1 and Y2 are given by the following equations:

$$Y1 = X + (1 - Z) \cdot Vq1 \dots\dots\dots(5)$$

$$Y2 = -Vq1 + (1 - Z) \cdot Vq \dots\dots\dots(6)$$

Where

X = Input data

Vq1= Quantization error of local quantizer Q1

Vq2= Quantization error of local quantizer Q2

Y1 = Output of single integration noise shaper

Y2 = Output of single

Y2 = Output of double intergration noise shaper

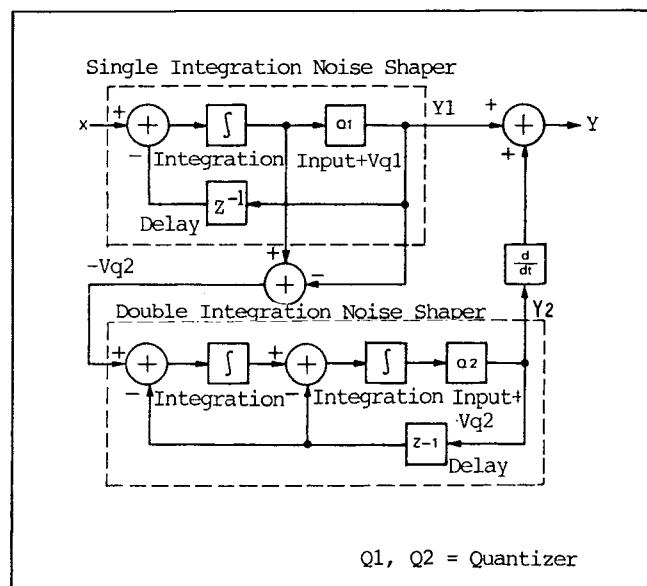


Figure 19-15 MASH Block Diagram

If we multiply both members of Eq.(5) by $(1 - Z)$ and add them to those of Eq. (6), we find that the quantization error is cancelled out.

Applying the output (Y2) of the second noise shaper to the differentiator and then adding the differentiator's output to the first noise shaper's output gives an overall MASH output, Y as represented by:

$$Y = X + (1 - Z) \cdot Vq2 \dots\dots\dots(7)$$

Eq.(7) indicates that the quantization error, Vq, is given in a third-order differentiation response (-18 dB/oct), which gives further reduction in quantization noise

Table 19-1 lists the quantization levels at the first and second local quantizers, Q1 and Q2.

Q1 OUTPUT	Q2 OUTPUT
33,792	11,264
22,528	
11,264	
0	0
-11,264	-11,264
-22,528	
-33,792	

Table 19-1 Quantization Levels at Local Quantizers

The first quantizer (Q1) quantizes the signal into 7 levels. and the second quantizer (Q2) quantizes it into 3 levels. Actually, the values which are obtained by normalizing by 11264. The output values listed in Table 19-1 appear at Y1 and Y2. i.e. values +3, +2, +1, and +0 appear at Y1 and values + and +0 appear at Y2. Since the output of the second local quantizer (Q2) is differentiated and added to the output of the first local quantizer (Q1), output Y yields 11 different values.

A high output level and wide dynamic range are thus achieved by quantizing the 16 bit, 4fs data from the digital filter at a sampling frequency 8 times as high (32fs).

The following chart contains the timings of the 11 types of digital data yielded by the MASH block.

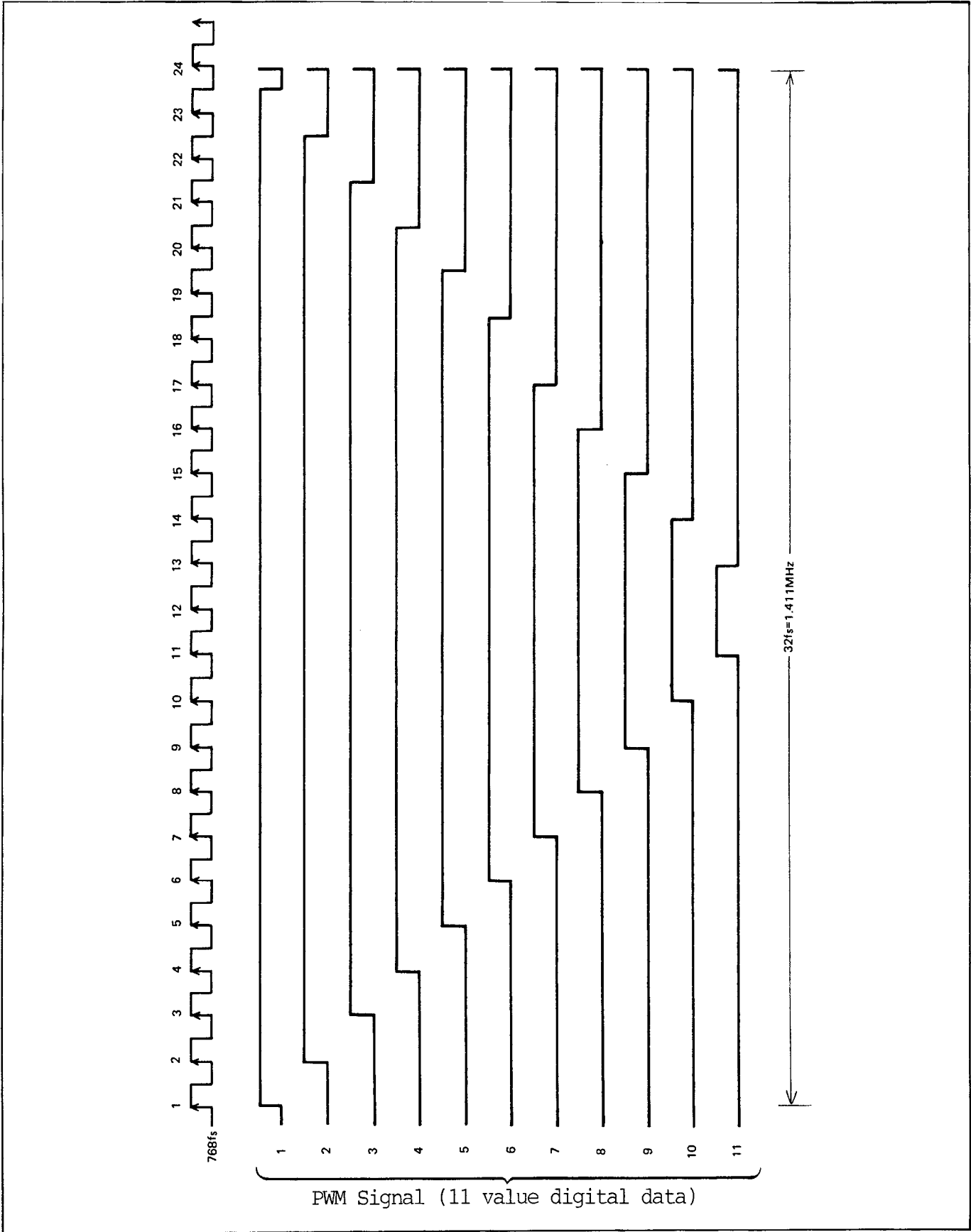


Figure 19-16 Output Data from MASH Part

PWM Block

The PWM block performs Pulse-Width Modulation on the 11 value digital data from the MASH block to convert into analog signals with 11 different pulse widths which are precision controlled by a crystal controlled oscillator.

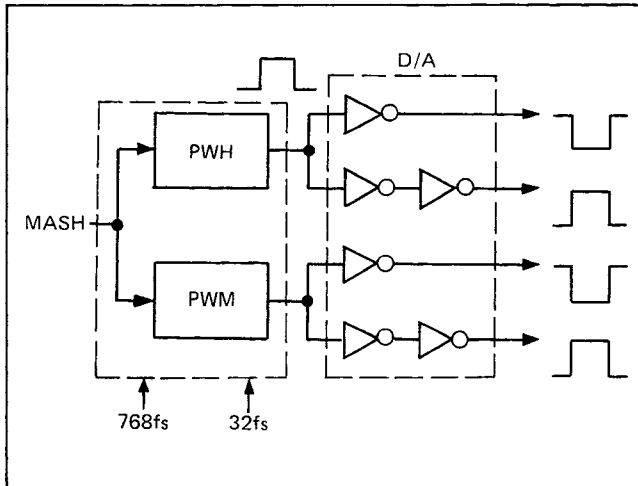


Figure 19-17 PWM Block Diagram

There are several different pulse modulation techniques available to convert analog signals into pulse signals.

With the PAM, PWM, and PPM techniques, the signal is sampled along the time axis but is not quantized along the amplitude axis. All of these techniques preserve the amplitude information exactly as it was the original analog signal. (see the following figure 19-18)

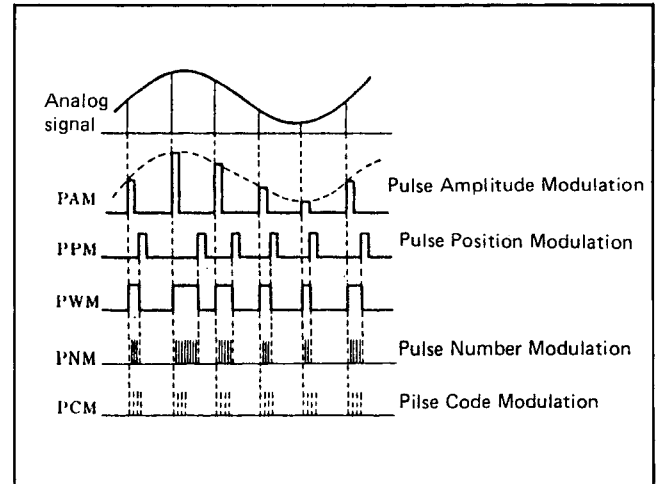
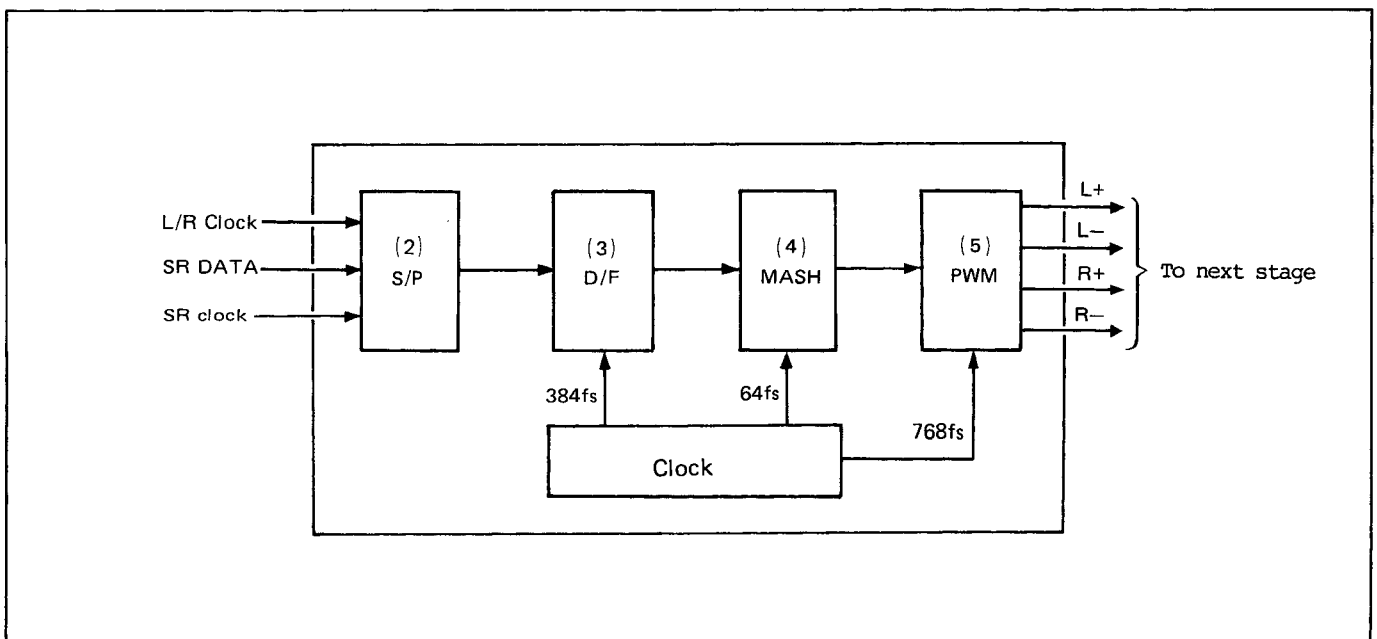


Figure 19-18

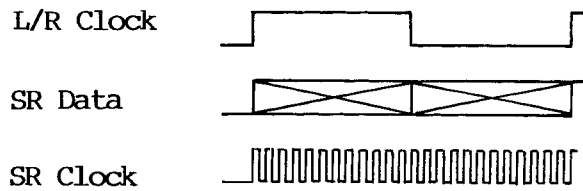
The pulse width modulation (PWM) samples the input signal along the time axis and yields pulses with pulse widths that correspond to sampled signal amplitudes. The resulting pulse signal contains frequency components similar to those obtained with the PAM technique.

Overall MN6623 Operations



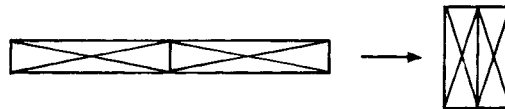
MN6623 Block Diagram

1. Input Signal



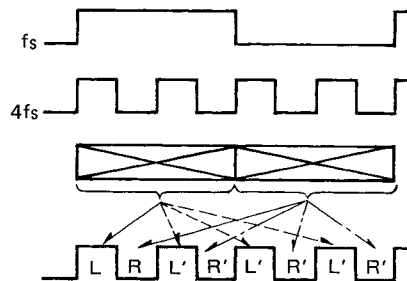
2. S/P Conversion

Converts serial L/R data into parallel L/R data



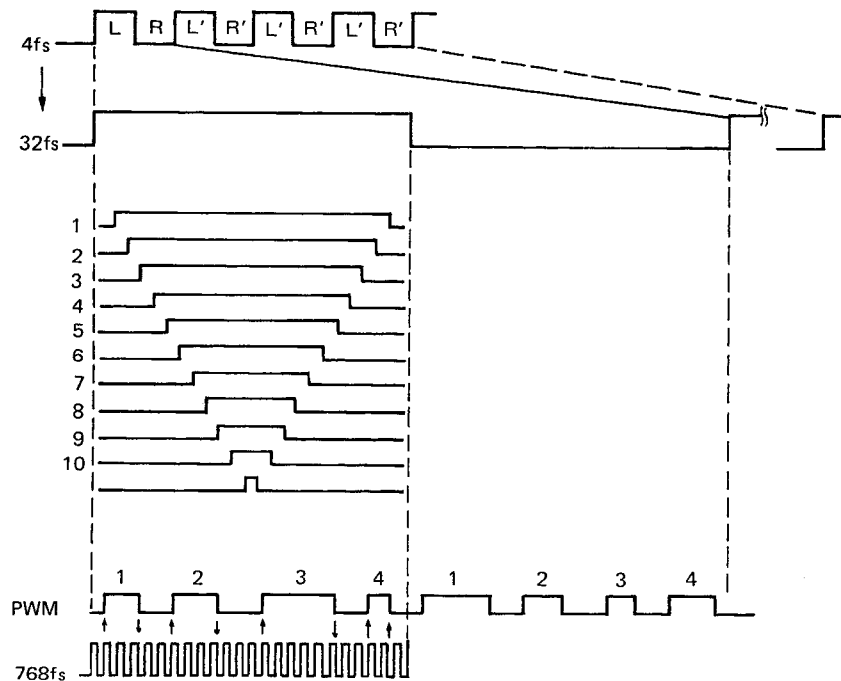
3. D/F Processing

4-hold oversampling



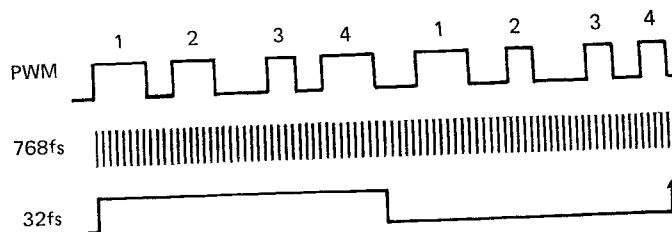
4. MASH Processing

Performs MASH processing on the L channel data from the first stage to determine 4 levels out of 11 levels and transfers them to the PWM block. Does the same processing for the R channel data.



5. PWM Processing

Pulse Width Modulates the 4 level data from the MASH block at a rate of 768fs, then separates the L and R channel data with 32fs clock. The L and R channel outputs of the PWM have the same phase, and are applied to the next stage at the rising edge of the 32fs clock.



10) Pin Description of MN6623A

● IC801: MN6623A (Digital Filter and D/A Converter)

Pin No.	Mark	I/O Division	Function	Pin No.	Mark	I/O Division	Function
1	MLD	I	Command load input (load: L)	22	AVDD1	I	Power supply (connected to +5V)
2	RSTB	I	Reset signal input (reset at "L")	23	AVDD1	I	Power supply (connected to +5V)
3	IE	I	I ² S select terminal ("H"=I ² S)	24	DVSS1	I	GND terminal (digital system)
4	TP1	—	TEST terminal (Not connected)	25	X2	O	Clock output
5	TP2	—		26	X1	I	Clock input
6	TEST1	I	TEST terminal 1 (connected to GND)	27	NC	—	Not connected
7	TEST2	I	TEST terminal 2 (connected to GND)	28	DVDD2	I	Power supply (connected to +5V)
8	NC	—	Not connected	29	DVSS2	I	GND terminal (digital system)
9	NC	—	Not connected	30	NSUB	I	Not used (connected to GND)
10	AVDD4	I	Power supply (connected to +5V)	31	768fs	O	768 fs (Not used)
11	OUTR(-)	O	Rch data output, (-) terminal	32	192fs	O	192 fs (Not used)
12	AVSS4	I	GND terminal	33	LRPOL	I	LR clock selector (Not used)
13	AVSS3	I	GND terminal	34	LR	I	L/R clock
14	OUTR(+)	O	Rch data output, (+) terminal	35	SRCK	I	Serial data best clock
15	AVDD3	I	Power supply (connected to +5V)	36	SRDATA	I	Serial data input (MSB first)
16	NC	—	Not connected	37	DVSS 3	I	GND terminal (digital system)
17	AVDD2	I	Power supply (connected to +5V)	38	NC	—	Not connected
18	OUTL(-)	O	Lch data output, (-) terminal	39	384 fs	O	384 fs (16.9344 MHz) output
19	AVSS2	I	GND terminal (analog system)	40	PD	I	Power down terminal
20	AVSS1	I	GND terminal (analog system)	41	MDATA	I	Command data input
21	OUTL(+)	O	Lch data output, (+) terminal	42	MCLK	I	Command clock input

20. Analog Circuit

The MA6623 (IC-801) uses independent +5 volts power supplies for its digital and analog processing block. Either channel of the analog audio signals (+) and (-) at the output of the D/A converter have the signal formats as shows in figure 20-1.

The positive and negative output of the D/A converter are coupled to the differential input of the comparator (IC-803), where they are amplified into a single analog signal with an amplitude twice as large. The output of the comparator is then waveform shaped by the following ternary LPF (IC-805) into a complete analog audio signal.

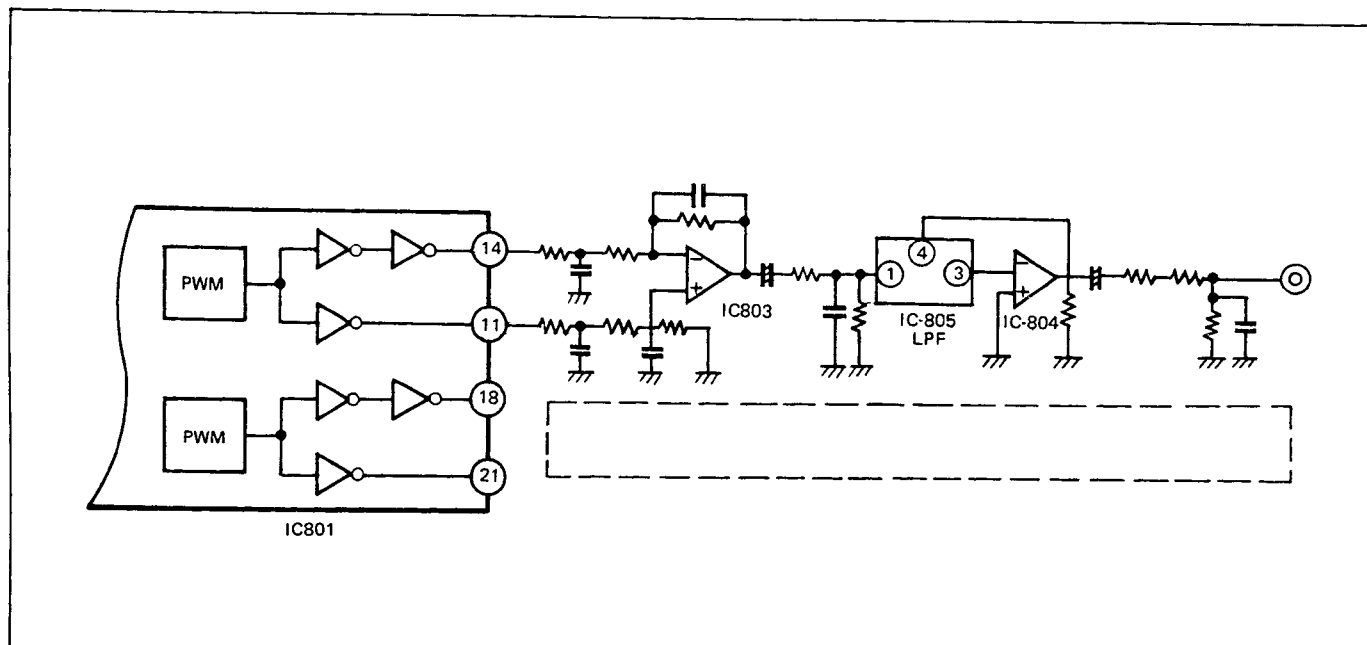


Figure 20-1

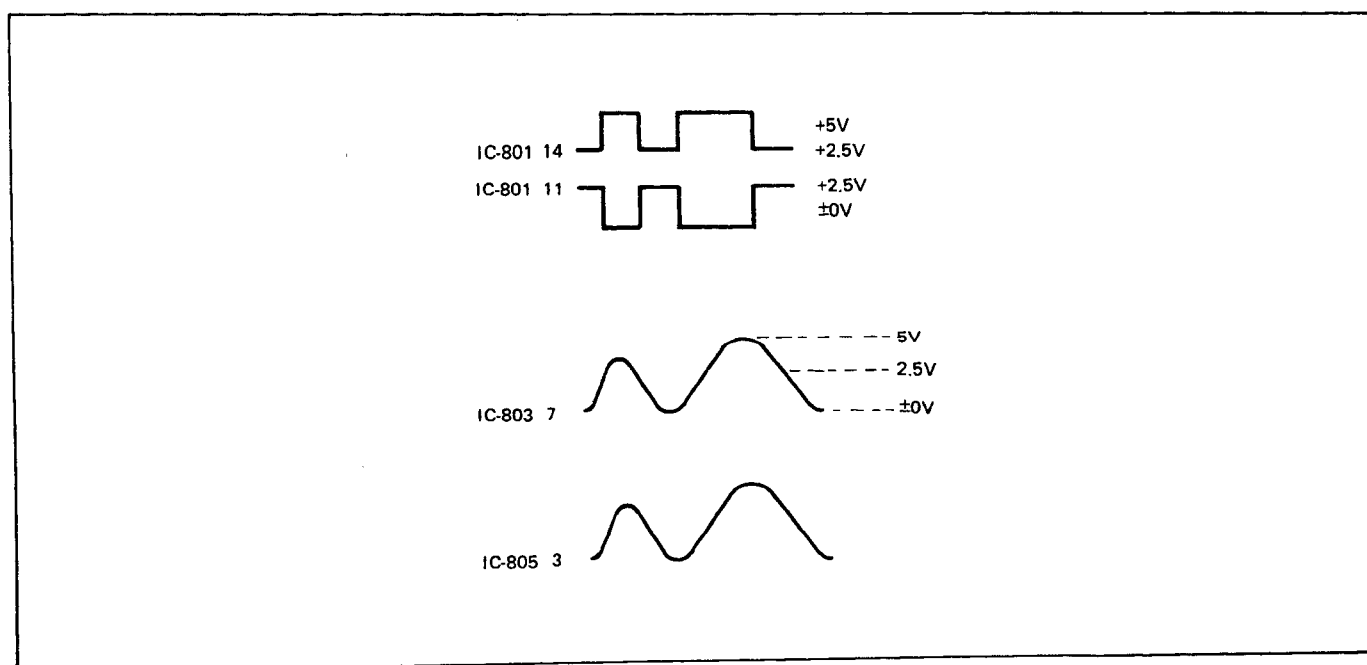


Figure 20-2

21. CD Changer's Operation

This section explains the SL-P405C CD changer's mechanism operation's and mechanism logic control operations (including SL-P400C etc).

1. General

The SL-P405C incorporates a recently-developed disc loading unit with an elevating mechanism which enables the user to load up to six compact discs in a magazine and play them in the desired sequence. The SL-P405C uses the One Beam system (FF-1MK2) for its optical (laser) pick-up.

The disc magazine allows loading and unloading of individual discs. The disc detection mechanism (provided only on selected models) provides disc search and displays the presence of discs on the player's display.

The mechanism section uses three DC motors for: Disc Loading, Elevator, and the Spindle. These motors are controlled by the system CPU (IC-401).

The SL-P405C signal processing logic and audio circuit are almost identical to those used in our comparable CD players.

2. Mechanism Orientation

The mechanism can be divided in three basic sections: The magazine holder, the Disc Loader, and the Optical Deck. See figure 21-1.

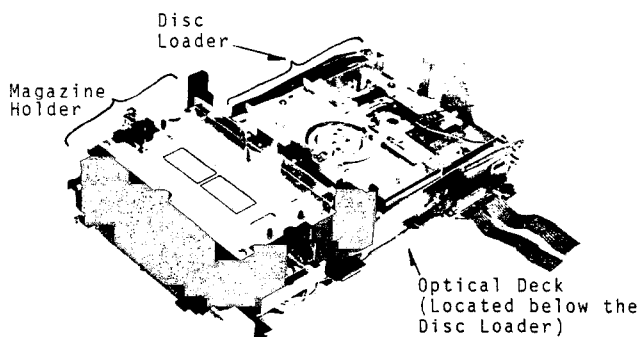


Figure 21-1 Mechanism

- * The magazine holder is located at the front-end of the mechanism. It holds the magazine in place, containing six discs (on trays), ready to be accessed by the disc holder.
- * The disc loader is located behind the magazine holder. Its function is to access the selected disc tray from the magazine and lower it on the optical deck's turntable.
- * The optical deck is located below the disc loader. Its function is to control the optics used to read the disc's data for Audio signal reproduction. The optical deck is the heart of the CD player. Its operation is explained on page 31.

2-1. Costruction of Disc Magazine

Figure 21-1 shows the construction of the disc magazine. The magazine can store up to six disc at a time.

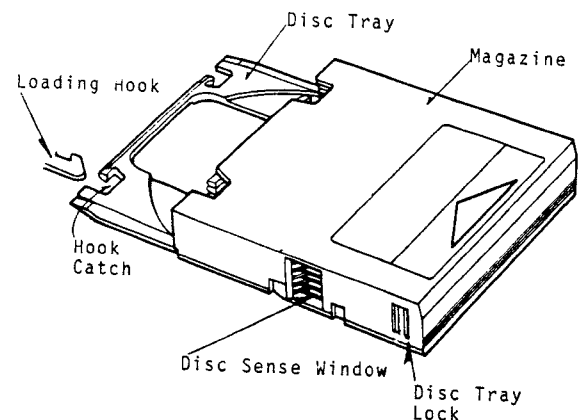


Figure 21-2 Disc Magazine

- (1). Hook Catch:
The disc loader accesses a tray by latching onto the HOOK CATCH of the disc tray.
- (2). Disc Sense Window:
This window is provided so that the six disc sensing switches (provided only on selected models) detect the presence or absence of a disc.
- (3). Disc Tray Lock:
The DISC TRAY LOCK retains the DISC TRAYS in place so that they will not drop out of the magazine accidentally. When replacing a disc, the user pushes this lock while pulling the DISC TRAY.

2-2. Construction and Operation of the Magazine Holder

A disc tray pulled out or returned to the magazine by the LOADING SLIDER which is driven by the LOADING MOTOR via the LOADING GEAR.

The tray to be transferred may be unlocked only at the moment when the tray is caught by the LOADING HOOK and pulled.

(1). The magazine is inserted into the magazine holder and pushed in the direction of arrow (1). see figure 21-3.

(2). The magazine first comes in contact with TAB (A) of the EJECT LEVER. see figure 21-3. At the same time PART (B) of the EJECT LEVER comes in contact with the LOADING SLIDER PIN and pushes the LOADING SLIDER in the direction of arrow (2). shows figure 21-4. PART (C) of the EJECT LEVER pushes LEVER (A) which causes the DISC SENSING SWITCH ASSEMBLY to move in the direction of arrow (3). When discs are loaded on any of the disc trays (1 through 6), the corresponding disc sensing switches are opened.

(3). As the LOADING SLIDER moves inward, PART (D) releases S452 (located on the bottom of the unit) and sets it to "OFF". This activates the LOADING MOTOR, which drives the LOADING SLIDER inward. The LOADING SLIDER PIN pushes the DISC TRAY LOCK RELEASE LEVER, moving it in the direction of arrow (3).

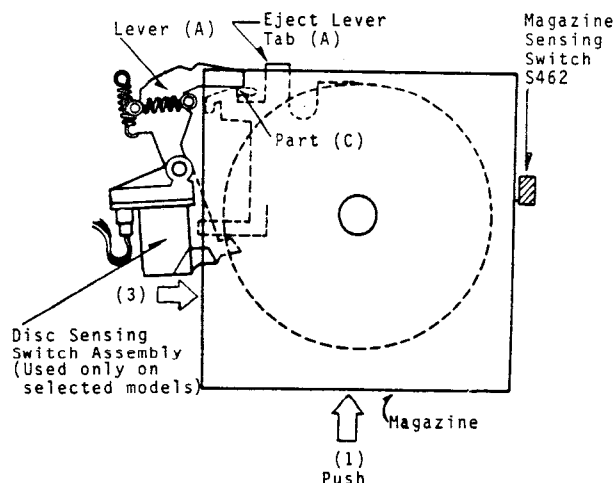


Figure 21-3

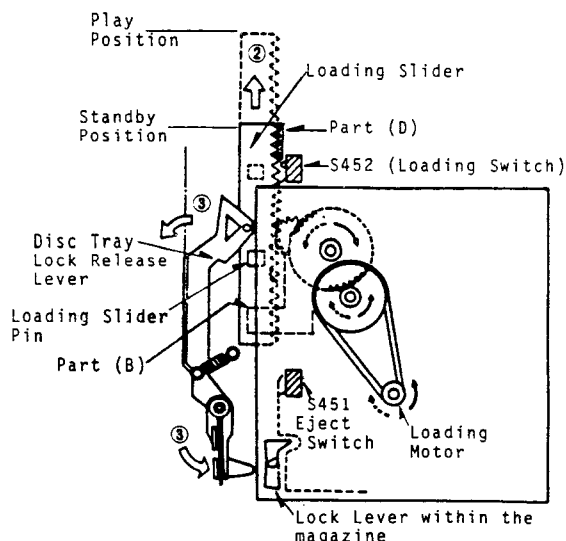


Figure 21-4

(4). On the right side of the magazine holder, you find the MAGAZINE SENSING SWITCH which is closed when the magazine is fully loaded.

Figure 21-4 shows the disc tray release mechanism within the magazine. The disc trays are locked so that they will not drop out of the magazine during transit. The trays must be unlocked before one of them can be transferred to the player's optical deck.

2-3. Construction and Operation of the Disc Loader

When selecting a disc, the LOADING MECHANISM must access the disc tray from the magazine and transfer it above the optical deck.

The CONNECTION ARM, part of the LOADING SLIDER, is coupled to the LOADING HOOK assembly. shows figure 21-5.

As the LOADING SLIDER moves in the LOADING SLIDER, is coupled to the LOADING HOOK moves in the direction of arrow (2).

Figure 21-6 describes the operation of the LOADING HOOK. As the LOADING SLIDER moves forward, The LOADING HOOK BASE moves in direction of arrow (3). The spring pulls the LOADING HOOK in the direction of arrow (2). As TAB (A) is released from PART (1), the LOADING HOOK then latches to the disc tray catch.

When the disc tray comes in contact with PART (B) of the TRAY SWITCH LEVER, PART (C) activates the DRAW SWITCH S453, detecting the disc tray's innermost position.

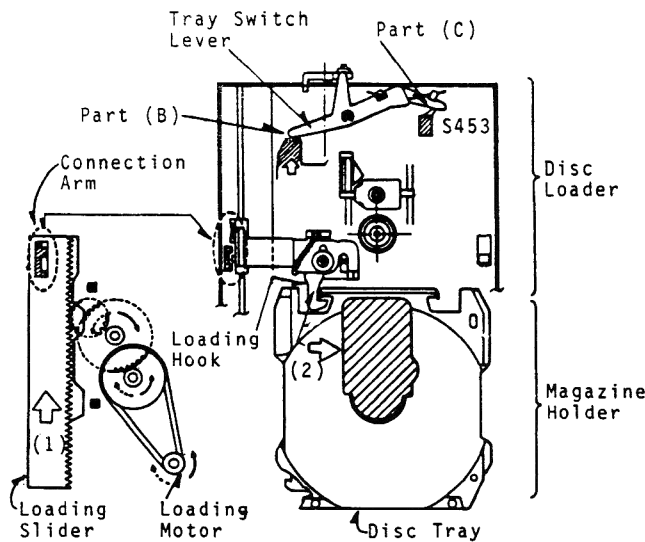


Figure 21-5

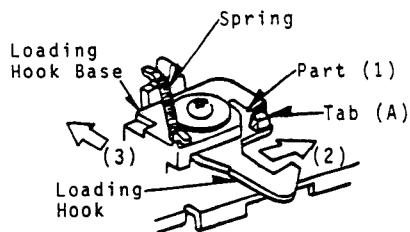


Figure 21-6

2-4. Disc Loader Elevation Mechanism

The function of the elevation mechanism is to drive the disc loader to the selected disc tray.

After a disc has been specified, the ELEVATOR MOTOR drives the ELEVATOR GEAR so that the LOADING UNIT is set at the height of the specified disc. shows figure 21-7.

The bottom of the ELEVATOR GEAR has a SHUTTER with two openings which operate the optical sensing circuit shows figure 21-8.

Two pulses are generated for each rotation of the ELEVATOR GEAR. The DISC LOADER is elevated from one disc to the next each time the ELEVATOR GEAR makes a single revolution.

The purpose of this sensing system is so that the system CPU can determine the location of the ELEVATOR UNIT.

A REST POSITION switch (S454) is provided to sense the lowest position of the ELEVATOR UNIT. This position corresponds to the position at which the disc rests on the turntable.

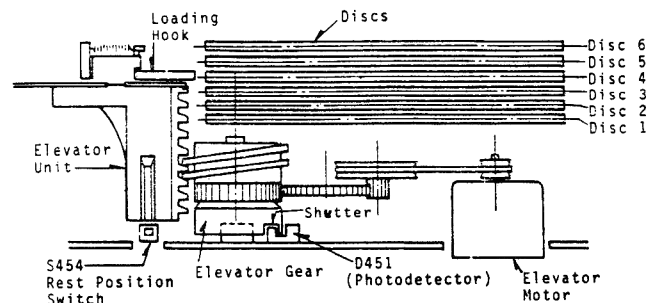


Figure 21-7 Side view of the elevator unit

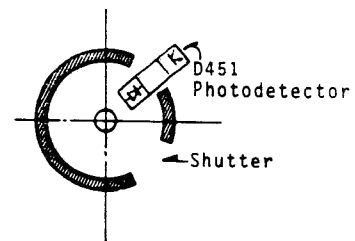


Figure 21-8 Bottom view of the elevator gear

3. Control Logic

3-1. Magazine Loading Operation

(1). The SL-P405C mechanism is equipped with five sensing switches and a height sensor. Four of the sensing switches along with the height sensor. Four of the sensing switches along with the height sensor contribute to the disc magazine loading operation shows figure 21-9. Their purpose is described as follows.

- A. S462 Magazine Detection Switch:
Normally open. Alerts the CPU that a magazine is being loaded.
- B. S451 Eject Switch:
Normally closed. Stops the loading motor operation and senses the "Eject" or (outermost) position of the magazine.
- C. S452 Loading Switch:
Normally closed. Detects the completion of the magazine insertion and activates the loading motor.
- D. S453 Draw Switch:
Normally open. Detects the disc tray's innermost position.

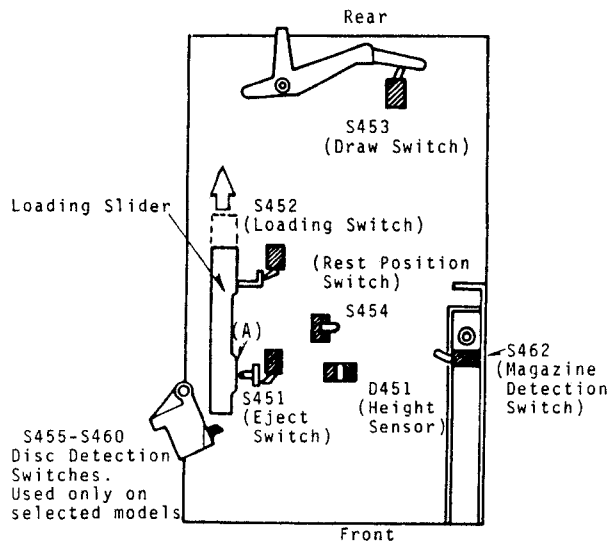


Figure 21-9 Sensing switches location

- E. S454 Rest Position:
Normally open. Detects the elevator unit's lowest position.
- F. D451 Height Sensor:
Detects the elevator's position.

The sensing switches are interfaced with the CPU (IC-401) input, pin 15. shows figure 21-10.

The CPU controls the loading motor and the elevator motor via its output pins 8 through 11, connected to the motor drive circuit of IC-401.

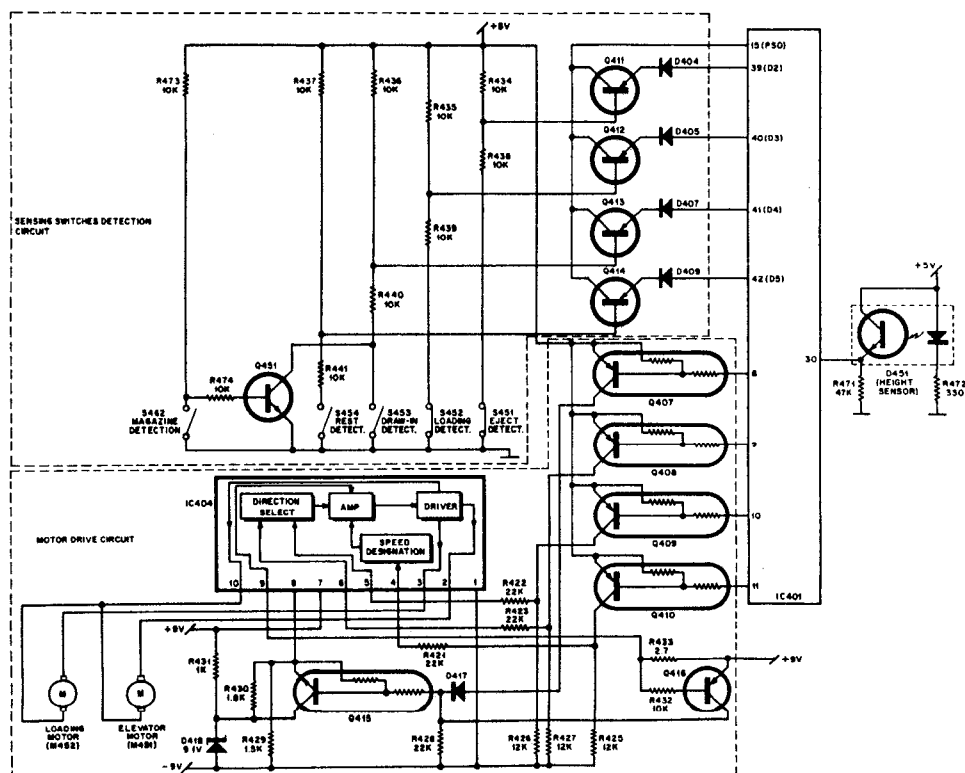


Figure 21-10 Switch sensing and motor drive circuit interface

- (2). As the magazine is loaded in its compartment, S462 is first activated alerting the CPU that the magazine is being loaded.

As it is further inserted, the LOADING SLIDER bar first activates S451 and then S452. This causes the loading motor to rotate clockwise and moves the loading slider forward. As the loading slider bar moves forward, PART (A) of the bar comes in contact with S451.

After a pause of approximately 0.24 sec. (Determined by the CPU, the CPU sets pins 8 through 11 so that the elevator motor is selected and rotates in a counterclockwise direction, moving the elevator unit downward).

The elevator unit hits S454 and sets it to the closed position, causing the elevator motor to stop shows figure 21-11.

After a second pause of approximately 0.24 sec, the CPU sets pins 8 through 11 so that the elevator motor is selected and rotates in a clockwise direction. The elevator unit moves fast in the upward direction for approximately 0.16 sec. and then slowly for about 0.08 sec, until the "standby" position (disc 1) is reached.

The above sequence completes the operation from magazine insertion to "standby" position.

MOTOR	MODE	MOTOR DRIVE IC OUTPUT (IC404)			MICROPROCESSOR OUTPUT (IC401)			
		2	3	10	8	9	10	11
Elevator	Up	L	H	H	L	H	H	L
	Down	H	L	L	L	L	H	L
Loading	Pull-out	L	H	L	H	L	L	H
	Return	H	L	H	L	H	L	H

Table 1 Truth table of microprocessor - motor drive circuit

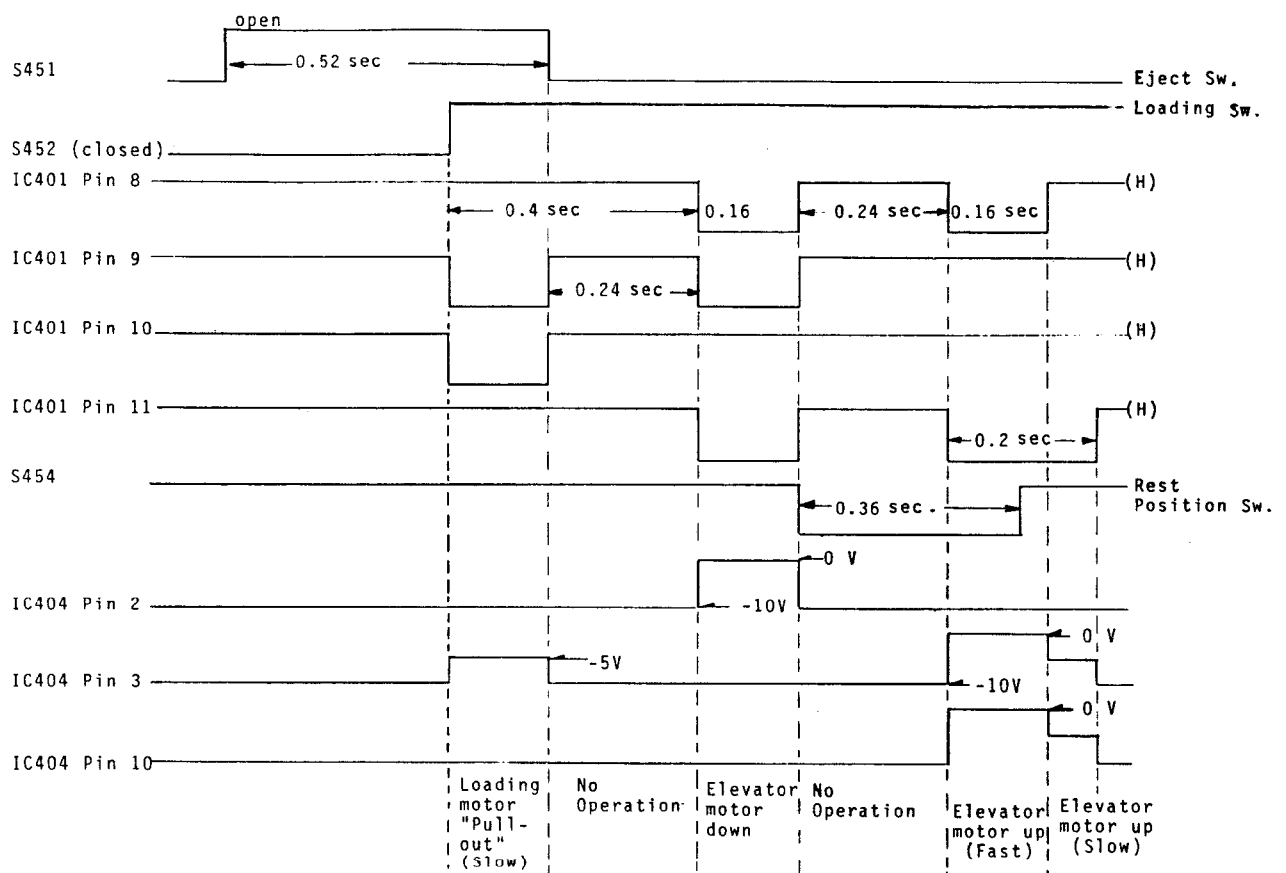


Figure 21-11 Mechanism operation sequence from magazine loading to "standby"

3-2. Disc Loading Operation

After the magazine has been loaded, the elevator unit remains in the "standby" mode, until a disc is selected. The following example will describe the operation sequence to select the third disc and play it shows figure 21-12.

- (1). Magazine loading to standby operation:

The magazine is inserted and the unit enters the "standby" mode as described in the previous section.

- (2). Disc loading operation:

When "Disc 3" is selected, the CPU (IC-401) energizes the elevation motor to lift the disc loader. At this time the elevator gear generates two pulses per rotation.

The CPU monitors the rising edge of the height sensor (D451) signal.

The disc loader elevates from one disc to the next each time the CPU detects a complete rotation. When the CPU detects the target pulse (Target - 1/2), it issues a command to reduce the supply voltage to the elevator motor (slow lift command). At the target disc the CPU shuts off the elevator motor current.

- (3). Disc tray "pull-out" sequence:

The loading motor is energized to pull out the third disc tray. When the CPU detects that the draw switch (S453) is closed, it reduces the supply voltage to the loading motor. When the CPU subsequently detects that the loading switch (S452) is closed, it shuts off the loading motor current.

- (4). Disc clamp operation:

Since the optical deck is located two disc span below the first disc, the loading units must be lowered to the C (clamp) position. To do this, the CPU energizes the elevator motor so that it will lower the disc loader. This time, the CPU counts the falling edge of the height sensor pulses. When the CPU counts 8 pulses, it brakes the elevation motor, and when it detects the falling edge of the 9th pulse, it stops the motor.

- (5). Play operation:

The disc clamper secures the disc on the turntable and the optical deck starts playing the disc.

Sensor switch and driving mechanism timing sequences from the standby state to the initiation of disc No. 3 play

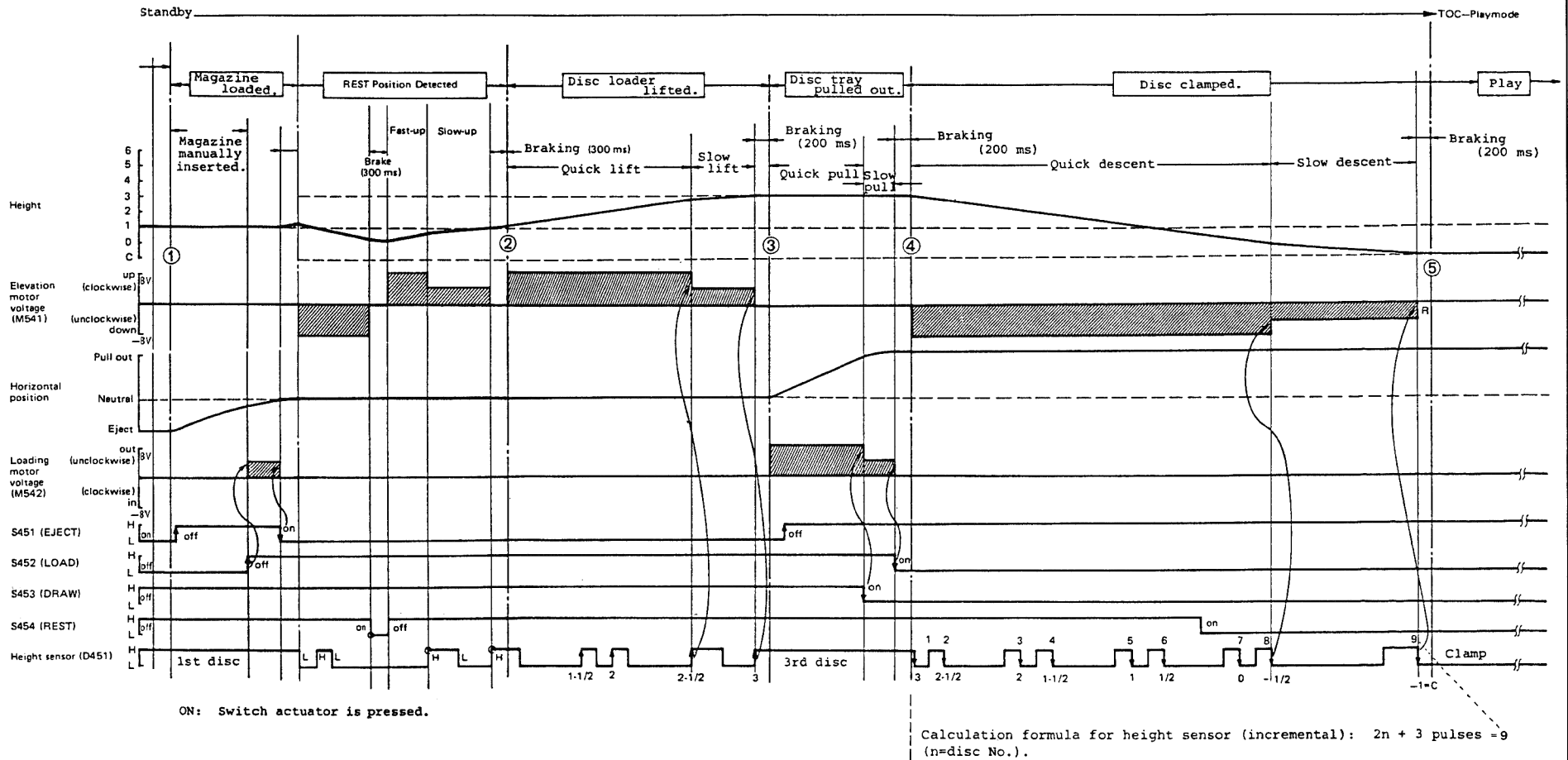


Figure 21-12 Sensing switches and driving motor timing sequence from "standby" to "play" of the third disc

3-3. Stop and Eject Operation

Figure 21-13 shows a timing of transition from play to stop and back to play modes. During the tray return sequence, the loading motor is driven in the reverse direction to release the tray loading hook from the hook catch.

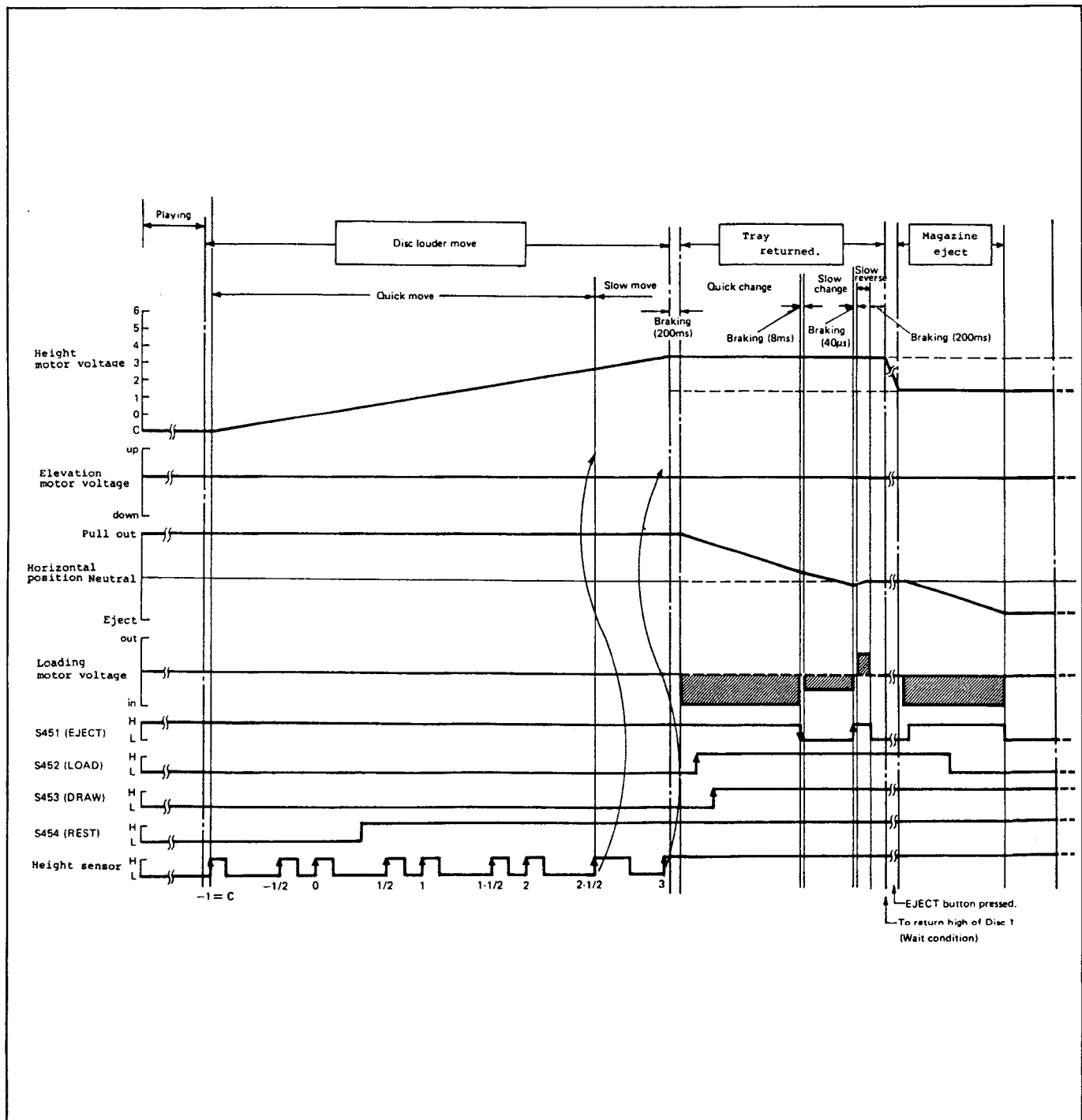
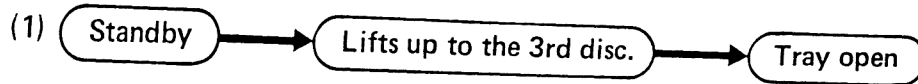


Figure 21-13 Timing chart



○ Circled edges indicates detection timing.

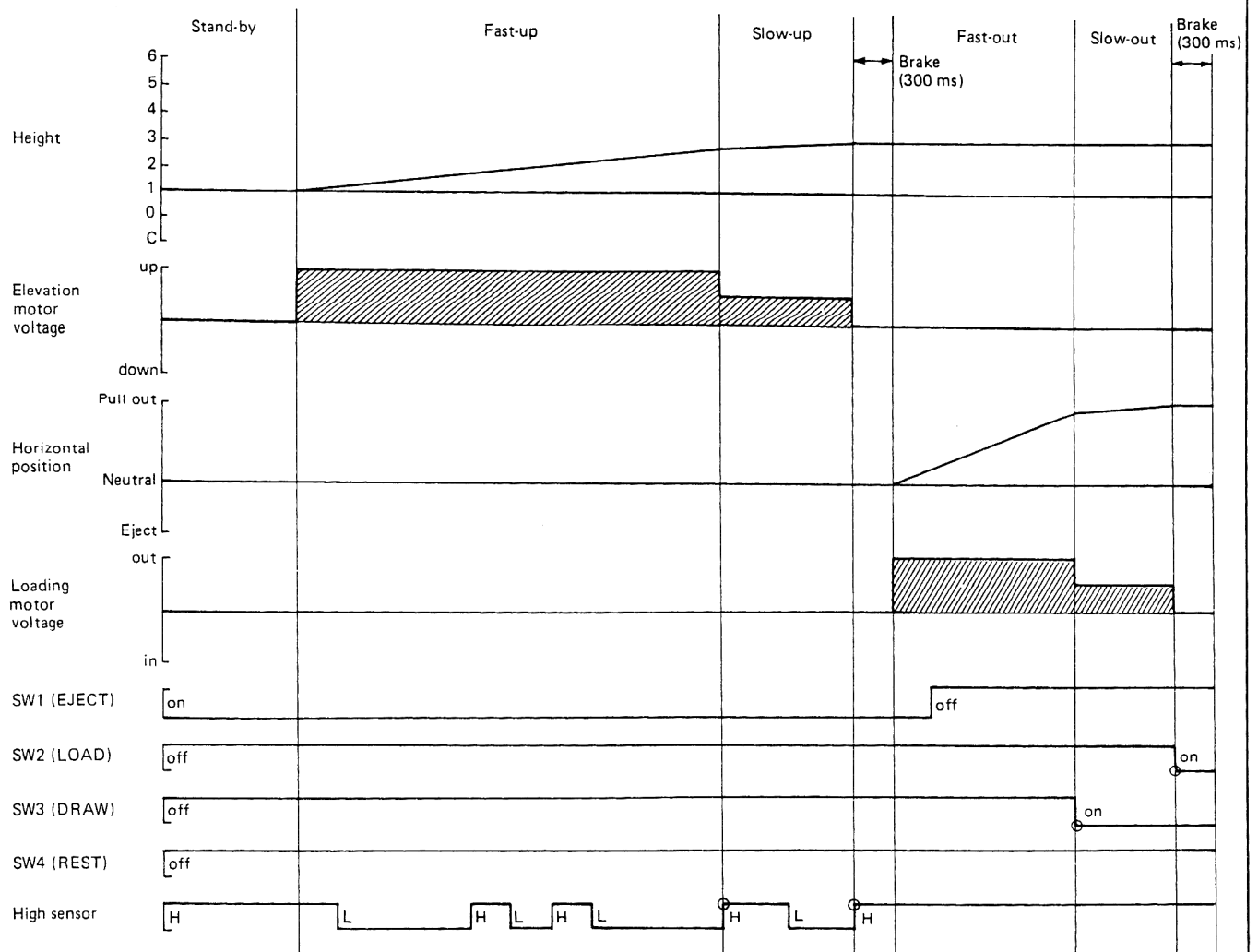


Figure 21-14 Timing Chart

(2) Open position

Clamp

○ Circled edges indicates detection timing.

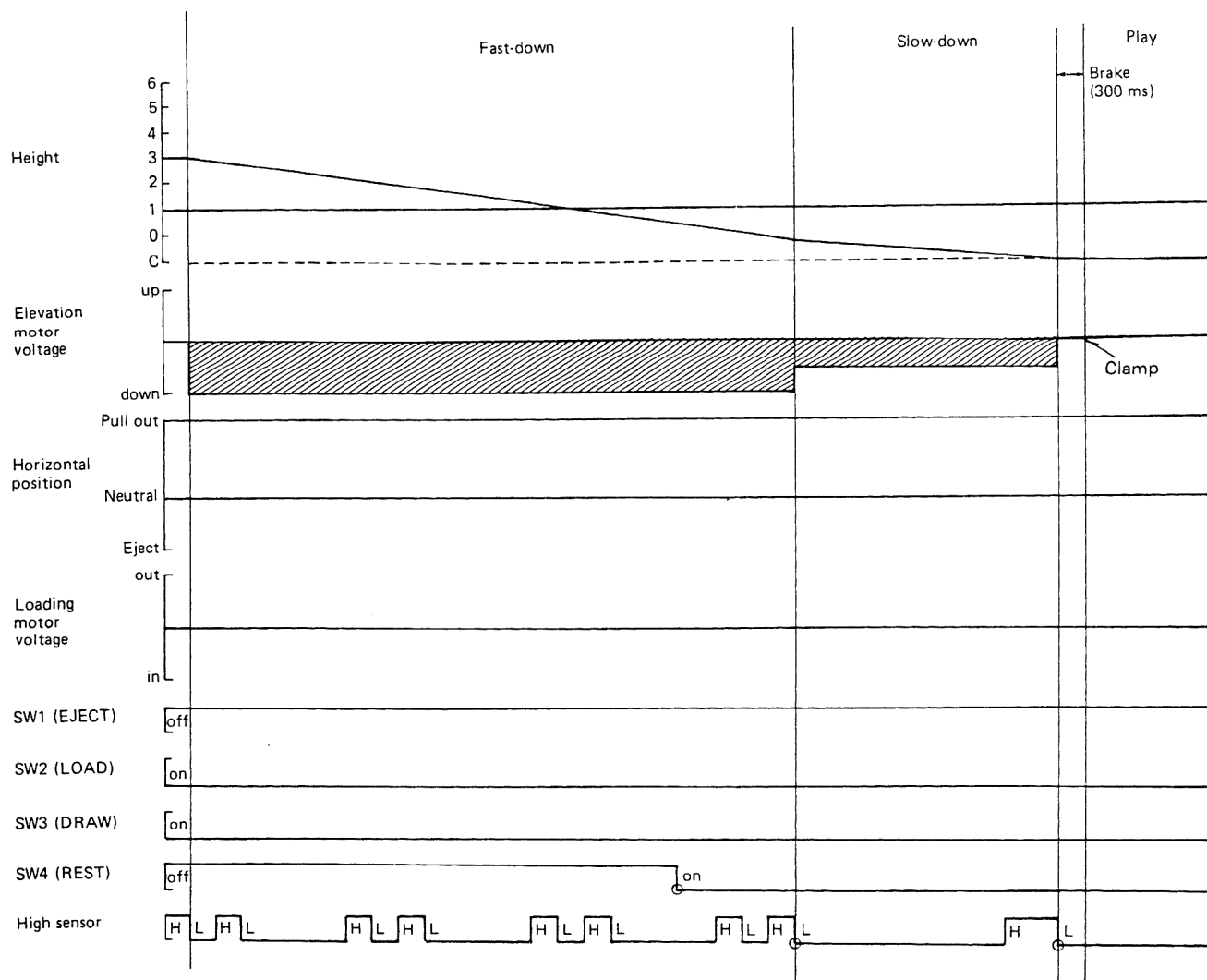
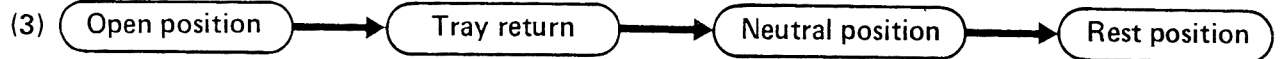


Figure 21-15 Timing Chart



○ Circled edges indicates detection timing.

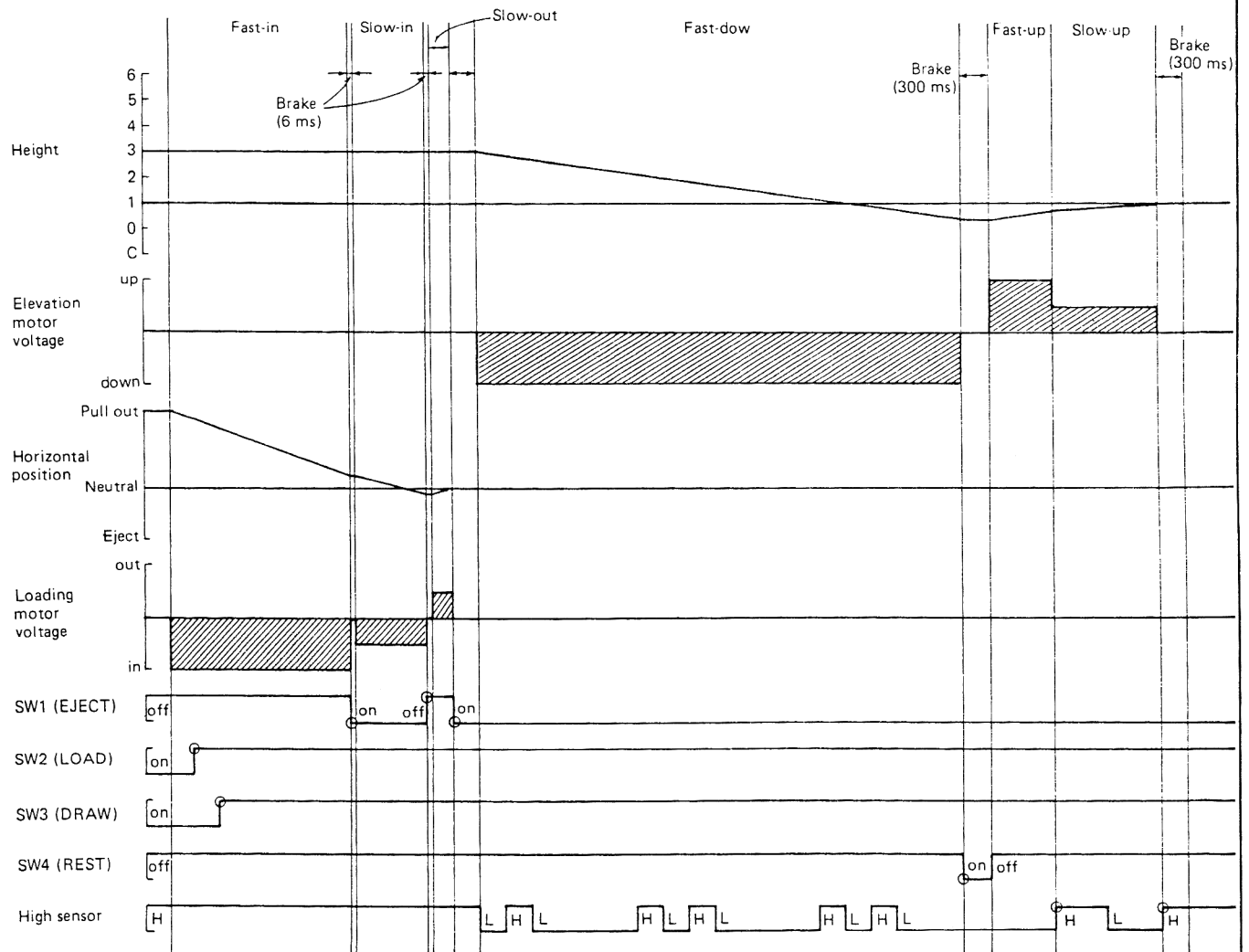
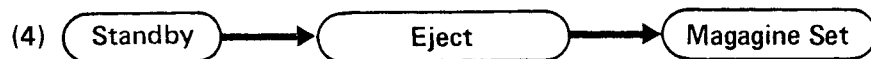


Figure 21-16 Timing Chart



○ Circled edges indicates detection timing.

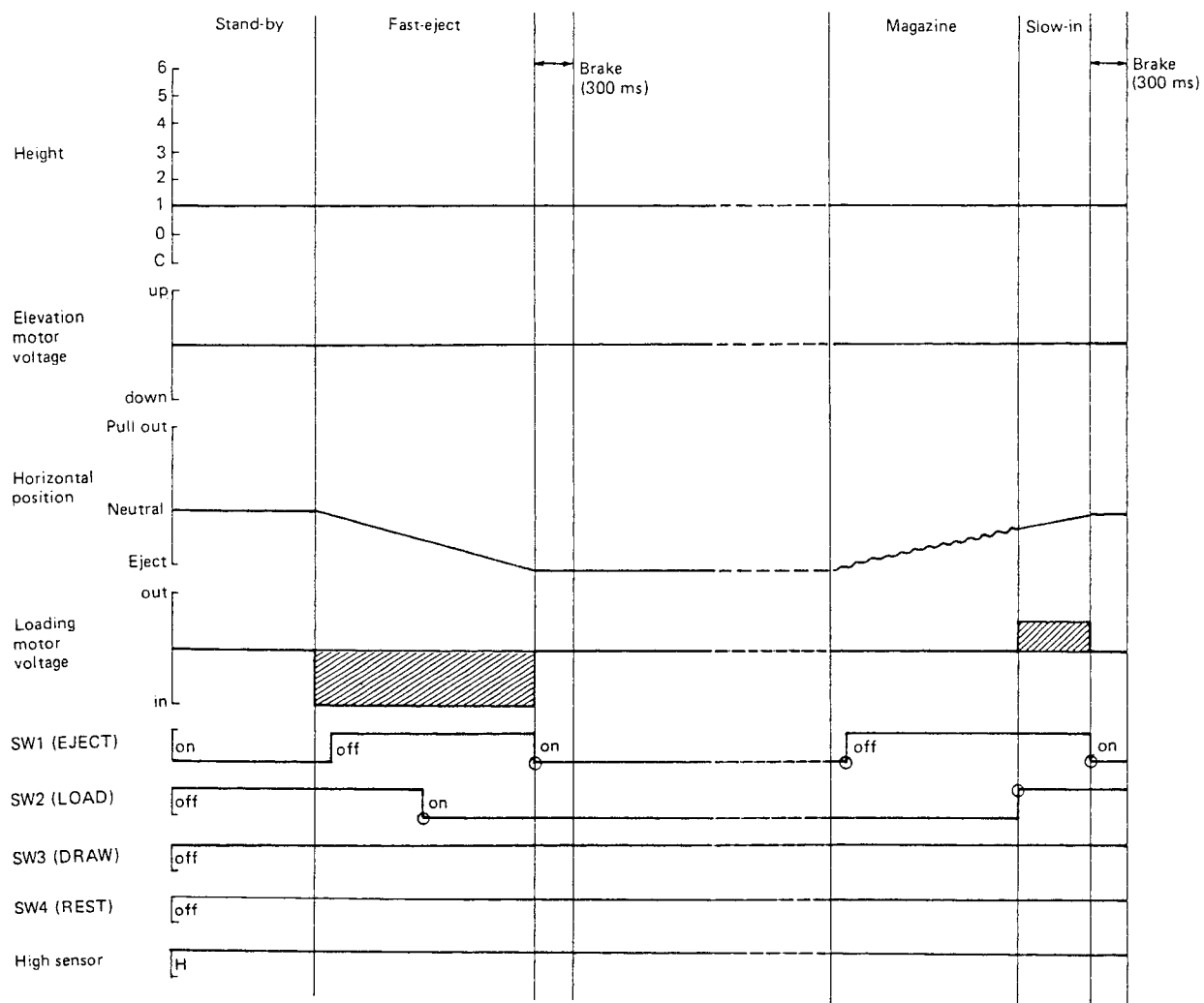


Figure 21-17 Timing Chart

4. Height Adjustment of the Elevator Unit

At rest (standby) position, the elevator unit must be positioned 1.6mm above the magazine holder surface.

The purpose of this adjustment is to set this relative position. When performing this adjustment, the shutter is adjusted so that the pulse count (generated by the photo detector and detected by the CPU) corresponds to the proper height. If the height is not properly set, the CPU does not detect the proper reference, causing the elevator unit to either miss the proper disc tray, access two disc trays, or access the wrong disc tray.

PROCEDURE

- (1). Remove the metal cover plate covering the magazine holder compartment.
- (2). Switch the power to "ON"
- (3). The left, front edge of the metal cover plate is provided with a 1.6mm gauge. Rest the gauge side of the plate on the right side surface of the magazine holder. Confirm the distance between the front edge of the elevator unit's chassis surface and the surface of the magazine holder.
- (4). If the distance is not within 1.6 $\pm$ 0.2mm, adjust the height by turning the HEIGHT ADJUSTMENT screw located on the top side of the ELEVATOR GEAR shows figure 21-18.
- (5). Insert the magazine in its compartment, and select the sixth disc tray, via the front panel. If the proper tray is not accessed, repeat steps 3 and 4 above.

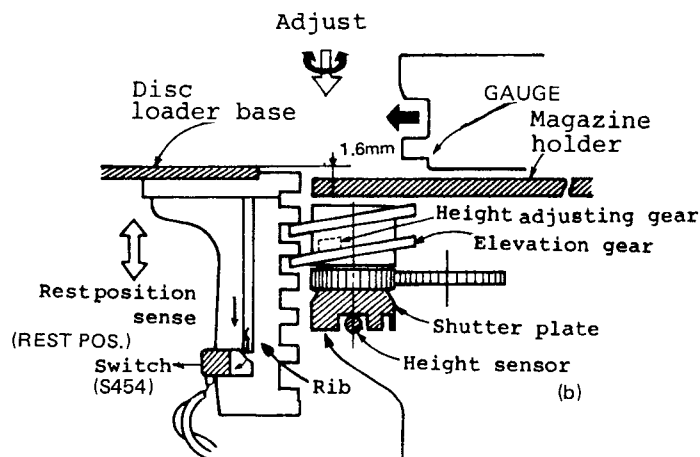


Figure 21-18 Positioning of the gauge

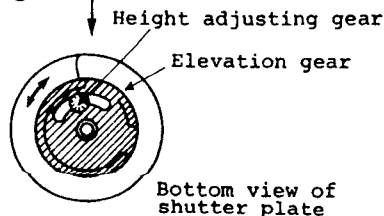


Figure 21-18 Top view of the elevator gear

5 Problem Solving

Symptom	Possible cause	Check point
At power on, the disc loader is lowered to the lowest position and remains there.	Rest switch is defective. Elevation motor has been stopped by an ON signal from the rest switch.	Check the Rest Position Sense switch (S454) for correct ON state.
At power on, the disc loader continues elevation for about 10 seconds.	Height sensor signal is not applied to the CPU. The CPU cannot compute disc loader height from the sensor pulses.	Check the sensor (D451) output while the elevation motor is rotating.
Disc magazine is ejected soon after it is inserted.	Magazine switch remains off.	Check the Magazine switch (S462) for proper ON state.
Disc tray is pulled out twice and "No Disc" indication appears. When the Play switch is pressed, message "-E" appears.	Draw switch remained off when the disc tray was pulled out. The CPU identifies it as no disc and returns the tray.	Press the Draw switch (S453) and check for correct ON state.
The same disc is pulled out then returned repeatedly at about 10 second interval.	(1) Mechanism operation is impeded in the pulling process. (2) Loading switch remains off at the fully pulled position.	(1) Manually rotate the loading motor to check for smooth mechanism movement. (2) Check the Loading switch (S452) for correct ON state.
When the magazine is ejected, the player displays loading indication for about 10 sec., then returns to the normal state.	(1) Mechanism operation is impeded in the eject process. (2) Eject or Loading switch remains off when the magazine is ejected.	(1) Manually rotate the loading motor to check for smooth mechanism movement. (2) Check the Eject switch (S451) and Loading switch (S452) for correct ON state.
At power on, all the display digits and indicators momentarily come on, then immediately go to the loading message. However, key entry is disabled after mechanism reset operation is completed.	Any of the key switches is left closed, causing CPU runaway.	Check for defective key switch.
"No Disc" message appears 3 seconds after a disc is clamped, then the disc is returned to the magazine.	Optical pickup is unfocused.	Check for optical deck misalignment. Rest position height error or disc clamp error due to height sensor signal error.

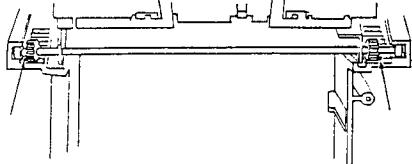
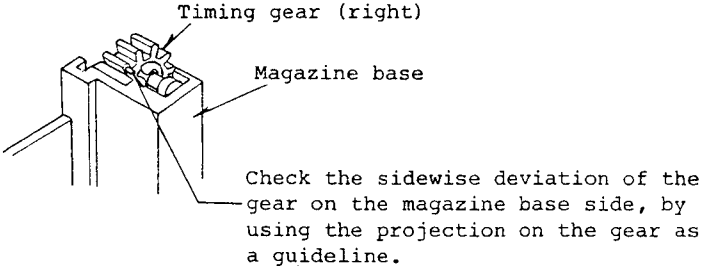
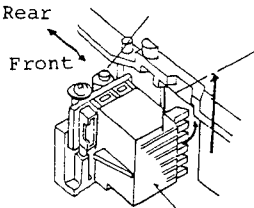
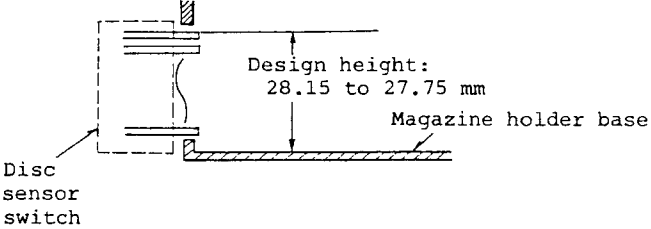
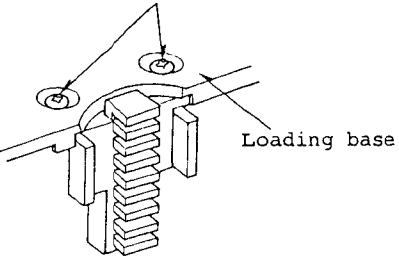
6 Apparent Fault

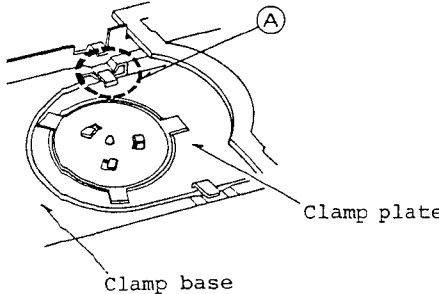
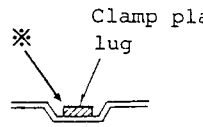
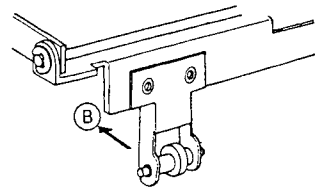
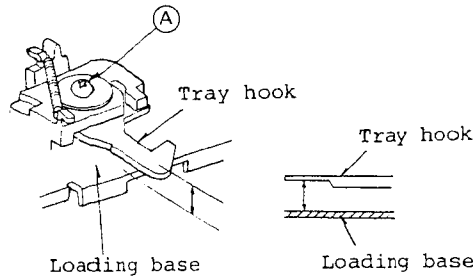
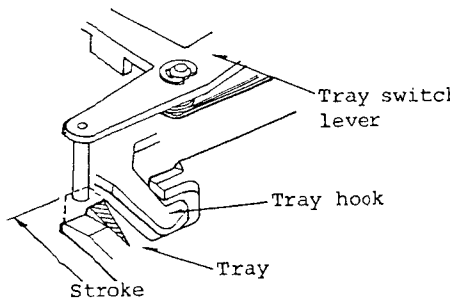
The changer's auto changing system may exhibit the following malfunctions but they may not be derived from mechanism or logic failure:

Apparent fault	Possible cause
1 Program numbers which do not exist on the discs can be programmed in the Program mode but nothing is played in Play mode. (e.g. the 25th track was programmed for a CD containing only 12 tracks.)	Other CD players first read the TOC just after a disc is loaded, then enter the Pause mode to wait for the next command. They can verify the validity of the programmed contents by comparing the number of tracks on the disc with that in the program. However, the SL-P600C takes so much time to read the TOCs on all of the six discs that it ignores invalid track numbers in the program.
2 Disc trays were loaded only in slots 1, 2, and 5 in the disc magazine, and the player was turned off when it was playing Disk 5. When the player was subsequently turned on, Disc 5 was returned to the tray for Disk 3 (disk position has changed in the magazine).	As long as the player is left turned on, the CPU memorizes which slot the disc now on the playing deck came from. However, once the player has been turned off, the CPU loses its memory and cannot determine to which slot the disc should be returned. In such a case, the CPU sequentially searches for a vacant slot from bottom to top, and returns the disc into the first vacant slot found. At an occupied slot, the Load switch remains off. The CPU checks the time until the Load switch is turned on after a return sequence was started (Draw switch S453 was turned off). If it is too long (more than ms), the CPU identifies that the slot is occupied. * This problem can be avoided if all of the six trays are always loaded in the magazine (discs need not be loaded in all of the six slots), or if trays are loaded from slot 1 with no intermediate vacant slot left in the magazine.

7 Mechanism Check Points for Servicing

When disassembling or reassembling the deck mechanism, observe the following cautions:

Item	Caution	Symptom caused by misalignment
1 Magazine base assembly (* Sidewise gear deviation) 	The magazine base gear should have no sidewise deviation from the timing gear on the loading base. 	Tray open/close operation failure.
2 Disc sensor switch assembly (S455-S460)  Disc sensor switch	If the disc sensor switches are skewed, they may fail to sense the discs. Swing shaft A in the longitudinal direction to the following design height. 	Disc sensor failure
3 Elevation cam retention  Retention screws Loading base	° If the retention screws are loose, malfunction or mechanism noise may occur due to insufficient elevation stroke. ° Be sure to tighten them firmly.	° Disc tray operation failure. ° Mechanism noise.

4	Insufficient disc clamp stroke  Clamp plate Clamp base	When the mechanism is lowered (in the Play position), the clamp plate "lug" must rest in the recess of the clamp base (marked with an asterisk).  Clamp plate lug Cross section at point A If there is any clearance between the two, tilt the spring on the clamp plate in the direction of arrow (B) to increase the stroke.  Diagonal rear view of clamp plate	° Disc does not rotate. ° Mechanism noise heard during disc rotation.
5	Unstable tray open operation  Tray hook Loading base	Major causes will include: 1) Tray hook skew. 2) Loading base height defect. Verify that the distance between the top surface of the tray hook and that of loading base is 4.3 mm. If necessary, adjust with (A) or replace the tray hook (S1RD75).	Disc tray fails to close.
6	Unstable tray loading (draw switch actuation error)  Tray switch lever Tray hook Tray Stroke	The tray switch lever activation stroke is mechanically limited. So it is likely that draw switch (S453) activation failure will be due to the tray switch end pins failing to catch the tray edge (deformed tray switch lever). Action: Replace tray hook, S1RD71.	Failing to enter Play mode (the mechanism attempts tray close drive twice for all of six disc trays, and finally stops in the neutral state).

8 Pin Description of MN15283PJB

• MN15283PER (System Control)

Pin No.	Mark	I/O	Function	Pin No.	Mark	I/O	Function
1	GND	I	GND terminal	26	CONT3	O	Optical servo IC control signal (KICKF: Kick direction [Forward] command)
2	XO	O	Not used, open	27	CONT2	O	Optical servo IC control signal (TRON: Tracking servo)
3	X1	I	Optical servo condition input	28	CONT1	O	Optical servo IC control signal (FOON: Focus servo)
4	M-RLY	I	Muting control	29	STAT	I	Key input strobe and processing status input from signal processing LSI travers position detector
5	MDATA	O	Command data output	30	HIGHT	I	Elevator height signal input
6	MCLK	O	Command clock output	31	FLOCK	I	Optical servo condition (Focus) input
7	MLD	O	Command load output	32	SENSE	I	Optical servo condition (Track cross) input
8	SLOW	O	Motor drive control	33	PAUSE	O	Not used, open
9	FOW/REV	O	Motor drive control	34	PLAY	O	Not used, open
10	LOADING	O	Motor drive control	35	DAC	O	Not used, open
11	HIGHT	O	Motor drive control	36	VPP	I	FL drive power supply (connected to -32V)
12	SYNC	I	Not used, open	37	DO	O	FL glide signal and key scan signal
13	RST	I	Reset signal input (reset at "L")	52	DF	O	
14	BLKCK	I	Sub-code block (Q-data) clock (75kHz) input	53	S8	O	FL anode signal
15	P50	I	Key scan input	61	S0	O	
16	~			62	VDD	I	Power supply (connected to +5V)
17	P53	I		63	OSC2	I	Clock terminal
19	CLOCK	I	Sub-code frame clock (7.35kHz)	64	OSC1	I	Clock input
20	SUBQ	I	Sub-code "Q" data input				
21	---	---	Not used, open				
22	---	---	Not used, open				
23	TRV-R	O	Traverse reverse command signal				
24	TRV-F	O	Traverse forward command signal				
25	CONT4	O	Optical servo IC control signal (KICKR: Kick direction [Reverse] command)				

• MN1550PEB (Remote Control Signal Processing)

Pin No.	Mark	Signal	I/O Division	Function
1	VDD	——	I	Power supply (connected to +5V)
2	OSC1	SMCK	I	Clock input
3	OSC2	——	——	Not used, open
4	P22	——	——	Not used, open
5	RST	RESET	I	Reset signal input
6	P21	——	——	Not used, open
7	P20	——	I	Remote control signal input
8	IRQ	——	I	Program enable/de-enable control

Pin No.	Mark	Signal	I/O Division	Function
9	P31	SYNC	O	Not used, open
10 └ 13	P10 └ P13	Data	I	} Key strobe
14 └ 17	P00 └ P03	Data		
18	VSS	GND		

22. Reference

1. Mechanism Adjustment

When the optical pickup and peripheral mechanism parts incorporated in the traverse deck are replaced or removed, such as the spindle motor, it is necessary to confirm the adjustment related to the optical pickup and then readjust as required.

A. Turntable Height Adjustment

a) Purpose

The output from the laser diode is focused on the signal side of the disc, via the focus lens. Focussing is

accomplished by moving the focus lens vertically. The focus servo is used to control such vertical movement.

Since the focus servo is designed to operate within a limited range, it is necessary to adjust the lens at the mechanical midpoint position of the servo's respective operating range. If this adjustment should be deviated, skipping may occur during play mode or in an extreme case, no turntable rotation, etc.

b) Positional relation between optical pickup and disc surface

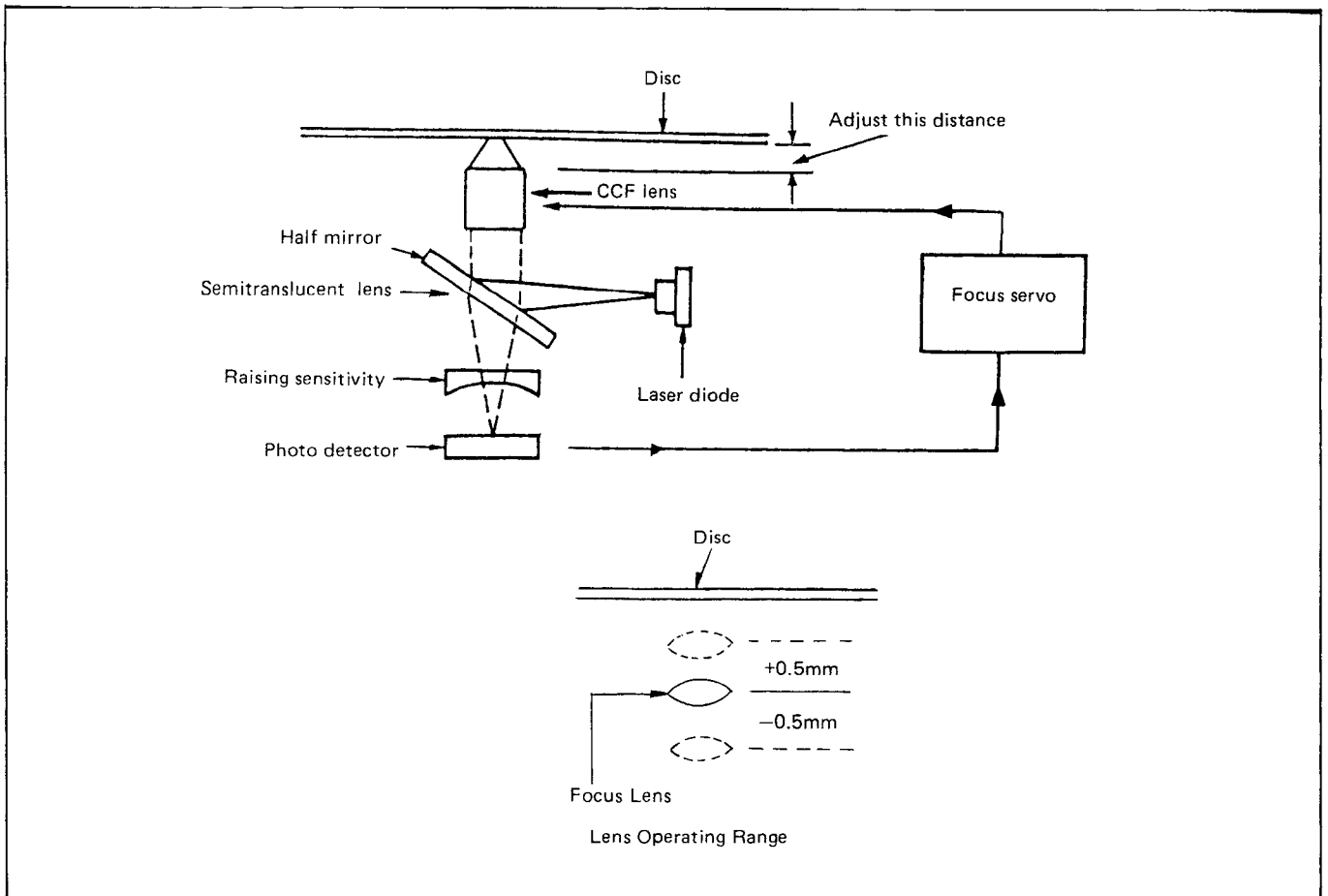


Figure 22-1 Beam Method

c)Basic principle of height adjustment of turntable height

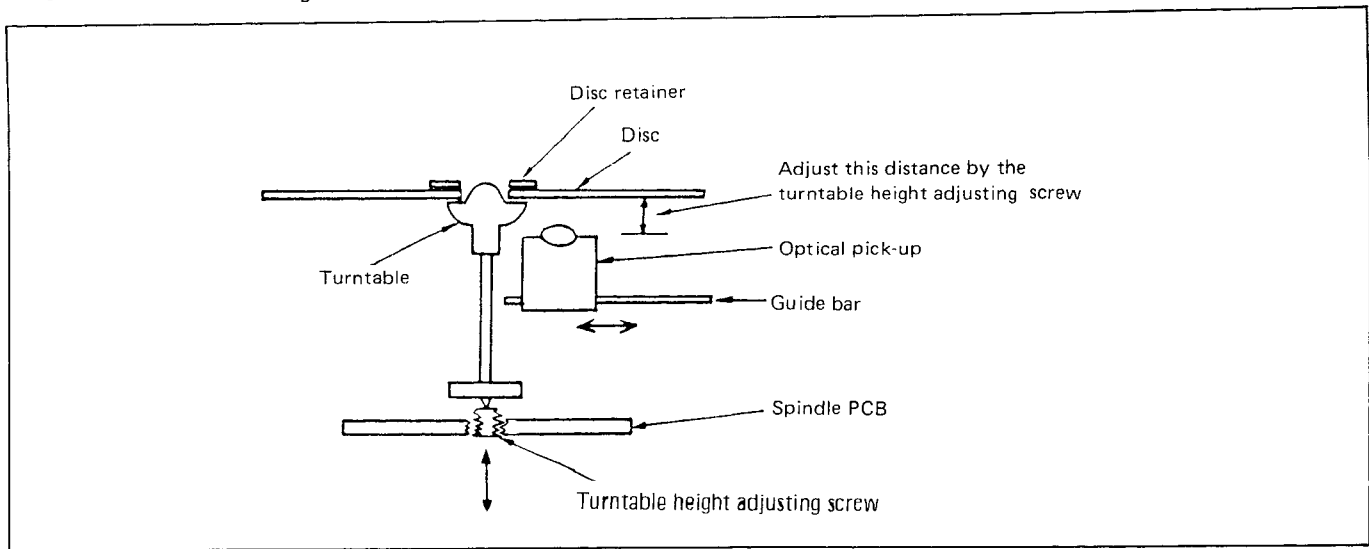


Figure 22-1 DD Motor

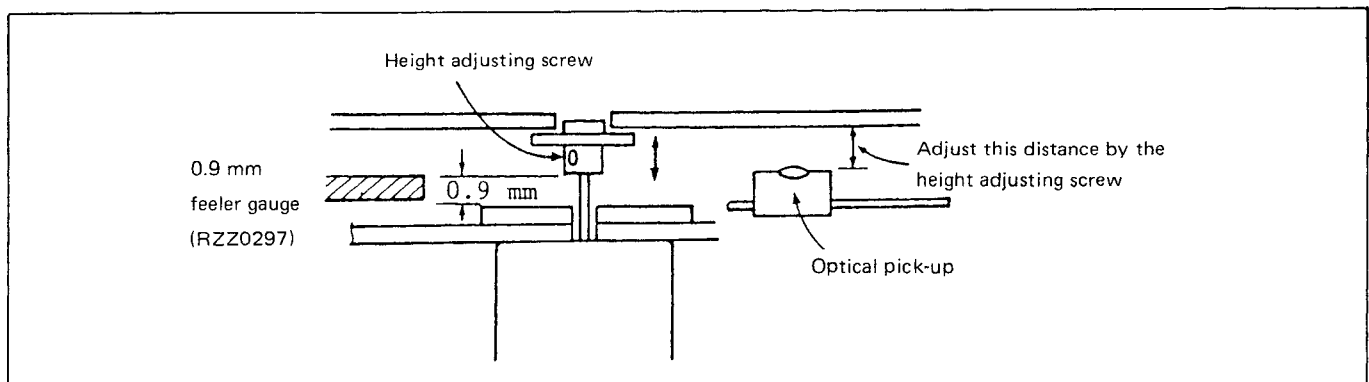


Figure 22-3 DC Motor

B.Mechanical Adjustment

a)Purpose

Unless the lens is perpendicular to the disc surface, the laser beam may not be focused. The mechanical alignment is therefore necessary to adjust the optical pickup lens so that

the lens is at an optimum perpendicular position.

b)Basic principle of mechanical adjustment

To accomplish the above condition, it is necessary to adjust Hex screws (A) and (B) as shows in figure 22-4.

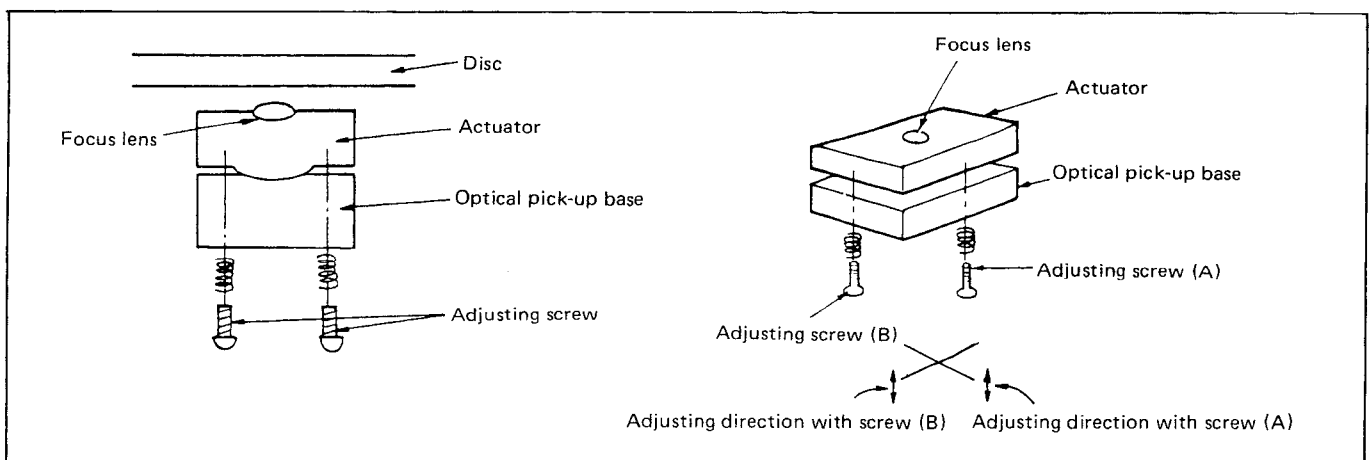


Figure 22-4

c) Basic principle of adjustment method

The warped TEST DISC (SZZP1056C) is used to simulate the non-perpendicular condition. While playing the disc, the RF amplitude's vertical fluctuation is to be minimized; at which point the most perpendicular point is set.

(1) RF waveform at use of the TEST DISC (SZZP1056C)

A thin piece of metal is mounted to the part where the disc touches the turntable. The RF waveform is fluctuated vertically (amplitude fluctuated).

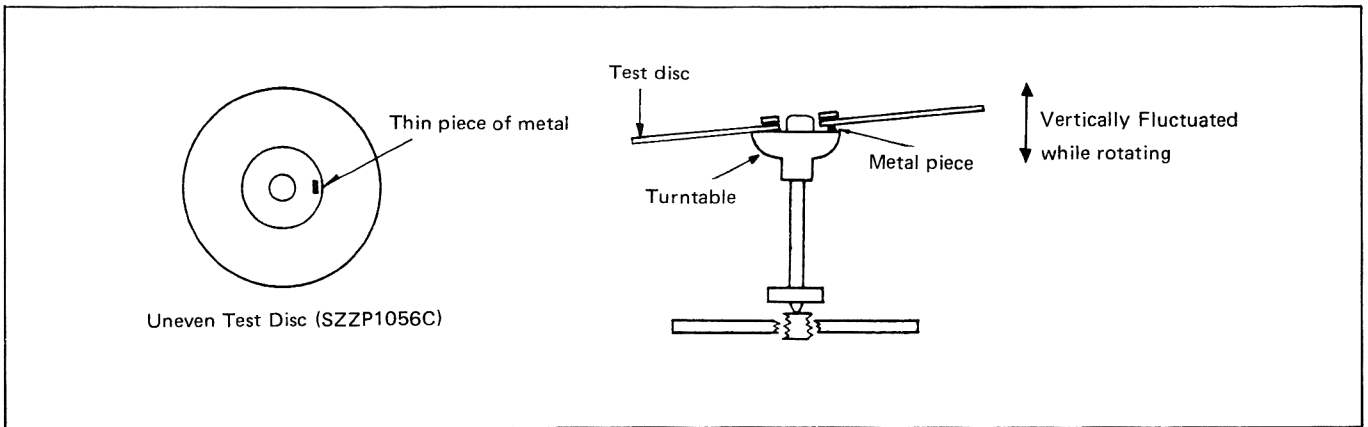


Figure 22-5

(2) RF waveform when properly aligned

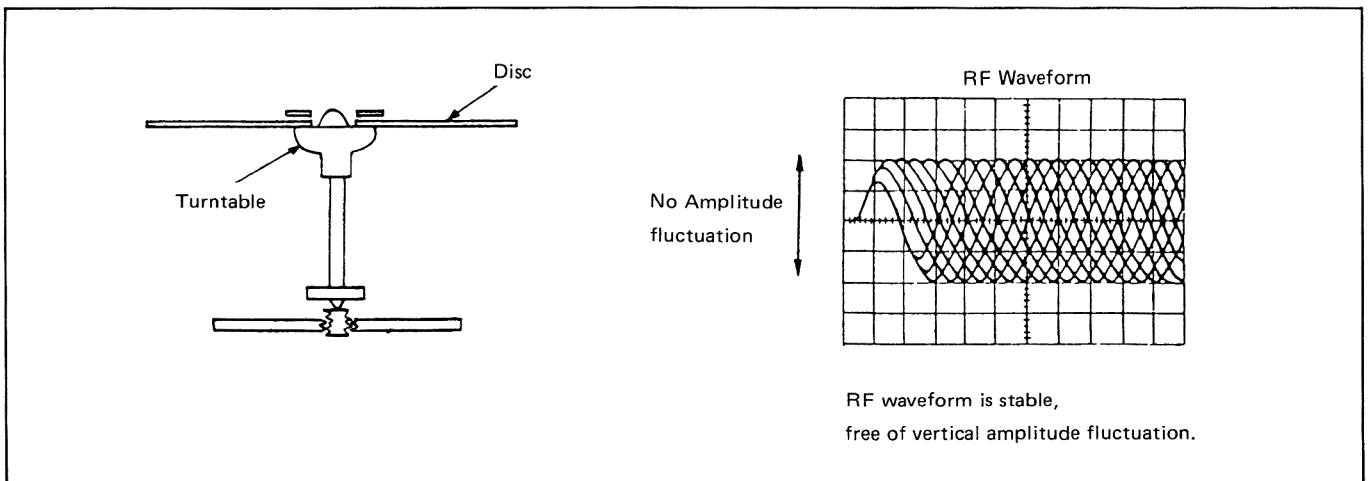


Figure 22-6

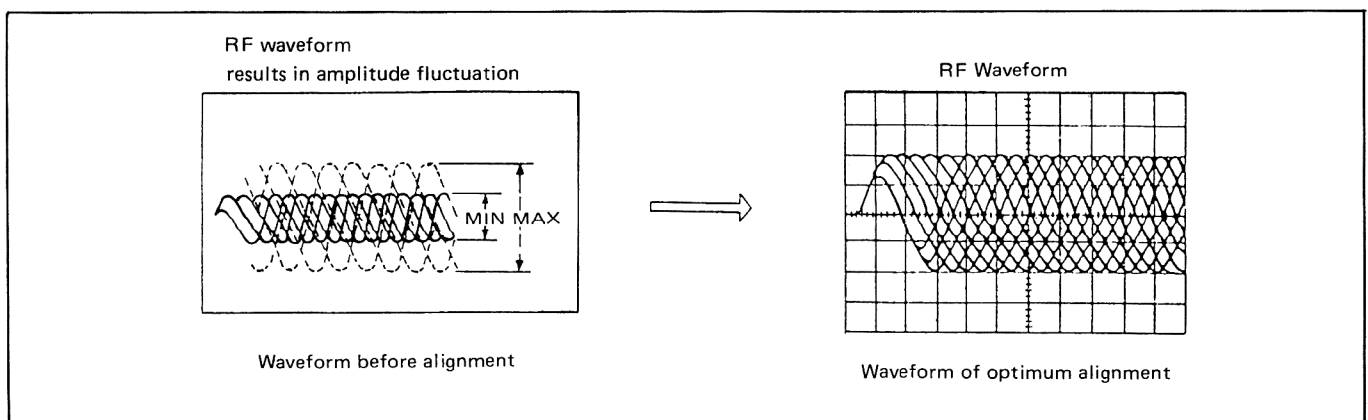


Figure 22-7

2. Electrical Adjustment

These adjustments are so important to the extent to the performance of the CD player depends mainly on their accuracy.

A. Best Eye (PD Balance) Adjustment

a) Purpose

Due to fluctuations of as described earlier, under "turntable height adjustment" the focus servo is designed to work within a certain

operating range. The purpose of this adjustment is to set the focus servo's reference voltage (zero); at which point the lens focusing depth is set.

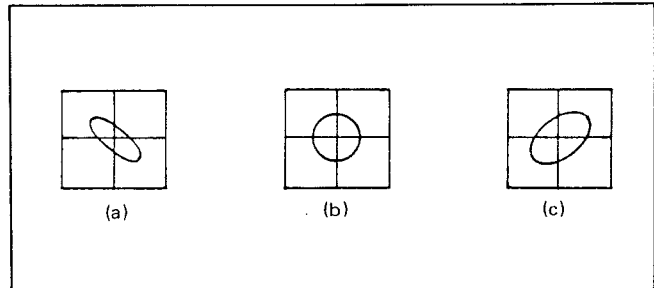


Figure 22-8 Beam Shapes Received by the Photodetector

b) Basic principle of best eye adjustment and best point

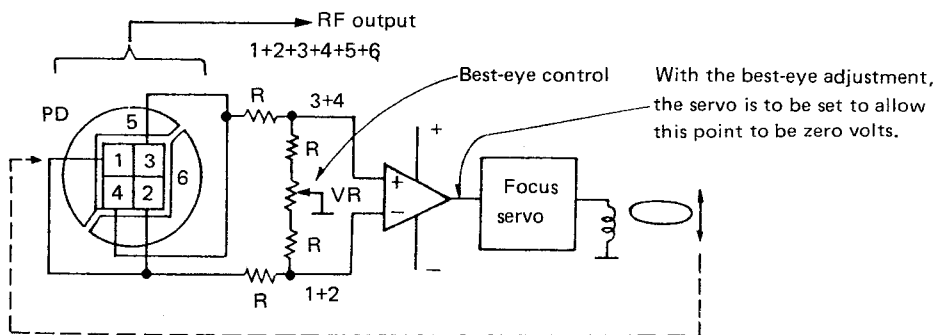
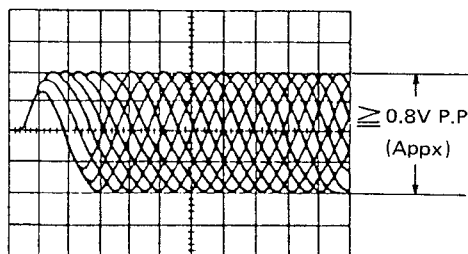


Figure 22-9 Focus Servo Loop



Maximum amplitude of the RF waveform is achieved when the best eye adjustment is properly set, as described above.

Figure 22-10 RF Waveform

c) Basic principle of adjustment

Due to distance variations between the disc surface and the lens caused by uneven discs, the laser beam covering a given pit varies in diameter. When

the lens is close to the disc the diameter is large and covers adjacent pits. The signal from adjacent pits causes interference and the RF becomes distorted.

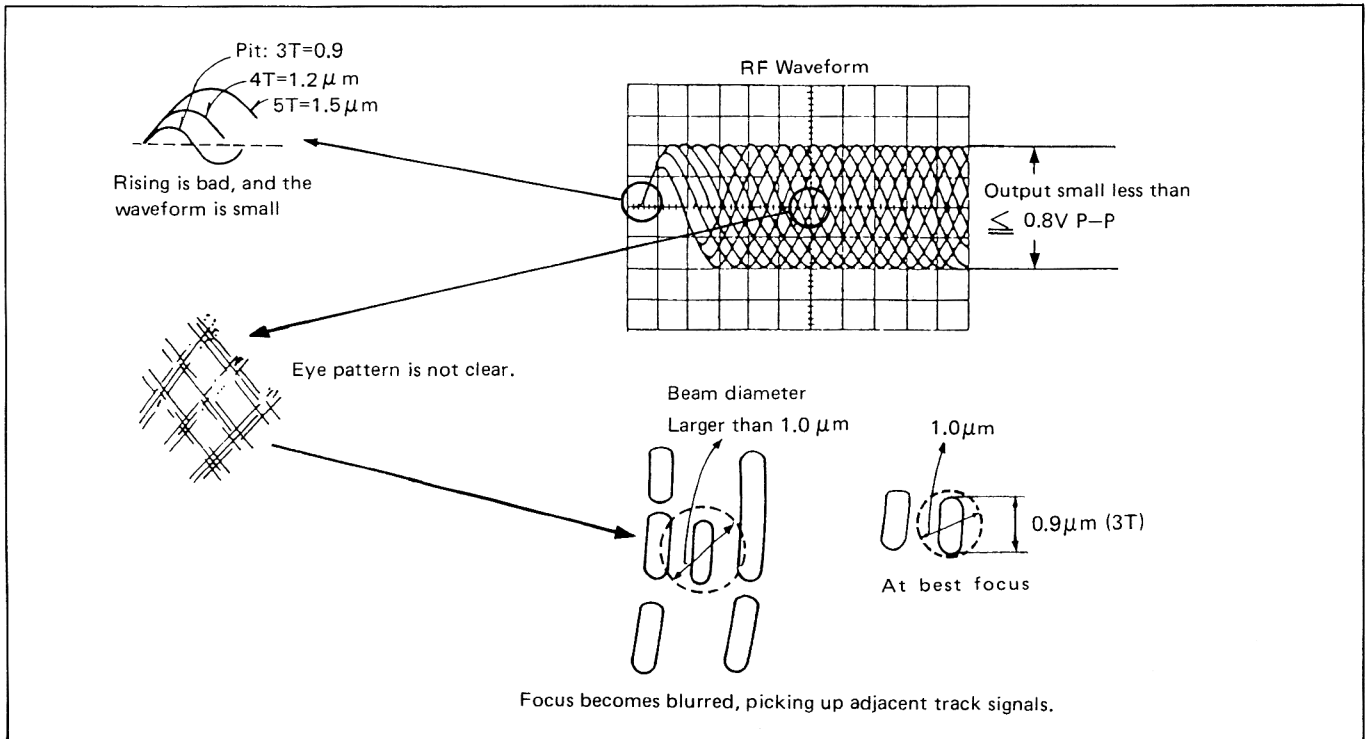


Figure 22-11

d) Adjustment method

It is required to adjust so that the RF waveform is maximum and the eye

pattern appears clearly.

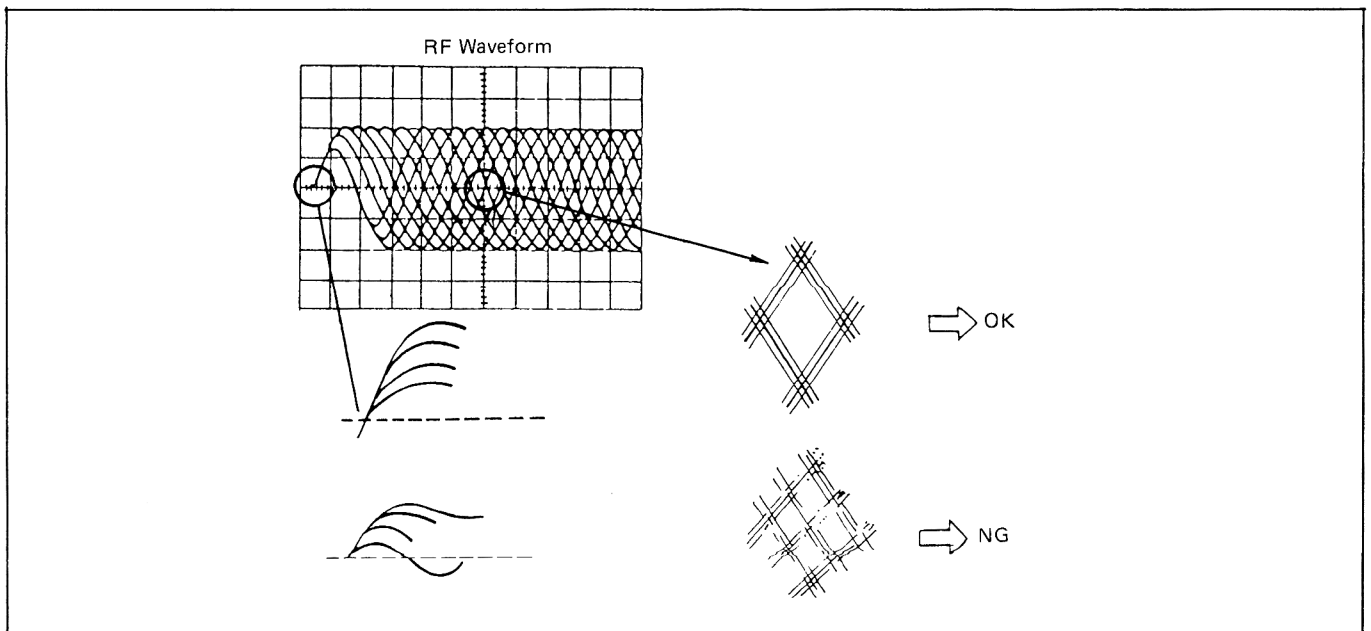


Figure 22-12

B. Focus Gain Adjustment

a) Purpose

Regardless of disc warpage ($\pm 0.4\text{mm}$) due to manufacturing tolerances, the focus servo must operate to constantly keep the lens focused on the disc's pits. The greater the servo's gain, the easier the lens can focus on the disc; however it becomes unstable due

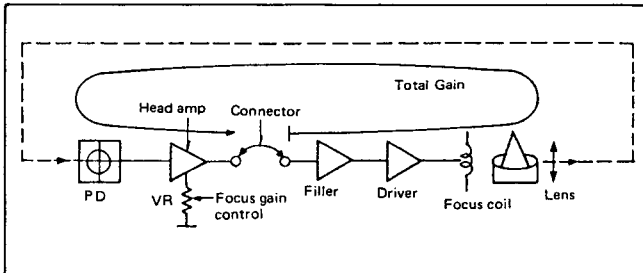


Figure 22-13 Focus Servo Loop

b) Principle of the focus gain adjustment

The focus servo loop is opened, at which point a 750 Hz signal is applied to simulate the acceptable condition. This point is TP2 on the "Servo Gain Adjuster".

The signal, once through the focus servo, appears at TP1. The focus gain

to unwanted signals. The gain is therefore set to unity ($0\text{ dB}=1$, 750Hz) at which point stability is acceptable and the gain is adequate.

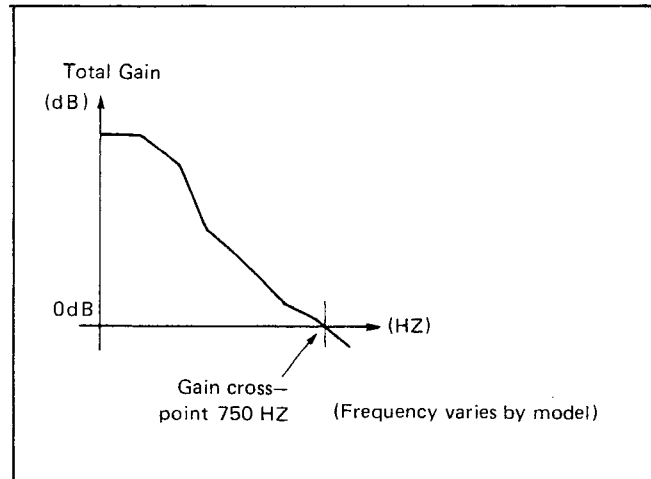


Figure 22-14 Focus Gain Characteristic

variable resistor is used to set the voltage at TP1 to equal the level at TP2; at which point the gain is one (0dB)

The reason why the focus servo must be adjusted using a 750 Hz tone is because this is the frequency that causes the gain to become unity ($0\text{dB}=1$)

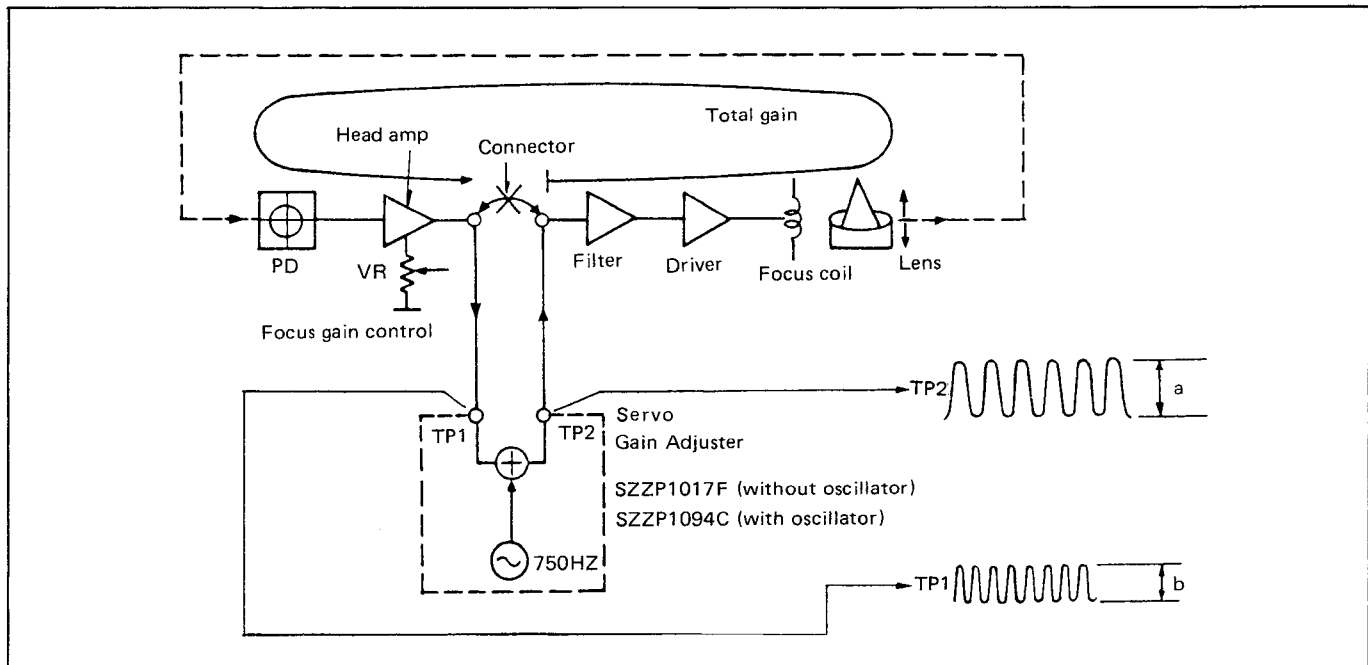


Figure 22-15

(Note)
Explanation was made based on 750 Hz gain cross-point frequency. This

frequency may differ depend on model. Please refer to the service manual of the corresponding model for frequency.

C. Tracking Gain Adjustment

a) Purpose

During disc manufacturing, disc eccentricity results due to tooling tolerances. Eccentricity also results if the tracking servo is designed to overcome these tolerances and keep the lens on track.

b) Principle of the tracking gain adjustment

The tracking gain loop is opened at which point a 1 KHz, 1.2 KHz, or 1.5 KHz signal is fed. This point is TP2 of the "Servo Gain Adjuster".

The signal, once through the servo, appears at TP1. The tracking gain control is to be adjusted so that the level of TP1 equals that of TP2, at which point the gain is one (0dB)

The reason why the focus servo to be adjusted using a 1 KHz, 1.2 KHz or 1.5 KHz tone is because these are the frequencies that cause the gain to become unity (0dB=1).

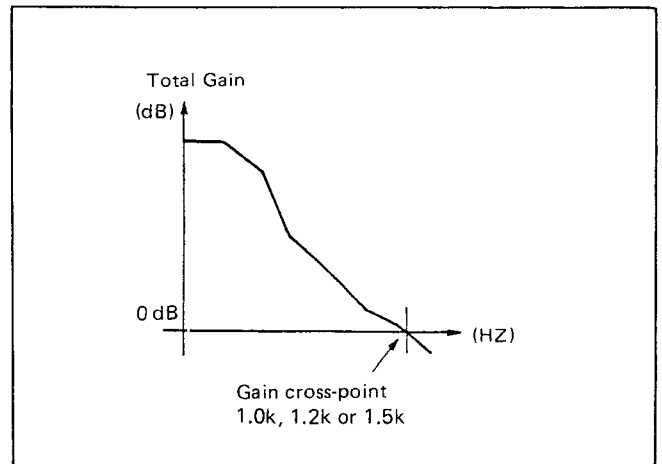


Figure 22-17 Tracking Gain Characteristic

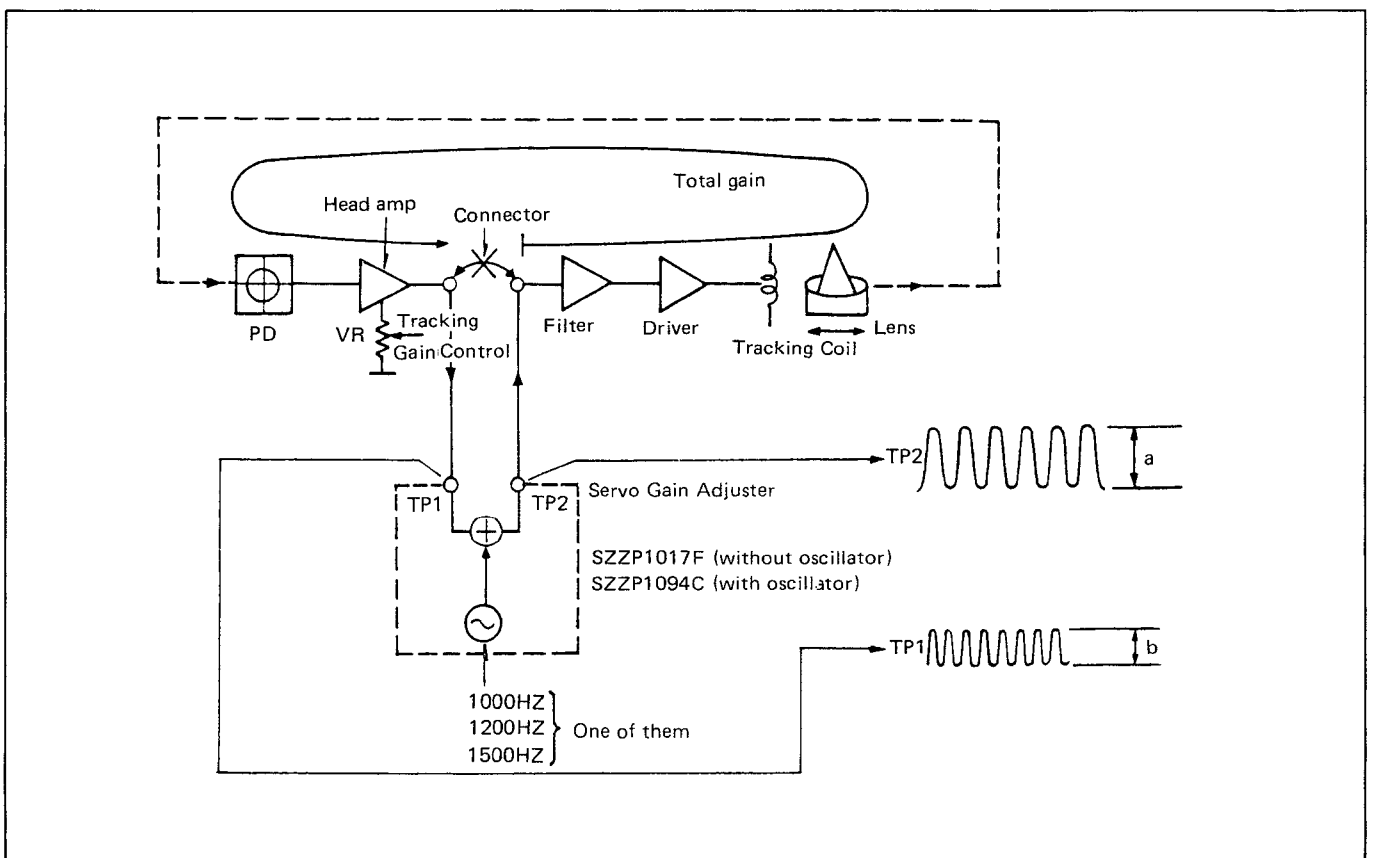


Figure 22-16

D. Focus Offset Adjustment

a) Purpose

The CD player must be able to compensate for manufacturing imperfections of disc and dust or any other contaminants on the signal side.

As the pickup traces the track, if dust or disc imperfections are present, the focusing depth changes.

Without proper focus the tracking servo is affected as well, causing the pickup to move to a different track.

When the lens detects imperfections as mentions above, the focus servo is enabled in order to maintain proper focusing within the operating range of the servo.

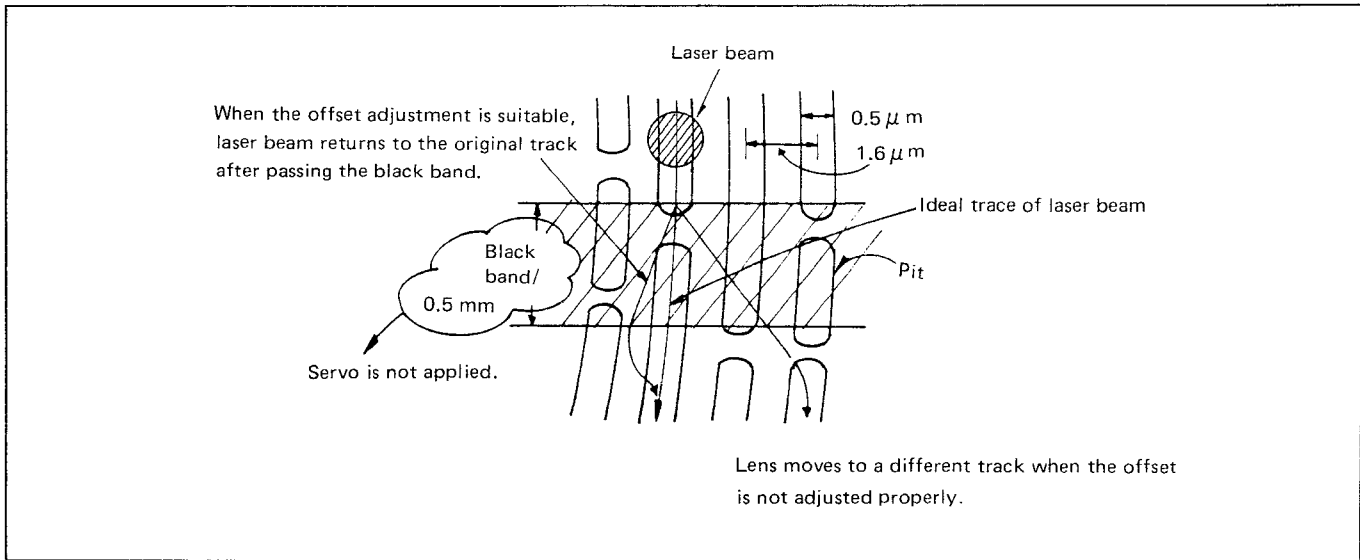


Figure 22-18

b) Test disc, RF waveform and FE signal
A black band disc is used to simulate disc imperfection or dust particles,

causing the RF to appear as in figure 22-21 (a) and (b).

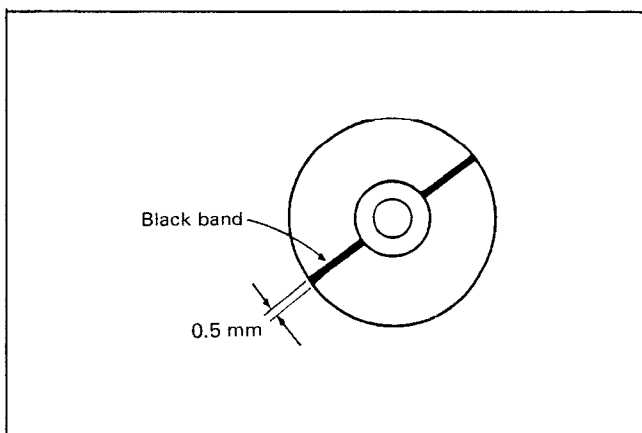


Figure 22-19 Test Disc (SZZP1057C)

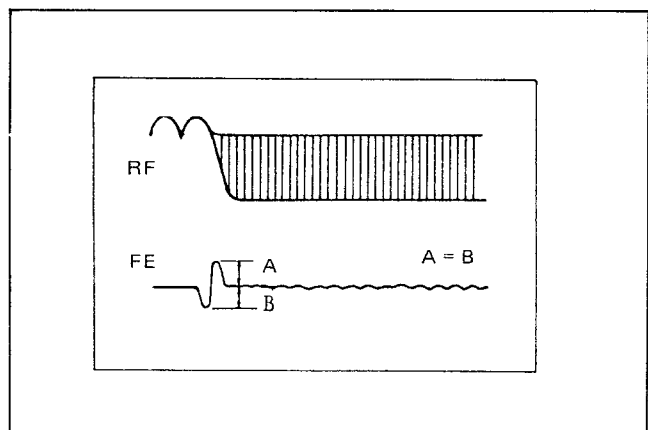


Figure 22-20 RF Waveform When Adjusted

c) Focus offset adjustment method

It should be adjusted so that the RF waveform dip is minimized or eliminated and the size of FE (a) and (b) is at minimum or balanced.

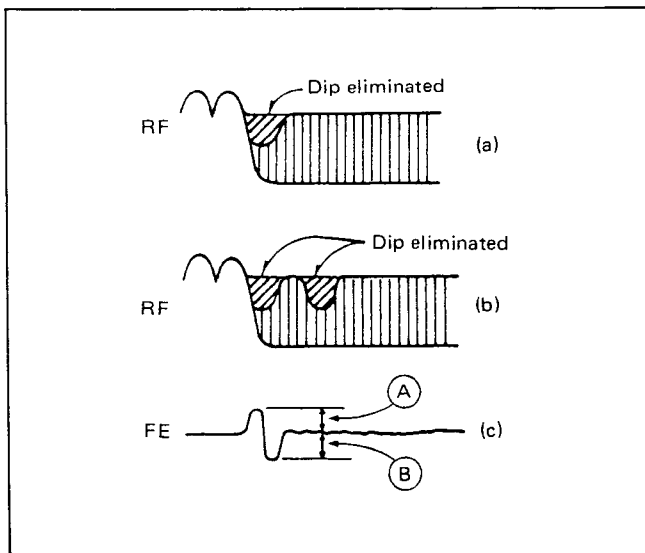


Figure 22-21

E. Tracking Offset Adjustment

a) Purpose

As with the focus offset adjustment, the tracking offset adjustment is necessary to set reference point of the tracking servo operating range.

The only difference is that the tracking servo compensate for horizontal control as opposed to vertical control by the focus servo.

b) Test disc, RF waveform and TE signal

The test disc, RF waveform, and TE (Tracking Error) signal waveform are the same as described in the focus offset adjustment. The only difference is the TE signal.

c) Tracking offset adjustment method

The RF waveform must be adjusted such that the dip's are minimized or eliminated as indicated below in figure 22-22 (a) and (b).

The TE waveform must be adjusted such that (a) and (b) are equal, as indicated in figure 22-22 (c) below.

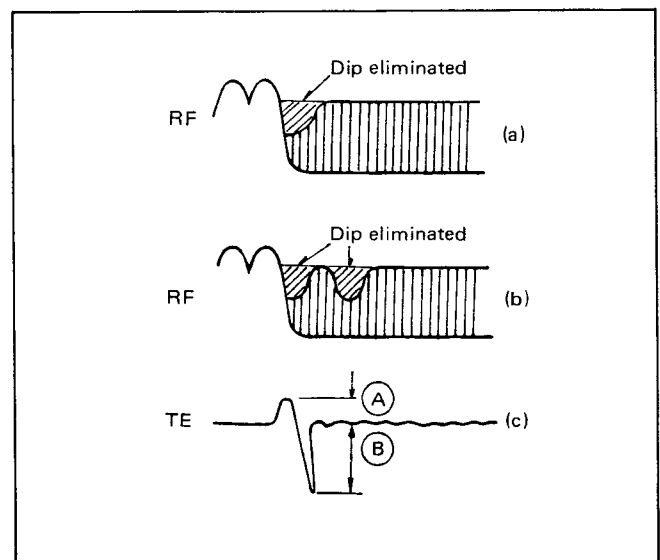


Figure 22-22

F. Tracking Offset Balance Adjustment

a) Purpose

During playback operation, the laser beam must follow the track exactly at its center. If it is not, proper tracking can not be accomplished, and skipping may occur during playback of the disc which has manufacturing imperfections and dust or any other contaminants on the signal side.

b) Basic principle of adjustment

A 1 KHz signal is applied to the tracking servo circuit to intentionally cause the beam to oscillate from side to side of the being traced (see figure 22-23).

This is done to simulate an error signal so that the tracking servo can be adjusted for minimum gain (minimum swing) at 1 KHz.

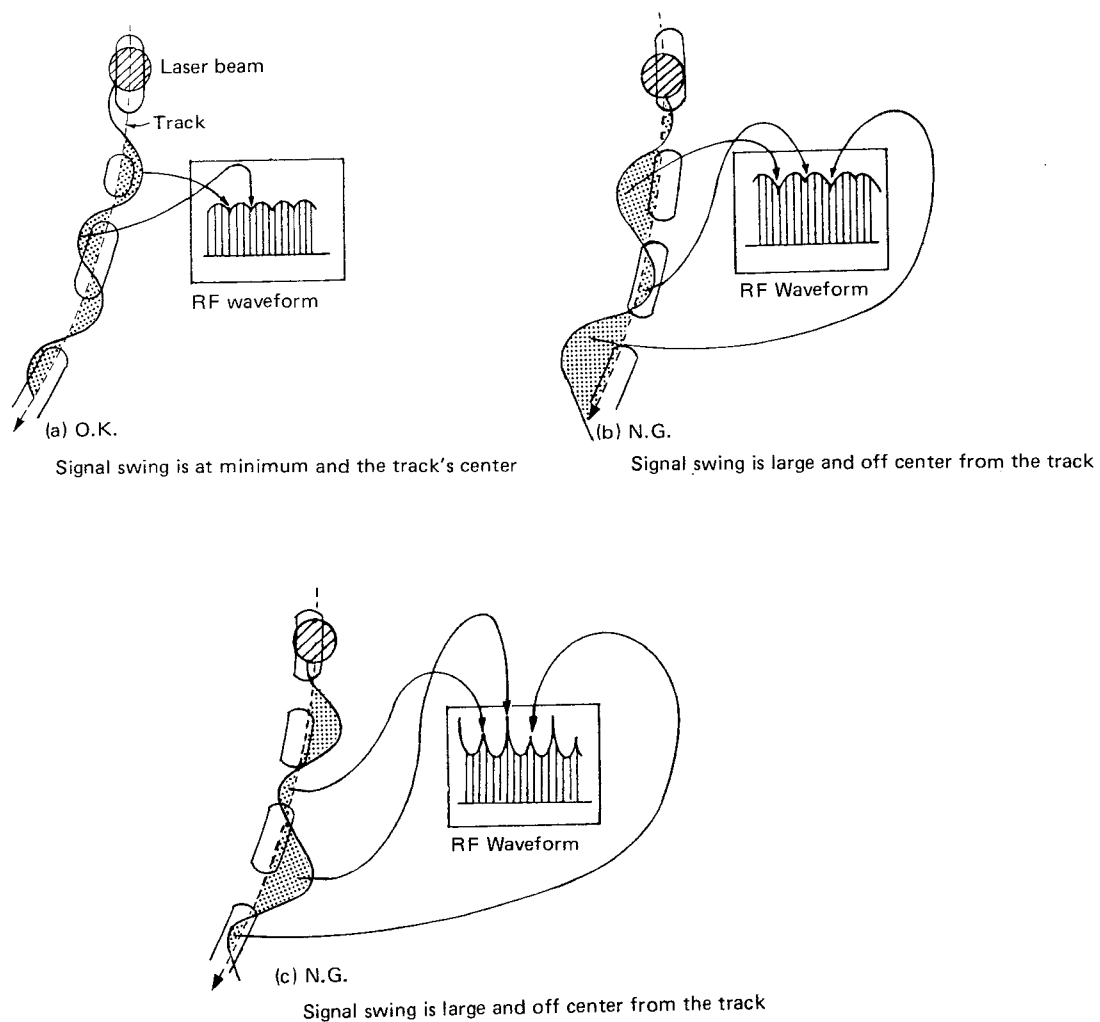


Figure 22-23 Beam Movement with Error Signal

c) Adjustment method

The 1 KHz signal is fed through the tracking servo. The tracking offset

balance control is to be adjusted for minimum and balanced level on the RF waveform.

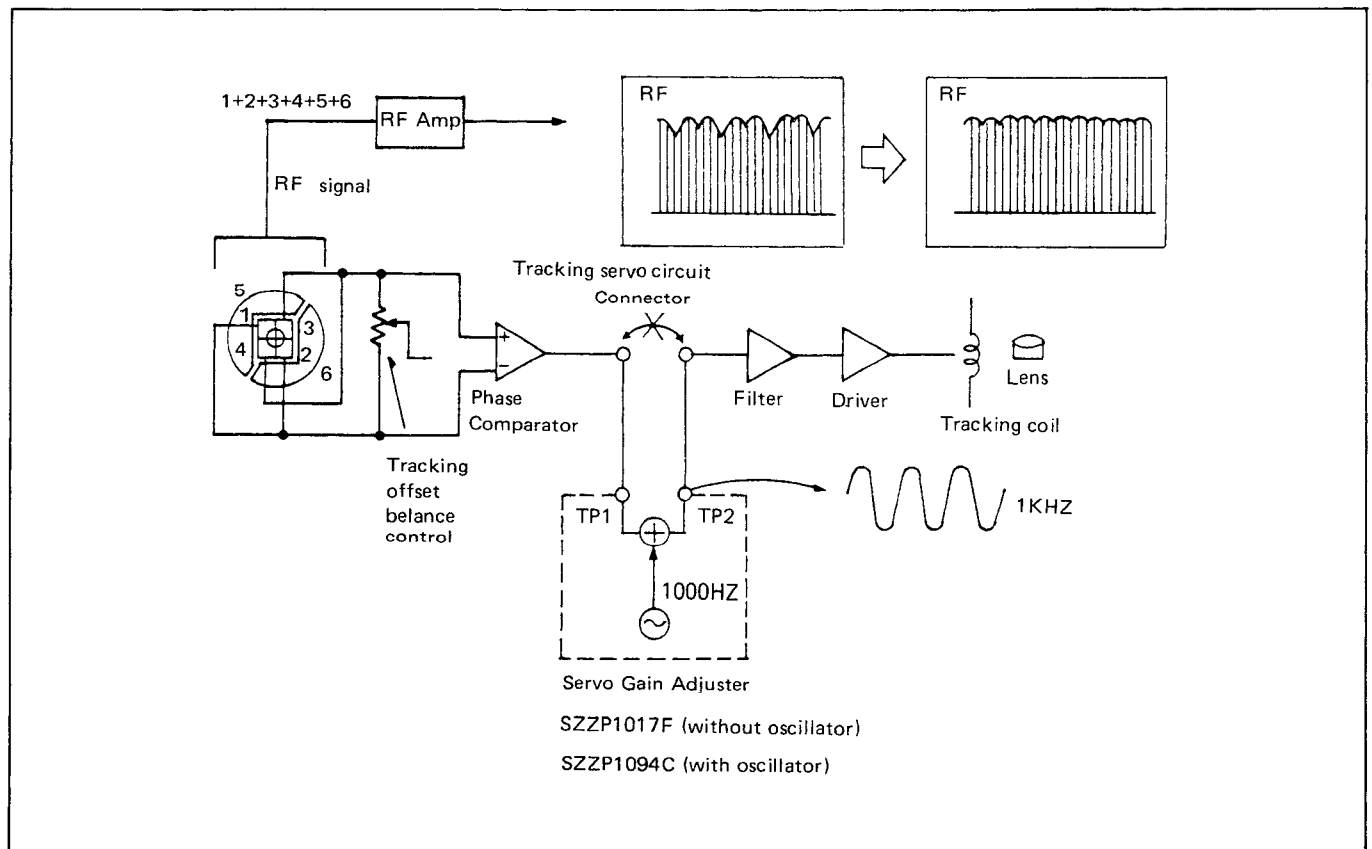


Figure 22-24

G. PLL Adjustment

a) Purpose

The purpose of the PLL adjustment is to set the VCO (Voltage Controlled Oscillator) circuit such that the 4.3218 MHz clock will lock within its operating range.

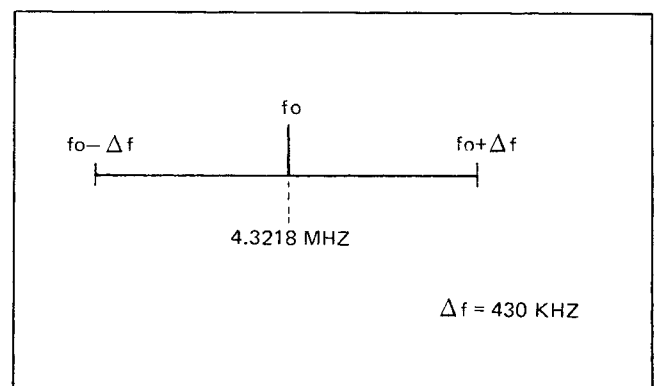


Figure 22-25

The PLL (Phase Locked Loop) reference frequency is constantly compare to the disc's sync data (extracted from SRF) representing a frequency of 4.3218 MHz. When these two frequencies are

equal the PLL is locked. If the PLL is not adjusted within its operating range it can cause incorrect disc rotation or dropout.

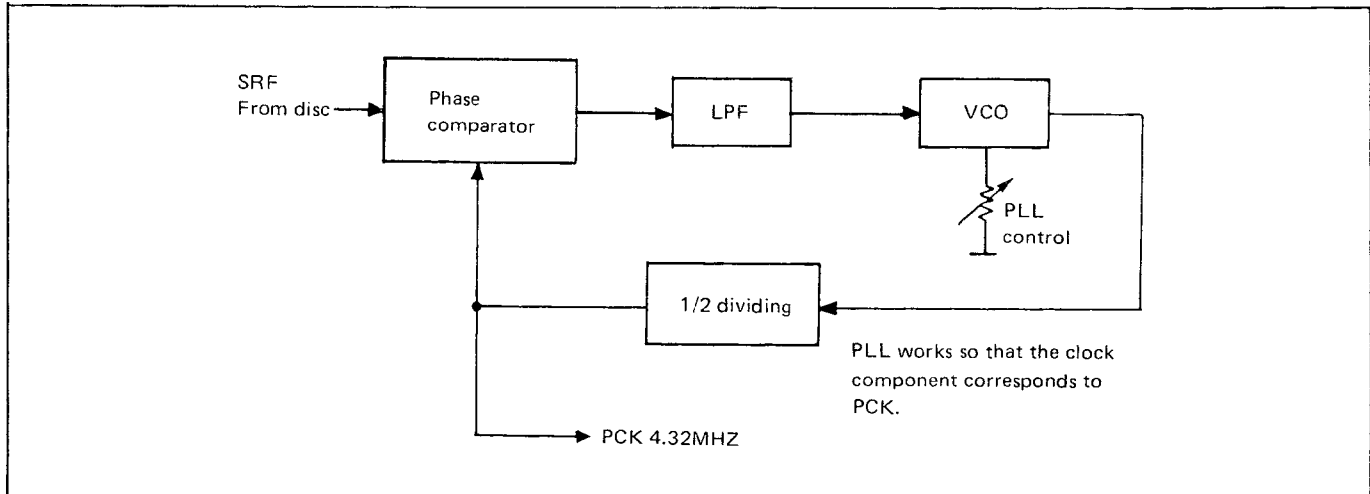


Figure 22-26

3. Various Defects of Disc & RF Waveform

Defects related to the disc are shows in figure 22-27 (A). Their respective RF

waveform are shows in figure 22-27 (B).

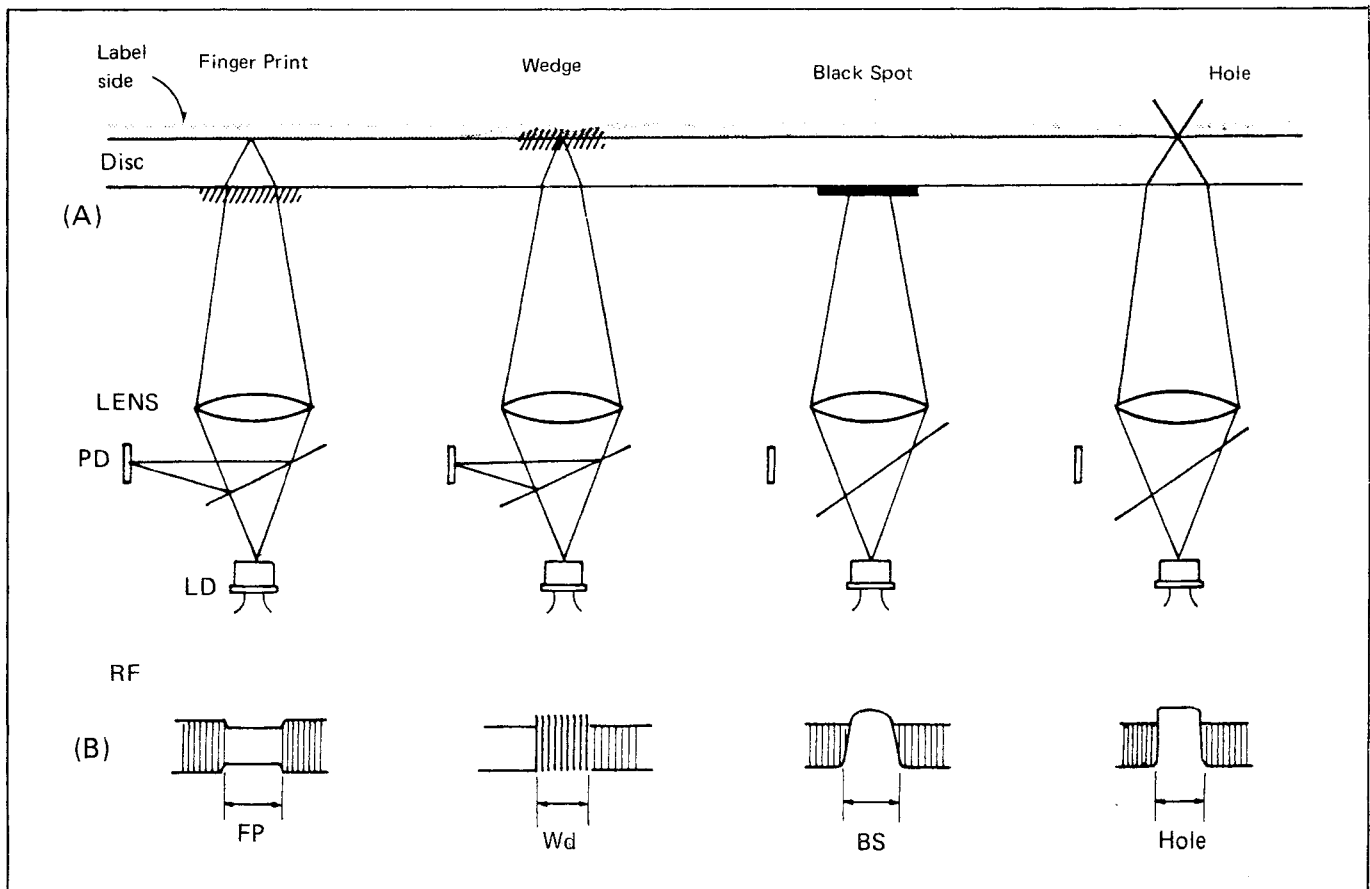


Figure 22-27

Information

We would like to inform you that we have developed new test fixture of **New Servo Gain Adjuster**. This new servo gain adjuster is improved version of former servo gain adjuster in order to adjust CD player quicker and accurately. To use this new servo gain adjuster, we believe CD player can be repaired much easier and repair time will decrease very much.

1. Main Feature

- (1). Employed internal oscillator and power supply:
 - It is not necessary to look for the power line on the main unit for connecting and prepare of external oscillator. The servo gain can be adjusted with just one connection of connector.
- (2). Equipped servo gain judgment circuit:
 - 3 LED indicate the condition of the servo gain and servo can be adjusted by monitoring those LED indication without connecting oscilloscope.
- (3). Can be used any of Technics and Panasonic CD player:
 - Including products beeing introduced in the future, all of CD players are applicable to this new servo gain adjuster (except SL-P10, SL-P7 and SL-P8)
- (4). Others:
 - 8 kind of different frequency are available. and equipped test points for signal waveform monitoring.

2. Name and Part Number

New Servo Gain Adjuster SZZP1094C

We are convinced that new test fixture is very useful and helpful for servicing of CD player and increasing

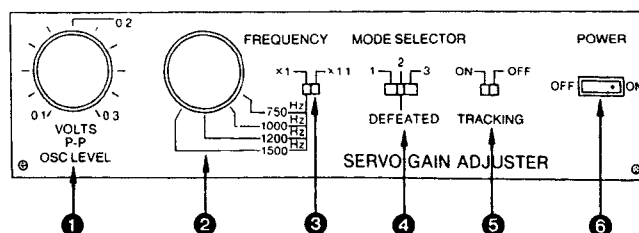


We would like ask you to study to use it in your service network and place an order necessary quantity to us through normal spare parts order route

Furthermore, we would like to supply this new servo gain adjuster in future instead of former servo gain adjuster (SZZP1017F). and it will be discontinued after consumed all of our stock.

3. Controls and Their Functions

Front View



① OSC Level Control

Used to adjust the output level of the built-in low frequency oscillator.

At minimum position (fully-counterclockwise rotated position):

A 0.1 Vp-p level signal is delivered.

At center position:

A 0.2 Vp-p level signal is delivered.

At maximum position (fully-clockwise rotated position):

A 0.3 Vp-p level signal is delivered.

② ③ Frequency Control and Switch x1 and x1.1

Used to select the oscillation frequency.

Frequency	Oscillation x1	Frequency x1.1
750Hz	750Hz	825Hz
1,000Hz	1,000Hz	1,100Hz
1,200Hz	1,200Hz	1,320Hz
1,500Hz	1,500Hz	1,650Hz

④ Mode Selector Switch

Used to set the modes for tracking and focus servo adjustments.

1. Setting for tracking servo adjustment.
2. DEFEATED (through)
3. Setting for focus servo adjustment.

NOTE

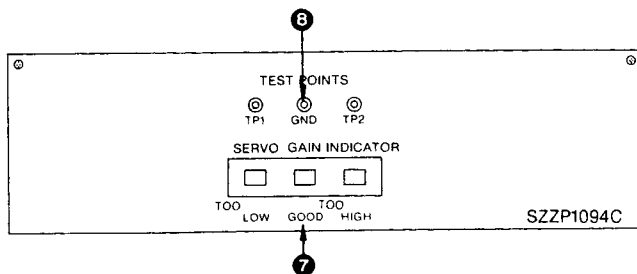
The adjustment settings (1 2 or 3) will depend on the model being used. Please refer to respective service manual.

⑤ Tracking Switch

Used to turn on and off the tracking servo.

⑥ Power Switch

Top View



⑦ Test Points

These are the in/output terminals for monitoring the servo amplifier. (Same T.P.s in the original servo gain adjuster).

⑧ Servo Gain Indicator

Displays the condition of the servo amplifier's gain during adjustment.

LED Status	Condition of the Servo Amplifier's Gain
LOW	Gain is low
GOOD	Gain is correct
HIGH	Gain is high

4. Servo Gain Adjustment

Procedures

1. Connect the 5-pin code supplied with this unit to the 5-pin connector on the rear panel.
2. Disconnect the short-jumper connected to the servo adjustment connector on a CD player and connect the other end of the 5-pin code to the servo adjustment connector.
3. Set the modes selector switch ④ on the unit panel to position "2" (DEFEATED) and turn on the CD player to read the TOC.

Focus Servo Gain Adjustment

1. Adjust the input level for the CD player with the OSC level controller ① on the unit panel. (Refer to the service manual for the input level).
2. Select the frequency of the oscillator with the frequency switches (② and ③). (Refer to service manual for the frequency).
3. Set the mode selector switch ④ on the panel of the unit to position "3" and adjust the semi fixed volume for focus gain control on the CD player so that the LED "GOOD" on the servo gain INDICATOR panel lights up.

NOTE

First adjust the volume for focus gain control so that LOW LED on the servo gain INDICATOR panel lights up and then make further adjustments until the GOOD LED lights up.

Tracking Servo Adjustment

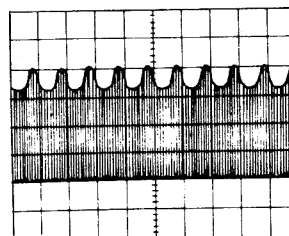
1. Adjust the input level for the CD player with the OSC level controller ① on the unit panel (Refer to the service manual for the input level).
2. Select the frequency of the oscillator with the frequency switches (② and ③) on the unit panel (Refer to the service manual for the frequency).
3. Set the mode selector switch ④ on the unit panel to position "1" (for tracking servo adjustment) and adjust the semi fixed volume for the tracking gain of the CD player until the GOOD LED of the servo gain indicator ⑧ lights up.

NOTE

First adjust the semi fixed volume for tracking gain control so that the LOW LED on the servo gain INDICATOR panel lights up and then make further adjustments until the GOOD LED lights up.

Tracking Balance Adjustment

1. Select the frequency of the oscillator with the frequency switches (② and ③) on the unit panel (Refer to the service manual for the frequency).
2. Adjust the input level for the CD player by adjusting the OSC level controller ① on the panel (Refer to the service manual for the input level).
3. Set the mode selector switch ④ to position "1". The frequency of the RF signal will be output as shows below. Adjust the tracking balance adjustment semi fixed volume so that amplitude X becomes minimum.

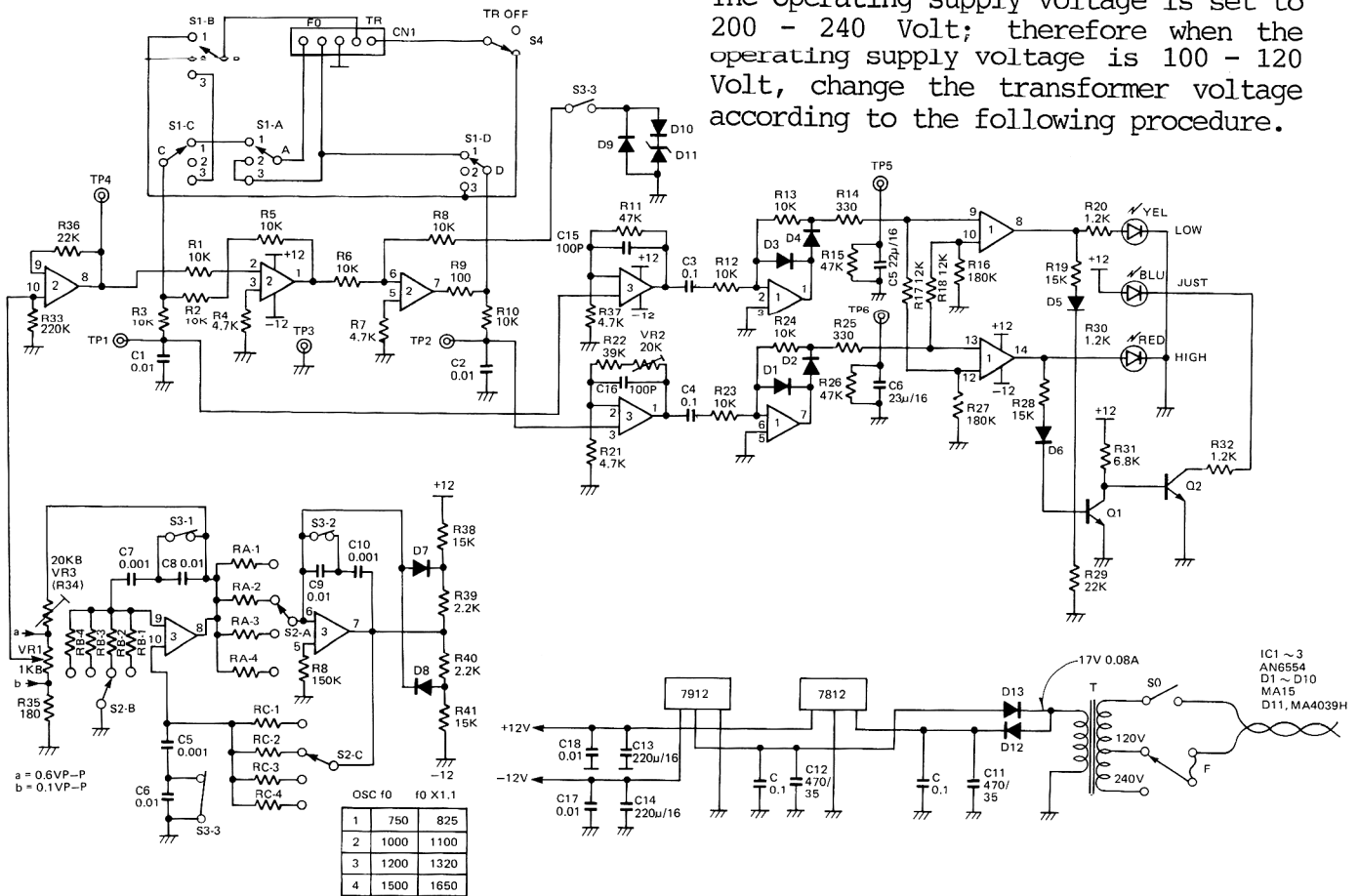


Keep this amplitude minimum.

Schematic Diagram (SZZP1094C)

CAUTION

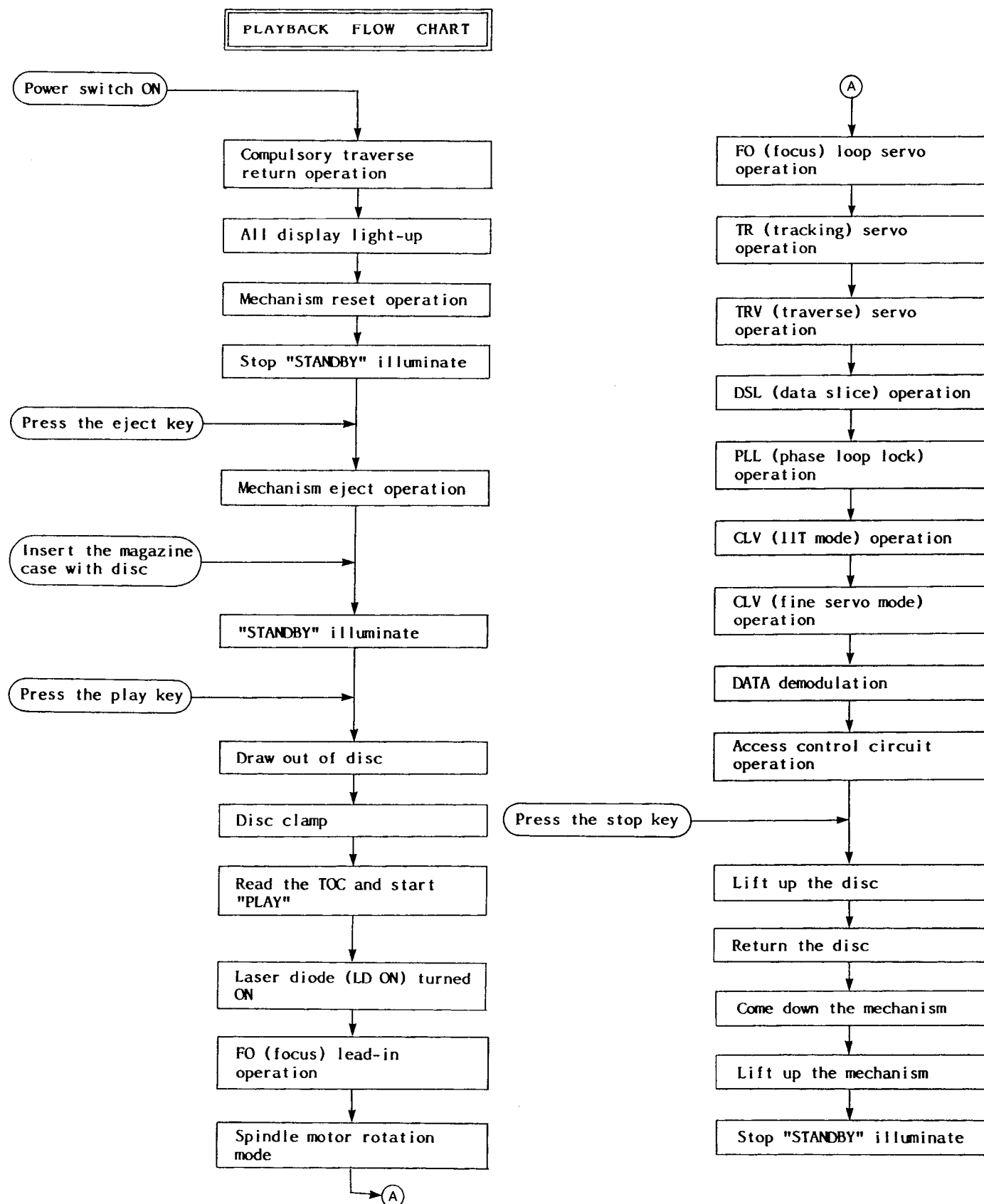
The operating supply voltage is set to 200 - 240 Volt; therefore when the operating supply voltage is 100 - 120 Volt, change the transformer voltage according to the following procedure.

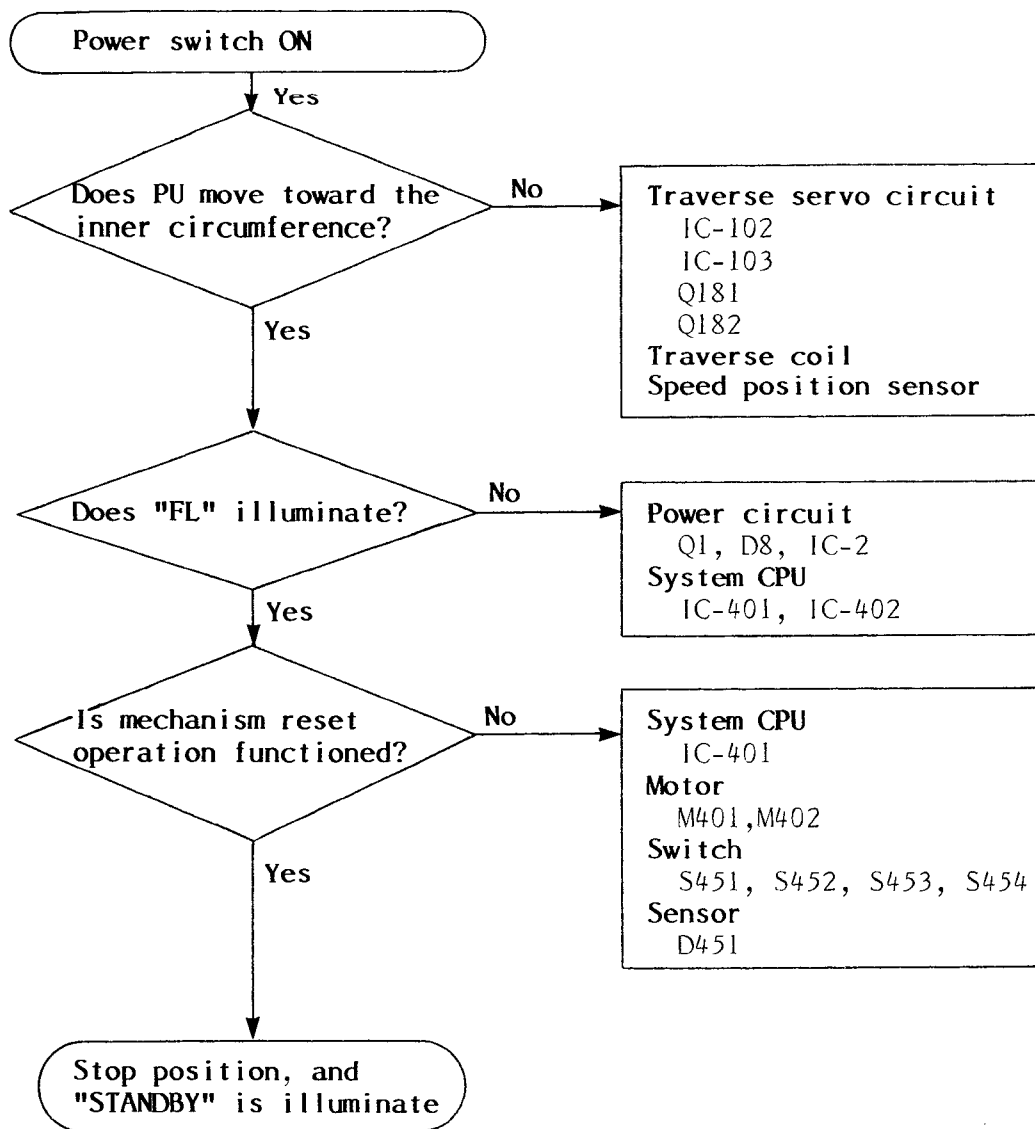


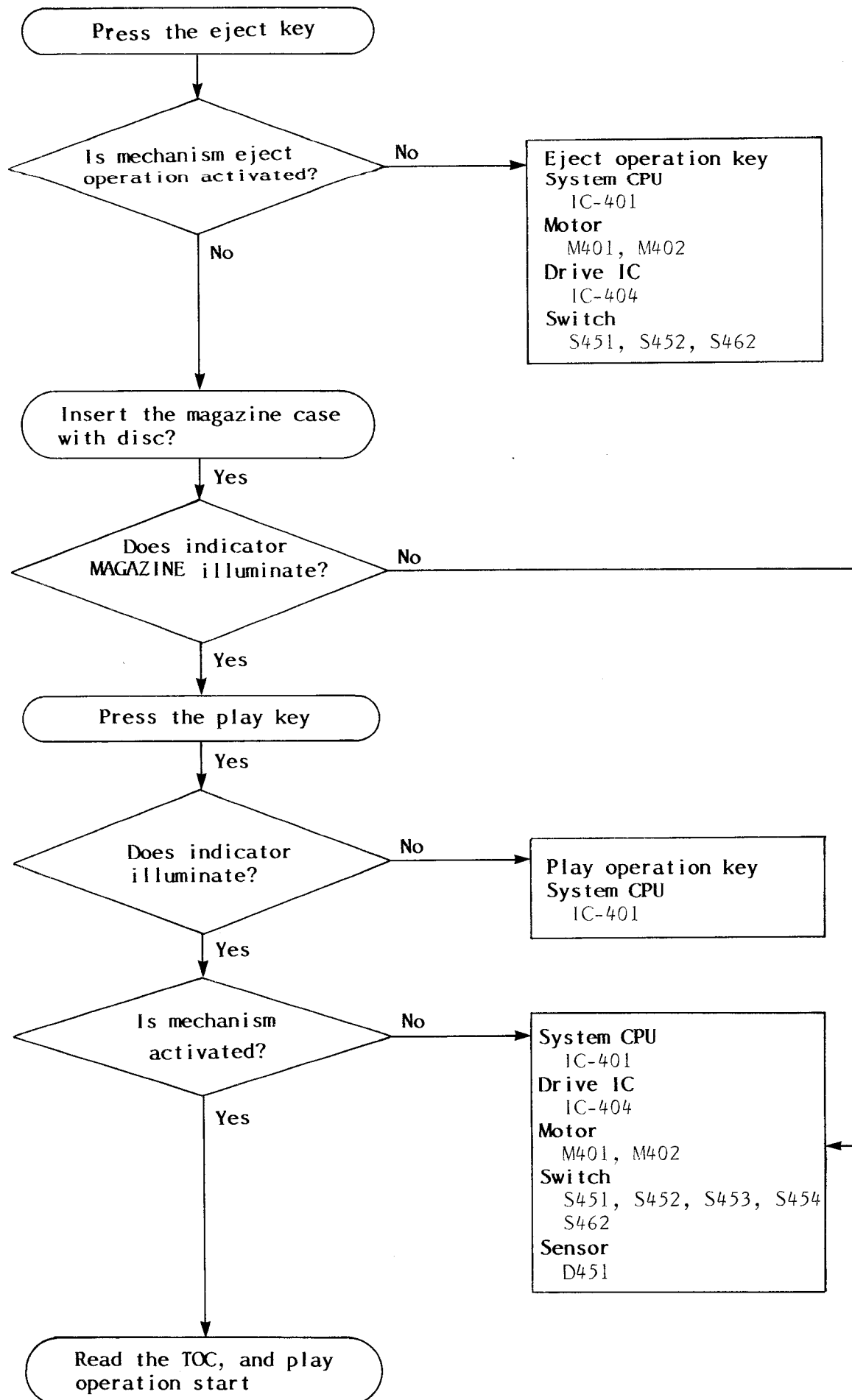
Parts List

Parts No	Description	Q	Ref No	Parts No	Description	Q	Ref No
1SR35-200A	Diode	2	D12, D13	ERD25PJ223	Resistor	2	R36, R29
AN7812	IC	1	7812	ERD25PJ224	Resistor	1	R33
AN7912	IC	1	7912	ERD25PJ335	Resistor	1	
5P	Connector	1	CN1	ERD25PJ393	Resistor	1	R22
2SC1685Q	TR	2	Q102	ERD25PJ472	Resistor	4	R4, R7, R21
AN6554	IC	3	IC1 - IC3	ERD25PJ473	Resistor	3	R11, R15
ECCF1H101J	C Capacitor	2	C15, C16	ERD25PJ682	Resistor	1	R31
ECEA1CS220	Capacitor	2	C5, C6	ERD25PJ331	Resistor	2	R14, R27
ECEA1CS221	Capacitor	2	C13, C14	ERD25CK105	Resistor	2	R14, RB4
ECEA1VS471	Capacitor	2	C11, C12	ER025CK1183	Resistor	1	RC4
ECQM1H103KZ	Polyester	4	C1, C2, C17	ER025CK1433	Resistor	1	RC3
			C18	ER025CK1583	Resistor	2	RA2 RB2
ECQM1H104KZ	Polyester	4	C3, C4, C0,	ER025CK1743	Resistor	1	RC2
ECQP1H102FZ	Polyester	3	C5, C7, C10	ER025CK2103	Resistor	2	RA1 RB1
ECQP1H103FZ	Polyester	3	C6, C7, C8	ER025CK2323	Resistor	1	RC1
ERD25PJ101	Resistor	1	R9	ER025CK1333	Resistor	2	RA3 RB3
ERD25PJ103	Resistor	11	R1 -	EVHGHA20-B13	VR	1	VR1
ERD25PJ122	Resistor	3	R20, R30	EVLTOA24	VR	2	VR2, VR3
ERD25PJ123	Resistor	2	R17, R18	LN252RPH	LED	1	H
ERD25PJ153	Resistor	4	R19-	LN352GPH	LED	1	J
ERD25PJ154	Resistor	1		LN452YPH	LED	1	L
ERD25PJ181	Resistor	1	R35	MA150	Diode	10	D1-D10
ERD25PJ184	Resistor	2	R16, R27	MA4039	Diode	1	D11
ERD25PJ222	Resistor	2	R39, R40				

23. CD Changer Troubleshooting Flow Chart







TOC READ OPERATION - PLAY OPERATION

